Inverse scope in *all…not* constructions: A cross-linguistic and diachronic investigation

Thesis for the degree of "Doctor of Philosophy"
by

Omri Amiraz

THE HEBREW UNIVERSITY OF JERUSALEM

Submitted to the Senate of the Hebrew University of Jerusalem

March 2025

This work was carried out under the supervision of Prof. Nora Boneh.

Acknowledgments

First and foremost, I am indebted to my advisor, Nora Boneh, for her consistent support and guidance. Throughout this work, her advice was essential in shaping the ideas in this dissertation and helping me communicate them clearly. I greatly benefited from her broad knowledge of syntax, which proved invaluable when confronting the challenges of analyzing cross-linguistic and diachronic corpus data. Her clear sense of priorities and steady guidance were crucial in bringing this dissertation to completion. Beyond the academic side, Nora stood by me through the ups and downs of these years, and I am deeply grateful for that as well.

I would also like to thank my PhD committee members—Eitan Grossman, Elitzur Bar-Asher Siegal, and Luka Crnič—for their thoughtful feedback and suggestions, and for many engaging and insightful conversations over the years, as well as other faculty members from the Department of Linguistics, including Malka Rappaport-Hovav and Aynat Rubinstein, for inspiring classes and discussions.

I am especially thankful to the organizers of the Negation Network—Chiara Gianollo, Chloé Tahar, and Elitzur Bar-Asher Siegal—for establishing such a lively forum to exchange ideas and learn about ongoing research on negation. Presenting parts of this work in that setting was invaluable, and I benefited greatly from the constructive feedback I received from both the organizers and other participants. I am particularly grateful to Chiara Gianollo, Hedde Zeijlstra, and Larry Horn for their thoughtful comments and suggestions, which helped sharpen several aspects of the analysis.

I have been fortunate to be part of several centers at the Hebrew University of Jerusalem. The Language, Logic and Cognition Center (LLCC) provided a vibrant environment in which to discuss ideas across the language sciences, take courses from leading visiting scholars, and organize student-focused events with generous support from the center. The Mandel Scholion Research Center and the Mandel School for Advanced Study in the Humanities brought me into sustained dialogue with researchers from across the humanities, broadened my methodological horizons, and offered valuable opportunities to develop professional skills and communicate my work to a wider audience. The Historical Linguistics and Formal Semantics research group, in particular, has played a central role in shaping me as a researcher in these fields.

I am thankful to the administrative and academic staff of these centers, as well as of the Department of Linguistics and the Authority for Research Students, whose guidance and assistance have been essential to my daily academic pursuits. I acknowledge with appreciation the financial support I received during my doctoral studies. Funding from the Language, Logic and Cognition Center, the Mandel Scholion Research Center, the Mandel School for Advanced Study in the Humanities, and the Israel Science Foundation (grant No. 2765/21) provided the resources and stability necessary to carry out this research.

I would also like to thank my colleagues for discussing the ideas developed in this dissertation at various stages and for generously answering my questions about their native languages or languages of expertise. Their feedback and observations were invaluable in shaping both the theoretical and empirical aspects of this work. Special thanks go to Akitaka Yamada, Alice Bondarenko, Angèle Bernard, Anne Mucha, Ashwini Deo, Bar Avineri, Benjamin Spector, Danfeng Wu, David Erschler, Deniz Özyıldız, Despina Oikonomou, Elena Anagnostopoulou, Enrico Flor, Eunsun Jou, Flóra Lili Donáti, Giacomo Bucci, Giorgos Spathas, Ishani Guha, Jakob Lesage, Jonathan Bobaljik, Julia Horvath, Justin Royer, Keny Chatain, Liina Lindström, Lilja Saeboe, Maria Vilkuna, Matti Miestamo, Milica Denić, Mira Ariel, Moysh Bar-Lev, Nikos Angelopoulos, Noa Bassel, Ömer Demirok, Omri Doron, Philippe Schlenker, Rajesh Bhatt, Roni Katzir, Steffen Haurholm-Larsen, Todd Snider, Vincent Rouillard, Yusuke Yagi, and Zahra Mirrazi.

Finally, I dedicate this dissertation with love to my wife, Chen, for being the best partner I could hope for, and to my parents, Gili and Eitay, for their support and devotion.

Abstract

This dissertation investigates the cross-linguistic variation in the use of inverse-scope constructions like *All that glitters is not gold*, where clausal negation takes wide scope over a preceding universal quantifier phrase (QP). These *all…not* constructions contrast with transparent-scope alternatives, such as *Not all that glitters is gold*, which reflect the intended scope relations through surface structure.

To distinguish between form and meaning, this dissertation adopts the following conventions. The term "*all…not* construction" refers to the structural pattern in which a subject universal QP precedes clausal negation within the same clause, regardless of its interpretation in a given language or context. In contrast, 'Not all X are Y' represents the meaning where negation takes wide scope over a universal quantifier [$\neg > \forall$], irrespective of how this meaning is expressed syntactically.

Given that scope transparency facilitates both processing and acquisition, and that syntactic economy principles predict a preference for minimizing mismatches between Logical Form (LF) and Phonetic Form (PF) (Bobaljik 1995; Diesing 1997; Reinhart 2005; Bobaljik and Wurmbrand 2012), a universal bias toward scope transparency is expected.

But is there cross-linguistic evidence that languages actually favor scope transparency? A reasonable expectation would be that if such a bias exists, languages should overwhelmingly prefer transparent-scope constructions and use inverse-scope constructions only as a last resort. However, as this study demonstrates, the cross-linguistic picture is considerably more nuanced: inverse-scope constructions are not only widespread but, in many languages, preferred over their transparent-scope alternatives.

To reconcile the apparent discrepancy between the hypothesized universal bias for scope transparency and the prevalence of inverse-scope constructions, this study explores whether scope transparency operates as a weak universal force (Seržant 2019)—one that systematically influences language change over time but is not strong enough to override competing pressures at the synchronic level.

The central hypothesis of this study is as follows:

(1) *The Scope Transparency Hypothesis*
Languages are shaped by a weak bias toward expressing scope relations transparently.

Given the expected weak effect of the bias for scope transparency, testing the Scope Transparency Hypothesis necessitates a large and diverse language sample. Since inverse-scope

constructions are seldom documented in reference grammars, this study relies on the collection of primary cross-linguistic data. To achieve this, a corpus-based approach is employed, analyzing New Testament translations in 217 languages from diverse language families and linguistic areas. Unlike previous studies, which have primarily investigated scope phenomena through grammaticality judgments, this approach leverages a large parallel corpus to systematically examine the distribution of inverse-scope and transparent-scope constructions across a wide range of languages. This large-scale dataset allows for statistical analyses that control for genealogical and areal biases, ensuring the robustness of the findings.

The dataset was constructed by curating New Testament translations from multiple sources, with particular emphasis on incorporating historical translations, including scanned books and manuscripts. Historical texts were available for 21 Germanic and Romance languages, offering a valuable resource for tracing diachronic developments in the interaction between negation and universal quantifiers.

The data analysis classifies constructions based on their observable syntactic properties, distinguishing five structural strategies for expressing 'Not all X are Y' propositions. To ensure that the identified constructions genuinely reflect native grammatical patterns rather than translation artifacts, the analysis prioritizes instances where translators deviate from the source text and cross-validates these findings by examining multiple independent translations into a given language.

To evaluate the Scope Transparency Hypothesis, this study formulates two more specific, empirically testable hypotheses, outlined in (2) and (3). The Synchronic Hypothesis (2) posits that although inverse-scope constructions are widely attested, their distribution is partially suppressed by the bias for scope transparency. As a result, they occur less frequently across languages than would be expected under the null hypothesis of no such bias.

Testing this prediction requires establishing a baseline against which the observed frequency of inverse-scope constructions across languages can be evaluated. This, in turn, requires an understanding of why languages employ inverse-scope constructions in the first place. I argue that their cross-linguistic prevalence stems primarily from the fact that, in the vast majority of languages, using the standard negation construction (the basic construction a language employs to negate verbal clauses, see Miestamo 2005) results in inverse scope. To avoid inverse scope, most languages need to employ a different negative construction or to allow alternative word order patterns that make the scope relations transparent.

This reasoning leads to the following prediction:

(2) *Synchronic Hypothesis*
Inverse-scope constructions occur less frequently cross-linguistically than would be expected based on the relative positions of subjects and clausal negation markers in different languages.

Under the null hypothesis of no bias for scope transparency, 'Not all X are Y' should pattern similarly to other negative sentences where the subject is not a scope-bearing element. That is, if a language's standard negation construction produces a particular word order in ordinary negative clauses—such as *The cookies didn't burn*—we would expect the same construction to be used when the subject is a universal QP, yielding the inverse-scope construction *All the cookies didn't burn.* By contrast, if there is a bias for scope transparency, languages should exhibit a tendency to diverge from the standard negation pattern and employ alternative strategies—such as constituent negation, as in *Not all the cookies burned*—to transparently reflect scope relations in the surface structure.

The findings of this study indicate that many languages systematically avoid inverse scope at the cost of deviating from the usual structural patterns, lending support to the Synchronic Hypothesis. Some languages develop constituent negation constructions specifically aimed at modifying subject QPs. Other languages repurpose existing constructions, typically those whose primary function relates to information structure.

The quantitative results provide partial support for the Synchronic Hypothesis. To control for genealogical and areal influences, the analysis was carried out separately for each macro-area, while also ensuring genealogical independence. In Africa and Eurasia, inverse-scope constructions are significantly less frequent than expected based on standard negation patterns, suggesting that a bias for scope transparency is shaping the observed distribution. However, no significant effect is found for North America, South America, or Papunesia, likely due to sample size limitations. This underscores that while scope transparency plays a role in shaping language structure, its influence is relatively weak.

The study also formulates the following Diachronic Hypothesis:

(3) *Diachronic Hypothesis*

(i) In a given language, the relative frequency of transparent-scope constructions tends to increase over time.

(ii) Conversely, inverse-scope constructions become more frequent only when the dominant transparent-scope alternative is lost due to broader grammatical changes, prompting the language to rely on the standard negation construction.

The Diachronic Hypothesis finds strong support in historical data from Germanic and Romance languages. The study identifies a recurrent cycle in which inverse-scope constructions decline as transparent-scope alternatives emerge:

(4) *Pathway of decline of inverse-scope constructions*

Stage 1 Inverse-scope constructions are frequently employed, possibly even the dominant strategy for expressing 'Not all X are Y' propositions.

Stage 2 The language innovates a transparent-scope strategy. Over time, the frequency of the novel construction increases, while the frequency of the older inverse-scope construction decreases.

Stage 3 The inverse-scope construction either falls out of use or becomes rare and mostly restricted to formulaic expressions.

Conversely, inverse-scope constructions (re-)emerge primarily when the dominant transparent-scope strategy is lost due to grammatical changes that extend beyond the domain of 'Not all X are Y' sentences, such as shifts in word order or the grammaticalization of a novel universal quantifier:

(5) *Pathway of emergence of inverse-scope constructions*

Stage 1 Inverse-scope constructions are rare or not in use.

Stage 2 The dominant transparent-scope strategy is lost due to broader grammatical changes.

Stage 3 The language resorts to using the standard negation construction, often resulting in inverse scope.

A key finding of this research is that the cross-linguistic variation in the availability of inverse scope is not entirely predictable based on a language's grammatical properties or its inventory of transparent-scope strategies. Instead, each language exists in a synchronic state reflecting its unique diachronic trajectory. Crucially, the timing of grammatical shifts plays a decisive role in shaping synchronic distributions through persistence effects, whereby older constructions remain in use even after new alternatives emerge.

These findings contribute to our understanding of how soft (violable) syntactic constraints influence language structure. While persistence effects may obscure the synchronic impact of these constraints, their influence becomes evident in cross-linguistic distributional asymmetries and recurrent diachronic pathways. By integrating both cross-linguistic and diachronic perspectives, this study demonstrates that universal biases can shape long-term language evolution without necessarily producing language-specific observable outcomes.

More broadly, this study sheds light on the nature of diachronic competition in the syntax-semantics interface, showing that it can occur even between complex constructions that differ substantially in their syntactic structure. While most constraint-based models of synchronic competition assume that the candidate set is restricted to PFs derived from the same LF, the evidence presented here indicates that diachronic competition does not follow the same restriction. Instead, constructions expressing the same meaning can block each other over time, even when they correspond to distinct LFs.

Overall, this dissertation provides empirical evidence that scope transparency functions as a weak universal force, subtly shaping both synchronic distributions and diachronic change, though its effects are counteracted by persistence effects and cycles of re-emergence.

The dissertation is structured into five parts. Part I establishes the theoretical foundations and outlines the main hypotheses (Chapters 1–3). Part II introduces the methodological approach, which employs a parallel corpus of New Testament translations, and evaluates its reliability (Chapters 4–6). Part III addresses the Synchronic Hypothesis by identifying and quantifying the distribution of syntactic strategies across languages, while controlling for genealogical and areal biases, and examining pathways toward scope transparency (Chapters 7–9). Part IV investigates the Diachronic Hypothesis by tracing historical changes in Germanic and Romance languages, revealing recurrent pathways in the evolution of inverse-scope and transparent-scope strategies (Chapters 10–11). Finally, Part V synthesizes these findings, underscoring the importance of integrating cross-linguistic and diachronic perspectives when investigating universal grammatical biases (Chapter 12).

Contents

List of Tables

List of Figures

List of glossing abbreviations

1	first person
2	second person
3	third person
ABL	ablative
ACC	accusative
ADE	adessive
AFF	affirmative
ALL	allative
AN	animate
AOR	aorist
ART	article
ASP	aspect
ASS	assertive
AT	attributive
AUG	augment
AUX	auxiliary
CLF	classifier
CNG	connegative
CNTR	contrastive
COMP	complementizer
COMPL	completive
COND	conditional
CONN	connective
COP	copula
DAT	dative
DECL	declarative
DEF	definite
DET	determiner
DIR	direct
ELA	elative
EMPH	emphatic
ERG	ergative

EX	existential
EXCL	exclusive
EXPL	expletive
F	feminine
FOC	focus
FUT	future
FV	final vowel
GEN	genitive
HYP	hypothetical
ILL	illative
IMP	imperative
IMPERF	imperfect
IMPERS	impersonal
INAN	inanimate
INF	infinitive
LOC	locative
M	masculine
MID	middle voice
N	neuter
NEG	negation
NMLZ	nominalizer
NOM	nominative
OBJ	object
OBL	oblique
OBV	obviative
PART	partitive
PASS	passive
PERF	perfect
PFV	perfective
PL	plural
POSS	possessive
PROG	progressive
PROH	prohibitive
PRS	present
PRT	particle
PST	past
PTCP	participle
QUOT	quotative
REFL	reflexive
REL	relative
SBJ	subject
SBJV	subjunctive

SG	singular
SS	same subject
SUP	supine
TRANSL	translative
VBLZ	verbalizer
VOC	vocative

Part I

Introduction

Chapter 1

Overview

1.1 The Scope Transparency Hypothesis

This dissertation investigates the cross-linguistic variation in the use of inverse-scope constructions like (6). These constructions exhibit a form-meaning mismatch: although the negative marker *not* follows the universal quantifier *all* in the surface word order, negation takes wide scope over the quantifier [$\neg > \forall$].[1]

(6) All that glitters is not gold.
≈ It is not the case that all that glitters is gold.

[1] In this study, I assume that natural language clausal negation corresponds to the propositional operator of negation in classical logic, which applies to full propositions. As an anonymous reviewer noted, under alternative analyses of negation, sentences like (6) might not be considered as instances of a form-meaning mismatch or appear "illogical". For example, in Aristotelian term logic, wide-scope negation is analyzed as predicate denial, in which the predicate is denied of the subject. In this framework, wide-scope negation is a mode of predication—an operation on subjects and predicates, contrasting with affirmation—rather than an operation on propositions. Horn (1989, 227) argues that this is the view implicitly assumed by Tobler (1902) in his discussion of sentences like (6): the predicate *is gold* is denied of the subject *all that glitters*. In other words, the property of being gold does not hold of all that glitters, yielding the appropriate interpretation. Crucially, this interpretation parallels predicate denial in cases where the subject is not a quantifier, as in:

(i) This glittering object is not gold.

However, the challenge posed by inverse-scope constructions does not disappear under the Aristotelian analysis of wide-scope negation as predicate denial: one still needs to explain why wide-scope negation appears to be sensitive to the type of subject quantifier and why there is cross-linguistic variation in the availability of inverse scope (as discussed below in this chapter). For example, in English, clausal negation typically does not take wide scope over a preceding subject headed by *many*: *Many things that glitter aren't gold* cannot mean 'Not many things that glitter are gold' (Jackendoff 1972). The broader issue of when and why clausal negation can take wide scope over a preceding subject quantifier thus remains a challenge for any theory of negation. For further discussion of non-propositional and ambiguist approaches to negation, see Horn (1989) and Bar-Asher Siegal (2025), among others.

To distinguish between form and meaning, this dissertation adopts the following conventions. The term "*all…not* construction" refers to the structural pattern in which a subject universal quantifier phrase (QP) precedes clausal negation within the same clause, regardless of its interpretation in a given language or context. In contrast, 'Not all X are Y' represents the meaning where negation takes wide scope over a universal quantifier [$\neg > \forall$], irrespective of how this meaning is expressed syntactically.[2]

All…not constructions like (6) constitute one of several structural strategies that languages employ for expressing 'Not all X are Y' propositions. In English, and in many other languages, such propositions are usually conveyed by constructions that transparently reflect the scope relations in the surface word order, including, but not limited to, negated QP constructions like (7).

(7) Not all that glitters is gold.

Do languages tend to favor transparent-scope constructions over inverse-scope constructions? There are several reasons to think they might.

First, scope transparency likely facilitates both language acquisition and sentence processing (see §3.2.1, §3.2.2). Inverse-scope constructions are assumed to involve covert syntactic movement, which must be inferred based on contextual cues or language-specific restrictions on scope possibilities. Transparent-scope constructions like (7), by contrast, allow their meaning to be read off the surface structure.

Second, from a theoretical perspective, economy principles in syntax predict a preference for transparent-scope constructions because they minimize mismatches between Logical Form (LF) and Phonetic Form (PF) (Bobaljik 1995; Diesing 1997; Reinhart 2005; Bobaljik and Wurmbrand 2012, among others).

Finally, *all…not* constructions are ambiguous in some languages between a transparent-scope and an inverse-scope interpretation, as demonstrated in (8).[3] If the intended meaning is the one where negation takes wide scope [$\neg > \forall$], a transparent-scope construction like (9) provides an unambiguous way to convey this meaning, which could be seen as another factor in favor of such constructions.

[2]Propositional meaning refers to the abstract semantic content of an utterance. While Logical Form (LF) represents an abstract level of syntactic structure where scope relations are explicitly encoded, it is still language-specific and closely tied to formal operations such as quantifier raising or covert movement (May 1985). By contrast, propositional meaning abstracts away from these formal representations, capturing the core truth-conditional content of a sentence in a way that is independent of any particular syntactic derivation. Importantly, two syntactic constructions can express the same proposition, even if their LF representations differ (see §3.1.1 for further discussion).

[3]Transparent scope is more commonly known as surface scope. However, I use the former term to reflect the notion of scope transparency, which is central to this dissertation. Another terminology, introduced by Carden (1970), distinguishes between NEG-Q [$\neg > \forall$] and NEG-V [$\forall > \neg$] readings, based on whether negation scopes over the quantifier (NEG-Q) or is limited to the verb phrase (NEG-V). Alternatively, the [$\forall > \neg$] reading is sometimes referred to as total negation, while the [$\neg > \forall$] reading is termed partial negation, since in the latter case, the predicate is denied only of a subset of the subject referents.

(8) All prices do not include shipping.
Transparent-scope reading: None of the prices include shipping. [$\forall > \neg$]
Inverse-scope reading: It is not the case that all of the prices include shipping. [$\neg > \forall$]

(9) Not all prices include shipping. [$\neg > \forall$]

But is there cross-linguistic evidence that languages actually favor scope transparency? A reasonable expectation would be that if such a bias exists, languages should overwhelmingly prefer transparent-scope constructions and use inverse-scope constructions only as a last resort. However, as this study will demonstrate, the cross-linguistic picture is considerably more nuanced: inverse-scope constructions are not only widespread but, in many languages, preferred over their transparent-scope alternatives.

On the one hand, some languages, such as present-day English, clearly favor transparent-scope constructions like (9) over inverse-scope constructions like (8) when expressing 'Not all X are Y' propositions. In these languages, inverse-scope constructions tend to be pragmatically marked and contextually restricted. Other languages, such as Hebrew and Mandarin, exhibit an even stronger preference toward scope transparency, lacking inverse-scope constructions altogether.

On the other hand, some languages exhibit the reverse pattern, favoring inverse-scope constructions over transparent-scope ones. In Swedish, for example, the inverse-scope construction in (10a) is not only more frequent than the transparent-scope construction in (10b) but is also preferred as a neutral statement (see §6.2.1). Finally, languages such as Turkish (Özyıldız 2017) and Hindi (Mahajan 2017) lack transparent-scope constructions like (9) altogether and rely almost exclusively on *all…not* constructions.

(10) Swedish (Alice Bondarenko, p.c.)

a. Alla barn gillade inte filmen.
all.PL children like.PST NEG movie.DEF.SG
'Not all the children liked the movie.'

b. Inte alla barn gillade filmen.
NEG all.PL children like.PST movie.DEF.SG
'Not all the children liked the movie.'

Notably, the issue is not merely that some languages deviate from the expected trend—rather, the findings of this dissertation show that inverse-scope constructions are, in fact, more widespread than any individual transparent-scope alternative. This raises a crucial question: if there is indeed a universal bias for scope transparency, why doesn't it manifest as a strong typological preference?

The answer lies in the nature of weak universal forces, as described by Seržant (2019). Unlike strong universal constraints, which create clear and widespread typological patterns, weak universal forces systematically influence language change over time but are not strong enough to override all other factors. As a result, their influence is often obscured at the synchronic level due to interactions with other grammatical constraints and inheritance effects from earlier stages of the language.

This perspective aligns with the account proposed by Bobaljik and Wurmbrand (2012), which treats scope transparency as a soft (violable) constraint—one that may be overridden by other grammatical constraints. However, framing scope transparency as a weak universal tendency shifts the focus from synchronic competition, where scope interpretations are determined by the interaction of syntactic constraints, to diachronic competition, where the effects of scope transparency emerge gradually over time.

This dissertation investigates whether scope transparency indeed operates as a weak universal force. If so, we might expect to find subtle synchronic effects, but likely not a strongly skewed cross-linguistic distribution favoring transparent-scope constructions. Instead, the influence of the hypothesized bias should be more readily observable in diachronic developments, where languages gradually shift away from inverse-scope constructions and toward transparent-scope constructions over time. Crucially, this bias does not necessarily result in a synchronic preference for transparent-scope constructions in every language. This hypothesis is articulated in (11).

(11) *The Scope Transparency Hypothesis*
Languages are shaped by a weak bias toward expressing scope relations transparently.

1.2 Domain of inquiry

The domain of inquiry is restricted to inverse scope in *all…not* constructions, including those with *every* and universal pronouns like *everyone*, as well as their counterparts in other languages.[4] Representative examples from the Corpus of Historical American English (COHA, Davies 2010) are given in (12).

(12) a. All politicians are not opposed to reform. [COHA: Newspaper; 1876]
b. Every thinker is not a philosopher. [COHA: Non-fiction; 1983]
c. Everybody is not willing to take a joke, you know. [COHA: Fiction; 1908]

All…not constructions are of particular interest in the context of the Scope Transparency Hypothesis (11) for two main reasons. First, these constructions represent a particularly challenging testing ground for the hypothesis, as they are widely attested cross-linguistically as a strategy for expressing 'Not all X are Y' propositions (as this study will demonstrate). Furthermore, in many languages, the inverse-scope interpretation is the preferred or even the only available reading of these constructions. If there is indeed a universal bias for scope transparency, it should manifest itself even in this domain, where, given the cross-linguistic picture, the null hypothesis of no bias seems entirely plausible.

Second, 'Not all X are Y' sentences constitute a well-defined domain for cross-linguistic comparison, since, presumably, all (or nearly all) languages possess the relevant building

[4] For clarity, *all* and *every* in italics refer specifically to the English lexemes, whereas 'all' and 'every' in quotation marks represent the cross-linguistic comparative concepts instantiated by these universal quantifiers.

blocks: negation (Miestamo 2005) and universal quantifiers (Paperno and Keenan 2017; Key and Comrie 2023). This is not always the case in other areas of grammar, as most grammatical categories are not universal in the sense of overt marking (Haspelmath 2021).

1.3 Research goals

Since the influence of scope transparency is likely obscured by competing factors at the synchronic level, detecting its effect requires a more fine-tuned approach. To operationalize the Scope Transparency Hypothesis (11), this study formulates two testable cross-linguistic hypotheses: one concerning the synchronic distribution of inverse-scope constructions, and the other addressing the diachronic processes underlying this distribution.

Synchronic predictions To investigate the synchronic dimension, this study examines whether inverse-scope constructions are less frequent cross-linguistically than they would be if there were no bias for scope transparency. Thus, while these constructions are widespread, their distribution would be even broader in the absence of such a bias.

To test this hypothesis, it is necessary to establish a baseline representing what we would expect under the null hypothesis of no bias, providing a point of reference for evaluating the observed frequency of inverse-scope constructions across languages. This baseline is derived from the following premise:

(13) *Premise*
If languages lacked a bias for scope transparency, they would typically use the standard negation construction in 'Not all X are Y' sentences, even if this results in inverse scope.

This idea relates to the question of why languages use inverse-scope constructions in the first place. I argue that their cross-linguistic prevalence stems primarily from the fact that, in the vast majority of languages, using the standard negation construction (the basic construction a language employs to negate verbal clauses) results in inverse scope. To avoid inverse scope, most languages need to employ a different negative construction or to allow alternative word order patterns that make the scope relations transparent.

For example, in English, standard negation is expressed by a construction where the negative particle *not* follows an auxiliary verb, as shown in (14). If the same negative construction were used to negate a sentence with a universal QP in subject position, the result would be an inverse-scope construction, as shown in (15). Avoiding inverse scope requires using a different negative construction, such as a negated QP construction, as in (16a). Crucially, this constituent negation construction is specifically used in situations where the subject is a QP, and it cannot be employed as a strategy for negating regular clauses, as shown in (16b).

(14) a. The kids liked the movie.
b. The kids didn't like the movie.

(15) a. All the kids liked the movie.
b. All the kids didn't like the movie.

(16) a. Not all the kids liked the movie.
b. *Not the kids liked the movie.

The tendency of subjects to precede the negative marker in standard negation characterizes most of the world's languages. According to Dryer (2013b), the most common patterns among SVO languages are SNegVO and SVONeg, while NegSVO is relatively rare. Similarly, among SOV languages, the dominant patterns are SONegV and SOVNeg, both of which are significantly more frequent than NegSOV.[5] The exception to this pattern is found in verb-initial languages, where the most common orders are NegVSO and NegVOS, meaning that negation precedes the subject rather than following it.

Given that verb-medial and verb-final languages are far more common than verb-initial ones (Dryer 2013a), it follows that in the majority of the world's languages, the subject typically appears before the negative marker in the basic word order. This general tendency toward subject-before-negation order has important consequences for scope transparency: most languages have to go the extra mile to avoid inverse scope in 'Not all X are Y' sentences by deviating from the usual word order.

We can thus establish a baseline to which we can compare the observed frequency of inverse-scope constructions. Under the null hypothesis of no bias for scope transparency, languages have no clear motivation to avoid inverse scope, although transparent-scope constructions might still arise for independent reasons—for instance, in languages where the negative marker precedes the subject in standard negation. Consequently, we would expect 'Not all X are Y' sentences to behave similarly to other negative sentences where the subject is not a scope-bearing element, such as a definite DP, as in (14). More precisely, we predict no systematic association between the type of subject (universal QP vs. definite DP) and the position of the negative marker relative to the subject.

While this null hypothesis might seem like a straw man, I contend that it is not. First, many languages often do treat universal QPs on a par with other subjects. This was demonstrated in Swedish, which typically employs the standard negation construction even when it results in inverse scope, as in (10a).

The second argument comes from a comparison between subject QPs and object QPs in negative clauses. Consider, for instance, SVO languages that position the negative marker before the verb (SNegVO). Unlike subjects, objects usually follow the negative marker in these structures. Consequently, using the standard negation construction does not lead to inverse scope when the object is a universal QP, as negation naturally takes wide scope over

[5]Dryer (2013b) considers several possible explanations for these tendencies. One observation is that in both SVO and SOV languages, the most common patterns involve placing the negative marker either directly before the verb (SNegVO and SONegV) or at the end of the clause (SVONeg and SOVNeg). However, while this describes the observed distribution, Dryer notes that it does not fully explain why these particular placements are preferred over others.

the object.[6] This can be seen in (17), where the presence of the quantifier does not introduce a motivation to deviate from the standard negation pattern, since the scope relations remain transparent.

(17) a. I ate (all) the cookies.
b. I didn't eat (all) the cookies.

To the best of my knowledge, languages usually do not treat universal QPs differently from other types of objects in negative clauses like (17b), unless inverse scope is involved. This suggests that there is no general constraint against the co-occurrence of QPs and standard negation.

Therefore, under the null hypothesis, we would expect subject QPs to behave similarly: just as object QPs in SNegVO languages pattern with regular DPs, subject QPs should likewise pattern with other subjects, leading to a preference for *all...not* constructions, as these involve standard negation and the usual word order in these languages. Thus, any systematic divergence from this pattern is likely driven by a bias for scope transparency rather independent grammatical properties of negation or universal quantifiers.

If instead a bias for scope transparency does exist, we should find cross-linguistic evidence that inverse-scope constructions are less frequent than expected given the relative positions of subjects and clausal negation markers. This leads to the following alternative hypothesis:

(18) *Synchronic Hypothesis*
Inverse-scope constructions occur less frequently cross-linguistically than would be expected based on the relative positions of subjects and clausal negation markers in different languages.

Diachronic predictions Even if the results support the Synchronic Hypothesis (which they largely do), this alone does not prove the existence of a bias for scope transparency. Given that inverse-scope constructions are clearly not rare in absolute terms, the argument hinges on the difference between their observed and expected frequency—the latter being derived from the premise in (13). But what if this premise is flawed? Without this assumption, the argument collapses.

To validate the premise in (13), we need to examine how languages actually come to develop inverse-scope constructions. If languages freely adopted or lost inverse-scope con-

[6] It remains an open question whether some SVONeg languages tend to depart from the standard negation construction when the object is a QP in order to avoid inverse scope. Notice, however, that while both subjects and objects precede the negative marker in these languages, the subject is typically positioned further away from negation, which could reinforce a dispreference for inverse scope. A similar pattern holds in verb-final languages, where the object is closer to the negative marker in both SONegV and SOVNeg orders. Some verb-final languages exhibit a gradient effect, where inverse scope becomes increasingly dispreferred as the distance between the QP and negation increases (see §2.4). However, the relevant measure of distance—whether in terms of intervening words, syntactic constituents, or prosodic domains—remains uncertain. This suggests a potential general effect of syntactic distance on the availability of inverse-scope readings, though further research is needed to determine whether this reflects a broader cross-linguistic tendency.

structions due to language contact, their cross-linguistic frequency could not be meaningfully compared to the baseline assumed in (18). This is because the Synchronic Hypothesis presupposes that inverse-scope constructions exist specifically due to their dependence on standard negation, rather than arising independently. Conversely, if languages develop inverse-scope constructions only when grammatical constraints force them to rely on standard negation, resulting in an *all…not* construction, this would bolster the argument that a bias for scope transparency is suppressing their cross-linguistic frequency.

This reasoning leads us to the following hypothesis:

(19) *Diachronic Hypothesis*

(i) In a given language, the relative frequency of transparent-scope constructions tends to increase over time.

(ii) Conversely, inverse-scope constructions become more frequent only when the dominant transparent-scope alternative is lost due to broader grammatical changes, prompting the language to rely on the standard negation construction.

The null hypothesis here is that languages adopt or lose inverse-scope constructions in the same way that transparent-scope constructions fall in or out of use. These changes might be due to language-internal factors or language contact, but there is no "invisible hand" pushing languages into one direction or the other.

What I propose instead is that there is an inherent asymmetry in how inverse-scope and transparent-scope constructions come to be used by a language. To illustrate, we can compare it to a snakes and ladders game: each square represents a possible synchronic balance between inverse-scope and transparent-scope constructions. The bottom square stands for a state where a language relies solely on inverse scope, while the top square represents one where only transparent-scope constructions are employed—though the game, metaphorically, never truly ends.

A bias for scope transparency gradually nudges the language toward the top square. Sometimes, this movement is slow, but other times it is fairly rapid, resembling the quick "climb" of a ladder. However, at any moment, the language may lose its dominant transparent-scope strategy due to various grammatical changes, causing it to "slide down a snake" and increase the relative frequency of inverse-scope constructions. Crucially, there is no opposite force driving languages toward the bottom square. That transition can still happen, but only as a byproduct of other, independently motivated changes.

This analogy implies that languages are not necessarily more likely to gain scope transparency over time than to lose it, as there is no a priori reason to assume that one type of transition occurs more often than the other. The key point, however, is that the causal factors driving each type of change are not symmetrical: scope transparency is reinforced by an ongoing bias, whereas changes that undermine it typically arise incidentally from unrelated grammatical developments.

The Diachronic Hypothesis therefore predicts a cyclic pathway of decline and re-emergence of inverse-scope constructions. The decline of an inverse-scope construction typically follows the innovation of a transparent-scope alternative rather than occurring as a direct consequence of broader grammatical shifts, as outlined in (20).

(20) *Pathway of decline of inverse-scope constructions*

Stage 1 Inverse-scope constructions are frequently employed, possibly even the dominant strategy for expressing ‘Not all X are Y’ propositions.

Stage 2 The language innovates a transparent-scope strategy. Over time, the frequency of the novel construction increases, while the frequency of the older inverse-scope construction decreases.

Stage 3 The inverse-scope construction either falls out of use or becomes rare and mostly restricted to formulaic expressions.

Importantly, changes can also occur in the opposite direction, as outlined in (21). However, while inverse-scope constructions decline due to the introduction and gradual adoption of transparent-scope strategies, the (re-)emergence of inverse-scope constructions tends to be triggered by the loss of a transparent-scope alternative. This loss is not due to changes specific to ‘Not all X are Y’ sentences but instead results from changes in other grammatical domains, such as shifts in word order or the grammaticalization of a novel universal quantifier.

(21) *Pathway of emergence of inverse-scope constructions*

Stage 1 Inverse-scope constructions are rare or not in use.

Stage 2 The dominant transparent-scope strategy is lost due to broader grammatical changes.

Stage 3 The language resorts to using the standard negation construction, often resulting in inverse scope.

1.4 Methodology

Drawing reliable conclusions about the cross-linguistic frequencies of inverse-scope and transparent-scope constructions requires a large, representative sample of languages, especially given the potential impact of both genealogical and areal influences.

Most of the data used in this study comes from corpora, with only a small portion drawn from secondary sources. There are two main reasons for relying on corpora. First, inverse-scope constructions are seldom described in grammars, even for otherwise well-documented languages, making primary cross-linguistic data essential. Second, corpora enable quantitative analyses of the relative frequencies of inverse-scope and transparent-scope constructions—information typically unavailable in secondary sources.

The principal data source for this study is a parallel corpus of New Testament translations into 217 languages. The language sample is essentially a convenience sample, constrained by the availability of suitable data and the feasibility of conducting a syntactic analysis of these data using grammars and dictionaries. A breakdown of the language families and their distribution across major macro-areas appears in Table 1.1.[7]

Macro-area	Language families	Languages
Africa	13	65
Australia	1	2
Eurasia	15	66
North America	13	29
Papunesia	9	32
South America	17	23
Total	**60**	**217**

Table 1.1: Overview of the areal and genealogical coverage of the language sample

Notably, historical translations were also accessible for 21 languages, offering a rich dataset for examining diachronic changes in the interaction between negation and universal quantifiers.

The data analysis is based on the observable structural properties of the relevant corpus examples. I distinguish five strategies for conveying propositions of the form ‘Not all X are Y’, schematically outlined in Table 1.2. Further subdivisions within each strategy, along with detailed examples, are provided in Chapter 7.

Strategy	Schematic example
A	*All the cookies didn't burn.*
B	*Not all the cookies burned.*
C	*Not burned all the cookies.*
D	*It's not all the cookies that burned.*
	It's not that all the cookies burned.
E	*The cookies did not all burn.*

Table 1.2: A schematic representation of five strategies for expressing ‘Not all X are Y’ propositions

Strategy A (*All the cookies didn't burn*) is an inverse-scope construction, whereas the remaining strategies employ different methods to maintain scope transparency. In Strategy B (*Not all the cookies burned*), the negative marker precedes the QP, often—though not always—through constituent negation. Strategy C (*Not burned all the cookies*) uses clausal negation but places the negated verb or auxiliary before the subject QP. Strategy D involves complex negative constructions, such as negated clefts (*It's not all the cookies that burned*) and other pragmatically marked forms of negation (*It's not that all the cookies burned*).

[7] The total number of language families does not match the aggregate from each macro-area due to language families that span multiple macro-areas.

Finally, in Strategy E (*The cookies did not all burn*), the quantifier appears in a lower position, with another constituent occupying the canonical subject position.

Although standard negation usually corresponds to Strategy A (*(All) the cookies didn't burn*), there are also languages where it corresponds to Strategies B (*Not (all) the cookies burned*) or C (*Not burned (all) the cookies*), or, less often, D (*It's not that (all) the cookies burned*). Strategy E stands apart in this respect, as it does not directly relate to the subject's position relative to the negative marker, as the quantifier does not function as the subject in that strategy.

In languages where the standard negation construction already aligns with a transparent-scope strategy, there is no basis to assume that the choice of construction reflects a bias for scope transparency. In such cases, the issue of scope transparency never even comes into play. Only when transparent-scope strategies differ from the standard negation construction can we argue that a transparent-scope strategy was chosen for the sake of scope transparency.

Simplifying somewhat, Strategies B, C, and E leverage different aspects of word order flexibility to avoid inverse scope. This flexibility may come from syntactic movement or from properties of the relevant elements. Strategy C (*Not burned all the cookies*) involves verb movement—and sometimes phrasal movement of the negative marker—whereas Strategy B (*Not all the cookies burned*) typically depends on multiple attachment sites for the negative marker rather than movement. Strategy E (*The cookies did not all burn*) relies on low attachment of the quantifier or its ability to remain in a lower position. By contrast, Strategy D does not make use of word order flexibility but instead relies on an inventory of negative constructions.

Because this study relies on translations of the New Testament (NT), ensuring that the strategies observed in the target languages are genuinely attested rather than artifacts of translation is particularly important. To address this, the analysis prioritizes cases where translators diverge from the Greek structure, as such deviations are more likely to reflect native grammatical patterns. If a translation follows the Greek structure closely, it is more difficult to determine whether the resulting construction is a natural feature of the target language or simply a direct rendering of the source text. By contrast, when a translator selects a different structure despite the availability of a more literal alternative, this strongly suggests that the chosen construction is part of the target language's grammatical repertoire.

Another safeguard against translation bias is the comparison of multiple translations. If the same construction appears across different translations—especially those produced independently—it is more likely to reflect a genuine linguistic pattern rather than the idiosyncratic choice of a single translator.

The Greek NT overwhelmingly employs Strategy B (12 out of 14 times in the relevant verses; see §5.5). For example, in Romans 10:16, the Greek text follows this pattern (22).

(22) Koine Greek

All'	ou	pantes	hypēkousan	tōi	euangeliōi.
but	NEG	all.NOM.PL.M	obey.AOR.3PL	the.DAT.SG.N	gospel.DAT.SG

'But not all obeyed the gospel.' [NA28]

Strategy	Text	Translation	Year
A	*But all have not duly hearkened to the good news*	Thomson	1808
B	*Nevertheless, to this gospel not all have hearkened*	Rutherford	1900
E	*But they haue not all obeyed the Gospel*	KJV	1611

Table 1.3: Representative examples of English translations of Romans 10:16

While some English translations follow this pattern, others do not, as shown in Table 1.3. For instance, Thomson employs an inverse-scope construction (Strategy A), diverging from the Greek structure. This suggests that Strategy A was independently available in English in this period.

By prioritizing cases where translations diverge from the Greek structure and cross-checking multiple translations, this study minimizes the risk of false positives—cases where a construction is attested but may be an artifact of translation. For a more detailed discussion of this methodology, see Chapter 5.

1.5 Results

Recall that the Synchronic Hypothesis (18) states that inverse-scope constructions occur less frequently cross-linguistically than one would predict based on the typical positions of subjects and clausal negation markers in different languages. In contrast, the null hypothesis suggests that languages lack a clear motivation to avoid inverse scope. Under this view, 'Not all X are Y' sentences should be structurally similar to other negative sentences where the subject is not a scope-bearing element.

What we actually find, however, is that many languages avoid inverse scope at the cost of deviating from the usual structural patterns. Some develop constituent negation constructions specifically aimed at modifying subject QPs—these are the negated QP constructions familiar from English and many other languages, as exemplified in (16a) and (10b).

Other languages repurpose existing constructions, typically those whose primary function relates to information structure. These constructions are employed because they happen to allow the negative marker to occupy a higher position than the QP, thereby maintaining scope transparency.

For example, in Serbian-Croatian-Bosnian, the basic word order is SV(O), as in (23a). In 'Not all X are Y' sentences, however, the neutral word order shifts to VS(O), as shown in (23b), corresponding to Strategy C. This language lacks negated QP constructions, and it leverages its word order flexibility to avoid inverse scope. Notably, the VSO pattern in (23b) is completely natural out of context, indicating that is not associated with its typical discourse function here. This is evidenced by the fact that if either negation or the universal quantifier is taken out of the picture, the VS(O) order no longer qualifies as neutral, and SVO becomes preferred, as in (23a).

(23) Serbian-Croatian-Bosnian (Milica Denić, p.c.)

a. Svi moji prijatelji jedu meso.
all.NOM.PL my.NOM.PL.M friend.NOM.PL eat.PRS.3PL meat.ACC
'All my friends eat meat.'

b. Ne jedu svi moji prijatelji meso.
NEG eat.PRS.3PL all.NOM.PL my.NOM.PL.M friend.NOM.PL meat.ACC
'Not all my friends eat meat.'

Similarly to Serbian-Croatian-Bosnian, Occitan lacks negated QP constructions. Instead, its dominant strategy for expressing 'Not all X are Y' propositions is Strategy D—specifically, a negated cleft construction where the QP serves as the cleft phrase, as shown in (24). Arguably, the reason for using a cleft in this context is to allow the negative marker to precede the QP, which it cannot otherwise do. This has little to do with the typical discourse functions of clefts, such as focus marking and contrastivity.

(24) Occitan

Mas es pas totes, ça que la, qu'-an obesit a la
but be.PRS.3SG NEG all.PL.M however who-have.PRS.3PL obey.PTCP to the.SG.F
Bona Novèla.
good news
'But not everyone obeyed the good news.' [Larzac, Rom. 10:16; 2016]

Cross-linguistically, the most common transparent-scope strategies are Strategies B and D, while Strategy C is less often the dominant strategy, and Strategy E almost never is. Despite the availability of these strategies in many languages, Strategy A is still attested in approximately half of the languages in each macro-area, with the notable exception of North America, where it is somewhat less common. Furthermore, when examining how often each strategy serves as the dominant one, Strategy A ranks either first or second across all macro-areas (see §8.1).

The quantitative results provide partial support for the Synchronic Hypothesis (18). For the sampled languages from Africa and Eurasia, the results were statistically significant, showing clear deviations from the expected baseline under the null hypothesis of no bias for scope transparency. By contrast, no significant effects were found for North America, South America, or Papunesia.

One possible explanation for these non-significant results is the smaller sample size in these macro-areas, which may have reduced statistical power, making it more difficult to detect an effect even if the hypothesized bias is present. Additionally, the conservative statistical approach taken in this study may have further contributed to the lack of statistical significance (see §8.2).

While these results do not allow us to confidently reject the null hypothesis under the statistical analysis conducted, the findings from Africa and Eurasia—the macro-areas best represented in the sample—suggest that the Synchronic Hypothesis is likely on the right track. However, the effect of the hypothesized bias for scope transparency is clearly weak. Even in Africa and Eurasia, where the prevalence of Strategy A is significantly lower than the established baseline, this strategy is still attested in more than half of the languages in these areas and is more geographically dispersed than any other strategy, each of which tends to be concentrated in smaller linguistic areas.

These findings support the idea that scope transparency functions as a weak universal force—one that influences language change but is not strong enough to override competing pressures at the synchronic level. While it may modestly reduce the cross-linguistic frequency of inverse-scope constructions, this effect appears to be obscured by two key factors: (i) cyclical re-emergence of inverse-scope constructions, as outlined in (21), can counteract the shift toward scope transparency, effectively undoing previous transitions; and (ii) persistence effects slow down the decline of inverse-scope constructions, allowing them to remain in a language long after the shift toward transparent-scope strategies has begun.

This brings us to the diachronic results. As predicted by the pathway in (20), the decline of inverse-scope constructions typically follows the introduction and gradual adoption of a transparent-scope alternative, rather than resulting directly from broader grammatical changes.

It is important to note, however, that the emergence of a transparent-scope strategy usually does not lead to an abrupt drop in the frequency of a pre-existing inverse-scope construction. For example, in English, negated QP constructions appeared as early as the 14th century, yet it was not until the 20th century that they became the dominant strategy. During this extended period, the older inverse-scope construction remained more commonly used (see §10.1). This means that for several centuries, English, much like present-day Swedish (10), did not exhibit a synchronic preference for scope transparency in terms of usage frequencies.

While the decline of inverse-scope constructions generally follows the predicted pathway in (20), changes can also occur in the opposite direction, as outlined in (21). The historical data reveal several types of transitions along this pathway.

One recurrent transition observed in the historical data is what I term the NEGATED QUANTIFIER PHRASE CYCLE. In this cycle, a language ceases to allow constituent negation of subject universal QPs and instead resorts to clausal negation. Over time, however, a new negated QP construction may emerge, completing the cycle.

This process is exemplified by the transition from Latin to the Medieval Romance languages. In Latin, the most frequent strategy for expressing 'Not all X are Y' propositions was a negated QP construction (Strategy B). As shown in (25), the primary universal quantifier in Latin was *omnis*.

(25) Latin
Sed non omnes obediunt Evangelio.
but NEG all.PL.NOM obey.PRS.3PL gospel.DAT
'But not all obey the gospel' [*Vulgate*, Rom. 10:16; late 4th century]

While this form was retained in some Romance varieties (e.g., Italian *ogni*), the basic universal quantifiers in most Romance languages are derived from the Latin adjective *totus* 'whole'. Notably, the negated QP construction was not initially extended to these novel quantifiers. As a result, Medieval Romance languages typically relied on clausal negation in 'Not all X are Y' sentences.

If the clause featured VS(O) word order, the result was a transparent-scope construction (Strategy C), as shown in (26). However, SV(O) word order—the most frequent pattern due to the verb-second tendency—produced an inverse-scope construction (Strategy A), as illustrated in (27).

(26) Old Spanish
Mas no obedecen todos a-la preygacion.
but NEG obey.PRS.3PL all to-the.SG.F preaching
'But not all obey the preaching.' [*Biblia prealfonsí*, Rom. 10:16; 13th century]

(27) Old Occitan
Mas tuit no obezisso a l'-avangeli.
but all NEG obey.PRS.3PL to the-gospel
'But not all obey the gospel'
[*Nouveau Testament de Lyon*, Rom. 10:16; 13th century]

The loss of the negated QP construction in the transition from Latin to Medieval Romance was likely not driven by a specific dispreference for this construction or a preference for another strategy. Instead, the grammaticalization of a universal quantifier from *totus* proceeded independently of any potential consequences for 'Not all X are Y' sentences. Crucially, these constructions represent a mere fraction of the contexts where universal quantifiers are used, meaning that the resulting shift in strategy for 'Not all X are Y' sentences was simply a byproduct of this broader grammatical change.

Over time, most Romance languages developed new negated QP constructions, which in some cases eventually displaced the earlier inverse-scope constructions, thereby completing the cycle.

A different possible route for the emergence of inverse-scope constructions involves changes in the negation system. Although this scenario is not clearly attested in the historical data (thought see Chapter 11 for a potential example in Swedish), it can be illustrated through a hypothetical example. In English, the negative adverb *never* is frequently used as a marker of clausal negation, stripped of its original quantificational meaning, as demonstrated in (28). However, *never* cannot replace *not* in constituent negation, as shown in (29). If *never* were to completely replace *not* as the standard marker of clausal negation, it is likely that negated QP constructions would disappear as a side effect, unless *not* were retained as a specialized form for constituent negation.

(28) A: Sorry. The five minutes is up.
B: That was never five minutes just now! [*Monty Python's Flying Circus*, E29, BBC]

(29) *Never everyone showed up.

Changes in word order present yet another possible scenario. Suppose, for instance, that Serbian-Croatian-Bosnian were to become a strict SVO language. In such a hypothetical scenario, the word order pattern in (23b) would no longer be an option. Since this language lacks a negated QP construction or any other transparent-scope alternative, it would likely default to using the standard negation construction with SVO word order, leading to inverse scope.

This is apparently what happened in the transition from Old English to Middle English. In Old English, the most frequent pattern in main negative clauses was NegVS(O)—a verb-initial pattern (since *ne* is considered a clitic). This construction produced a transparent-scope construction in 'Not all X are Y' sentences (Strategy C), as shown in (30). However, in Middle English, this pattern was gradually replaced, and the negated verb typically followed the subject. As a result, inverse-scope constructions (Strategy A) became the dominant strategy for 'Not all X are Y' sentences, as exemplified in (31).

(30) Old English
Ne under-foð ealle men þis word
NEG receive.PRS.3PL all men this word
'Not everyone receives this word' [*Wessex Gospels*, Matt. 19:11; about 990]

(31) Middle English
alle men ne take not þis word
'Not everyone takes this word' [*Life of Soul*, about 1400]

To summarize, the pathways outlined in (20) and (21) differ in their underlying causal factors but not necessarily in how often they occur across languages. Notably, there are no clear cases of inverse-scope constructions increasing in frequency without broader grammatical changes that undermine the dominant transparent-scope alternative.

The combination of the synchronic and diachronic results lends credence to the Scope Transparency Hypothesis in (11), according to which languages are shaped by a universal bias for scope transparency. Despite the prevalence of inverse-scope constructions—both cross-linguistically and in terms of usage frequency in many languages—there is a persistent pressure driving languages toward expressing scope relations transparently.

1.6 Dissertation outline

This dissertation is organized into five parts. The first part introduces the research goals and main hypothesis (Chapter 1) and establishes the empirical and theoretical domain of the study (Chapter 2). This includes the rationale for focusing on 'Not all X are Y' propositions (§2.1) and key theoretical and methodological considerations for investigating inverse scope cross-linguistically (§2.2-§2.4). This part concludes with a review of previous research on inverse scope and its relevance to the present study (Chapter 3).

The second part details the materials and methodology used in this study. Chapter 4 describes the language sample and the sampling method. Chapter 5 introduces the methodology, which relies on a parallel corpus of New Testament translations, and discusses its advantages and limitations. Chapter 6 assesses the reliability of the corpus by comparing the distribution of inverse-scope and transparent-scope constructions in New Testament translations against data from online news corpora, using nine present-day Germanic languages as a case study.

The third part addresses the Synchronic Hypothesis (18). Chapter 7 constitutes the exploratory portion of this study, surveying the syntactic strategies languages use to express 'Not all X are Y' propositions. Chapter 8 presents a quantitative analysis of the cross-linguistic distribution of these strategies, examining their relative frequency and geographic distribution while assessing whether the observed patterns align with the predictions of the Synchronic Hypothesis. Chapter 9 builds on these findings, offering a theoretical analysis of the data. It argues that some constructions are repurposed to maintain scope transparency independently of their primary discourse functions, while others are specifically innovated to avoid inverse scope.

The fourth part tackles the Diachronic Hypothesis (19). Chapter 10 focuses on diachronic changes in Germanic and Romance languages, which are the best-represented groups in the historical dataset. Chapter 11 discusses these results, arguing that they largely support the Diachronic Hypothesis.

The fifth part concludes the dissertation, discussing its broader implications, highlighting the methodological contributions, and outlining prospects for future research.

Chapter 2

How to study inverse scope cross-linguistically

Inverse scope is a broad phenomenon that varies across syntactic constructions and discourse contexts. Since mapping all possible sources of variation is impractical, a more targeted approach is necessary. Narrowing the investigation makes it possible to focus on cases that are most informative for testing the hypotheses of this study while keeping the scope of inquiry manageable. This chapter outlines the empirical and theoretical domain of the study and motivates the approach taken.

Section 2.1 explains why ‘Not all X are Y’ propositions provide an optimal testing ground for investigating the Scope Transparency Hypothesis. A crucial motivation is that cross-linguistically, inverse scope is more likely to occur in these cases than in other quantificational constructions. If a bias for scope transparency exists, its effects should be observable even where inverse scope is most prevalent. Furthermore, languages overwhelmingly lack dedicated lexical items for ‘not all’, requiring syntactic strategies to express this meaning. This makes it a particularly revealing case for understanding how languages navigate conflicting syntactic constraints.

Section 2.2 addresses how to approach the study of quantifier scope, contrasting form-based and meaning-based perspectives. A form-based approach examines what interpretations are available for a given syntactic structure, while a meaning-based approach investigates how a language expresses a particular meaning. This study adopts the latter perspective, as the presence or absence of inverse scope is influenced by the availability of alternative strategies for expressing the same meaning.

Section 2.3 identifies contexts that exceptionally facilitate inverse scope. These include (i) echo denial, (ii) certain embedded environments, and (iii) a specific class of collective predicates. These contexts are excluded from the analysis to ensure that it captures the basic cross-linguistic variation.

Section 2.4 discusses the methodological challenge of defining inverse scope. In verb-medial languages like English, hierarchical relations and linear order align: the subject both c-commands and precedes negation, making inverse scope straightforward to identify. However, in verb-final languages like Turkish and Hindi-Urdu, where negation follows the subject regardless of its hierarchical position, determining whether an *all…not* construction involves inverse scope depends on whether scope is defined in terms of word order or hierarchical relations. This study adopts a definition based on linear precedence rather than syntactic hierarchy due to both methodological and empirical considerations.

2.1 Why focus on 'Not all X are Y' propositions?

If a universal bias for scope transparency exists, its effects should be observable across different types of quantificational constructions. But which cases provide the most informative test? The key methodological concern is ensuring that the choice of data does not inadvertently favor the hypothesis. If we focus only on constructions where inverse scope is already rare cross-linguistically, we may find evidence that inverse scope is dispreferred in those contexts—but this would not rule out the possibility that languages are indifferent to scope transparency in other types of constructions, potentially undermining the hypothesis.

To avoid this, we should instead examine the cases where inverse scope is most likely to occur cross-linguistically. If scope transparency emerges as a constraint even in constructions where inverse scope is most common, this provides much stronger evidence that it reflects a general linguistic tendency rather than a property of specific constructions. By focusing on the hardest case, we maximize the chance of encountering counterevidence that could challenge the hypothesis, making the test more rigorous.

A particularly relevant case for testing scope transparency is inverse scope in negative clauses, which appears to follow a strong implicational pattern across languages: if a language allows inverse scope at all, it is more likely to do so when the subject is a universal QP than when it is an indefinite pronoun or a QP headed by 'many'.[1] This pattern is captured in the following universal:

(32) *The Inverse-Scope Universal*
Any language that allows inverse scope in negative clauses when the subject QP is headed by 'many' or is a non-negative indefinite pronoun (e.g., a weak negative polarity item (NPI) or a polarity-insensitive item) also allows inverse scope when the subject is a universal QP. However, the reverse does not necessarily hold: a language may permit inverse scope with a universal QP while still disallowing it with 'many' or indefinite pronouns.

This universal is illustrated through several representative languages. Hebrew, for instance, does not permit inverse scope in any of the cases described above—whether involving universal quantifiers, 'many', or indefinite pronouns—as shown in (33).[2]

[1] Similar generalizations are proposed by Horn (1989, 228) and Mayr and Spector (2010), focusing primarily on the contrast between universal and existential quantifiers. However, their accounts frame this distinction as categorical rather than an implicational universal.

[2] Testing the availability of an inverse-scope reading with indefinite pronouns can be challenging, as they are often polarity-sensitive, precluding certain scope configurations. In Hebrew, *mišehu* 'someone' is a

(33) Hebrew

a. kulam lo heʔeminu li.
everyone NEG believe.PST.3PL me
‘Nobody believed me.’ [∀ > ¬ ; *¬ > ∀]

b. harbe anašim lo heʔeminu li.
many people NEG believe.PST.3PL me
‘Many people didn’t believe me.’ [‘many’ > ¬ ; *¬ > ‘many’]

c. mišehu xašuv lo higia.
someone important NEG arrive.PST.3SG
‘Someone important didn’t come.’ [∃ > ¬ ; *¬ > ∃]

Unlike Hebrew, which prohibits inverse-scope readings across the board, Swedish readily permits inverse scope with a universal QP, and the construction in (34a) is even the most neutral way of expressing this meaning (see §6.2.1). Notice that the inverse-scope reading is the only available interpretation of (34a), and it is relatively common cross-linguistically for a language not to allow universal quantifiers to take wide scope over negation [∀ > ¬] (see §2.2). By contrast, an inverse-scope reading is unavailable with ‘many’, as shown in (34b). Similarly, the indefinite pronoun *någon* ‘some(one), any(one)’, which is acceptable in the scope of negation when it appears post-verbally, cannot scope below negation from subject position (Haspelmath 1997, 215), as in (34c).[3]

(34) Swedish (Alice Bondarenko, p.c.)

a. Alla barn gillade inte filmen.
all.PL children like.PST NEG movie.DEF.SG
‘Not all the children liked the movie.’ [*∀ > ¬ ; ¬ > ∀]

b. Många barn gillade inte filmen.
many children like.PST NEG movie.DEF.SG
‘Many children didn’t like the movie.’ [‘many’ > ¬ ; *¬ > ‘many’]

c. Någon gillade inte filmen.
someone like.PST NEG movie.DEF.SG
‘Someone didn’t like the movie.’ [∃ > ¬ ; *¬ > ∃]

positive polarity item (PPI), but it is often acceptable in the scope of negation when it is modified, as in (33c). Even so, in (33c), the indefinite pronoun cannot scope below negation, similarly to other quantifiers. Importantly, when the subject appears post-verbally, a reading where negation takes wide scope over the existential quantifier becomes acceptable, indicating that the lack of an inverse-scope reading in (33c) is not due to polarity sensitivity.

[3]English behaves similarly to Swedish in this respect, although inverse-scope constructions like (ia) are relatively rare in present-day English (see §6.1.1). Like in Swedish, neither *many* (Jackendoff 1969, 223) nor the weak NPI *any* can scope below negation from subject position, as shown in (ib)-(33).

(i) a. All the arrows didn’t hit the target. [∀ > ¬ ; ¬ > ∀]
b. Many arrows didn’t hit the target. [*many* > ¬ ; *¬ > *many*]
c. *Any arrow didn’t hit the target. [*∃ > ¬ ; *¬ > ∃]

Other languages are even more permissive with respect to inverse scope. In Estonian, not only universal QPs but also ‘many’ and indefinite pronouns can scope below negation from subject position, as shown in (35). Notice that universal QP must scope below negation in this construction, whereas *paljud* ‘many’ and *keegi* ‘someone, anyone’ only tend to do so.[4]

(35) Estonian (Pook and Lindström 2024, 193 and Liina Lindström, p.c.)

a. Kõik ei tea sellest.
everyone NEG know.CNG it.ELA
‘Not everyone knows about this.’ $[{*}\forall > \neg\ ;\ \neg > \forall]$

b. Paljud inimesed ei tea sellest.
many people NEG know.CNG it.ELA
‘Not many people know about this.’ or:
‘Many people don’t know about this.’ [‘many’ $> \neg$; $\neg >$ ‘many’]

c. Keegi ei tea sellest.
someone NEG know.CNG it.ELA
‘No one knows about this.’ or:
‘Somebody doesn’t know about this.’ $[\exists > \neg\ ;\ \neg > \exists]$

In light of the implicational universal in (32), focusing on inverse scope with universal QPs sets the bar high for proving the existence of a universal bias for scope transparency. By examining the domain where it is most difficult to show that inverse scope is ultimately dispreferred, we can make a compelling case for the existence of this bias.

Another reason why ‘Not all X are Y’ sentences are particularly intriguing is their connection to the generalization, proposed by Horn (1972), that languages do not lexicalize the meaning of ‘not all’. Horn frames this observation within the context of the Aristotelian square of opposition, presented in Figure 2.1. The square of opposition illustrates the logical relations between the four types of categorical propositions in Aristotelian logic—propositions that affirm or deny that all or some members of one category (the subject term) belong to another category (the predicate term).

According to Horn (1972, 1989), many languages (including English) have dedicated lexical forms for the A-, I-, and E-corners of the square of opposition, but no language lexicalizes the meaning of ‘not all’.

While this claim has been widely accepted, this study systematically tested its validity across a diverse sample of 217 languages as part of a broader investigation into how languages express ‘Not all X are Y’ propositions. The findings largely confirm Horn’s generalization

[4]The transparent-scope reading in (35c) is somewhat difficult to access, but it becomes more salient when this interpretation is contextually plausible, as in (i). Notice, however, that the inverse-scope remains the preferred reading even in this context.

(i) Estonian (Liina Lindström, p.c.)
Keegi ei pannud ust lukku.
someone NEG put.PST.CNG door.PART lock.ILL
‘Nobody locked the door.’ or: ‘Somebody didn’t lock the door.’

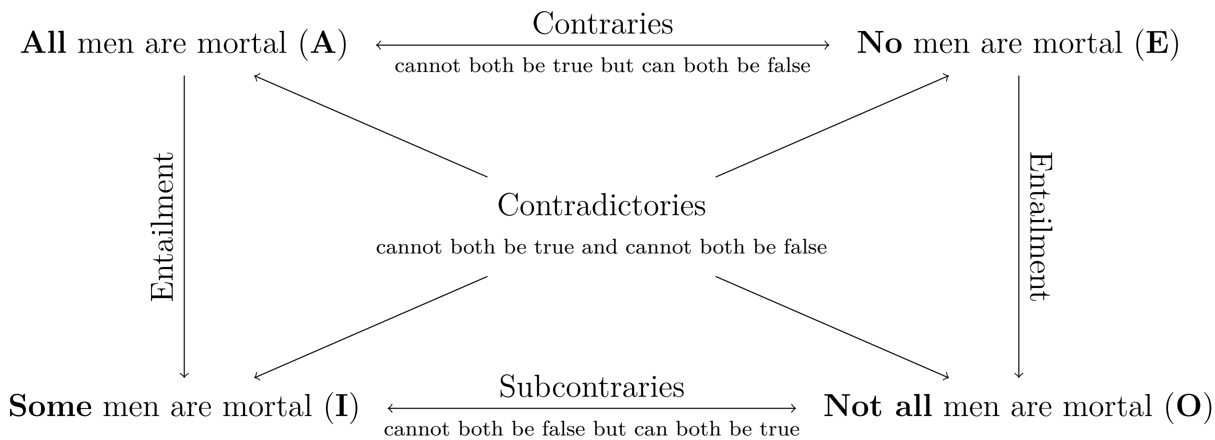


Figure 2.1: The Aristotelian square of opposition

but reveal a single counterexample, suggesting that the non-lexicalization of 'not all' is not an absolute universal but rather a near-absolute one (see §9.3 for further details).

This near-absolute universal is significant for the present study because it underscores that languages normally do not solve the problem of scope transparency in 'Not all X are Y' sentences by simply lexicalizing a form for 'not all'. Instead, they must rely on syntactic strategies to express this meaning.

This limitation is one of the factors behind the universal in (32). It becomes particularly striking when compared to the lexicalization of 'not many' and 'no'. While 'not all' resists lexicalization, many languages do have lexical alternatives for 'not many', such as *few* in English, or for the E-corner, such as *no* in English.

This contrast has important consequences for how languages manage scope transparency. Cross-linguistically, inverse scope in negative clauses is more common when the subject is a universal quantifier than with 'many' or an indefinite pronoun. This suggests that the availability of lexical alternatives—such as *few* for 'not many'—reduces the need for inverse scope. By contrast, the absence of a dedicated lexical form for 'not all' makes avoiding inverse scope more challenging.

French provides a clear illustration of how lexicalization can help circumvent a violation of scope transparency. Standard French does not allow constituent negation of subject QPs (see §10.2). How, then, can a sentence like (36a) be negated? Given that (36b) is not an option, one might expect French to resort to inverse scope, as in (36c). However, this is not what we find. Instead, French tends to avoid the issue altogether by using *peu* 'few', as in (36d). Crucially, this kind of lexical workaround is not available with a universal QP.

(36) Standard French

a. Beaucoup de mes amis le savent.
many of my.PL friends it know.PRS.3PL
'Many of my friends know this.'

b. *Pas beaucoup de mes amis ne le savent.
NEG many of my.PL friends NEG it know.PRS.3PL
'Many of my friends know this.'

c. Beaucoup de mes amis ne le savent pas.
many of my.PL friends NEG it know.PRS.3PL NEG
'Many of my friends don't know this.'

d. Peu de mes amis le savent.
few of my.PL friends it know.PRS.3PL
'Few of my friends know this.'

Similarly, many languages express 'No X are Y' propositions (the E-corner) using a dedicated negative indefinite or negative quantifier, such as *no* in English. This allows them to avoid inverse-scope constructions like (37a), which require a non-negative indefinite pronoun to scope below negation from the subject position. Instead, they use a lexical alternative, as shown in (37b).

(37) a. *Any arrow didn't hit the target.
b. No arrow hit the target.

That is not to say that languages lexicalize the meanings of 'few' or 'no' for the purpose of avoiding inverse scope. However, the fact remains that the existence of lexical items that incorporate negation into the quantificational expression makes it possible to avoid "inverse-scope constructions" like (36c) or (37a) by lexical, rather than syntactic, means. By contrast, there are no comparable lexical items that can be used for expressing the meaning of 'not all'.

Admittedly, the problem of inverse scope in 'Not all X are Y' propositions can be sidestepped by leveraging their logical equivalence to 'Some X are not Y'. However, these forms are not always completely interchangeable. For instance, (38a) is a reasonable statement, whereas (38b) sounds like a gross understatement. This is because (38b) tends to trigger a scalar inference that the stronger alternative in (38c) is false, which conflicts with world knowledge. While (38c) fares better as a paraphrase of (38a), there remains a subtle difference: (38c) evokes an alternative set that includes (38a) and possibly also *Few people can be astronauts*. By contrast, (38a) does not evoke such alternatives and thus remains neutral as to just how many people are capable of being astronauts (although world knowledge may fill this gap). Consequently, although the felicity conditions of 'Not all X are Y' and 'Some X are not Y' statement partially overlap, they are not identical.

(38) a. Not everyone can be an astronaut.
b. Some people can't be astronauts.

c. {Many / Most} people can't be astronauts.

This contrasts with the choice between 'not many' and 'few', which appear to be more interchangeable, as illustrated in (39). Both forms seem to evoke the same set of alternatives, making them felicitous in the same contexts.

(39) a. Not many people can be astronauts.
b. Few people can be astronauts.

This discussion highlights that languages are only equipped with their syntactic toolkit when tackling the challenge of expressing 'Not all X are Y' propositions. This offers us a rare opportunity to observe how languages engage in problem-solving within a clearly defined grammatical domain. Some languages bite the bullet and settle for inverse scope, while others resort to various syntactic gymnastics to avoid it.

2.2 Two complementary perspectives in semantic typology

The previous section established why 'Not all X are Y' propositions are particularly useful for investigating the Scope Transparency Hypothesis (11). The next step is to consider how to study them.

In cross-linguistic semantics, there are two complementary perspectives for analyzing the relationship between form and meaning. A form-based approach (also known as semasiology) starts from linguistic expressions and examines their possible interpretations. It asks, "What does form X mean in this language?" By contrast, a meaning-based approach (onomasiology) starts from a specific meaning and investigates how different languages express it. It asks, "How does this language express meaning X?"

A form-based approach to quantifier scope would take an *all…not* construction like (40) and examine whether it permits both a transparent-scope and an inverse-scope reading.[5]

(40) All politicians are not opposed to reform.
Transparent-scope reading: No politicians are opposed to reform. [$\forall > \neg$]
Inverse-scope reading: Not all politicians are opposed to reform. [$\neg > \forall$]

In contrast, a meaning-based approach would take a particular meaning as the starting point. Instead of asking what interpretations an *all…not* construction allows, this approach examines the broader range of strategies a language employs to express 'Not all X are Y' propositions. This means considering multiple possible constructions rather than focusing on a single form.

[5] *All…not* constructions can also have a collective interpretation in certain contexts (Taglicht 1985; Tottie and Neukom-Hermann 2010). See §3.3.1 for further details.

While much of the existing research on quantifier scope has taken a form-based approach (see Chapter 3), this study adopts a meaning-based perspective. The reason for this is that the availability of inverse-scope constructions is not determined by whether *all…not* constructions can have a transparent-scope interpretation corresponding to 'No X are Y'. Rather, it is shaped by competition with alternative, transparent-scope strategies for expressing 'Not all X are Y'. By examining the full range of strategies a language employs for this meaning, we can assess whether inverse-scope constructions emerge due to a lack of transparent-scope alternatives, or whether they coexist with other options.

A form-based approach is useful for describing differences in scope possibilities across languages. However, it does not explain why some languages permit inverse scope while others disallow it. To illustrate this limitation, consider a classification of languages based on the possible interpretations of *all…not* constructions.

Broadly speaking, languages fall into four categories. First, some languages allow both a transparent-scope and an inverse-scope interpretation, making these constructions structurally ambiguous. English belongs to this category, as seen in (40). In Kam (Niger-Congo), *all…not* constructions are the most natural strategy for expressing 'Not all X are Y' propositions, yet these constructions also permit a transparent-scope interpretation, as illustrated by (41).[6] Second, some languages permit only the transparent-scope reading, ruling out an inverse-scope interpretation. This is the case in Hebrew (33) and Standard Dutch (Zeijlstra 2017). Third, other languages only permit the inverse-scope reading, disallowing a transparent-scope interpretation. For instance, Hemforth and Konieczny (2019) found that French speakers tend to reject a transparent-scope reading of sentences like (42). This pattern is also evident in other languages where inverse-scope constructions are frequently employed, such as Swedish (34), Estonian (35), and Turkish (Öztürk 2005; Kelepir 2001; Kamali and Zeijlstra 2024). Finally, a fourth category consists of languages where neither reading is available, rendering *all…not* constructions generally unacceptable. While no clear example of this pattern is known to me, its existence remains a possibility.

(41) Kam (Jakob Lesage, p.c.)
ʧì jō ŋwāb kím ḿfàm hǹ.
child PL all eat meat NEG
'Not all the children ate meat.' or: 'None of the children ate meat.'

(42) French (Hemforth and Konieczny 2019, 169)
Tous mes amis ne sont pas allés au dernier film de Marvel.
all.PL.M my.PL friends NEG be.PRS.3PL NEG go.PTCP.PL to.the last film of Marvel
'Not all my friends went to the last Marvel movie.' [$*\forall > \neg$; $\neg > \forall$]

A cross-linguistic survey is likely to reveal an inverse correlation between the availability of transparent-scope and inverse-scope interpretations. That is, languages that allow a transparent-scope reading tend to disfavor an inverse-scope reading, and vice versa. How-

[6] In (41), the quantifier *ŋwāb* 'all' is an adnominal quantifier that follows the noun phrase, and the negative marker *hǹ* is a clause-final particle (Lesage 2020, 2025).

ever, this correlation alone does not explain the underlying factors that determine whether a language permits inverse scope. The crucial question is: which of the two readings tends to block the other?

I argue that there is a fundamental asymmetry between these two readings. In languages where inverse-scope constructions are frequently employed, the inverse-scope reading often prevents access to a transparent-scope interpretation. Conversely, it is unlikely that the transparent-scope reading actively blocks the inverse-scope interpretation in other languages, as the transparent-scope reading is itself pragmatically marked and typically restricted to specific contexts.

The limited distribution of the transparent-scope reading of *all…not* constructions becomes evident when considering the broader landscape of strategies for expressing 'No X are Y' propositions (the E-corner in the square of opposition, see Figure 2.1). While I have not systematically investigated this, I am not aware of any language where *all…not* constructions serve as the dominant strategy for conveying this meaning. This stands in sharp contrast to their use under an inverse-scope interpretation. While the meaning of 'No X are Y' is never strongly associated with *all…not* constructions, these constructions often emerge as the dominant strategy for expressing 'Not all X are Y' propositions across languages.[7]

This asymmetry suggests that if one reading blocks the other, the effect is likely unidirectional, with the inverse-scope interpretation suppressing the transparent-scope reading. Given that the transparent-scope reading of *all…not* constructions is typically marked and contextually restricted, it is unlikely to exert a strong blocking effect on inverse scope.[8]

Thus, while a form-based approach can describe how different languages interpret *all…not* constructions, it does not account for the causal factors shaping the distribution of inverse-scope constructions. The key factor determining the availability of inverse scope is com-

[7]It seems that languages employ *all…not* constructions to express the meaning of 'No X are Y' primarily when the domain of quantification is restricted to a specific group. Tottie and Neukom-Hermann (2010) observe that corpus instances of *all…not* constructions with a transparent-scope reading often involve demonstratives or the determiner *other*, as exemplified in (i). They propose that in such cases, *all* functions as an intensifier, enforcing a maximal interpretation.

(i) (Tottie and Neukom-Hermann 2010, 165-166)
  a. […] all these kids didn't know what two pence was, didn't know what twenty pence was and ten pence and fifty pence […]!
  b. I think that's what makes Curve stand out from the rest of the pack, asserts Alex, because all these other indie bands don't seem to be interested in guitar music.

This finding relates to the observation that universal quantifiers tend to be dispreferred in positions directly above negation (Beghelli and Stowell 1997). For example, (iia) is perceived as unnatural under neutral intonation when intended to mean 'No kid received a present this year'. However, the sentence becomes more acceptable when a demonstrative is added, as in (iib).

(ii) a. ?Every kid didn't receive a present this year. [$?\forall > \neg$]
  b. Every one of these kids didn't receive a present this year. [$\forall > \neg$]

[8]If the focus were instead on the transparent-scope interpretation, a form-based approach would be appropriate, as it would account for its competition with the inverse-scope reading.

petition with alternative, transparent-scope strategies that convey the same meaning. If a language has a dedicated transparent-scope construction for this meaning, it may disprefer or even lack inverse-scope constructions. Conversely, languages that frequently use inverse-scope constructions often do so because alternative structures are either unavailable or disfavored for pragmatic reasons.

This underscores the advantages of adopting a meaning-based perspective: rather than asking, "What does an *all…not* construction mean in this language?", the more informative question is, "How does this language express 'Not all X are Y' propositions?" Answering this requires shifting the focus from what meanings a single construction allows to the how different languages encode the meaning under question. This study, therefore, adopts a meaning-based approach, examining the broader range of strategies languages employ for expressing 'Not all X are Y' propositions. By taking this perspective, it provides a more explanatory account of the cross-linguistic variation in the availability of inverse scope.

2.3 Contexts excluded from the analysis

This study focuses on inverse scope in main declarative clauses. It excludes three specific contexts that are particularly conducive to inverse-scope interpretations, since including these cases would obscure the cross-linguistic variation by inflating the availability of inverse scope. The excluded contexts are: (i) echo denial, (ii) certain embedded environments, and (iii) constructions involving a certain type of collective predicates. These environments have been shown to facilitate inverse scope, even in languages that otherwise disallow it. As such, they are not reliable indicators of whether a given language truly permits inverse scope.

Echo denial occurs when a speaker rejects a prior statement by repeating it with an added negative marker, as illustrated in (43) from the Corpus of Contemporary American English (COCA, Davies 2008).

(43) Polly Williams: They left the public schools, go into the private school, and the children are just like 100 percent changed. Children are happy. They love going to school. Their grades have improved. The parents are involved. **Everybody's happy**.
Art Hackett: **But everybody is not happy**, perhaps least of all Wisconsin Superintendent of Public Instruction Herbert Grover.
[COCA: Spoken; 1991, boldface added]

It is not always the case that the rejected statement appears verbatim in the preceding discourse; sometimes, it is only strongly implied. This is illustrated in (44), originally in Hebrew, taken from an interview on a reality TV show (most of the conversation is translated for readability). In this exchange, the claim that all teenage boys break into places is implied rather than explicitly stated, yet this is considered an instance of echo denial.

(44) Contestant: When I was young, I used to do a lot of stupid things, cause a lot of trouble.

Interviewer 1: Why? What kind of stupid things?
Contestant: My friends and I would break into all kinds of places.
Interviewer 2: Just the usual stupid things, the kind all teenage boys pull.
Contestant: Yeah.
Interviewer 1: You even told her "yeah". What do you mean by "yeah"?
Contestant: Yeah, of course. Why not?
Interviewer 1:

kol	ha-near-im	lo	porcim.
all	the-teenage.boy-PL.M	NEG	break.in.PRS.PL.M

'Not all teenage boys break into places.'

Determining what qualifies as echo denial in cases where the prior statement is not explicitly stated presents a conceptual challenge. If the rejected assertion does not need to be overtly articulated for its negation to function as an echo denial, where do we draw the line? This issue is further complicated by the fact that negative sentences are typically used when the corresponding affirmative is salient in the discourse context (Givón 1978). However, not all rejections of a salient assumption constitute echo denial. Crucially, echo denial specifically rejects an assertion made by another speaker, whether it was stated directly or implied through their wording. In contrast, a negative sentence that merely responds to a background assumption—even a salient one—does not qualify as echo denial if the assumption was never actually articulated by the other speaker.

Echo denial tends to preserve the lexical choices and syntactic structure of the preceding utterance. As a result, it exhibits properties that make it distinct from ordinary clausal negation. Notably, inverse scope is often permitted in echo denial contexts even in languages that otherwise disallow it. This has been observed in languages such as Standard Dutch and Hebrew, where *all…not* constructions do not normally allow inverse scope, yet such readings become available in echo denial (Zeijlstra 2017; Amiraz 2019).

Similarly, PPIs can appear in the scope of negation in echo denial (Baker 1970, 169), as illustrated with the PPI *already* in (45).

(45) No, **Trump has not already won the election**, and it is deeply irresponsible for him to say he has. [web example[9]]

The exceptional ability of negation to take wide scope in both cases likely reflects the same phenomenon, tied to the interactional function of echo denial rather than the general syntax of scope-taking in a given language.

The second distinctive property of echo denial is that 'Not all X are Y' sentences often do not trigger the scalar inference 'not all (but some)' in this context. In (46) from the iWeb Corpus (Davies 2018), the assertion simply rejects the idea that everything is alright,

[9] Shapiro, Ben. *Twitter*, 3 November 2020, `twitter.com/benshapiro/status/1323890527767416834`

without implying that some things are alright. This makes it challenging to distinguish between an inverse-scope and a transparent-scope interpretation.[10]

(46) I once had to pull a car over because of my child's behavior. I put my hazards on to deal with what was happening in the back seat. The next thing you know, I had a cop at my window making sure everything was alright. That got my children's attention. I wanted to shout, "No! **Everything is not alright**. My kids are driving me crazy." [iWeb Corpus; boldface added]

Another context, which bears some resemblance to echo denial but is still included in this study, involves contrastive sentences of the form 'All X are Y, but all X are not Z'. This is exemplified in (47). Although the *all…not* construction in the second clause mirrors the structure of the first clause, this context is considerably less accommodating for inverse-scope interpretations than echo denial. Hebrew, for instance, disallows inverse scope in this context, unlike in echo denial.

(47) "All men must die," said a voice quite close at hand; "**but all are not condemned to meet a lingering and premature doom**, such as yours would be if you perished here of want." [Brontë, *Jane Eyre*, chap. 28; 1847]

Another excluded context involves clauses embedded in Strawson-downward-entailing environments (von Fintel 1999; Crnič 2014, 2019) or cases where certain elements scopally intervene between negation and the universal quantifier. As with echo denial, PPIs can appear in the scope of negation in these contexts, a phenomenon known as PPI rescuing or shielding (Szabolcsi 2004). Similarly, inverse-scope interpretations of *all…not* constructions become available in these contexts in languages that typically disallow them (Zeijlstra 2017; Amiraz 2019).

Given these considerations, instances of inverse scope in echo denial or certain embedded environments should not be treated as evidence that a language allows inverse scope. By analogy with PPIs, the availability of inverse scope is best assessed through the examination of main declarative clauses, while phenomena like echo denial and rescuing in embedded environments represent distinct objects of study (though see §3.1.2).

The third excluded context involves clauses with certain types of collective predicates. These predicates are unique in that the presence of a universal quantifier does not alter the meaning of the sentence. In general, plural predicates (whether distributive or collective)

[10] The uncertainty regarding the precise truth-conditional meaning of the boldfaced sentence in (46) aligns with the view of Ariel (2002a), who argues against the assumption that each sentence has a unique, fixed truth-conditional meaning. Instead, Ariel (2002b, 2008) introduces the concept of privileged interactional interpretation, which refers to the meaning that interlocutors take as the speaker's minimal and necessary commitment in discourse. Crucially, privileged interactional interpretation is not strictly determined by linguistic meaning but is shaped by what is most interactionally relevant in a given context. In the case of echo denial in (46), the primary communicative function is rejection, rather than the articulation of a quantificational statement. As a result, whether the QP takes scope above or below negation is beside the point from this perspective.

exhibit homogeneity effects—a phenomenon where the affirmative and negative counterparts of a sentence do not have fully complementary truth conditions (Križ 2016, 2019; Bar-Lev 2021, among others). This is illustrated in (48), where the affirmative interpreted as 'All X are Y', while the negative conveys 'No X are Y', rather than 'Not all X are Y'.

(48) a. The suitcases are in the trunk.
(≈ All of the suitcases are in the trunk.)
b. The suitcases are not in the trunk.
(≈ None of the suitcases are in the trunk.)

Adding a universal quantifier eliminates this homogeneity effect: in (49), the truth conditions of the affirmative and negative are fully complementary.

(49) a. The suitcases are all in the trunk.
b. The suitcases are not all in the trunk.

While many collective predicates are homogeneous (Križ 2016), a distinct class of collective predicates behaves differently, exhibiting non-homogeneity (Amiraz 2020). For instance, the predicate *fit in the trunk* is non-homogeneous on its collective reading. The affirmative in (50a) is true only if all of the suitcases fit in the trunk; if even one does not, the sentence is false, and its negation (50b) is true. This means that *fit in trunk* is non-homogeneous, since the truth conditions of the opposing sentences are complementary. Crucially, in contrast to homogeneous predicates, adding *all* to these sentences does not change their meaning, as there is no homogeneity to remove. This is illustrated by the paraphrase in (50b).

(50) a. The suitcases fit in the trunk.
(≈ All of the suitcases fit in the trunk.)
b. The suitcases don't fit in the trunk.
(≈ Not all of the suitcases fit in the trunk.)

According to Amiraz (2019), this class of predicates permits inverse-scope interpretations in Hebrew. This is illustrated by (51a), taken from a children's birthday song, and the naturally-occurring example in (51b), overheard at an airport. Other predicates in this category include expressions involving *same*, such as *think the same way*, *arrive at the same time* (Amiraz 2019, 2020). For clarity, I will refer to these as *same*-type predicates, based on the most frequent member of this group.

(51) Hebrew
a. ma naim u-ma nexmad še-kol ben ve-kol bat lo noldu
what pleasant and-what nice that-every boy and-every girl NEG born.PST.3PL
be-yom exad – rak be-yom huledet!
in-day one only in-day.of birth
'How pleasant and lovely it is that not every boy and girl was born on the same day—only on their birthday!'

b. kol ha-anašim ha-ele lo nixnasim l-a-otobus ha-ze.
all the-people the-these NEG fit.PRS.3PL to-the-bus the-this
'Not all of these people can fit on this bus.'

Another related context involves a possibility modal intervening between negation and the universal quantifier. In (52), the intended interpretation is that it is not possible for everyone to eat like Americans [$\neg > \diamond > \forall$]. The intervening modal precludes distributive inferences: the sentence does not imply that there are specific individuals who cannot eat like Americans. Instead, it asserts that there is no accessible world where everyone eats like Americans. As with *same*-type predicates, this context is particularly amenable to inverse-scope interpretations.[11]

(52) The world is eating more and more like Americans. Higher calorie diets with more meat. **But everyone can't eat like Americans**. There actually isn't enough water in the world. [*Explained*, S01E19, Netflix]

2.4 Defining inverse scope: Linear order vs. hierarchical relations

As outlined in the previous sections, this study investigates the cross-linguistic variation in the use of *all…not* constructions for expressing 'Not all X are Y' propositions. While these constructions share a common surface pattern—where a universal QP in subject position precedes clausal negation—their underlying syntactic structures may differ across languages. Crucially, languages vary in where subjects and negation are positioned within the clausal spine. As a result, a subject may be structurally lower than negation even when it appears before the negative marker in linear order.

This variation raises important questions. On a practical level, when negation takes wide scope in an *all…not* construction, should this always be classified as inverse scope, or does the classification depend on the underlying syntactic structure of the clause? More fundamentally, does scope transparency favor an alignment between meaning and word order, or between meaning and hierarchical structure? And when these two factors diverge, which one takes precedence?

In verb-medial languages like English, these questions do not arise: the subject is both structurally higher and linearly precedes the negation marker in *all…not* constructions. This means that when negation takes wide scope, we can confidently analyze the sentence as

[11] Example (52) has another possible reading, though not intended in this case, which states that not everyone has the capability of eating like Americans [$\neg > \forall > \diamond$]. This configuration is less likely to allow inverse scope, since the predication is distributive, and the presence of the modal has not effect. As for the intended reading of (52), note that the enabling effect of the possibility modal on inverse scope is unrelated to the phenomenon of rescuing discussed earlier. This is evidenced by the fact that a possibility modal does not act as an intervener for polarity-sensitive items. For instance, it does not "shield" PPIs, as seen in the unacceptability of **You can't touch something.*

involving inverse scope, since the subject QP both precedes the negative marker and must reconstruct to its base position at LF to derive this interpretation.[12]

By contrast, in verb-final languages like Turkish and Hindi-Urdu, the linear order of the subject and the negative marker often does not straightforwardly reflect their hierarchical relationship. Specifically, if negation is positioned to the right of vP, it is unclear whether the subject is syntactically higher or lower than it. This uncertainty arises because the subject is base-generated at the left edge of vP, and any movement to a higher position (e.g., [Spec, IP]) would not alter the word order. As a result, a subject may precede negation linearly but still be structurally lower than it, creating a potential mismatch between word order and syntactic hierarchy.

How, then, do we determine whether a given *all…not* construction involves inverse scope? The answer depends on whether inverse scope is defined in terms of word order or hierarchical structure. Under a word-order-based definition, any case where negation takes wide scope in and *all…not* construction is considered inverse scope, regardless of whether scope reconstruction is actually required.

Under a hierarchy-based definition, however, *all…not* constructions may or may not involve inverse scope. If negation is structurally higher than the subject, then a wide-scope reading of negation is considered transparent scope, since the observed interpretation simply reflects the syntactic structure.

This issue is illustrated by example (53) from Turkish. Öztürk (2005, 131) argues that in Turkish, a non-scrambled subject remains in its base position, where it is c-commanded by negation, rather than raising to [Spec, IP], as subjects do in languages like English. Thus, in this *all…not* construction, the QP precedes negation in word order, but negation is higher than the QP. Whether this construction qualifies as an instance of inverse scope, then, depends on whether inverse scope is defined in terms of word order or hierarchical structure.

(53) Turkish (Öztürk 2005, 131)
[$_{IP}$ [$_{vP}$ bütün çocuklar o test-e t_1] gir-me-di$_1$]
all children that test-DAT take-NEG-PST
'Not all of the children took that test.' [$*\forall > \neg$; $\neg > \forall$]

A similar issue arises in Hindi-Urdu. Homer and Bhatt (2020, 10) argue that in this language, the subject raises from its base position to [Spec, AspP], which they identify as the canonical subject position. However, negation occupies an even higher position in the structure. In

[12] Another theoretical possibility is that negation undergoes covert movement at LF, analogous to Quantifier Raising (QR). However, following Ladusaw (1980), this type of movement is generally considered impossible, though see Boeckx (2001) for an alternative view. A key argument against this analysis is that it incorrectly predicts unavailable readings. For instance, in examples like (ia), the scope of negation is fixed: (ia) can only be interpreted as [*often* $> \neg$] and cannot convey the same meaning as (ib), where negation takes wide scope over the adverb. If negation could undergo covert raising, we would expect it to be able to take wide scope over the adverb as well, contrary to fact.

(i) a. Children often don't listen to their parents. [*often* $> \neg$]
b. Children don't often listen to their parents. [$\neg >$ *often*]

(54), the subject QP does not need to reconstruct to its base position to derive an LF where negation takes wide scope, as the subject is already hierarchically lower than negation at PF. Once again, whether this construction is considered as inverse scope depends on the criterion used to define it.

(54) Hindi-Urdu (Homer and Bhatt 2019)
[$_{TP}$ [$_{NegP}$ [$_{AspP}$ har laṛke=ne$_1$ [$_{vP}$ t_1 vo kitaab t_2]] nahĩː paṛhiː$_2$]]
every boy-ERG that book NEG read.PFV.SG.F
'Not every boy read that book.' [?∀ > ¬ ; ¬ > ∀]

If inverse scope is defined based on hierarchical relations at PF, then the structures in (53) and (54) would be classified as exhibiting transparent scope, since the wide scope of negation directly reflects the syntactic structure without requiring reconstruction. In this case, the observed alignment between syntactic structure and scope interpretation would appear unremarkable, and scope transparency would not be a relevant factor, as no mismatch exists.

This dissertation defines inverse scope in terms of linear precedence rather than hierarchical structure. This choice is driven by practical, conceptual, and empirical considerations, as detailed in the discussion that follows. Under this definition, no distinction is made between two possible types of *all…not* constructions: (i) the subject is hierarchically higher than negation at PF but undergoes scope reconstruction at LF; and (ii) negation is hierarchically higher than the subject at PF, making scope reconstruction unnecessary. Since the universal quantifier linearly precedes the negative marker in both (53) and (54), these constructions are classified as instances of inverse scope.

From a practical point of view, defining inverse scope based on word order rather than hierarchical structure facilitates the analysis of cross-linguistic data. Unlike hierarchical structure, which often must be inferred, word order is an observable syntactic property. This is particularly crucial given the lack of a universally applicable diagnostic for determining the relative hierarchical positions of the subject and negation, as will be demonstrated shortly.[13]

Beyond this practical advantage, a hierarchy-based definition faces a deeper conceptual challenge. In languages where word order does not transparently reflect hierarchical relations, the structural position of negation is often inferred from scope preferences. If this inferred structure is then used to explain the observed scope preferences, the reasoning becomes circular, as this analysis presupposes the very clausal structure that scope preferences are supposed to reveal.

[13]Another potential theoretical motivation for prioritizing word order comes from the theory of Antisymmetry of Syntax (Kayne 1994), which posits that hierarchical structure uniquely determines word order. Under this view, negation in languages like Turkish and Hindi-Urdu occupies the same structural position as in English—to the left of *v*P—while the surface SOV order is derived through leftward movement of the subject and object. If correct, this would imply that the subject is structurally higher than negation even in these languages, eliminating any potential mismatch between hierarchical structure and linear order. However, this is one possible syntactic analysis among others, and I remain agnostic on whether verb-final languages should be analyzed in this way.

This issue arises in Turkish, where negation obligatorily takes wide scope over the subject in sentences like (53). Based on this observation, Öztürk (2005, 131) concludes that the subject typically remains in situ (unless it raises to [Spec, TP] for agreement). However, since this conclusion is itself derived from scope preferences, it cannot then be used to explain why negation obligatorily takes wide scope in (53).

Furthermore, different syntactic diagnostics can sometimes lead to conflicting conclusions about the hierarchical relationship between the subject and negation. In Hindi-Urdu, for instance, while both Homer and Bhatt (2020, 9) and Kumar (2006, 120) agree that negation is positioned above AspP (Aspect Phrase), they diverge on the placement of the subject. Homer and Bhatt (2020, 4–8) argue that the canonical subject position is [Spec, AspP], which is lower than negation. Their analysis is based on scope interactions, particularly how negation interacts with adverbs and subject NPIs.[14] In contrast, Kumar (2006, 120) proposes that the subject instead occupies [Spec, FP (Finiteness Phrase)]. In his analysis, the negated verb moves to T^0 but remains below the subject. Notably, Kumar's analysis relies on structural properties such as word order and agreement rather than scope relations. This discrepancy underscores the difficulty of determining the relative positions of subjects and negation independently of scope scope-based evidence.

These practical and conceptual considerations support defining inverse scope in terms of word order rather than hierarchical structure. However, an important question remains: do wide-scope negation readings in *all…not* constructions represent a unified phenomenon across languages? If, according to certain analyses, these readings can arise through different mechanisms—some involving scope reconstruction and others not—are we not comparing apples to oranges? In the remainder of this section, I turn to empirical evidence, showing that these constructions are subject to the same constraints cross-linguistically, reinforcing the view that they should be analyzed as part of the same phenomenon.

The first argument supporting this view is based on intonation patterns. Kroch (1979, 125) observes that inverse scope in English is blocked when an intonation break occurs between the universal quantifier and negation. This often happens when a parenthetical clause is inserted, as illustrated in (55).

[14] Homer and Bhatt (2020) focus on examples like (i), where the subject NPI *ek-bhiː laṛke-ne* 'any boy' is licensed by clausal negation, indicating that it occurs within its scope at LF.

(i) Hindi-Urdu (Homer and Bhatt 2020, 5)

ek-bhiː	laṛke-ne	hameshaa	mehnat	nahĩː	kiː.
one-even	boy-ERG	always	handwork.F	NEG	do.PFV.F

'Not even one boy worked hard all the time.' [¬ > 'any' > 'always']

According to Homer and Bhatt (2020), the subject has to be below negation at PF to derive this reading. If the subject were higher than negation, it would need to reconstruct to its base position for NPI licensing, but this would place it below the adverb *hameshaa* 'always' at LF [¬ > 'always' > 'any'], since *hameshaa* is attached higher than the position where the subject is generated. Since universal quantifiers like *hameshaa* act as interveners blocking NPI licensing, this configuration would render the sentence unacceptable, contrary to fact. Therefore, Homer and Bhatt (2020) argue that the canonical subject position is below negation, eliminating the need for reconstruction in (i). Although their analysis focuses on NPI licensing, the preference for wide scope of negation observed in sentences like (54) and (i) may be related phenomena (see §2.1).

(55) a. All of the soldiers, when the captain mumbled, didn't hear the order. (Kroch 1979, 125)

b. All of these benefits, whilst clear within the Bank, were not widely known by our customers. (Tottie and Neukom-Hermann 2010, 166)

Another context that disrupts inverse scope through an intonation break is when the subject QP is modified by a long relative clause. For instance, (56) is a naturally occurring example of inverse scope, where the relative clause is short. However, in the constructed example (57), the addition of a long relative clause makes it harder to squeeze the entire sentence into a single intonation unit, thereby disfavoring an inverse-scope reading.

(56) I do think one of the messages of A.I. is that most people make middling work, and middling work is easy to replace. **Every email I write is not a great work of art**, you know.
[*The Ezra Klein Show*, "Will A.I. Break the Internet? Or Save It?", 5 April 2024]

(57) Every email I write to colleagues, friends, or customer support representatives is not a great work of art. [$?\neg > \forall$]

The interpretation of *all…not* constructions in Hindi-Urdu seems to be subject to the same constraint. In (58), the reading where negation takes wide scope is, at best, marginal. This appears to be due to an intonation break after the relative clause, which blocks the inverse-scope reading. Crucially, if scope relations were solely determined by hierarchical structure, the presence of the relative clause would not affect the interpretation, as it does not alter the subject's structural position.[15]

(58) Hindi-Urdu (Ashwini Deo and Rajesh Bhatt, p.c.)[16]

har	bacchaa	jis=ne	kitaab	paṛhiː,	skuul	nahĩː	aayaa
every	child.NOM	who.REL.OBL=ERG	book	read.PERF.F.SG	school	NEG	come.PERF.M.SG

'No child who read the book went to school.' [$\forall > \neg$; $?\neg > \forall$]

[15]Interestingly, the inverse-scope reading becomes more accessible when the subject is modified by a participial phrase rather than a finite relative clause, as in (i). In this case, the transparent-scope reading is also available and appears more salient than in examples where the QP is unmodified, such as (54). The difference between (i) and (58) might be attributed to distinct intonation patterns, but further research is required to confirm this hypothesis.

(i) Hindi-Urdu

kitaab	paṛhne-vaalaa	har	bacchaa	skuul	nahĩː	aayaa
book	read-PTCP.NOM.M.SG	every	child.NOM	school	NEG	come.PERF.M.SG

'Not every child who read the book went to school.' or:
'No child who read the book went to school.' [$\forall > \neg$; $\neg > \forall$]

[16]I am grateful to Ashwini Deo and Rajesh Bhatt for their judgments and valuable insights on the Hindi-Urdu data. Unless otherwise noted, all examples in this language are based on personal communication with them.

The second piece of evidence supporting the importance of linear precedence concerns scope preferences with other types of quantifiers. In Hindi-Urdu, while a subject universal quantifier typically takes scope below negation (54), the quantifier *bahut* 'many' tends to take scope above negation in an otherwise identical configuration, as illustrated in (59).

(59) Hindi-Urdu

bahut	bacchon=ne	kitaab	nahĩː	paṛhiː.
many	child-ERG	book	NEG	read.PERF.F.SG

'Many children didn't read the book.' ['many' > ¬ ; ?¬ > 'many']

Importantly, *bahut* is not a PPI. The reading where *bahut* scopes below negation becomes the preferred reading when the object is scrambled (60a) or occurs post-verbally (60b).[17]

(60) Hindi-Urdu

a.

kitaab	bahut	bacchon=ne	nahĩː	paṛhiː.
book	many	child-ERG	NEG	read.PERF.F.SG

'Not many children read the book.' ['many' > ¬ ; ¬ > 'many']

b.

bahut	bacchon=ne	nahĩː	paṛhiː	kitaab.
many	child-ERG	NEG	read.PERF.F.SG	book

'Not many children read the book.' ['many' > ¬ ; ¬ > 'many']

The contrast between universal quantifiers (54) and *bahut* 'many' (59) aligns with the Inverse-Scope Universal in (32), which states that languages are more likely to permit inverse scope in *all…not* constructions than in cases where the subject is headed by 'many'. As discussed in §2.1, some languages avoid inverse scope with 'many' by leveraging their lexical inventory—a strategy not attested for expressing 'Not all X are Y' propositions. In Hindi-Urdu, propositions of the form 'Not many X are Y' are often expressed using the NPI *zyaadaa* 'more' instead of *bahut* 'many', as demonstrated in (61).[18] Taken together, these data suggest that scope preferences likely cannot be explained solely by hierarchical relations.

[17] If the constructions in (60a) and (60b) indeed involve inverse scope, as this dissertation posits, these word order patterns may serve to reduce the distance between the quantifier and negation. This could potentially mitigate the processing difficulty associated with inverse scope. Testing this hypothesis, however, lies beyond the scope of the present study.

[18] *Zyaadaa* is considered an NPI because it cannot mean 'many' in affirmative contexts, where it instead signifies 'more'. This distinction is demonstrated by the following contrast:

(i) Hindi-Urdu

a.

aaj	zyaadaa	log	aaye	haĩ
today	more	people	come.PFV.M.PL	be.PRS.PL

'More people have come today.'

b.

aaj	zyaadaa	log	nahĩː	aaye	haĩ
today	more	people	NEG	come.PFV.M.PL	be.PRS.PL

'Not many people have come today.'

(61) Hindi-Urdu
zyaadaa bacchon=ne kitaab nahĩː paṛhiː.
more child-ERG book NEG read.PERF.F.SG
'Not many children read the book.'

In summary, there are compelling reasons to treat *all…not* constructions as a unified phenomenon, irrespective of the hierarchical relations between canonical subjects and negation within the clausal structure. The evidence suggests that linear precedence plays a significant role in shaping scope preferences, even in languages where negation is argued to occupy a higher structural position than the subject. Therefore, this study adopts a definition of inverse scope based on linear word order rather than hierarchical relations.

Chapter 3

Previous studies

This chapter provides an overview of previous research on inverse scope in *all...not* constructions and other types of structures involving scope ambiguity. It examines theoretical approaches that have been proposed to explain scope possibilities, factors influencing scope preferences, and empirical findings from corpus studies and language acquisition research on the use of *all...not* constructions. While much of this research has focused on English and a handful of other well-documented languages, the broader cross-linguistic picture remains uncertain, as comparatively little is known about how these constructions behave in other languages.

Section 3.1 discusses different theoretical perspectives on scope possibilities, particularly as they relate to *all...not* constructions. Subsection 3.1.1 discusses constraint-based accounts, which propose that inverse scope is generally dispreferred unless syntactic restrictions work against the transparent-scope alternative. These accounts highlight the role of competition in scope interpretation and argue that greater word order flexibility reduces the need for inverse scope. While these approaches share the assumption of a universal bias for scope transparency, they primarily focus on synchronic competition between different syntactic configurations. In contrast, this dissertation investigates how scope transparency operates as a diachronic force, influencing the long-term evolution of inverse- and transparent-scope constructions across languages.

Subsection 3.1.2 examines an alternative perspective, which treats scope restrictions as a manifestation of polarity sensitivity. Under this view, some universal quantifiers function as positive polarity items (PPIs) and resist scoping below negation, whereas others behave like negative polarity items (NPIs), exhibiting the opposite pattern. Subsection 3.1.3 examines the role of intonation, highlighting how prosodic cues help signal scope interpretations in languages such as English and German.

Section 3.2 examines empirical studies on scope ambiguity and processing. Subsection 3.2.1 discusses experimental research on scope ambiguity resolution, evaluating the relative contributions of syntactic, prosodic, and pragmatic factors in determining scope preferences. Subsection 3.2.2 reviews psycholinguistic studies on the processing of inverse scope. While some findings suggest that inverse scope is inherently more challenging to process, others argue that processing difficulty arises primarily when syntactic or contextual cues are absent.

Section 3.3 investigates the usage patterns of *all…not* constructions. Subsection 3.3.1 presents corpus-based findings on the relative frequencies of different interpretations of *all…not* constructions across languages. Subsection 3.3.2 examines how young children acquire these constructions, revealing that they sometimes produce inverse-scope constructions even when such patterns are absent from their linguistic input.

3.1 Theoretical perspectives on scope possibilities

3.1.1 Constraint-based accounts

Constraint-based accounts seek to explain restrictions on inverse scope by appealing to competition between alternative forms and a syntactic constraint that favors scope transparency.

One such account is proposed by Horn (1989, 228), who highlights the asymmetry in the availability of inverse-scope readings across different quantifier-negation combinations. While *all…not* constructions often allow inverse readings, *some…not* constructions typically do not (see §2.1). Horn attributes this asymmetry to competition between forms. In 'No X are Y' propositions, the negative quantifier *no*, as in *No one left*, blocks the truth-conditionally equivalent inverse-scope reading of *Somebody didn't leave*. By contrast, in 'Not all X are Y' propositions, no such blocking occurs, as 'not all' lacks a dedicated lexical form. Negated QP constructions, such as *Not everyone left*, are morphologically and syntactically more complex than negative quantifiers like *no*. Consequently, in the competition between *Not everyone left* and the inverse-scope construction *Everyone didn't leave*, the scales are less strongly tipped in favor of the transparent-scope alternative, allowing both forms to coexist.

Horn (1989, 498-499) also introduces a weak constraint, Nexal Not, which enables the inverse-scope reading of *all…not* constructions. The term "nexal negation", borrowed from Jespersen (1917), refers to clausal negation situated at the link between the subject and the predicate (the "nexus"). The Nexal Not constraint reflects a preference for the negation particle to appear in its unmarked syntactic position, as in *Everyone didn't leave*, rather than modify a QP, as in *Not everyone left*. Crucially, this preference conflicts with the general dispreference for inverse scope, leading to a tension between competing constraints. However, this tension does not extend to lexically incorporated negative quantifiers, such as *nobody*, where the Nexal Not constraint does not come into play.[1]

While Horn's account focuses on the interaction between negation and quantifiers, other scholars have examined inverse scope more broadly. Like Horn, these studies assume that inverse scope is generally dispreferred and that transparent-scope alternatives are favored whenever possible. According to these accounts, inverse scope arises as a last-resort strategy when a language lacks sufficient word order flexibility to express the intended interpretation

[1] Horn (p.c.) suggests that this analysis can be cast in terms of Optimality Theory, as a conflict between Markedness (favoring the unmarked position of negation) and Faithfulness (favoring scope transparency).

transparently or when other grammatical constraints interfere (Beck 1996; Szabolcsi 1997; Bobaljik and Wurmbrand 2012).

These accounts have primarily focused on sentences with multiple QPs, such as (62), where an object QP can take wide scope in some languages.

(62) At least one teacher talked to every student.
Transparent-scope reading: There is at least one (specific) teacher who talked to every student. [$\exists > \forall$]
Inverse-scope reading: Every student is such that at least one teacher talked to them (but not necessarily the same teacher). [$\forall > \exists$]

Constraint-based approaches posit that languages prefer to express scope relations transparently through the surface word order whenever possible. For example, in English, the inverse-scope reading of (62) is permitted because the transparent word order in (63) is impossible. In contrast, languages with more flexible word order, such as German, Hungarian, and Chinese, tend to avoid inverse scope through scrambling.

(63) *To every student at least one teacher talked.

Bobaljik and Wurmbrand (2012) argue that scope freezing (situations where only a transparent-scope interpretation is available) is construction-specific. That is, there is no scope rigidity parameter: a language may exhibit scope freezing with respect to one construction and scope flexibility with respect to another, since the availability of inverse scope depends on competition between alternative constructions, and in each case there is a different candidate set, and a different set of constraints are relevant.

To capture this, Bobaljik and Wurmbrand propose the Scope Transparency (ScoT) constraint in (64), which penalizes mismatches between scope at LF and word order at PF. The symbol '$\gg$' represents scope at LF and linear precedence at PF, the latter taken to be a proxy for c-command relations.

(64) *Scope Transparency* (ScoT) (Bobaljik and Wurmbrand 2012)
If the order of two elements at LF is $A \gg B$, the order at PF is $A \gg B$.

ScoT is a soft constraint, meaning that it can be violated when it interacts with another constraint. In other words, inverse scope is only permitted in cases where the transparent-scope alternative violates a different constraint. For example, (63) violates a constraint on scrambling, which is a hard constraint in English, rendering the inverse-scope interpretation in (62) acceptable. In other situations, ScoT interacts with another soft constraint, in which case both the inverse-scope candidate and the transparent-scope candidate are permitted as expressions of the same LF.[2]

[2]Notably, several experimental studies challenge a strong correlation between word order flexibility and scope rigidity. For example, Oikonomou et al. (2020) find that inverse scope is generally acceptable in Greek, despite its relatively flexible word order.

Similarly to the model proposed by Bobaljik and Wurmbrand (2012), this dissertation is predicated on the notion that inverse-scope constructions compete with alternative transparent-scope constructions that convey the same meaning. However, it diverges from Bobaljik and Wurmbrand (2012) in relaxing the assumptions about which constructions constitute the set of competitors.

Most constraint-based models of synchronic competition assume that the candidate set is restricted to PFs derived from the same LF (Müller and Sternefeld 2001). According to Bobaljik and Wurmbrand, the candidate set is further constrained to include only PFs derived from the same LF and numeration (i.e., the set of lexical items and their instances available for computation). For example, (62) and (63) belong to the same candidate set because they are both derivations of an LF where the universal QP takes wide scope, differing only in the overt or covert movements they involve. In contrast, sentences with the same meaning but derived from distinct numerations, such as active and passive constructions, are not considered part of the same candidate set. For example, the active sentence *At least one teacher interviewed every student* is argued not to compete with its passive counterpart, *Every student was interviewed by at least one teacher*.

Under this view, what other constructions can be considered part of the same candidate set as (65a) on its inverse-scope interpretation? The negated QP construction in (65b) does not qualify, as it involves constituent negation and is therefore derived from a distinct LF. Similarly, the cleft construction in (65c), with its more complex syntactic structure, would also fall outside the relevant candidate set. This leaves only a construction involving clausal negation with VS word order, as in (65d), as a plausible competitor derived from the same LF as (65a). However, since such a construction is not available in English, the set of competing alternatives appears to be notably restricted.

(65) a. Everyone couldn't pull this off.
b. Not everyone could pull this off.
c. It's not everyone who could pull this off.
d. *Couldn't everyone pull this off.

Thus, the notion that transparent-scope constructions like (65b) and (65c) directly compete with inverse-scope constructions like (65a) is incongruous with the model proposed by Bobaljik and Wurmbrand (2012). Relaxing the conditions on the candidate set to include any construction that conveys the same meaning risks an inflation of permissible candidates within this model, potentially predicting unattested inverse-scope readings.

In sum, the constraint-based approach is not directly applicable to the notion of competition envisaged in this dissertation. Instead, restrictions on inverse-scope interpretations of *all...not* constructions can be understood as emerging from diachronic competition between alternative ways of expressing the same meaning. Unlike synchronic constraints, which govern the range of available structures at a given point in time, diachronic competition reflects the gradual shift in usage patterns, where one form becomes increasingly preferred over another, eventually leading to systematic changes in acceptability and frequency.

Alternatively, one can deny the role of competition in certain cases of scope restrictions, and instead treat the unavailable scope interpretation as a polarity sensitivity phenomenon, as outlined in the next subsection.

3.1.2 Polarity sensitivity and scope restrictions

As discussed earlier, some languages restrict quantifier scope possibilities, allowing only transparent-scope or inverse-scope readings (see §2.2). Zeijlstra (2017, 2022) and Kamali and Zeijlstra (2024) propose to view such restrictions as part of the phenomenon of polarity sensitivity: universal quantifiers that obligatorily take wide scope over negation are treated as positive polarity items (PPIs), while those that scope below negation are negative polarity items (NPIs).

Zeijlstra (2017, 2022) observes that Dutch *iedereen* 'everyone' takes obligatory wide scope over clausal negation in main declarative clauses where it precedes negation, as shown in (66a). However, inverse scope becomes available in specific configurations: (i) when the clause is embedded under an NPI licenser, as in (66b); (ii) when an intervening element like *altijd* 'always' takes scope between *iedereen* and negation; or (iii) in echo denial (see §2.3).

(66) Dutch (Zeijlstra 2017, 15-17)

a. Iedereen vertrok niet.
everybody left NEG
'Nobody left.' [$\forall > \neg$; $*\neg > \forall$]

b. Het verbaast me dat iedereen niet blijft.
it surprises me that everybody NEG stays
'It surprises me that not everybody stays.' [$\neg > \forall$]

According to Zeijlstra, *iedereen* cannot reconstruct below negation because it is a PPI. However, it is an usual type of PPI, exhibiting polarity sensitivity only when it precedes negation. When *iedereen* follows negation, it can freely occur within its scope, as demonstrated in (67).

(67) Dutch (Zeijlstra 2022, 373)
Niet iedereen komt.
NEG everybody comes
'Not everybody comes.'

Zeijlstra proposes that the ability of *iedereen* to scope below negation depends on the attachment site of an exhaustivity operator that is responsible for the polarity sensitivity of *iedereen*. According to the exhaustification-based model of polarity sensitivity (Chierchia 2006, 2013; Crnič 2019), an NPI like *any* introduces subdomain alternatives, which need to be exhaustified by an **O** operator, which is the covert counterpart of *only*. In upward-entailing contexts, such as plain affirmative sentences, this exhaustification results in a contradiction, making the sentence unacceptable.

Given the duality of existential and universal quantifiers, a universal quantifier with similar properties would behave as a PPI. According to Zeijlstra, *iedereen* indeed functions as a PPI but can still scope below negation under certain conditions. Specifically, while NPIs are licensed when a DE operator intervenes between the exhaustifier and the NPI (68a), PPIs are generally unacceptable in this configuration (68b). However, PPIs become acceptable when the exhaustifier attaches below the DE operator, creating a licit configuration (68c).

(68) a. ✓ EXH > DE > NPI
b. ✗ EXH > DE > PPI
c. ✓ DE > EXH > PPI

When *iedereen* follows negation, as in (67), the exhaustifier can attach between negation and the PPI, resulting in the licit configuration in (68c). In contrast, when *iedereen* precedes negation, the exhaustifier must attach above negation, yielding the illicit configuration in (68b) if reconstruction occurs. The critical constraint accounting for the position of the exhaustifier is as follows:

(69) In any sentence with an NPI/PPI that requires checking by a covert exhaustifier, this exhaustifier must occupy a position where it c-commands the NPI/PPI at surface structure. (Zeijlstra 2017, 12)

When *iedereen* follows negation, the exhaustifier can attach low enough to c-command *iedereen* while remaining below negation. However, when *iedereen* precedes negation, the exhaustifier must attach higher, above negation, to satisfy the c-command constraint. This structural difference explains the obligatory wide scope of *iedereen* in sentences like (66a).

For universal quantifiers that obligatorily take narrow scope below negation, such as Turkish *her* 'every' and *bütün* 'all'—exemplified in (70)—Kamali and Zeijlstra (2024) propose an alternative explanation.

(70) Turkish (Kamali 2011, 188)
Herkes o test-e gir-me-di.
everyone that test-DAT enter-NEG-PST
'Not everyone took that test.' [$*\forall > \neg$; $\neg > \forall$]

Their account builds on previous analyses of "superweak NPIs", such as Greek *típota* 'anything' and Mandarin *shenme* 'what, any', which are licensed not only in DE contexts but also in other non-veridical environments, such as under possibility modals (Giannakidou 1998; Lin 1998). According to these analyses, superweak NPIs cannot appear in contexts that entail the existence of a referent satisfying the predicate. For example, the sentence *Maybe John bought something to eat* does not entail the existence of something that John actually bought to eat, allowing the licensing of a superweak NPI in position of the indefinite pronoun. By contrast, weak NPIs like *any* require a DE environment for licensing and would not be permitted in this context.

Kamali and Zeijlstra propose that Turkish universal quantifier lexically encode the opposite condition of superweak NPIs: they cannot appear in contexts that entail the *non*-existence of a referent possessing the relevant property. For example, the transparent-scope reading in (70) is unavailable because it entails the non-existence of a student who will walk. On the other hand, the inverse-scope reading is possible because it is compatible with the existence of such a student.

In summary, the polarity sensitivity approach offers a framework for analyzing restrictions on scope interpretations. This approach can be seen as complementary to the perspective of this dissertation, which focuses on the synchronic and diachronic factors driving these restrictions and the broader cross-linguistic patterns they reveal.

3.1.3 The role of intonation in scope interpretation

In English, at least in certain dialects, the inverse-scope reading is characterized by a rise-fall-rise intonation contour (Jackendoff 1972, 352), as illustrated in (71b). In contrast, the transparent-scope reading typically features a standard final fall, as in (71a), which is characteristic of assertions. Similarly, in German, the inverse-scope reading is associated with a rise-fall contour, also known as the Hat Contour (Jacobs 1984; Büring 1997), as shown in (72b). Experimental studies have confirmed that these intonation contours reliably guide listeners toward the inverse-scope interpretation in both English (Syrett et al. 2014) and German (Yatsushiro et al. 2019).[3]

(71) a. /ALL\ the men didn't go\. [$\forall > \neg$]
b. /ALL\ the men didn't go/. [$\neg > \forall$]

(72) German (Büring 1997)
a. /ALLE\ Politiker sind nicht korrupt\.
all politicians are NEG corrupt
'No politician is corrupt.' [$\forall > \neg$]
b. /ALLE Politiker sind NICHT\korrupt.
'Not all politicians are corrupt.' [$\neg > \forall$]

Following Jackendoff (1972), Büring (1997) argues that the rise accent on the quantifier in (72b) signals a contrastive topic—an aboutness topic that contains a focus. In Alternative Semantics (Rooth 1985, 1992), focus indicates the relevance of alternatives. For instance, in (73), the focus on *John* implies the presence of an alternative topic, such as Bill. Importantly, a sentence with a contrastive topic can only be felicitously used when it does not resolve the QUD (Büring 2003, 2016). For instance, B's answer in (73) is merely a partial answer to A's question.

[3] More precisely, Yatsushiro et al. (2019) argue that the neutral intonation contour in (72a) is, in fact, ambiguous, whereas the rise-fall contour in (72b) unambiguously signals the inverse-scope interpretation.

(73) A: Where were John and Bill at the time of the robbery?
B: Well, I know that $[[\text{John}]_F]_T$ was at $[\text{the barbershop}]_F$.

Büring (1997) argues that in (72b), the alternatives for the contrastive topic are other quantifiers, such as *most*, *many*, *some*, or *no*. This gives rise to the alternative set in (74). The transparent-scope reading ('No politician is corrupt') is ruled out because it fully resolves the QUD, as it either entails or contradicts all the other alternatives. In contrast, the inverse-scope reading ('Not all politicians are corrupt') leaves the QUD unresolved, allowing for follow-up questions like, "But how many politicians are, in fact, corrupt?"

(74) Alternative set: {All politicians are (not) corrupt, Most politicians are (not) corrupt, Many politicians are (not) corrupt, Some politicians are (not) corrupt}

By contrast, Krifka (1998) points out that the rise-fall contour in German is not particular to *all…not* sentences but is also observed in inverse-scope constructions involving multiple QPs, where residual uncertainty does not seem to play a role. According to Krifka, the contrastive topic marking is not employed to indicate an unsettled QUD but to enable overt movement of a QP to a high position at [Spec, CP].[4]

Furthermore, it is important to note that not all languages employ special intonation contours in inverse-scope constructions. For example, most speakers of Turkish only allow an inverse-scope reading in sentences like (75a) (Öztürk 2005; Kelepir 2001; Kamali and Zeijlstra 2024). Yet, Kamali (2011, 188-189) observes that if the universal quantifier bears a pitch accent, the sentence becomes unacceptable (75b). This suggests that the inverse-scope reading requires neutral intonation in Turkish. Interestingly, the quantifier can bear a pitch accent in the corresponding affirmative sentence (75c), indicating that the contour itself is not inherently disallowed. Rather, accenting the quantifier in (75b) seems to signal an attempt to enforce a transparent-scope reading, which is unavailable in Turkish.

(75) Turkish (Kamali 2011, 188-189)

a. Herkes o test-e gir-me-di.
everyone that test-DAT enter-NEG-PST
'Not everyone took that test.' [$*\forall > \neg$; $\neg > \forall$]

b. ?HERKES o test-e gir-me-di.
everyone that test-DAT enter-NEG-PST
(Intended: 'Not EVERYONE took that test.') [$*\forall > \neg$; $*\neg > \forall$]

c. HERKES o test-e gir-di.
everyone that test-DAT enter-PST
'EVERYONE took that test.'

[4] See Büring (2003); Sauerland (2005); Wagner (2008) for analyses of the German rise-fall contour that align with the approach of Büring (1997), as well as Constant (2012) for a related analysis of the English rise-fall-rise contour in (71b). By contrast, see Wurmbrand (2008) and Bobaljik and Wurmbrand (2012) for additional arguments in support of the syntactic view.

A similar phenomenon is observed in Czech. In this language, inverse scope is the preferred but not exclusive reading of *all…not* constructions. As in Turkish, the quantifier is unaccented in the inverse-scope reading. However, unlike Turkish, accenting the quantifier in Czech makes the transparent-scope reading accessible (Grygar-Rechziegel 1980, 114–115).

These data indicate that contrastive topic marking is not an essential feature of inverse-scope constructions. Furthermore, as Horn (1989, 231) argues, it is unlikely that the inverse-scope interpretation arises from conscious reasoning based on the meaning of the rise-fall(-rise) contour. Instead, some level of conventionalization is likely at play, suggesting that any comprehensive account of the inverse scope phenomenon must consider factors beyond the contribution of intonational patterns.

It's also worth noting that inverse-scope constructions are relatively rare in English, and even more so in German (Tottie and Neukom-Hermann 2010; Neukom-Hermann 2016, also see §6.1.1-6.1.2). In these languages, a special intonation contour might be needed to highlight a less salient reading. Alternatively, as Horn (1989, 496) suggests, such intonation contours may primarily serve specific discourse functions, such as echo denial, which are prominent contexts for inverse-scope constructions in English and German. Supporting this view, Horn notes that no special intonation is required in formulaic expressions like *All is not lost*, implying that the rise-fall-rise contour is likely tied to the typical discourse functions of inverse-scope constructions in English rather than to the inverse-scope interpretation itself.

While these studies highlight the role of intonation in scope disambiguation in certain languages, this dissertation does not focus on intonational patterns, primarily because the data come from a corpus of written texts. As a result, the influence of prosodic cues on inverse-scope constructions across languages remains an open question.

3.2 Empirical studies on scope ambiguity and processing

3.2.1 Resolving scope ambiguity

Experimental studies on scope ambiguity resolution have sought to disentangle the effect of syntactic and pragmatic factors in shaping scope preferences.

Conroy et al. (2008) show that English-speaking adults exhibit a preference for a transparent-scope interpretation when time-constrained, suggesting that this is the default interpretation in real-time sentence processing. Further supporting the role of syntax, Syrett et al. (2014) demonstrate that prosody can effectively disambiguate between transparent-scope and inverse-scope interpretations within the same context. They argue that this disambiguation is possible because the two interpretations correspond to distinct syntactic parses, rather than being solely derived from the contextual plausibility of these interpreations.[5]

[5] As Chen and van Tiel (2021, 262) point out, however, listeners can use prosody to infer the likely QUD, suggesting that the disambiguation effect of prosody might alternatively be viewed as pragmatic in nature.

Other studies emphasize the critical role of pragmatic factors in resolving scope ambiguity. Scontras and Pearl (2021) propose a computational model of ambiguity resolution in which syntactic and pragmatic elements are treated as distinct yet interacting components. The model assumes that the interpretation of *all…not* constructions is determined by three factors: (i) the syntactic parse; (ii) prior expectations about the state of the world; and (iii) the QUD. Their simulation results suggest that pragmatic factors play a more substantial role in ambiguity resolution than grammatical processing biases toward specific syntactic parses.

Building on this, Chen and van Tiel (2021) tested the model by comparing participants' ratings in a truth-value judgment task that manipulated prior probabilities and the salience of different QUDs. Their findings support the view that pragmatic considerations, particularly prior expectations about the state of the world, are the primary drivers of scope ambiguity resolution.

As Chen and van Tiel (2021, 261) point out, however, syntactic constraints must also play a significant role in resolving scope ambiguity, especially given the cross-linguistic variation in this domain. For instance, Chen and van Tiel (2021) report that English speakers show a preference for the inverse-scope reading.[6] By contrast, Song et al. (2021) found that Mandarin speakers consistently reject the inverse-scope interpretation, regardless of manipulations to the priors or QUD. This cross-linguistic difference suggests that scope ambiguity—or its absence—is fundamentally rooted in syntax. Pragmatic factors do not create the ambiguity; they only guide listeners toward one of the interpretations licensed by syntax.

Another line of investigation in this area concerns how children interpret scope ambiguity and whether their preferences align with those of adults. Research on scope ambiguity resolution has revealed systematic differences between children and adults, particularly in their ability to access inverse-scope interpretations. For example, while adults can interpret sentences like *Every horse didn't jump over the fence* to mean either that none of the horses succeeded (transparent scope) or that not all of them succeeded (inverse scope), children often struggle to access the latter interpretation.

Processing limitations have been proposed as a key factor in children's difficulty with inverse scope. Musolino (1998, 2006) suggests that the transparent-scope interpretation is easier to process, making it more accessible. According to this view, children tend to prefer this interpretation due to their relatively limited processing ability.

Pragmatic factors also play a significant role in shaping children's scope preferences. Studies have shown that experimental manipulations influencing the QUD can significantly alter children's scope interpretation (Gualmini 2004; Musolino and Lidz 2006; Gualmini 2008; Viau et al. 2010). For example, when the context highlights the QUD "Did every horse jump

[6] The distinction between *every* and *all* may influence scope preferences. Chen and van Tiel (2021) focused on sentences involving *every*, such as *Every essay hasn't been graded*, while Hemforth and Konieczny (2019) examined sentences with *all*, reporting a preference for the transparent-scope interpretation. This suggests that the choice of universal quantifier might affect the relative salience of scope interpretations (see Beghelli and Stowell 1997). However, as discussed in §1.2, this study does not systematically differentiate between quantifiers corresponding to 'all' and 'every', as both typically exhibit similar patterns regarding the availability of inverse scope.

over the fence?", children are more likely to endorse an inverse-scope interpretation of the sentence *Every horse didn't jump over the fence*. Scontras and Pearl (2021) demonstrate that children's non-adult-like scope interpretations largely stem from challenges in managing the pragmatic context, such as identifying the salient QUD or adapting to experimental setups.

Thus, it remains an open question whether children's scope preferences are indicative of a bias for scope transparency or are primarily shaped by prior assumptions about the plausibility of certain types of statements.

3.2.2 Processing inverse scope

Psycholinguistic evidence offers mixed support for the claim that there is a bias for scope transparency in sentences with multiple QPs. In self-paced reading tasks, sentences with inverse-scope interpretations, such as (62), show slower reading times, indicating greater processing difficulty (Tunstall 1998; Anderson 2004; Bott and Schlotterbeck 2015). Wurmbrand (2018) further argues that inverse scope becomes progressively less acceptable as quantifier raising (QR) spans more syntactic domains. For instance, QR within simple predicates is relatively easy, but still costly, whereas inverse scope across clausal boundaries is significantly harder to process.

However, not all studies find inverse scope to be inherently costly. Brasoveanu and Dotlačil (2015) demonstrate that processing difficulty can be mitigated when overt cues in the sentence signal scope relations. For instance, in (76a), the non-anaphoric reading of *same* (known as the internal reading) requires the presence of a wide-scope, semantically plural noun phrase. This provides the parser with an early cue about an upcoming, wide-scope QP. Indeed, Brasoveanu and Dotlačil did not find evidence for processing difficulty in such cases. By contrast, there is no similar cue in (76b), making this construction more difficult to process.

(76) (Brasoveanu and Dotlačil 2015)
  a. The same student saw every movie.
  b. A student saw every movie.

Brasoveanu and Dotlačil argue, following Fodor (1982), that the difficulty of inverse scope arises only when it requires revising an already established discourse model. In an incremental processing of (76b), for instance, the indefinite *a student* initially introduces a discourse referent by default. When this indefinite is later understood to be bound by the universal quantifier, the discourse model must be revised, incurring a processing cost. Conversely, in constructions like (76a), no such revision is necessary because the parser anticipates a wide-scope QP.

Inverse-scope in *all…not* constructions received less attention in self-paced reading studies. Lee (2009) finds that Korean speakers prefer the transparent-scope interpretation in a truth-value judgment task, and that contexts favoring the inverse-scope interpretation lead to slower reaction times and reading pace. The same pattern is found for English speakers, contrasting with other studies that report a preference for the inverse-scope interpretation in English (see §3.2.1).

To establish a stronger case that inverse scope in *all…not* constructions is inherently associated with processing difficulty, it would be necessary to demonstrate similar effects in languages where inverse-scope constructions are more frequently employed. Without such evidence, the slower reading times observed in Lee (2009) might simply reflect the marginal acceptability of the constructions in question rather than a universal processing difficulty. To the best of my knowledge, no study has yet investigated this phenomenon in languages where inverse-scope constructions are more prevalent. Thus, it remains premature to conclude that inverse scope in *all…not* constructions is intrinsically difficult to process.

3.3 *All…not* constructions in language use

3.3.1 Corpus studies of *all…not* constructions

Corpus-based studies of *all…not* constructions remain relatively scarce. Languages that have been examined include British English (Tottie and Neukom-Hermann 2010; Neukom-Hermann 2016), German (Neukom-Hermann 2016), Hebrew (Amiraz 2019), and Dutch (van Kampen 2023). Unlike this dissertation, these studies take a form-based (semasiological) approach, focusing on the interpretation of *all…not* constructions rather than studying the choice between inverse-scope and transparent-scope constructions for expressing 'Not all X are Y' proposition.

Tottie and Neukom-Hermann (2010) study the use of *all…not* constructions in the British National Corpus. Following Taglicht (1985), they demonstrate that these constructions can yield three distinct readings: inverse scope, transparent scope, and a collective reading (e.g., *All the bills don't amount to $50*, where the sum of the bills taken together fails to reach this amount).[7] Their analysis reveals that inverse scope is the most frequent interpretation (54%), followed by the collective reading (28%), and the transparent-scope reading (18%).

[7] As an anonymous reviewer pointed out, the collective reading resembles the inverse-scope reading in that negation takes wide scope over the quantified proposition. This can be seen in the following paraphrases of the classic examples discussed by Tottie and Neukom-Hermann (2010):

(i) All the king's horses and all the king's men
Couldn't put Humpty together again.
≈ It is not the case that [all the king's horses and all the king's men could (collectively) put Humpty Dumpty together again].

(ii) All the perfumes of Arabia will not sweeten this little hand.
≈ It is not the case that [all the perfumes of Arabia will (collectively) sweeten this little hand].

At the same time, the collective reading typically implies that the distributive transparent-scope reading is true as well. For instance, (i) implies that nobody could (individually) put Humpty Dumpty together again. In this respect, it differs from the distributive inverse-scope reading ($\neg > \forall$), which is logically weaker than the transparent-scope reading ($\forall > \neg$). The collective interpretation therefore stands in a complex relation to the other two. This may also explain why this interpretation remains available even in languages that

Tottie and Neukom-Hermann identify several factors influencing scope interpretation: (i) the syntactic function of *all* (pronominal vs. predeterminer); (ii) the complexity of noun phrases quantified by *all*; and (iii) the formulaicity of expressions. Pronominal *all* strongly favors inverse-scope readings, while predeterminer *all* exhibits greater ambiguity. Additionally, contextual factors, including discourse cues and contrastive markers like *but* or *however*, play a critical role in disambiguating scope. Notably, the inverse-scope and collective readings are often used for emphatic denial (including echo denial) or self-correction.

Despite the relatively high proportion of inverse-scope interpretations, a substantial number of these instances (55%) occur in formulaic expressions, such as *All is not well*. This stands in sharp contrast to the competing negated QP construction *not all*, which is rarely formulaic: "[The British National Corpus] does not contain a single instance of *not all is/was lost* [...] [s]imilarly, there are no instances of *not all is/was well*" (Tottie and Neukom-Hermann 2010, 162-163).

They suggest that historical factors may have contributed to the prevalence of formulaic *all...not* constructions with inverse scope. This hypothesis is supported by the results of the present study: inverse-scope constructions predate negated QP constructions by several centuries in the history of English (see §10.1). As Tottie and Neukom-Hermann point out, idioms such as *All that glitters is not gold* entered the language at least as early as Chaucer's time (late 14th century), and historical persistence likely accounts for the continued presence of formulaic expressions like *All is not lost* in present-day English.

Neukom-Hermann (2016, 208) compares the prevalence of different readings of *all...not* construction in British English with their frequencies in two German corpora (Korpus C4 and deWaC). Her findings reveal a significant cross-linguistic difference: in German, the inverse-scope interpretation accounts for only about 10% of the examples, while the transparent-scope reading is far more common at 57%—a considerably higher proportion than in English. In contrast, the collective reading shows no major differences in frequency between the two languages.[8]

Amiraz (2019) studies *all...not* constructions in Hebrew using several corpora of written language. He finds that the transparent-scope reading is by far the most frequent one (82%), followed by the collective reading (11%), and the inverse-scope reading (5%), with the remaining cases classified as unclear. Furthermore, the results indicate that the inverse-scope reading is restricted to specific contexts, as previously discussed in §2.3: when the clause is embedded in a DE context or contains an intervener, in echo denial, and with *same*-type collective predicates. These findings demonstrate that Hebrew typically avoids inverse-scope constructions, employing them only in well-defined contexts.

normally disallow the inverse-scope reading, such as Hebrew (Amiraz 2019). Further exploration of this type of interpretation is left for future research.

[8]It is worth noting that Neukom-Hermann's statistics include instances where the universal quantifier precedes negation but is not in subject position, such as topicalized object QPs in clause-initial position. This might explain why inverse-scope constructions are more frequent in German in her results compared to the findings of the present dissertation (see §6.1.2).

Similarly, van Kampen (2023) demonstrates that adult speakers of Dutch predominantly use *all…not* constructions with a transparent-scope interpretation. Her study focuses on the divergence between adult and child language use (see §3.3.2), drawing data primarily from the CHILDES database, which largely represents child-directed speech. Notably, the examples do not include instances of inverse scope, consistent with existing claims in the literature (Zeijlstra 2017). As van Kampen points out, while inverse-scope constructions are attested in many Dutch dialects (Barbiers et al. 2008), they are absent from the speech of the caregivers represented in the CHILDES database.

In sum, previous corpus studies reveal notable differences in the interpretation of *all…not* constructions across languages. These findings indicate that, in some (but not all) languages, inverse-scope constructions tend to be associated with specific contexts, such as formulaic expressions, contrast with a preceding utterance, embedded environments, and certain types of predicates. This suggests that the availability of inverse scope is not a binary property but rather exists on a continuum.[9]

3.3.2 Acquisition of *all…not* constructions

While much of the research on scope ambiguity in acquisition has focused on comprehension (see §3.2.1), van Kampen (2023) shifts the focus to production, analyzing children's spontaneous speech. This work examines the production of *all…not* constructions in child and adult Dutch, using data from the CHILDES database. It reveals a striking difference between children and adults. In standard adult Dutch, *all…not* constructions always exhibit a transparent-scope interpretation (see Zeijlstra 2017). In contrast, children frequently produce utterances that reflect an inverse-scope interpretation, such as (77). Importantly, this inverse-scope preference in child Dutch is not mirrored in the input they receive from caregivers, suggesting that it is not simply modeled on adult speech.

(77) Dutch (van Kampen 2023, 12)
Alle kinderen lusten toch geen pepernoten.
all children like PRT no pepernoots
'Not all children like Dutch ginger nuts.' (age 4;06)

In total, children produced 20 transparent-scope constructions and 18 inverse-scope constructions. Van Kampen does not specify the syntactic position of the quantifier in these cases, making it unclear how often the quantifier appears in subject position—the primary focus of this dissertation. Nevertheless, the data suggest no strong preference for scope transparency in the production of 'Not all X are Y' constructions.

Anecdotal evidence suggests a similar pattern in Hebrew, a language where *all…not* constructions consistently exhibit a transparent-scope interpretation in adult usage (Amiraz 2019). Unlike adults, children occasionally produce inverse-scope constructions, such as in (78).

[9] See §5.4 for a discussion of the correlation between usage frequency and acceptability judgments.

(78) Hebrew

a. hakol lo nigmar.
everything NEG finish.PST.3SG
(Pointing to leftover food on a plate:) 'It's not all gone.' (age 2;10)

b. kol ha-kaki lo yaca.
all the-poop NEG come.out.PST.3SG
'Not all the poop has come out.' (age 3;03)

c. kulam lo meaxor-ay, xelek milfan-ay, xelek meaxor-ay.
everyone NEG behind-1SG part in.front-1SG part behind-1SG
'Not everyone is behind me: some are in front of me, some behind me.' (age 4:10)

Children's tendency to produce inverse-scope constructions, even when these are absent in their input, may reflect a reliance on the standard negation construction as the default strategy for forming negative clauses. This likely occurs because they have not yet fully acquired the negated QP construction, the most frequent transparent-scope construction in both Dutch and Hebrew. Instead, children resort to the standard negation construction, which is more accessible due to its high frequency. Thus, their use of inverse scope does not necessarily indicate a lack of bias for scope transparency; rather, it reflects the limitations of their developing syntactic resources in consistently avoiding inverse scope.

In sum, while children are more reluctant than adults to interpret *all…not* constructions with an inverse-scope interpretation in comprehension tasks, they often produce inverse-scope constructions in spontaneous speech. This suggests that inverse-scope constructions may, in some cases, be easier to produce because they rely on the highly accessible standard negation construction. Conversely, transparent-scope constructions may facilitate comprehension due to the correspondence between form and meaning.

3.4 Summary

This chapter reviewed previous research on inverse scope in *all…not* constructions and other types of scope-ambiguous structures. Constraint-based accounts assume a general dispreference for inverse scope (§3.1.1), arguing that languages favor constructions where surface structure transparently reflects scope relations. This assumption is largely supported by processing studies, which often find that inverse-scope readings impose greater cognitive load, reinforcing the idea that scope transparency is the default preference (§3.2.2). Similarly, research on the comprehension of *all…not* constructions indicates that children favor transparent-scope interpretations, possibly suggesting that the availability of inverse-scope readings is a late-acquired property (§3.2.1).

However, acquisition studies on the production of *all…not* constructions present a more complex picture. Children sometimes produce inverse-scope constructions even when these are rare or even absent in their linguistic input. This suggests that scope preferences are influenced not only by an inherent bias for transparency but also by structural factors, such as the syntactic accessibility of negation constructions and the relative ease of forming inverse-scope or transparent-scope structures in spontaneous speech (§3.3.2).

Taken together, these findings suggest that while scope transparency influences both scope preferences and real-time processing, its strength and universality remain uncertain. This dissertation fills these gaps by adopting a broad cross-linguistic perspective and incorporating a diachronic dimension.

Part II

Materials and methodology

Chapter 4

Language sample

This chapter provides the necessary background for interpreting the cross-linguistic findings in subsequent chapters, clarifying how the language sample was constructed and what methodological constraints should be considered when evaluating the results.

Section 4.1 outlines the composition of the language sample, which consists of two distinct datasets: a synchronic sample, comprising NT translations into 217 languages published after 1950, and a diachronic sample, which includes historical translations into Germanic and Romance languages dating back to the Middle Ages. The synchronic sample serves as the primary dataset for the statistical analysis in Chapter 8, while the diachronic sample provides insights into historical shifts in strategy usage, which are the focus of Chapter 10.

Section 4.2 details the sampling methodology, which is designed to balance genealogical and areal diversity while minimizing potential biases in the dataset. Since linguistic patterns are often shaped by genealogical inheritance and language contact, the selection of languages must ensure broad representation across language families and macro-areas. Following Dryer (1989), this study uses genera—subgroupings of languages within families that share a common ancestor within the past 3,500–4,000 years—as the primary unit of comparison. To further account for areal effects, the distribution of strategies is analyzed separately within each macro-area. This approach reduces the risk of overrepresenting large language families while ensuring that findings are not skewed by areal diffusion. The limitations of the sample, particularly the underrepresentation of certain linguistic areas due to data availability, are also discussed.

4.1 Composition of the language sample

Drawing reliable conclusions about the cross-linguistic frequencies of inverse-scope and transparent-scope constructions requires a large, representative sample of languages. This is particularly crucial due to the potential influence of both genealogical and areal factors, which can shape the distribution and prevalence of these constructions.

This study relies on two distinct language samples to investigate the cross-linguistic frequencies of inverse-scope and transparent-scope constructions: a synchronic sample and a diachronic sample. Both samples are drawn from translations of the NT, supplemented in some cases by secondary sources.

The synchronic sample consists of NT translations into 217 languages, defined as translations published after the year 1950.[1] This sample forms the basis of the statistical analysis in Chapter 8. It is essentially a convenience sample, constrained by the availability of suitable data and the feasibility of conducting syntactic analysis using accompanying grammars and dictionaries. Despite these limitations, the sample is genealogically and areally diverse, covering 60 language families distributed across six macro-areas. Table 4.1 provides an overview of the genealogical and areal composition of the sample.[2]

Macro-area	Language families	Languages
Africa	13	65
Australia	1	2
Eurasia	15	66
North America	13	29
Papunesia	9	32
South America	17	23
Total	**60**	**217**

Table 4.1: Overview of the areal and genealogical coverage of the language sample

Australia is underrepresented in the sample, with data from only two English-lexified creole languages. This limited representation is primarily due to the nature of the data available. Translations of the NT into Australian languages are often paraphrases rather than direct translations, which poses significant challenges for identifying the relevant constructions. Consequently, while Australia is mentioned in Table 4.1 for completeness, these languages are excluded from the statistical analysis.

Table 4.2 provides a hierarchical breakdown of the synchronic sample by macro-area, top-level family, language, and Glottocode.

Macro-area	Top-level family	Language	Glottocode
Africa	Afro-Asiatic	Amharic	amha1245
		Borana-Arsi-Guji Oromo	bora1271
		Hausa	haus1257
		Moroccan Arabic	moro1292
		Somali	soma1255
		Tawallammat Tamajaq	tawa1286
		Tunisian Arabic	tuni1259
	Atlantic-Congo	Bemba (Zambia)	bemb1257
		Bulu (Cameroon)	bulu1251

[1] The sole exception is Cherokee, where the existing translation was published in 1860. It is included to enhance the representation of languages from northern North America.

[2] The total number of language families does not match the aggregate from each macro-area due to language families that span multiple macro-areas.

Macro-area	Top-level family	Language	Glottocode
		Chokwe	chok1245
		Ewondo	ewon1239
		Ganda	gand1255
		Gogo	gogo1263
		Hausa States Fulfulde	nige1253
		Igbo	nucl1417
		Kikuyu	kiku1240
		Kimbundu	kimb1241
		Kinyarwanda	kiny1244
		Kituba (Democratic Republic of Congo)	kitu1246
		Kuanyama	kuan1247
		Lingala-Bangala	ling1269
		Luba-Lulua	luba1249
		Lunda	lund1266
		Luvale	luva1239
		Makaa	maka1304
		Makhuwa	makh1264
		Makonde	mako1251
		Nande	nand1264
		Ndonga	ndon1254
		Nyakyusa-Ngonde	nyak1260
		Nyamwezi	nyam1276
		Nyanja	nyan1308
		Pular	pula1262
		Sango	sang1328
		Sena	nucl1396
		Shona	shon1251
		Sukuma	suku1261
		Supyire Senoufo	supy1237
		Swahili	swah1253
		Tetela	tete1250
		Tumbuka	tumb1250
		Umbundu	umbu1257
		Xhosa	xhos1239
		Yoruba	yoru1245
		Zimbabwean Ndebele	nort2795
		Zulu	zulu1248
	Austronesian	Plateau Malagasy	plat1254
	Central Sudanic	Ngiti	ngit1239
	Ijoid	Kalabari	kala1381
	Indo-European	Afrikaans	afri1274
		Krio	krio1253
		Morisyen	mori1278
		Nigerian Pidgin	nige1257
		Seselwa Creole French	sese1246
	Khoe	Nama (Namibia)	nama1264
	Koman	Uduk	uduk1239
	Mande	Guinea Kpelle	guin1254
	Nilotic	Acoli	acol1236
		Kumam	kuma1275

Macro-area	Top-level family	Language	Glottocode
		Lango (Uganda)	lang1324
		Luo (Kenya and Tanzania)	luok1236
	Songhay	Koyraboro Senni Songhai	koyr1242
		Zarma	zarm1239
	Surmic	Didinga	didi1258
	Ta-Ne-Omotic	Wolaytta	wola1242
Australia	Indo-European	Kriol	krio1252
		Torres Strait-Lockhart River Creole	torr1261
Eurasia	Afro-Asiatic	Maltese	malt1254
	Austroasiatic	Central Khmer	cent1989
		Khasi	khas1269
		Nuclear Waic	waaa1245
		Santali	sant1410
		Vietnamese	viet1252
		Central Malay	mala1479
		Jarai	jara1266
		Standard Malay	stan1306
	Dravidian	Malayalam	mala1464
	Hmong-Mien	Hmong Daw	hmon1333
	Indo-European	Belarusian	bela1254
		Bengali	beng1280
		Breton	bret1244
		Bulgarian	bulg1262
		Catalan	stan1289
		Corsican	cors1241
		Czech	czec1258
		Danish	dani1285
		Dutch	dutc1256
		Eastern Armenian	nucl1235
		English	stan1293
		French	stan1290
		Friulian	friu1240
		Galician	gali1258
		German	stan1295
		Icelandic	icel1247
		Irish	iris1253
		Italian	ital1282
		Latvian	latv1249
		Lithuanian	lith1251
		Luxemburgish	luxe1243
		Macedonian	mace1250
		Modern Greek	mode1248
		Northern Tosk Albanian	tosk1239
		Norwegian	norw1258
		Occitan	occi1239
		Polish	poli1260
		Portuguese	port1283
		Romanian	roma1327
		Russian	russ1263
		Scottish Gaelic	scot1245

Macro-area	Top-level family	Language	Glottocode
		Serbian-Croatian-Bosnian	sout1528
		Slovak	slov1269
		Slovenian	slov1268
		Spanish	stan1288
		Swedish	swed1254
		Ukrainian	ukra1253
		Welsh	wels1247
		West Low German	west2357
		Western Farsi	west2369
	Isolates	Basque	basq1248
	Japonic	Japanese	nucl1643
	Kartvelian	Georgian	nucl1302
	Koreanic	Korean	kore1280
	Mongolic-Khitan	Halh Mongolian	halh1238
	Sino-Tibetan	Burmese	nucl1310
		Hmar	hmar1241
		Mandarin Chinese	mand1415
		Mizo	lush1249
	Tai-Kadai	Thai	thai1261
	Turkic	North Azerbaijani	nort2697
		Tatar	tata1255
	Uralic	Estonian	esto1258
		Finnish	finn1318
		Hungarian	hung1274
North America	Algic	Northwestern Ojibwa	nort2961
	Athabaskan-Eyak-Tlingit	Navajo	nava1243
		Western Apache	west2615
	Chibchan	Ngäbere	ngab1239
	Huavean	San Mateo del Mar Huave	sanm1287
	Indo-European	Belize Kriol English	beli1260
		Haitian	hait1244
		Hutterite German	hutt1235
		Jamaican Creole English English	jama1262
		Pennsylvania German	penn1240
		Plautdietsch	plau1238
		Saint Lucian Creole French	sain1246
		San Andres Creole English	sana1297
		Sea Island Creole English	gull1241
	Iroquoian	Cherokee	cher1273
	Mayan	K'iche'	kich1262
		Yucatec Maya	yuca1254
	Misumalpan	Mískito	misk1235
	Mixe-Zoque	Coatlán Mixe	coat1238
	Otomanguean	Cajonos Zapotec	cajo1238
		Chayuco Mixtec	chay1249
		Jamiltepec Mixtec	jami1235
		Zoogocho Zapotec	zoog1238
	Tarascan	Western Highland Purepecha	west2631
	Totonacan	Upper Necaxa Totonac	uppe1275
	Uto-Aztecan	Central Guerrero Nahuatl	guer1241

Macro-area	Top-level family	Language	Glottocode
		Central Huasteca Nahuatl	cent2134
		Eastern Huasteca Nahuatl	east2538
		Huichol	huic1243
Papunesia	Austronesian	Acehnese	achi1257
		Balinese	bali1278
		Batak Karo	bata1293
		Fijian	fiji1243
		Hiri Motu	hiri1237
		Iban	iban1264
		Kupang Malay	kupa1239
		Madurese	nucl1460
		Malayic Dayak	mala1480
		Manado Malay	mala1481
		Minangkabau	mina1268
		Motu	motu1246
		North Moluccan Malay	nort2828
		Sasak	sasa1249
		Sundanese	sund1252
		Tagalog	taga1270
		Tuvalu	tuva1244
		Wuvulu-Aua	wuvu1239
	Border	Amanab	aman1265
	Indo-European	Bislama	bisl1239
		Chavacano	chav1241
		Hawai'i Creole English	hawa1247
		Pijin	piji1239
		Tok Pisin	tokp1240
	Isolates	Kuot	kuot1243
	Nuclear Torricelli	Bukiyip	buki1249
	Nuclear Trans New Guinea	Enga	enga1252
		Iyo	iyoo1238
		Kuman	kuma1280
	Sentanic	Sentani	nucl1632
	Sepik	Mende (Papua New Guinea)	mend1268
	South Bougainville	Naasioi	naas1242
South America	Araucanian	Mapudungun	mapu1245
	Arawakan	Garifuna	gari1256
	Aymaran	Central Aymara	cent2142
	Barbacoan	Awa-Cuaiquer	awac1239
	Cariban	Galibi Carib	gali1262
	Chicham	Aguaruna	agua1253
	Chocoan	Northern Emberá	nort2972
	Guaicuruan	Mocoví	moco1246
	Indo-European	Aukan	ndyu1242
		Papiamento	papi1253
		Saramaccan	sara1340
		Sranan Tongo	sran1240
	Isolates	Camsá	cams1241
		Páez	paez1247
	Mataguayan	Güisnay	wich1264

Macro-area	Top-level family	Language	Glottocode
		Weenhayek	wich1262
	Nadahup	Nadëb	nade1244
	Nuclear-Macro-Je	Xavánte	xava1240
	Quechuan	Bolivar-North Chimborazo Highland Quichua	chim1302
		South Bolivian Quechua	sout2991
	Tucanoan	Tucano	tuca1252
	Tupian	Eastern Bolivian Guaraní	east2555
		Paraguayan Guaraní	para1311

Table 4.2: Synchronic language sample

It is important to note that there are additional languages with partial NT translations or with data available only from secondary sources. These languages are not included in the synchronic sample because they do not provide the full datasets required for quantitative analysis of the relative frequencies of the different strategies. However, such languages are occasionally referenced in the text when they provide relevant qualitative insights, even though they are excluded from the statistical analysis.

In addition to the synchronic sample, a diachronic sample was compiled to examine historical patterns in the interaction between negation and universal quantifiers. This diachronic sample includes NT translations into Germanic and Romance languages published before 1950. This sample enables the analysis of how the relevant constructions have evolved over time within specific language families. While smaller in scope compared to the synchronic sample, the diachronic data provide valuable insights into historical trends and the development of these syntactic phenomena.

Table 4.3 lists the 21 languages included in the diachronic sample, along with the range of years for which translations are available. This list also includes languages with only partial NT translations. However, translations that were heavily dependent upon prior translations into the standard varieties, such as those made into various Romance varieties of Italy and France during the 19th century, were excluded to maintain the reliability of the data.

4.2 Sampling method

The estimation of cross-linguistic frequencies requires careful consideration of how languages are sampled. A common approach in many fields is proportional stratified sampling, where the sample reflects the prevalence of different groups based on a specific criterion. For instance, suppose we wanted to estimate the global proportion of left-handed people. Using proportional stratified sampling, we might select participants based on the population sizes of different regions or ethnic groups. Larger populations, such as those in South Asia, would contribute more participants to the sample than smaller populations, such as those in the Pacific Islands. This approach ensures that the sample mirrors the global population distribution.

In linguistic typology, a similar approach might involve estimating how frequent different word order patterns—such as SOV and SVO—are across the world's languages. To do this,

Language family	Language	Years
Germanic	Danish	1550–1997
	Dutch	13th century–2013
	English	990–2011
	Faroese	1823–1937
	High German	9th century–2015
	Icelandic	1584–2007
	Low German	9th century–2019
	Norwegian	1921–2018
	Scots	1904–2016
	Swedish	1526–2015
Romance	Catalan	1832–2000
	Corsican	1861–2015
	French	13th century–1987
	Italian	14th century–2008
	Occitan	13th century–2016
	Piemontese	1834
	Portuguese	1681–2000
	Romanian	1668–1998
	Romansh	1560–1861
	Spanish	13th century–1973
	Venetian	13th century–1859

Table 4.3: Diachronic language sample

we could stratify our sample by language families or geographic regions, selecting more languages from larger families or areas with many languages, such as Indo-European or Niger-Congo. For example, several Indo-European languages might be included because of their wide distribution, alongside fewer languages from smaller families like Uralic. This method reflects the global distribution of languages across genealogical and areal groups.

While proportional stratified sampling works well for descriptive statistics—like determining the global proportion of left-handed people or the relative frequencies of SVO and SOV word orders—it is less suited for inferential statistics, which seek to uncover underlying causes or tendencies. As Dryer (1989) explains, proportional sampling in linguistic typology often reflects historical factors, such as the spread of large language families, rather than linguistic principles. For example, Bantu languages, which are predominantly SVO, are overrepresented in a proportional sample because they are part of a large, widespread family. As a result, the observed frequency of SVO in the sample may simply mirror the historical expansion of Bantu languages rather than revealing a universal linguistic preference for SVO.

In other words, proportional sampling can provide a reliable description of how common certain linguistic patterns are among the world's languages (descriptive statistics). However, it cannot determine whether these patterns are due to linguistic factors, such as cognitive or structural preferences, or to non-linguistic factors, like historical migrations and demographic dominance (inferential statistics). To address this limitation, typologists often adopt alternative sampling methods that reduce the influence of genealogical and areal biases. One common method involves non-proportional stratified sampling, which controls for these biases in the sampling process by giving equal weight to smaller and less widespread language families or isolates, regardless of their size. This approach enables researchers to better isolate the linguistic factors that shape cross-linguistic patterns, emphasizing diversity and independence over the dominance of large families or linguistic areas.[3]

Dryer (1989) proposes a sampling procedure to reduce genealogical and areal biases by using genera as the primary unit of analysis. A genus is a subgroup of phylogenetically related languages within a larger language family. Unlike language families, which can vary greatly in size and time depth, genera are designed to represent more consistent genealogical groupings, as they reflect a comparable time depth, usually about 3,500–4,000 years. By focusing on genera instead of individual languages, this approach reduces the overrepresentation of closely related languages within large families.

To address areal biases, Dryer organizes genera into large macro-areas, which are continent-sized regions designed to limit the impact of language contact and diffusion. These macro-areas are chosen based on physical boundaries, such as oceans, which historically restricted interactions between regions. Thus, genera are used to mitigate genealogical biases, while macro-areas address areal influences.

In this method, genera are sampled for the presence of linguistic patterns—such as VO or OV word order—with genera that exhibit multiple patterns counted for each. The hypothesis being tested is considered supported only if the pattern is observed in all macro-areas, ensuring robust conclusions that are less influenced by genealogical or areal dominance.

In this study, I follow Dryer's method by using genera, rather than individual languages, to estimate cross-linguistic frequencies of different strategies for expressing 'Not all X are Y' propositions. As discussed in §4.1, the language sample is unbalanced, primarily due to an overrepresentation of European and Bantu languages. However, this is not necessarily a limitation, since it is possible to use an unbalanced sample without discarding any languages (Guzmán Naranjo and Becker 2022). A more significant issue—without an easy remedy—is that many language families remain unrepresented in the sample, particularly in Australia, due to a lack of available data. Consequently, the sample only supports partial stratified sampling, which may limit the generalizability of the results to other language families.

[3]Returning to the left-handedness example, proportional stratified sampling can estimate the global proportion of left-handed people by reflecting the real-world distribution of population groups. However, while this approach tells us how many people are left-handed (a descriptive statistic), it cannot reveal whether humans have an inherent tendency toward right-handedness, as the results may be skewed by larger groups sharing genetic or cultural traits. Non-proportional stratified sampling addresses this by giving equal weight to different population groups, regardless of their size. For instance, we might select the same number of participants from South Asia and the Pacific Islands, ensuring that smaller groups are not overshadowed. Similarly, in linguistic typology, this approach reduces biases from large language families or linguistic areas, whose prevalence stems from non-linguistic factors.

In light of these constraints, I did not adopt the more fine-grained stratification approach proposed by Miestamo et al. (2016), which refines Dryer's method by ensuring that genera are sampled in a way that more proportionally reflects genealogical diversity within macro-areas. Their approach seeks to prevent macro-areas from being dominated by a single widespread family by adjusting the selection of genera to better balance linguistic diversity across different regions. While such an approach would be ideal for a larger and more evenly distributed language sample, implementing it in this study was not feasible due to the uneven availability of corpus data.

Instead, Dryer's method remains the most practical choice, as it is both straightforward to implement and well-suited for a medium-sized, globally distributed language sample. See §8.1 for further details regarding the implementation of this method and a comparison to alternative approaches.

Chapter 5

Using New Testament translations as a parallel corpus

This chapter outlines the rationale and methodology behind the use of NT translations as a parallel corpus for investigating the cross-linguistic variation in the expression of 'Not all X are Y' propositions. While typological studies usually rely on grammars, typological databases, and questionnaires, corpus-based typology has emerged as an alternative, particularly when quantitative data on usage frequencies and patterns in individual languages are needed. The NT corpus is particularly well-suited to this study, as it allows for systematic comparisons across a large number of languages and historical periods. However, the use of translated texts comes with certain methodological challenges, such as the potential for source-language interference. These issues are carefully considered in this chapter.

This chapter is organized as follows. Section 5.1 situates the study within the broader field of corpus-based typology, highlighting the advantages of using corpora to investigate grammatical phenomena that are often underreported in descriptive grammars. Section 5.2 then addresses reliability concerns, focusing on potential sources of error, including false positives (cases where a construction is attested but may be an artifact of translation) and false negatives (cases where a construction is absent in the corpus but may exist in the language).

Building on this, Section 5.3 examines the representativeness of the NT corpus. While it is not a randomly sampled corpus of natural discourse, comparisons with non-translated corpora suggest that its results align well with broader patterns in formal written language. Section 5.4 further contextualizes the methodological approach by comparing corpus data with acceptability judgments, outlining the strengths and limitations of each for studying scope phenomena.

Finally, Section 5.5 presents the relevant NT verses analyzed in this study, providing both the Greek source text and representative translations. This section serves as a reference point for the comparative analysis in later chapters and offers an initial glimpse into the cross-linguistic and diachronic variation observed in the corpus.

5.1 Corpus-based typology and the choice of a parallel corpus

Typological research has traditionally relied on grammars, dictionaries, and typological databases such as WALS (Dryer and Haspelmath 2013), AUTOTYP (Bickel et al. 2023), and Grambank (Skirgård et al. 2023). These resources provide valuable cross-linguistic data, yet they often lack detailed information on specific grammatical phenomena, particularly those that are not considered a core issue of a general descriptive grammar. To address these gaps, some studies turn to expert questionnaires, as in the two-volume *Handbook of Quantifiers in Natural Language* (Keenan and Paperno 2012; Paperno and Keenan 2017), which provides in-depth descriptions of quantificational expressions across diverse languages. While this approach allows the systematic collection of detailed and nuanced grammatical information, it is not always feasible to coordinate such a large-scale collaborative effort.

In recent years, corpus-based typology has gained ground as an alternative approach (Levshina 2022; Schnell and Schiborr 2022). Rather than relying on secondary sources, corpus-based studies analyze primary cross-linguistic data. This approach is particularly useful when frequency-based information is needed or when investigating linguistic phenomena that tend to be underreported in grammars.

This study investigates how languages express propositions of the form 'Not all X are Y'. Descriptive grammars rarely address this question and even less frequently specify which strategies are unavailable, either because 'Not all X are Y' statements occur infrequently in natural discourse or because scope phenomena in general receive limited attention in comprehensive grammatical descriptions. A corpus-based approach is therefore essential, as it enables both the identification of attested constructions and an assessment of their relative frequencies.

Unlike most studies in corpus-based typology, which typically focus on basic linguistic features—such as word order flexibility (Futrell et al. 2015; Levshina 2019), word order tendencies (Östling 2015; Naranjo and Becker 2018), and the correlation between word length and frequency (Bentz and Ferrer-i-Cancho 2016; Stave et al. 2021)—this study investigates a relatively rare grammatical phenomenon. As a result, while it still aims to estimate the relative frequencies of different syntactic strategies, the available data are far sparser than in studies of phenomena like word order, where nearly every sentence in a corpus can be analyzed.

Furthermore, rather than relying on automated tools such as word alignment, which have been effective in large-scale studies of word order (Östling 2015), this study relies on manual syntactic analysis of individual examples to ensure a reliable classification of syntactic strategies. While word alignment can potentially identify the relative order of the universal quantifier and the negative marker within the clause, it is insufficient for capturing the structural properties of complex syntactic constructions, necessitating a detailed examination of sentence structure. Given these methodological requirements, this study takes a different approach to corpus-based typology, prioritizing structural analysis over large-scale automated processing.

The syntactic analysis of cross-linguistic data required consulting existing grammars and dictionaries. The selection of languages for this study took into account the availability of such resources, ensuring that sufficient descriptive information was accessible for accurate analysis (see §4.1).

A crucial aspect of this methodological approach is selecting a suitable corpus. Corpus-based typology can rely on either non-parallel or parallel corpora. Non-parallel corpora contain original, non-translated texts, whether written or spoken. These provide naturalistic data and are particularly useful for studying phenomena that can be identified through form-based queries or automatic syntactic parsing. However, they are unsuitable for meaning-based (onomasiological) studies, where the goal is to examine how different languages express a specific meaning (see §2.2). Since the researcher does not know in advance what structures or lexical items a language might use to convey a given meaning, form-based searches may fail to retrieve all relevant examples.

Parallel corpora, in contrast, consist of translated texts, making them ideal for meaning-based cross-linguistic comparisons. A major advantage of a parallel corpus is that it allows researchers to query for specific meanings without needing to pre-identify all possible syntactic realizations. This is particularly valuable for the present study, where the goal is to compare how different languages convey 'Not all X are Y' propositions. For instance, determining whether any language has a dedicated lexical item for 'not all' requires searching for how this meaning is expressed rather than looking for specific word forms. As discussed in Section 9.3, this study finds only a single counterexample to Horn's lexicalization universal. Crucially, we can only be confident that this restriction is a near-absolute universal because the data collection was not biased toward identifying any particular syntactic form.

The dataset used in this study is a parallel corpus of contemporary and historical translations of the NT. Bible translations are particularly useful for typological research because they are available in thousands of languages, ensuring broad linguistic coverage, and they exist in multiple historical versions for many languages, allowing for diachronic analysis.

Previous studies have used NT translations to investigate a range of typological phenomena, including the position of locative phrases in imperatives and non-imperatives (Wälchli 2009), motion verbs (Wälchli and Cysouw 2012), knowledge predication (Sjöberg 2023), and colexification patterns (Wälchli 2014; Östling 2016; Liu et al. 2023; Wälchli and Sjöberg 2024; Rognan and Beekhuizen 2025, among others). Notably, Östling (2015) extracted word order patterns from NT translations in 986 languages, finding a strong correlation between the tendencies observed in the corpus and the categorical classifications in WALS. These studies demonstrate that NT translations can serve as a valuable resource for cross-linguistic research, even though they represent a specific register and genre.

Bible translations have also proven useful in historical linguistics for studying language change. Rosemeyer (2012) argues that they provide a controlled environment for diachronic analysis by preserving the same discourse contexts across different periods, thereby reducing the variability found in general-purpose historical corpora. Applying this approach to Old Spanish translations of the Latin Vulgate, Rosemeyer systematically tracks changes in auxiliary selection, comparing how the same Latin structures were rendered across different historical translations. This method allows for the identification of both continuity and change while accounting for potential source-language influence. Similarly, Rosemeyer and Enrique-Arias (2016) use the same corpus to examine competition between possessive constructions in Old Spanish, further demonstrating the potential of Bible translations for studying grammatical competition between alternative constructions expressing the same meaning in a stable textual environment.

This dissertation draws on NT translations for both typological and historical purposes, combining the advantages of broad cross-linguistic coverage with the ability to trace diachronic developments in the strategies used to express 'Not all X are Y' propositions.

To assemble the corpus for both the synchronic and diachronic language samples (see §4.1), NT translations were curated from multiple sources, such as Bible.com (YouVersion), Google Books, and library collections, with particular emphasis on incorporating historical translations to track diachronic developments. This study did not use an existing collection of NT translations, such as the verse-aligned corpus of (Mayer and Cysouw 2014), which includes translations into over 1,400 languages. The main reason for this decision was the need to incorporate historical translations, many of which are only available as scanned books or manuscripts. A list of the NT translations cited in this work is found in Appendix (586).

5.2 Assessing reliability: False positives and false negatives

A key methodological challenge in using translated texts for typological research is ensuring that the data reliably reflect the grammar of the target language rather than influence from the source language. Since translations can reflect both the structure of the original text and the conventions of the target language, there is a risk of false positives, where a construction appears in the corpus but is actually a calque, and false negatives, where an attested construction in the language does not appear due to translation choices or its low frequency.

Translated texts are susceptible to calques and other, more subtle traces of source-language influence, creating a genuine risk of false positives—cases where a construction appears in the corpus but is merely a reflection of the source text rather than an independently existing pattern in the target language. This issue is not unique to the present study. Although translated texts are not typically the primary resource for studying present-day languages, they often constitute a major portion of the extant textual evidence for certain historical periods. Cichosz et al. (2016) demonstrate that one can use translated texts to study not only morphology and other linguistic domains that are less susceptible to source-language interference but also syntactic patterns such as word order frequencies, provided that the influence of the source text is carefully controlled. Thus, this approach builds on a well-established tradition in historical linguistics.

Because this study relies on translations of the NT, mitigating the risk of false positives is particularly important. To address this, the analysis prioritizes instances where translators diverge from the Greek structure, as such deviations are more likely to reflect native grammatical patterns. Additionally, while translation bias remains a concern, cross-checking multiple translations helps confirm whether a given construction is genuinely attested in the target language.

Another way to assess the influence of the source text is by comparing translated texts to non-translated material in the same language. This provides an additional layer of evidence for determining whether a construction is genuinely attested in the target language or introduced through translation effects. Gianollo (2011) applies this method in her study of adnominal genitives in the Latin *Vulgate*, demonstrating that while the Latin translation closely follows the Greek syntax, it also aligns with broader syntactic changes in Late Latin. By systematically comparing the Vulgate with earlier and contemporary Latin texts, she shows that the shift toward post-nominal genitives was not merely a calque but part of an independent grammatical development. This dissertation similarly uses non-translated texts as a secondary source of evidence, complementing the primary method of identifying divergences between the translation and its Greek or Latin source. Together, these methods help minimize the risk of false positives.

The classification of strategies in this study is based on the structural properties of the corpus examples. As a reminder from §1.4, I identify five primary strategies for expressing ‘Not all X are Y’ propositions, each characterized by distinct syntactic patterns. These strategies are schematically outlined in Table 5.1. The classification serves as an analytical framework for comparing languages, and it is elaborated in greater detail in Chapter 7, where additional subdivisions and representative examples are discussed.

Strategy	Schematic example
A	*All the cookies didn’t burn.*
B	*Not all the cookies burned.*
C	*Not burned all the cookies.*
D	*It’s not all the cookies that burned.*
	It’s not that all the cookies burned.
E	*The cookies did not all burn.*

Table 5.1: A schematic representation of five strategies for expressing ‘Not all X are Y’ propositions

The Greek NT overwhelmingly employs Strategy B (12 out of 14 times in the relevant verses; see §5.5). When a translation instead uses a different strategy, this strongly suggests that the alternative strategy is genuinely available in the target language. This approach significantly reduces the risk of false positives, as it is unlikely that a translator would both diverge from the Greek structure and, at the same time, introduce a construction that is not naturally used in the target language. As a result, for inverse-scope constructions (Strategy A), which are the primary focus of this study, the risk of false positives is relatively low.

In the specific case of biblical translations, the extent of source-language influence may partly depend on the translation approach. The methodologies of formal equivalence and dynamic equivalence, terms originally coined by Nida (1964), represent two divergent strategies for interpreting ancient texts for modern audiences. This distinction highlights the contrast between prioritizing fidelity to the lexical choices and grammatical structure of the source text (formal equivalence) and conveying the underlying meaning of the original (dynamic equivalence).

Formal equivalence is committed to maintaining the exact contextual meaning of the original text without adding or subtracting from its content. This approach respects the historical and cultural nuances embedded in the original language, often preserving idiomatic expressions and ancient metaphors. However, this can sometimes result in translations that are rigid or difficult for contemporary readers to grasp due to the culturally specific nature of the source texts.

On the other hand, dynamic equivalence prioritizes conveying the text's meanings in a way that is linguistically and culturally accessible to the target language's audience. This method allows the translator more flexibility to interpret the context and the intent behind the original phrases, adapting cultural references and idiomatic expressions into equivalents that are more immediately understandable to contemporary readers. While this approach enhances readability and relatability, it also introduces the risk of straying from the precise wording of the source text, potentially overlaying the translator's interpretations onto the original meanings.

To illustrate the distinction between formal and dynamic equivalence, consider two representative renderings of 1 Corinthians 10:23. The *New Revised Standard Version Updated Edition* predominantly employs formal equivalence. It closely follows the Greek source (139) without explicitly clarifying that Paul is presumably responding to a statement attributed to the Corinthians (see §5.5), except for the addition of the quotation marks. By contrast, the *New International Reader's Version*, which predominantly employs dynamic equivalence, adds words that help clarify the context of this verse. Modern Bible translations often strive for a balance between these methodologies, attempting to respect the integrity of the source material while ensuring that the translation remains engaging and clear for its intended readership.

(79) "All things are permitted," **but not all things are beneficial**. "All things are permitted," **but not all things build up**. [NRSVue, 1 Cor. 10:23; 2021]

(80) You say, "I have the right to do anything." **But not everything is helpful**. Again you say, "I have the right to do anything." **But not everything builds us up**. [NIrV, 1 Cor. 10:23; 1996 (2014)]

Beyond the conventional methods of formal and dynamic equivalence, paraphrases represent a distinct approach. It involves rewording and often reinterpreting the text to make it resonate more powerfully or clearly with contemporary audiences. For instance, *The Message* paraphrases 1 Corinthians 10:23 without even preserving the 'Not all X are Y' proposition of the source text.

(81) Looking at it one way, you could say, "Anything goes. Because of God's immense generosity and grace, we don't have to dissect and scrutinize every action to see if it will pass muster." But the point is not to just get by. We want to live well, but our foremost efforts should be to help others live well.
[*The Message*, 1 Cor. 10:23-2; 1993 (2018)]

In principle, translations adhering to formal equivalence are more prone to dependence on the source text compared to translations that mostly employ dynamic equivalence and paraphrases. In practice, however, even self-proclaimed "literal" translations, such as Robert Young's *Literal Translation* ("Literally and Idiomatically Translated out of the Original Languages"), often diverge from the Greek word order in the relevant verses. The preface to the second edition of Young's translation (1887) states:

> If a translation gives a *present tense* when the original gives a *past*, or a *past* when it has a *present*; a *perfect* for a *future*, or a *future* for a *perfect*; an *a* for a *the*, or a *the* for an *a*; an *imperative* for a *subjunctive*, or a *subjunctive* for an *imperative*; a *verb* for a *noun*, or a *noun* for a *verb*, it is clear that verbal inspiration is as much overlooked as if it had no existence. THE WORD OF GOD IS MADE VOID BY THE TRADITIONS OF MEN.
> [Young, *Literal Translation* (2nd ed.), "Preface"; 1887, emphases in original]

And yet, Young employs Strategy A in 1 Corinthians 10:23 instead of following the Greek word order, which employs Strategy B.

(82) All things to me are lawful, but all things are not profitable; all things to me are lawful, but all things do not build up. [Young, 1 Cor. 10:23; 1862]

It seems that modern translations aspiring for (even extreme) formal equivalence typically focus on lexical and semantic equivalence rather than on syntax and word order. Consider, for instance, another self-proclaimed literal translation, composed by Julia Evelina Smith. Smith literally translates Greek *chōreō* 'have room for, contain' in Matthew 19:11, and yet she employs Strategy A instead of Strategy B in Greek. Thus, modern translations adhering to formal equivalence do not necessarily slavishly follow the Greek syntax.

(83) And he said to them, All have not room for this word, but to whom it is given.
[Smith, Matt. 19:11; 1876]

By contrast, some translations composed in antiquity follow the Greek source word-for-word. The Latin Vulgate is a prime example of this approach, as demonstrated in §5.5. Independent evidence indicates that Latin did, in fact, employ Strategy B (see §10.2.3), but this cannot be inferred solely from the *Vulgate* due to its heavy reliance on the Greek source.

In some cases, early translations warrant even greater skepticism. The *Gothic Bible* (also known as the *Wulfila Bible*), a 4th-century translation of the NT into Gothic (East Germanic), also closely follows the Greek in the relevant verses. For instance, in (84), the negative marker *ni* precedes the universal QP, forming a Strategy B construction, just as in Greek (117). Although Gothic is not closely related to other attested Germanic languages, Old West Germanic languages apparently lacked Strategy B (see §10.1.3). Notably, their cognates of *ni* functioned as pre-verbal negative clitics that could not attach to other constituents. Given these facts, it remains unclear whether the use of Strategy B in the *Gothic Bible* reflects an independent construction in Gothic or simply a calque from Greek. Due to this uncertainty, such translations were excluded from the analysis.

(84) Gothic
akei ni allai ufhausidedun aiwaggeljon
but NEG all.PL obey.PST.3PL gospel.DAT
'But not all have obeyed the good news' [*Gothic Bible*, Rom. 10:16; 4th century]

Similarly, the *Classical Armenian Bible* (about 434), the *Old Georgian Bible* (5th century), adhere closely to the Greek source, making it difficult to determine whether the structures they employ reflect independent grammatical patterns in these languages.

By contrast, some early translations diverge more significantly from the Greek. The Syriac *Peshitta* (5th century) predominantly employs Strategy D, a translation into Sahidic Coptic (2nd century) occasionally uses Strategy A, and the *Ge'ez Bible* (4th century) primarily adopts Strategy A. These cases demonstrate that not all early translations slavishly follow the Greek text, highlighting the need for a case-by-case assessment rather than writing off early translations altogether.

In the medieval period, Bible translations into western European languages were generally independent of the Latin *Vulgate*, on which they are based. This independence is evident in a Middle French translation of Matthew 19:11, which employs Strategy A instead of B and even includes an explanatory note in parentheses (85).

(85) Old French
"Touz ne prennent mie," (c'est a dure n'entendent
all.PL.M NEG take.PRS.3PL NEG this-be.PRS.3SG to say.INF NEG-understand.PRS.3PL
mie,) "ceste parole
NEG this.SG.F word
'All do not take (that is to say, do not understand) this word'
[*Old French Bible*, Matt. 19:11; 1235-1260]

A notable exception is a complete Middle English translation of the Bible, traditionally attributed to John Wycliffe. Two major versions have been identified: the Early Version, which adheres closely to the Latin text in a word-for-word manner, and the Later Version, which allows for somewhat greater linguistic adaptation. Despite these differences, both versions closely follow the Latin word order in the relevant verses, setting them apart from other extant Middle English translations (see §10.1.4). Given this high degree of dependence on the *Vulgate*, this translation was excluded from the analysis. Overall, very few translations had to be excluded altogether (see Chapter 10 for further discussion of historical translations).

While controlling for the dependence on the source text significantly reduces the risk of false positives (calques), there still remains a risk of false negatives, where a strategy is unattested in NT translations but is, in fact, available in the language. This risk is particularly high for strategies with low but nonzero usage frequency. Given the relatively small number of relevant instances available for each language, there is a considerable chance that infrequent constructions will be absent from the sample purely by chance. This issue is inherent to the sampling process and cannot be entirely eliminated. However, in Chapter 6, I attempt to assess the reliability of the NT corpus by comparing its results with those from a larger corpus in several selected languages.

Beyond empirical evaluation, the probability of false negatives can also be theoretically estimated based on the number of observed data points, treating them as independent trials—though this is, of course, a simplification. A Bayesian approach is particularly suitable for this purpose, as it facilitates the integration of both prior knowledge and observed (or simulated) data to estimate the actual frequency of a given strategy. The objective of this theoretical analysis is to assess the likelihood that a strategy that was not observed in a small sample might actually be more common than the data suggests. By assuming independent observations—a condition seldom met in actual corpus data—–this analysis yields a theoretical upper bound on the confidence one can reasonably have regarding the true frequency of a strategy that does not appear in the sample. Since true independence among data points is unlikely, these estimates represent a best-case scenario.

The graph presented in Figure 5.1 illustrates the theoretical probabilities that the actual frequency of a particular strategy, denoted by p, falls below specific thresholds (0.01, 0.05, and 0.1) based on the number of independent trials in which the strategy is not observed. The y-axis represents the probability that p, the actual frequency of the strategy, is below the specified thresholds, while the x-axis indicates the number of trials where this strategy is not observed.

This model adopts a non-informative prior, specifically a Beta(1,1) distribution, which implies minimal prior knowledge regarding the actual frequency of the strategy. This prior choice allows the posterior estimates to be predominantly informed by the data gathered. The model assumes a binomial distribution for the outcomes, appropriate for scenarios where each trial is an independent observation with two possible results: the strategy being used or not used.

The model demonstrates that as the number of trials where the strategy is not observed increases, the confidence that the actual frequency p of the strategy is below specific thresholds likewise increases. Notably, the probability of p being below lower thresholds (e.g., below 0.01) increases more slowly compared to higher thresholds (e.g., below 0.1). This pattern suggests that more trials are required to be reasonably confident that the actual frequency p is extremely low (e.g., below 0.01) compared to moderately low (e.g., below 0.1). For instance, after 10 trials of not observing the strategy, the theoretical probabilities are $P(p < 0.1) = 0.69$, $P(p < 0.05) = 0.43$, and $P(p < 0.01) = 0.1$. After 30 trials, the probabilities are $P(p < 0.1) = 0.96$, $P(p < 0.05) = 0.8$, and $P(p < 0.01) = 0.27$.

One can further quantify the number of trials required to achieve a 95% confidence level, aligning with common statistical practice for strong evidence against the null hypothesis in hypothesis testing. To be 95% confident that $p < 0.1$, at least 28 trials are required; for $p < 0.05$, 58 trials are needed; and for $p < 0.01$, as many as 298 trials are required.

On the positive side, this theoretical analysis suggests that even a relatively small number of data points (e.g., $n = 28$) is enough to be reasonably confident that the actual frequency of a strategy is at best low (e.g., below 0.1), under the assumption that the observations are independent. On the negative side, there still remains a distinct possibility that the strategy is just extremely rare (e.g., below 0.01), but not altogether absent. In practice, a small corpus like the one used in this study cannot reliably differentiate between a strategy being rare or absent.

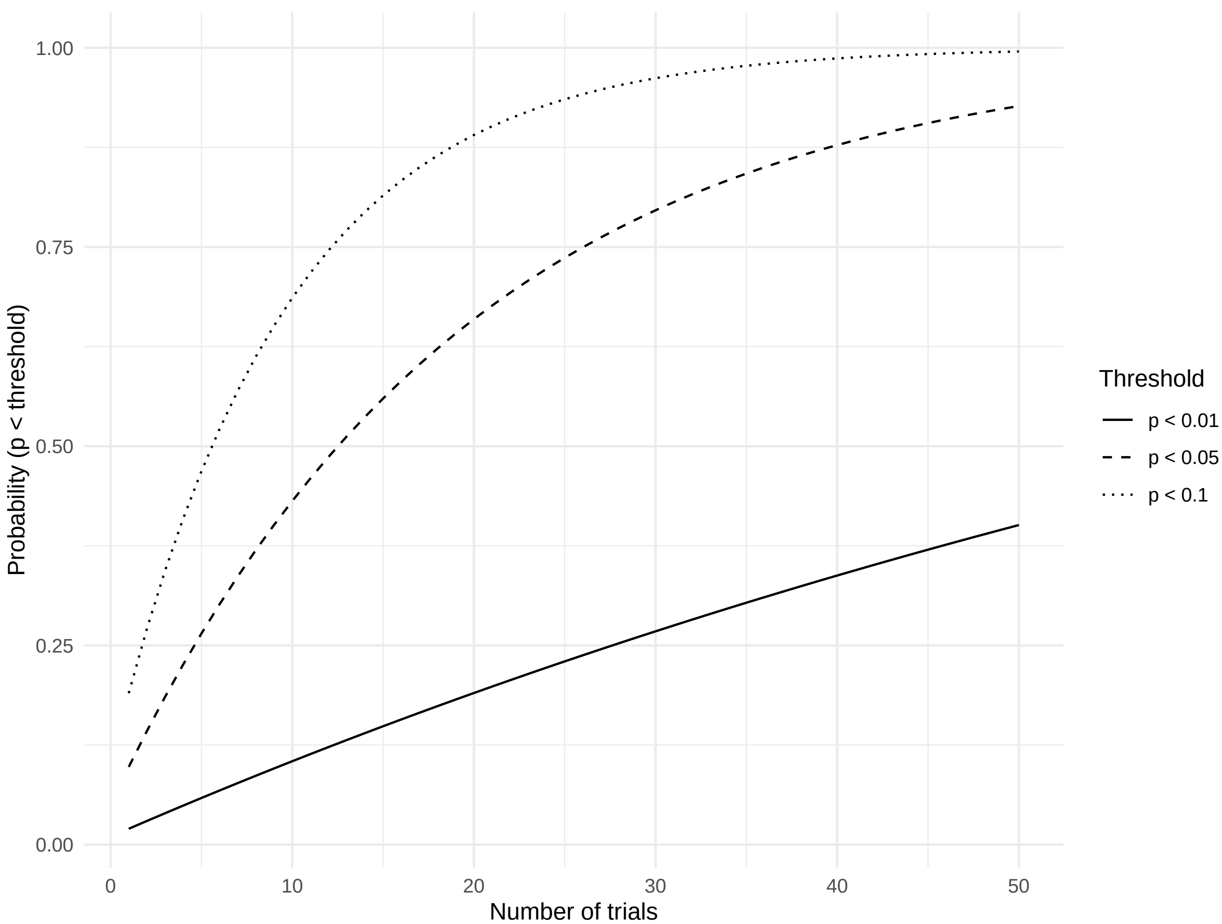


Figure 5.1: The increasing probability that the actual frequency of a linguistic strategy falls below predefined thresholds (0.01, 0.05, and 0.1) as the number of trials in which the strategy is not observed increases, illustrating the Bayesian update of beliefs based on simulated data.

5.3 Assessing representativeness: Genre and dialectal variation

Another issue related to the risk of false positives and negatives is representativeness. What can we infer from NT translations about the language at large? Strictly speaking, corpus statistics make predictions about the population of texts from which the corpus was (presumably) randomly sampled. For instance, the *British Component of the International Corpus of English* contains informal face-to-face conversations between British adults. The corresponding population is “English informal face-to-face conversations between educated British adults in the years 1990 to 1992” (Wallis 2020, 28). Any conclusion drawn about English at large or even spoken British English in the 1990s is at the researcher’s own risk, since the corpus was not randomly sampled from the corresponding populations.

From this strict perspective, the NT corpus, at best, allows us to make predictions about the population of NT translations produced by translators with similar linguistic and cultural backgrounds during the same historical periods. Drawing broader generalizations about the language as a whole requires additional assumptions about the stability of syntactic patterns across dialects, genres, and registers.

To assess the extent to which the NT corpus aligns with other textual sources, Chapter 6 compares the results from the NT corpus to those from the online news section of the Leipzig Corpora Collection in several languages. While this dataset does not necessarily reflect linguistic patterns found in fiction or face-to-face conversations either, a strong correlation between the results suggests that the NT corpus is likely representative of formal written language beyond the specific genre of biblical translations.

Another potential concern is that inverse-scope constructions might be more common in speech than in writing—or vice versa. While this possibility cannot be ruled out, there is no clear evidence to support it. Tottie and Neukom-Hermann (2010) studied the use of *all…not* constructions in British English using the British National Corpus. Their findings indicate that inverse-scope constructions are slightly more frequent in writing than in speech, though this difference appears to be driven by the greater prevalence of formulaic expressions like *All is not lost* in written language (Tottie and Neukom-Hermann 2010, 158-159; 162-164).

By contrast, Neukom-Hermann (2016) tentatively suggests that inverse-scope constructions in German may be more common in speech than in writing. However, the examples she provides involve either a topicalized object QP, as in (86)-(87), or a modal verb intervening between negation and the quantifier—neither of which qualifies as evidence for Strategy A in the present study (see §2.3). Thus, I am not aware of any language where there is a clear, systematic difference between written and spoken language in the frequency of inverse-scope constructions.

(86) Swiss German (Neukom-Hermann 2016, 101)

und	alles	kann	er	nicht	auf	die	politischen	Gegner	abschieben.
and	all	can	he	not	on	the	political	opponent	shift.on

‘and he cannot shift everything on the political opponent.’

(87) Swabian German (Neukom-Hermann 2016, 101)

weil	alle	Folgen	vom	Michel	hammer	noch	nicht	guckt.
because	all	parts	of	Michel	have.we	yet	not	watched

'because we haven't watched all parts of Michel yet.'

These findings suggest that written language provides a reliable basis for investigating inverse-scope constructions, particularly in the absence of systematic differences between written and spoken usage. More broadly, written language is an integral part of linguistic competence, not merely a reflection of spoken language. The ability to produce and comprehend complex written texts requires a deep internalization of linguistic rules, even if the acquisition process differs from that of a spoken language. Moreover, as Bar-Asher Siegal (2021) argues, even literary languages—languages primarily used in written contexts—develop new grammatical patterns and undergo processes of regularization. Thus, rather than being an artificial or secondary representation of language, written language serves as a valuable window into the structural properties of linguistic systems.

Beyond the issues of genre and register, the corpus data are produced by writers who have a certain native dialect, and their language might not be representative of other dialects of the same language. This issue is, of course, ubiquitous in linguistic research, whether the data source is acceptability judgments or a corpus. Nonetheless, it is common practice to speak of "English" or "Spanish" without specifying the particular dialect(s) represented in the data. This practice is justified when the relevant feature is shared across all dialects of a language (e.g., in English, the article precedes the noun).

In other cases, the linguistic data represent a prestige dialect, such as Mainstream American English, or a standardized variety, such as Standard French. For example, in Mainstream American English, negative quantifiers do not co-occur with clausal negation (e.g., *Nobody saw me*), whereas some dialects of American English, such as African American English, exhibit negative concord (*Nobody didn't see me*). Importantly, speakers are typically aware of this dialectal variation, and it is widely recognized that negative concord is considered non-standard in present-day English.

However, standardization does not extend to all domains of grammar. To the best of my knowledge, no prescriptive set of rules dictates whether inverse-scope constructions may or may not be used in English. Thus, inverse-scope constructions are neither standard nor non-standard, making it meaningless to ask whether Standard English "permits" inverse scope in the same way that it excludes negative concord. Instead, variation in the use of inverse scope should be understood as dialectal rather than standard/non-standard.

The difficulty lies in the fact that the dialectal variation in this domain is not as sociolinguistically salient as the variation in the domain of negative concord. In the latter case, speakers are typically aware of the variation, and they may choose whether or not to adjust their speech in certain communicative situations. On the other hand, it appears that speakers whose dialect allows inverse scope are typically unaware that this construction is not prevalent in other dialects. Consequently, it seems that speakers do not adjust their speech to the dominant dialect in this specific domain, even when addressing a diverse audience.

To illustrate this point, consider the *Common English Bible* (2011), a translation designed for easy readability for English readers worldwide, according to its official website.[1] Despite this goal, the translation employs Strategy A multiple times throughout the text, as shown in (88). Speakers of Mainstream American English usually find these examples unnatural, and the same probably applies to speakers of many other varieties. Since this translation was produced collaboratively by a large team of translators, it is not possible to pinpoint the specific dialect(s) reflected in these instances. However, it does not appear that the translators deliberately aimed to use a dialectal or non-standard form. Instead, it is more likely that they were simply unaware that these constructions would sound unnatural to many English speakers.[2]

(88) a. But everyone hasn't obeyed the good news. [Common English Bible, Rom. 10:16; 2011]
b. Everything is permitted, but everything isn't beneficial. Everything is permitted, but everything doesn't build others up. [ibid, 1 Cor. 10:23]
c. Pray too that we will be rescued from inappropriate and evil people since everyone that we meet won't respond with faith. [ibid, 2 Thess. 3:2]

Similarly, the French *La Bible en français courant* (1982 [1997]) employs a cleft (Strategy D) in two verses (89), a strategy that is common in Quebec French but rare in Metropolitan French (data from other varieties is unavailable; see §10.2.6 for more details). It is likely that the translators of these particular verses—who may have been Canadian—were unaware that they were using a construction that was not prevalent in other dialects. This seems especially plausible given that the translation is intended for a broad Francophone audience, as stated on its official website.[3]

(89) French

a. Ce ne sont pas tous ceux qui me disent: "Seigneur, Seigneur", qui entreront dans le Royaume des cieux
it NEG are NEG all those who to.me say Lord Lord who will.enter in the kingdom of.the heavens
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven' [*Bible en français courant*, Matt. 7:21; 1982 [1997]]

b. Car ce n'-est pas tout le monde qui accepte de croire.
because it NEG-is NEG all the world who agree of believing
'For not everyone agrees to believe.' [ibid, 2 Thess. 3:2]

[1] `https://www.commonenglishbible.com/Explore/AbouttheCEB/tabid/196/Default.aspx`

[2] Labov (1975, 33) makes a similar point about positive *anymore*, arguing that "it is a matter of unconscious usage for most speakers in the midwest", but once it is consciously identified as a regional form, it is considered non-standard and might be stigmatized.

[3] `https://nfc.bible/la-bible-en-francais-courant/`

I believe that the lack of standardization in this domain stems from the low usage frequency of 'Not all X are Y' propositions. These constructions are not common enough for specific forms to become consciously associated with particular dialects. As a result, even a relatively homogeneous corpus of NT translations likely reflects a mix of dialects rather than a single standardized variety.

Nevertheless, for simplicity, I will refer to "English" and "French" as idealized languages, while acknowledging dialectal variation when relevant data is available. In most cases, however, I do not have detailed information about the specific dialects represented in a given text. Thus, when I state that "French does not employ Strategy B", this should be understood as meaning that the dialects of French reflected in the corpus do not use this strategy, though other dialects might. Despite these limitations, this approach is in line with standard linguistic practice, as research often generalizes based on available data rather than accounting for every dialectal (or even idiolectal) variation.

5.4 Corpus frequency vs. acceptability judgments

The previous sections addressed the reliability and representativeness of corpus data, focusing on the risks of false positives (source-language interference) and false negatives (low-frequency constructions going unobserved). A related question is how corpus findings align with acceptability judgments, which have traditionally been the standard methodology for studying scope phenomena. While corpus data provide direct evidence of language use, acceptability judgments offer insight into speakers' intuitions about grammaticality. This section explores the relationship between the two, arguing that they often converge but sometimes diverge in ways that require explanation.

In a traditional Generative framework, grammatical rules determine whether a sentence is well-formed or ungrammatical, independent of how frequently it is used. From this perspective, the key question is whether a given language (or dialect) permits inverse scope as a possible interpretation, while its actual usage frequency is considered a matter of performance rather than competence. In other words, grammar may allow a certain construction, yet speakers might rarely produce it in practice due to processing constraints, discourse preferences, or other factors.

Still, acceptability judgments are typically correlated with usage frequency. Hemforth and Konieczny (2019) examined the interpretation of *all…not* constructions in English, French, and German. Each target sentence was accompanied by two disambiguating sentences, one compatible with a transparent-scope reading and the other with an inverse-scope reading. Participants rated how well each of the disambiguating sentences fits with the target *all…not* sentence on a Likert scale. They found that French speakers prefer an inverse-scope interpretation, whereas English and German speakers prefer a transparent-scope interpretation. Moreover, in French, the mean rating of the inverse-scope interpretation was about 4/5, whereas in English it was about 2.5/5, and in German it was about 1.5/5.

As the authors suggest, these results are correlated with the usage frequency of inverse-scope constructions in the different languages: they are less common in English than French, and even less so in German (Neukom-Hermann 2016) (an observation corroborated by the present study, see §6.1.2). This alignment between frequency and acceptability supports the idea that corpus data can provide meaningful insights into scope preferences.

However, it is often argued that usage frequency alone cannot fully account for grammaticality judgments. A common criticism of placing too much emphasis on usage frequency is that certain structures are judged acceptable despite being exceedingly rare or absent from corpus data. For instance, Nissenbaum (2000, 12) argues that example (90), which is put forth in support of a certain theory of parasitic gaps, is "relatively acceptable" (although he marks it with a '?'). One could reasonably contend that the absence of such examples in a corpus does not necessarily undermine their grammaticality, especially given their structural complexity.

(90) ?Which senator did you persuade to borrow which car after getting an opponent of to put a bomb in? (Nissenbaum 2000, 12)

The apparent divergences between corpus frequency and acceptability judgments can be reconciled by considering how usage frequency is defined. A crucial distinction, highlighted by Wallis (2020, 7), is between exposure rate and choice rate. Exposure rate refers to how often a given word or construction appears per million words in a corpus. For instance, the definite article *the* has an extremely high exposure rate, as it appears frequently in English text. Choice rate, on the other hand, measures how often a construction is selected when the opportunity arises to use it. For example, one can calculate how often Strategy A is selected in English when a speaker or writer wishes to express a proposition of the form 'Not all X are Y'.

Importantly, it is choice rate, not exposure rate, that is most relevant to acceptability. A structure like (90) has an extremely low exposure rate, but it is possible that the choice rate of this construction is not low, given the rare opportunity of expressing this particular meaning.

From this perspective, it would be surprising if the choice rate of a given construction were not correlated with its acceptability. It seems unlikely, for instance, that an inverse-scope construction would be judged fully acceptable in a language yet almost never be selected when speakers have the opportunity to express a 'Not all X are Y' proposition. Conversely, if inverse-scope constructions have a substantial choice rate, we would expect them to be considered acceptable. Thus, a strong correspondence between usage frequency—when framed as choice rate—and acceptability ratings is expected.

However, some discrepancies between corpus data and acceptability judgments may still arise. One reason is that acceptability judgment tasks typically rely on constructed examples designed to control for specific factors. As a result, these examples do not necessarily belong to the same population of utterances from which corpus data are drawn, making direct comparisons between the two sources of evidence less straightforward.

More fundamentally, the issue of productivity presents a challenge for the view that grammaticality is independent of usage patterns. A construction that is not fully productive may yield mixed acceptability judgments, with some instances considered acceptable and others rejected. For example, (91) is a naturally-occurring example of Strategy A in the speech of a Mainstream American English speaker, suggesting that inverse scope is at least possible in this variety. However, speakers of the same dialect typically reject (92), except in an echo denial context (see §2.3). Without delving into the specific conditions under which Strategy A is productive in English (see §6.1.1), this contrast is difficult to reconcile with a view that treats usage preferences as external to grammar. If English permits inverse scope in principle, as suggested by the acceptability of (91), does it follow that (92) is grammatical despite being typically rejected by speakers?

(91) I do think one of the messages of A.I. is that most people make middling work, and middling work is easy to replace. **Every email I write is not a great work of art**, you know.
[*The Ezra Klein Show*, "Will A.I. Break the Internet? Or Save It?", 5 April 2024]

(92) ?Everyone isn't hungry.

A final issue concerns the interpretation of informal judgments. Suppose that participants in the study by Hemforth and Konieczny (2019) had been asked to provide a binary acceptability judgment for the inverse-scope interpretation. It is likely that most, if not all, French speakers would have accepted these sentences (mean rating of approximately 4/5), while the majority of German speakers would have rejected them (mean rating of about 1.5/5). But how would English-speaking participants (mean rating of around 2.5/5) respond in a binary task? Despite the relatively low variation in individual ratings, it is conceivable that a binary choice could split speakers into two groups, creating the impression of two distinct dialects.

In light of this discussion, what does it mean to say that (93) allows an inverse-scope interpretation, as is often claimed in the literature?

(93) All the men didn't go.

Suppose the variety in question is Mainstream American English. The results of Hemforth and Konieczny (2019) suggest that similar sentences are only marginally acceptable. While this might be sufficient to avoid marking them with a '?', there is a sharp contrast between Mainstream American English and Metropolitan French in the acceptability of these constructions. Treating both languages as equivalent in this regard would be misleading. Instead, gradient acceptability judgments and usage frequency provide a more nuanced and informative picture.

For this reason, I am cautious about taking informal judgments reported in the literature at face value unless they are explicitly described as robust or supported by corpus evidence. When eliciting informal judgments from consultants, I asked whether they would naturally use an inverse-scope construction in conversation rather than whether they found such a construction acceptable under an inverse-scope interpretation. This formulation likely yields more reliable results, as it places consultants in the speaker's shoes rather than the listener's.

Presumably, speakers have a clearer sense of what they themselves would say than of what others in their speech community would naturally produce.

5.5 The relevant New Testament verses

This section presents all 13 New Testament verses that feature 'Not all X are Y' statements, along with general observations about how they are typically translated. Particular attention is given to influential translations that have either served as source texts or significantly influenced later translations. These include: (i) the Latin *Vulgate*, which formed the basis for medieval translations in the Western tradition and influenced many post-Reformation Catholic translations; (ii) the Early New High German translation by Martin Luther (NT: 1522), which played a key role in shaping other Reformation-era translations and even served as the source text for some Dutch and Scandinavian translations; and (iii) the English translation by William Tyndale (NT: 1526) and *King James Version* (1611), which have had a lasting impact on subsequent English translations.

The primary goal of this section is to provide a reference point for comparing translated examples in later chapters against the original source texts. Additionally, it offers a preliminary glimpse into the cross-linguistic and diachronic variation observed in different translations, demonstrating the corpus's potential for investigating the phenomenon at hand. To illustrate this variation, I include representative English translations of each verse at the end of the passage. Finally, given that each of these verses appears multiple times in different languages and versions throughout this dissertation, readers may find it helpful to familiarize themselves with their meaning and context in advance.

The Greek text is based on the 28th edition of the Nestle–Aland *Novum Testamentum Graece* (NA28). Greek is transliterated following the *Chicago Manual of Style*. Unless otherwise noted, English translations of NT passages are taken from the *New Revised Standard Version Updated Edition* (2021), abbreviated as NRSVue. Literal translations from Greek and other languages are my own.

For an overview of negation and word order in the Greek NT, see Gianollo (2024).

Matthew 7:21 This verse, part of Jesus' Sermon on the Mount, highlights the importance of genuine discipleship, a central theme in Jesus' teachings. It challenges the assumption that verbal profession of faith alone is sufficient for salvation. Instead, Jesus emphasizes that true obedience to God's will is essential.

> 21 "**Not everyone who says to me, 'Lord, Lord,' will enter the kingdom of heaven**, but only the one who does the will of my Father in heaven. 22 On
> that day many will say to me, 'Lord, Lord, did we not prophesy in your name,
> and cast out demons in your name, and do many mighty works in your name?'
> 23 Then I will declare to them, 'I never knew you; go away from me, you who
> behave lawlessly.' [NRSVue, Matt. 7:21-23]

The Greek text employs Strategy B in this verse (94).

(94) Koine Greek
Ou pas ho legōn moi: kyrie
NEG all.NOM.SG.M the.NOM.SG.M say.PRS.PTCP.NOM.SG.M 1SG.DAT Lord.VOC.SG
kyrie, eiseleusetai eis tēn basileian tōn ouranōn
Lord.VOC.SG enter.FUT.3SG into the.ACC.SG.F kingdom.ACC.SG.F the.GEN.PL heaven.GEN.PL
'Not everyone who says to me: Lord, Lord, will enter into the kingdom of heaven [but he who does the will of my father who is in heaven.]' [NA28]

The *Vulgate* largely follows the Greek source but contains another clarifying clause at the end (95), which might have been found in a variant of the Greek text.

(95) Latin
Non omnis qui dicit mihi Domine Domine
NEG all.NOM.SG.M who.NOM.SG.M say.PRS.3SG 1SG.DAT Lord.VOC.SG Lord.VOC.SG
intrabit in regnum caelorum
enter.FUT.3SG into kingdom.ACC.SG heaven.GEN.PL
'Not everyone who says to me Lord Lord will enter into the kingdom of heaven [but he who does the will of my father who is in heaven, he will enter into the kingdom of heaven.]' [*Vulgate*]

Matthew 7:21 is the verse that is the least often translated using Strategy A and the one most commonly rendered with Strategy B. This preference is likely due to the length of the subject, which includes a relative clause. When Strategy A is used, the distance between the quantifier and negation increases, potentially making such sentences more challenging to process than other instances of Strategy A. Nonetheless, examples of Strategy A can still be found, as in the Middle English version in (96).

(96) Middle English
al þilke þat seyȝen to me lord lord ne schul not entren in to þe
all those that say.PRS.PL to me Lord Lord NEG shall NEG enter.INF in to the
kyngdome of heuene
kingdom of heaven
'Not everyone who says to me, "Lord, Lord," will enter into the kingdom of heaven'
[*Life of Soul*; about 1400]

In Tyndale's version (97) and the *King James Version* (98), this is the only verse where Strategy B appears.

(97) Not all they thatt say unto me, Master, Master, shall enter in to the kyngdome off heven: but he that fulfilleth my fathers will which ys in heven. [Tyndale; 1526]

(98) Not euery one that saith vnto me, Lord, Lord, shall enter into the kingdome of heauen: but he that doth the will of my father which is in heauen. [KJV; 1611]

Strategy	Text	Translation	Year
A	***al*** *þilke þat seyȝen to me lord lord* ***ne*** *schul* ***not*** *entren in to þe kyngdome of heuene* (see gloss in (96))	Life of Soul	1400
B	***Not euery*** *one that saith vnto me, Lord, Lord, shall enter into the kingdome of heauen*	KJV	1601
C	***Ne*** *gæð* ***ælch*** *þæra on heofene riche þe cwyþ to me drihten drihten* (see gloss in (428))	Wessex Gospels	990
D	*It is* ***not every*** *one who says to me "Master! Master!" who will enter the Kingdom of Heaven*	TCNT	1904

Table 5.2: Representative examples of English translations of Matthew 7:21

On the other hand, Luther, who avoids Strategy B altogether, employs an expletive inversion construction (Strategy E, see §7.5.2, §9.1.2), as shown in (99). Some contemporary translations into other languages follow Luther in this verse.

(99) Early New High German
Es werden nicht alle/ die zu mir sagen/ Herr herr/ yñ das hymel reych
it will.3PL NEG all who to me say.3PL Lord Lord into the heaven kingdom
komenn
come.INF
'Not everyone who says to me, "Lord, Lord," will enter into the kingdom of heaven'
[Luther; 1522]

Table 5.2 provides representative examples of the various ways Matthew 7:21 has been translated into English.

Matthew 19:11 This verse occurs during an exchange between Jesus and the Pharisees regarding the lawfulness of divorce. After Jesus outlines his stringent view that divorce should only occur in cases of marital unfaithfulness, his own disciples suggest that it might be better not to marry at all if such strictures are in place. In response, Jesus introduces the concept of celibacy as a special calling, acknowledging that the ability to live a celibate life is a gift ("given") and is not expected of all believers.

> 9 And I say to you, whoever divorces his wife, except for sexual immorality, and marries another commits adultery, and he who marries a divorced woman commits adultery." 10 The disciples said to him, "If such is the case of a man with his wife, it is better not to marry." 11 But he said to them, "**Not everyone can accept this teaching**, but only those to whom it is given. 12 For there are eunuchs who have been so from birth, and there are eunuchs who have been made eunuchs by others, and there are eunuchs who have made themselves eunuchs for the sake of the kingdom of heaven. Let anyone accept this who can."
> [NRSVue, Matt. 19:10-12]

The Greek text employs Strategy B in this verse (100).

(100) Koine Greek

ou	pantes	chorousin	ton	logon	[touton].
NEG	all.NOM.PL.M	contain.PRS.3PL	the.ACC.SG.M	word.ACC.SG	this.ACC.SG.M

'[But he said to them:] Not all make room for [this] word, [but those to whom it has been given].' [NA28]

Translators have grappled with several difficulties in this verse. First, the verb *chōreō* 'have room for, contain' is used figuratively in an uncommon sense. Second, it is not entirely clear what the antecedent is of *ton logon* [*touton*] 'this word/saying' (the demonstrative *touton* is missing in some manuscripts). Does it refer to what the disciples said, that it is better to remain single if divorce is not an option ("If such is the case of a man with his wife, it is better not to marry."), or to what Jesus had said earlier, that marriage is unbreakable ("what God has joined together, let no one separate.")? The intended meaning is probably the former, as the next verse asserts that only some men may remain celibate, either by external circumstances or by choice, but some translations also understood it to mean that it is marriage that is not for everyone.[4]

The *Vulgate* closely follows the Greek source (101). It uses the verb *capiō* 'grasp, contain, understand' to translate *chōreō* 'have room for, contain'. Later translations into other languages often similarly picked a word that can have both a physical and a mental sense, such as *grasp*.

(101) Latin

non	omnes	capiunt	verbum	istud
NEG	all.NOM.PL.M	grasp.PRS.3PL	word.ACC.SG	that.ACC.SG.N

'[who said:] Not all grasp this word [but to whom it is given].' [*Vulgate*]

On the other hand, Tyndale opts for an idiomatic translation (*away with* means 'tolerate, endure'), as shown in (102). The *King Jame Version* translates *chōreō* as 'receive' (103), a choice picked up by later translations into English, but less often in other languages. Both translations employ Strategy A in this verse.

[4] The ambiguity between these two interpretations can be illustrated by two different paraphrases into contemporary English:

(i) Jesus replied, "Not everyone can accept the idea of staying single. Only those who have been helped to live without getting married can accept it. Some men are not able to have children because they were born that way. Some have been made that way by other people. Others have chosen to live that way in order to serve the kingdom of heaven. The one who can accept this should accept it."
[*New International Reader's Version*; 1994 (2014)]

(ii) But Jesus said, "Not everyone is mature enough to live a married life. It requires a certain aptitude and grace. Marriage isn't for everyone. Some, from birth seemingly, never give marriage a thought. Others never get asked—or accepted. And some decide not to get married for kingdom reasons. But if you're capable of growing into the largeness of marriage, do it."
[Peterson, *The Message*; 1993 (2018)]

Strategy	Text	Translation	Year
A	***All*** *men can**not** receiue this saying*	KJV	1611
B	***Not all*** *men can receive this saying*	ASV	1901
C	***ne*** *under-foð* ***ealle*** *menn þis word* (see gloss in (428))	Wessex Gospels	990
D	*It is* ***not every*** *man who can receive this teaching*	Weymouth	1903
Paraphrase	*This teaching does* ***not*** *apply to* ***everyone***	GNB	1976

Table 5.3: Representative examples of English translations of Matthew 19:11

(102) He sayde unto them: all men cannot awaye with that saynge: but they to whom it is geven. [Tyndale; 1526]

(103) But hee said vnto them, All men cannot receiue this saying, saue they to whom it is giuen. [KJV; 1611]

Luther employs Strategy C with XVS word order, the object being fronted to clause-initial position (104). Some contemporary translations into other languages follow Luther in this verse.

(104) Early New High German

das	wortt	fasset	nit	yderman
the.ACC.SG.N	word.ACC.SG	grasp.PRS.3SG	NEG	everyone

'Not everyone grasps this word' [Luther; 1522]

Table 5.3 provides representative examples of the various ways Matthew 19:11 has been translated into English.

John 13:11 This verse appears in the scene of Jesus washing the disciples' feet at the Last Supper, symbolizing servitude and spiritual cleansing. Despite knowing that Judas would betray him, Jesus includes him in this ritual.

> 5 Then he poured water into a basin and began to wash the disciples' feet and to wipe them with the towel that was tied around him. 6 He came to Simon Peter, who said to him, "Lord, are you going to wash my feet?" 7 Jesus answered, "You do not know now what I am doing, but later you will understand." 8 Peter said to him, "You will never wash my feet." Jesus answered, "Unless I wash you, you have no share with me." 9 Simon Peter said to him, "Lord, not my feet only but also my hands and my head!" 10 Jesus said to him, "One who has bathed does not need to wash, except for the feet, but is entirely clean. And you are clean, though not all of you." 11 For he knew who was to betray him; for this reason he said, "**Not all of you are clean.**" [NRSVue, John 13:5-11]

The Greek text employs Strategy B in this verse (105a). Notice that John 13:10 contains a similar sentence, but with NEG-stripping (105b). The negative particle *ouchi* is an emphatic variant of *ou* (Thayer 1889).

(105) Koine Greek

a. ouchi pantes katharoi este.
NEG all.NOM.PL.M clean.NOM.PL.M be.PRS.2PL
'[For he knew the one betraying him; because of this, he said:] Not all of you are clean.' [NA28, John 13:11]

b. kai hymeis katharoi este, all' ouchi pantes.
and 2PL.NOM clean.NOM.PL.M be.PRS.2PL but NEG all.NOM.PL.M
'And you are clean, but not all.' [ibid, John 13:10]

The *Vulgate* uses a different construction, classified as C/E (see §7.6.2), as shown in (106).

(106) Latin
non estis mundi omnes.
NEG be.PRS.2PL clean.NOM.PL.M all.NOM.PL.M
'[For he knew indeed who it was that would betray him; therefore he said:] You are not all clean.' [*Vulgate*]

Later translations also often employ Strategies E or C/E in this verse, as these are common strategies when the subject is a referential personal pronoun. Thus, Tyndale, the *King James Version*, and Luther all employ Strategy E in this verse (107)-(109).

(107) For he knewe his betrayer. Therfore sayde he: ye are not all clene. [Tyndale; 1526]

(108) For he knew who should betray him, therefore said he, Ye are not all cleane. [KJV; 1611]

(109) Early New High German
yhr seyt nit alle reyn.
2PL.NOM be.PRS.2PL NEG all.NOM.PL clean
'You are not all clean.' [Luther; 1522]

Table 5.4 provides representative examples of the various ways John 13:11 has been translated into English.

Romans 9:6-7 In these verses, Paul addresses the complex issue of God's promises to Israel and their fulfillment. It is part of Paul's broader theological argument concerning the nature of true Israel and the role of faith in God's covenant. Paul emphasizes that the promises made by God are not tied merely to ethnic lineage or biological descent but to spiritual lineage—those who have faith like Abraham. By distinguishing between physical and spiritual descent, Paul reshapes the understanding of what constitutes God's people, focusing on faith as the crucial element. This redefinition allows for the inclusion of Gentiles

Strategy	Text	Translation	Year
B	***Not all*** *of you are clean*	The Living Bible	1971
C	***ne*** *sende ge* ***ealle*** *clæne* (see gloss in (428))	Wessex Gospels	990
E	*Ye are* ***not all*** *cleane*	KJV	1611
Paraphrase	*For Jesus knew his betrayer and that is why he said, "though* ***not all*** *of you"*	Phillips	1958

Table 5.4: Representative examples of English translations of John 13:11

into the covenant, highlighting a shift from a purely ethnic to a faith-based community of believers.

> 6 It is not as though the word of God has failed. **For not all those descended from Israel are Israelites**, 7 **and not all of Abraham's children are his descendants**, but "it is through Isaac that descendants shall be named for you." 8 This means that it is not the children of the flesh who are the children of God, but the children of the promise are counted as descendants.
>
> [NRSVue, Rom. 9:6-8]

In Romans 9:6, the Greek text employs Strategy B (110). Notice that the conjunction *gar* 'for, because' occurs in between negation and the quantifier.

(110) Koine Greek

ou	gar	pantes	hoi	ex	Israēl	houtoi	Israēl.
NEG	for	all.NOM.PL.M	the.NOM.PL.M	from	Israel	these.NOM.PL.M	Israel

'[Not as though the word of God has failed.] For not all who are from Israel, these are Israel.' [NA28]

On the other hand, in Romans 9:7, negation is introduced by the negative conjunction *oud'* 'nor', so this verse is not classifiable (111).[5]

[5]Romans 9:7 is interesting in terms of the interaction between negation and the reason clause. A classic example of this sort of interaction is (i): in Reading 1, the reason clause is outside the scope of negation, whereas in Reading 2, negation has narrow focus on the reason clause.

(i) John didn't marry Sue because she's rich.
Reading 1: The reason John didn't marry Sue was that she was rich.
Reading 2: Sue's wealth was not the reason John married her.

Romans 9:7 is more similar to Reading 2 in that the causal relation is denied. Yet, there is a crucial difference: Reading 2 of (i) implies that John *did* marry Sue, whereas Romans 9:7 implies that the Israelites are *not* all Abraham's children. The difference stems, I believe, from the meaning of the reason clause: in (i) the reason clause expresses a motivation for an action, while in Romans 9:7 the reason clause expresses a premise for a logical conclusion. Thus, Romans 9:7 can be paraphrased in contemporary English as *Just because they are Abraham's seed doesn't mean they are all his children.*

(111) Koine Greek
oud' hoti eisin sperma Abraam pantes tekna.
nor because be.PRS.3PL seed.NOM.SG.N Abraham.GEN all.NOM.PL.M child.NOM.PL
'Nor because they are Abraham's seed [they are] all [his] children.' [NA28]

The *Vulgate* follows the Greek source in the two verses (112).

(112) Latin

a. non enim omnes qui ex Israhel hii sunt Israhel
NEG for all.NOM.PL.M who.NOM.PL.M from Israel these.NOM.PL.M be.PRS.3PL Israel
'[But not that the word of God has failed.] For not all who are from Israel, these are Israel.'

b. neque quia semen sunt Abrahae omnes filii
nor because seed.NOM.SG.N be.PRS.3PL Abraham.GEN all.NOM.PL.M child.NOM.PL
'Nor because they are Abraham's seed [they are] all [his] children.'

Tyndale (113) and the *King James Version* (114) employ Strategy E in Romans 9:6 and a negative conjunction in Romans 9:7. For the purpose of classification, I consider instances with a negative conjunction as paraphrases, even though these actually follow the Greek and Latin sources. However, this is the default category for instances that cannot be classified into any of the strategies.

(113) a. I speake not these thynges as though the wordes of God had toke none effecte. For they are not all israhelites which cam off Israel, [Tyndale, Rom. 9:6; 1526]

b. Nether are they all children strayghtway because they are the seede of Abraham. [ibid, Rom. 9:7]

(114) a. Not as though the word of God hath taken none effect. For they are not all Israel which are of Israel: [KJV, Rom. 9:6; 1611]

b. Neither because they are the seed of Abraham are they all children: [ibid, Rom. 9:7]

Luther likewise employs Strategy E in Romans 9:6 and a negative conjunction in Romans 9:7. Notice, however, that (115b) could conceivably be parsed differently, with *nicht* modifying the QP instead of being part of the complex conjunction *auch nicht* 'nor'. In that case, (115b) would be classified as a B construction. This analysis is unlikely since Luther does

Strategy	Text	Translation	Year
A	*for **all** who have sprung from Israel do **not** count as Israel*	Weymouth	1903
B	*For **not all** descendants of Israel are truly Israel*	NEB	1970
D	*It is **not all** who are sprung from Israel that are Israel*	Rutherford	1900
E	*For they are **not all** Israel which are of Israel*	KJV	1611
Paraphrase	*For you can**not** count **all** "Israelites" as the true Israel of God*	Phillips	1958

Table 5.5: Representative examples of English translations of Romans 9:6

not employ Strategy B anywhere else in this text. To be on the safe side, I classify similar instances in other languages as paraphrases.

(115) Early New High German

a. denn es sind nicht alle Jsraeliter/ die von Jsrael sind/
for it be.PRS.3PL NEG all.NOM.PL Israelites who.NOM.PL from Israel be.PRS.3PL
'For they (lit. 'it') are not all Israelites who are from Israel.'
[Luther, Rom. 9:6; 1522]

b. auch nicht alle die Abrahams samen sind/ sind darũb auch kinder.
also NEG all.NOM.PL who.NOM.PL Abraham.GEN seed.NOM.SG be.PRS.3PL, be.PRS.3PL therefore also child.NOM.PL
'Nor all who are Abraham's seed are therefore also children.' [ibid, Rom. 9:7]

Even though many later translations similarly use a negative conjunction in Romans 9:7, there are also some that do not, exemplified in (116).

(116) And they are not all children because they are the seed of Abraham
[*Bible in Basic English*; 1941 (1965)]

Tables 5.5 and 5.6 provide representative examples of the various ways Romans 9:6 and 9:7 have been translated into English, respectively.

Romans 10:16 This verse addresses the mixed responses to the gospel among the Jews. This verse is part of a larger passage emphasizing that faith, rather than ethnic origin or adherence to the law, is the key to righteousness. By referencing Isaiah, Paul underscores the prophetic anticipation that not all Israelites would accept the gospel. This reference serves to affirm that such disbelief among some Israelites aligns with God's overarching sovereign plan, rather than contradicting it.

Strategy	Text	Translation	Year
B	*Not all of Abraham's children are called Abraham's descendants*	CEB	1970
E	*they are not all children of Abraham because they are descended from Abraham*	Moffatt	1922
Paraphrase	*Neither because they are the seed of Abraham are they all children*	KJV	1611

Table 5.6: Representative examples of English translations of Romans 9:7

> 11 The scripture says, "No one who believes in him will be put to shame." 12 For
> there is no distinction between Jew and Greek; the same Lord is Lord of all and
> is generous to all who call on him. 13 For "everyone who calls on the name of the
> Lord shall be saved." 14 But how are they to call on one in whom they have not
> believed? And how are they to believe in one of whom they have never heard?
> And how are they to hear without someone to proclaim him? 15 And how are
> they to proclaim him unless they are sent? As it is written, "How beautiful are
> the feet of those who bring good news!" 16 **But not all have obeyed the
> good news**, for Isaiah says, "Lord, who has believed our message?"
>
> [NRSVue, Rom. 10:11-16]

The Greek text employs Strategy B in this verse (117).

(117) Koine Greek

All'	ou	pantes	hypēkousan	tōi	euangeliōi.
but	NEG	all.NOM.PL.M	obey.AOR.3PL	the.DAT.SG.N	gospel.DAT.SG

'But not all obeyed the gospel.' [NA28]

The *Vulgate* follows the Greek source in this verse (118).

(118) Latin

sed	non	omnes	oboedierunt	evangelio
but	NEG	all.NOM.PL.M	obey.PERF.3PL	gospel.DAT.SG

'But not all obeyed the gospel.' [*Vulgate*]

Tyndale, the *King James Version*, and Luther employ Strategy E in this verse, using a personal pronoun with anaphoric reference to clarify that the universal quantification applies specifically to the Jews rather than all people.

(119) But they have nott all obeyed to the gospell. [Tyndale; 1526]

(120) But they haue not all obeyed the Gospel. [KJV; 1611]

Strategy	Text	Translation	Year
A	*But **all** have **not** obeyed the Gospel*	Purver	1764
B	*But **not all** the Israelites accepted the good news*	NIV	2011
E	*But they haue **not all** obeyed the Gospel*	KJV	1611

Table 5.7: Representative examples of English translations of Romans 10:16

(121) Early New High German

Aber	sie	sind	nit	alle	dem	Euangelio	gehorsam
but	they.NOM	be.PRS.3PL	NEG	all.NOM.PL	the.DAT.SG.N	gospel.DAT.SG	obedient

'But they are not all obedient to the gospel.' [Luther; 1522]

Not all translations use a personal pronoun, of course, as demonstrated in (122), which employs Strategy A.

(122) But al do not obey the Gospel. [*Douay–Rheims Bible*; 1582]

Table 5.7 provides representative examples of the various ways Romans 10:16 has been translated into English.

Romans 12:4 This verse uses the metaphor of the human body to illustrate the concept of the Christian community. Paul emphasizes that diversity within the church is essential and intentional, highlighting the interconnectedness and interdependence of Christians.

> 3 For by the grace given to me I say to everyone among you not to think of yourself more highly than you ought to think but to think with sober judgment, each according to the measure of faith that God has assigned. 4 For as in one body we have many members **and not all the members have the same function**, 5 so we, who are many, are one body in Christ, and individually we are members one of another. [NRSVue, Rom. 12:3-5]

The Greek text employs Strategy A in Romans 12:4 (123). Notice that this example involves a *same*-type predicate, so it is not considered a genuine instance of Strategy A (see §2.3).

(123) Koine Greek

ta	de	melē	panta	ou	tēn	autēn
the.NOM.PL.N	but	member.NOM.PL	all.NOM.PL.N	NEG	the.ACC.SG.F	same.ACC.SG.F
echei	praxin.					
have.PRS.3SG	function.ACC.SG					

'[For just as in one body we have many members,] but all the members do not have the same function' [NA28]

The *Vulgate* follows the Greek source in this verse (124).

Strategy	Text	Translation	Year
A	*and **all** members haue **not** the same office*	KJV	1611
B	*and **not all** members have the same function*	Montgomery	1924
E	*and these members do **not all** have the same function*	NIV	2011
Paraphrase	*and **all** the parts have different functions to perform*	TCNT	1904

Table 5.8: Representative examples of English translations of Romans 12:4

(124) Latin
omnia autem membra non eundem actum
all.NOM.PL.N but member.NOM.PL NEG same.ACC.SG.M function.ACC.SG
habent
have.PRS.3PL
'[For just as in one body we have many members,] but all the members do not have the same function' [*Vulgate*]

Tyndale and the *King James Version* similarly employ Strategy A.

(125) As we have many members in one body: and all members have not one office: [Tyndale; 1526]

(126) For as we haue many members in one body, and all members haue not the same office: [KJV; 1611]

Luther also employs Strategy A here, and nowhere else in the text.

(127) Early New High German
aber alle gelider nicht eynerley gescheffte haben
but all.NOM.PL member.NOM.PL NEG same activity.ACC.SG have.PRS.3PL
'But all members do not have the same activity.' [Luther; 1522]

Table 5.8 provides representative examples of the various ways Romans 12:4 has been translated into English.

1 Corinthians 6:12 This verse addresses the theme of Christian liberty and its limits. It begins with the phrase "All things are permissible for me", which is presumably an assertion made by the Corinthians or a prevalent attitude among them, to which Paul is responding. Paul emphasizes that while Christians are free from the strictures of the Jewish law, this freedom should not lead to behaviors that are detrimental or enslaving.

> 9 Do you not know that wrongdoers will not inherit the kingdom of God? Do not be deceived! The sexually immoral, idolaters, adulterers, male prostitutes, men who engage in illicit sex, 10 thieves, the greedy, drunkards, revilers, swindlers—none of these will inherit the kingdom of God. 11 And this is what some of

> you used to be. But you were washed, you were sanctified, you were justified in the name of the Lord Jesus Christ and in the Spirit of our God. 12 “All things are permitted for me,” **but not all things are beneficial**. “All things are permitted for me,” but I will not be dominated by anything.
>
> [NRSVue, 1 Cor. 6:9-12]

The Greek text employs Strategy B in this verse (128).

(128) Koine Greek

Panta	moi	exestin	all’	ou	panta	sympherei.
all.NOM.PL.N	1SG.DAT	be.allowed.PRS.3SG	but	NEG	all.NOM.PL.N	be.useful.PRS.3SG

‘Everything is allowed to me, but not everything is useful.’ [NA28]

The *Vulgate* follows the Greek source in this verse (129).

(129) Latin

omnia	mihi	licent	sed	non	omnia	expediunt
all.NOM.PL.N	1SG.DAT	be.allowed.PRS.3PL	but	NEG	all.NOM.PL.N	be.useful.PRS.3PL

‘Everything is allowed to me, but not everything is useful.’ [*Vulgate*]

Tyndale (130) and the *King James Version* (131) employ Strategy A.

(130) All thynges are lawfull unto me: but all thynges are not profitable. I maye do all thynges: but I will be brought under nomans power. [Tyndale; 1526]

(131) All things are lawfull vnto mee, but all things are not expedient: all things are lawfull for mee, but I will not bee brought vnder the power of any. [KJV; 1611]

Luther employs Strategy E (132). It is not clear, however, whether the subject pronoun *es* ‘it’ is expletive or anaphorically refers to the various sins mentioned in the preceding verses. Note that Luther departs quite substantially from the Greek source in the first part of the sentence. Some contemporary translations follow Luther’s wording in this case, too.

(132) Early New High German

Jch	hab-s	alles	macht/	es	nutzt	myr	aber
I	have.PRS.1SG-it	all.GEN.SG.N	power.ACC.SG	it	be.useful.PRS.3SG	me.DAT	but

nicht	alles
NEG	all.NOM.SG.N

‘I have power over everything; it is not all, however, useful for me.’ [Luther; 1522]

Table 5.9 provides representative examples of the various ways 1 Corinthians 6:12 has been translated into English.

1 Corinthians 8:7 This verse is concerned with the varying levels of understanding and conviction among Christian believers regarding practices associated with idolatry. Paul is responding to a debate within the Corinthian church about whether it is permissible for

Strategy	Text	Translation	Year
A	*but **all** things are **not** expedient*	KJV	1611
B	*but **not all** things are expedient*	ASV	1901
Paraphrase	*but that does **not** mean that **everything** is good for me*	Phillips	1958

Table 5.9: Representative examples of English translations of 1 Corinthians 6:12

Christians to eat meat that has been sacrificed to idols. While some believers, endowed with a strong understanding of Christian freedom, recognize that an idol has no real existence and therefore see no issue with eating such meat, others, whose consciences are still sensitive from their previous pagan practices, feel guilty when they eat it. Paul uses this situation to teach that the exercise of Christian freedom should be sensitive to the consciences of other believers. This passage encourages more mature believers to consider the impact of their actions on those who are less mature.

> 1 Now concerning food sacrificed to idols: we know that "all of us possess knowledge." Knowledge puffs up, but love builds up. 2 Anyone who claims to know something does not yet have the necessary knowledge, 3 but anyone who loves God is known by him. 4 Hence, as to the eating of food offered to idols, we know that "no idol in the world really exists" and that "there is no God but one." 5 Indeed, even though there may be so-called gods in heaven or on earth—as in fact there are many gods and many lords— 6 yet for us there is one God, the Father, from whom are all things and for whom we exist, and one Lord, Jesus Christ, through whom are all things and through whom we exist. 7 **It is not everyone, however, who has this knowledge.** Since some have become so accustomed to idols until now, they still think of the food they eat as food offered to an idol, and their conscience, being weak, is defiled. [NRSVue, 1 Cor. 8:1-7]

The Greek text employs Strategy B in this verse (133). Notice that the QP occurs inside a PP in this case.

(133) Koine Greek
All' ouch en pasin hē gnōsis.
but NEG in all.DAT.PL.M the.NOM.SG.F knowledge.NOM.SG
'But not in all is the knowledge.' [NA28]

The *Vulgate* follows the Greek source in this verse (134).

(134) Latin
sed non in omnibus est scientia
but NEG in all.DAT.PL.M be.3SG knowledge.NOM.SG
'But not in all is (the) knowledge.' [*Vulgate*]

Strategy	Text	Translation	Year
A	*But **all** have **not** this knowledge*	Thomson	1808
B	*But **not everyone** possesses this knowledge*	NIV	2011
D	*Still, it is **not every** one that has this knowledge*	TCNT	1904
Paraphrase	*Howbeit there is **not** in **euerie** man that knowledge*	KJV	1611

Table 5.10: Representative examples of English translations of 1 Corinthians 8:7

Tyndale (135) employs Strategy A in this verse, whereas the *King James Version* (136) employs an existential construction, which is classified as a paraphrase. Another common type of paraphrase is where *the knowledge* is the grammatical subject, demonstrated in (137).

(135) But every man hath not knowledge. [Tyndale; 1526]

(136) Howbeit there is not in euerie man that knowledge [KJV; 1611]

(137) However, this knowledge is not in all [*The Living Oracles*; 1826]

Luther (138) employs an expletive inversion construction (Strategy E, see §7.5.2, §9.1.2).

(138) Early New High German

Es	hat	aber	nicht	yderman	das	wissen
it	have.3SG	but	NEG	everyone	the.ACC.SG.N	knowledge

'But not everyone has the knowledge.' (lit. 'It has not, however, everyone the knowledge.') [Luther; 1522]

Table 5.10 provides representative examples of the various ways 1 Corinthians 8:7 has been translated into English.

1 Corinthians 10:23 This verse revisits the theme of Christian freedom and its ethical implications, a topic Paul also explores in 1 Corinthians 6:12, whose first part is almost identical to the first part of 1 Corinthians 10:23. Paul maintains that while Christians might be free to engage in many activities without violating any explicit command, this freedom comes with a responsibility to consider the impact of their actions on others and on the community as a whole.

> "All things are permitted," **but not all things are beneficial**. "All things are permitted," **but not all things build up.** [NRSVue, 1 Cor. 10:23]

The Greek text employs Strategy B in both instances in this verse (139).

(139) Koine Greek
Panta exestin all' ou panta sympherei; panta
all.NOM.PL.N be.allowed.PRS.3SG but NEG all.NOM.PL.N be.useful.PRS.3SG all.NOM.PL.N
exestin all' ou panta oikodomei.
be.allowed.PRS.3SG but NEG all.NOM.PL.N build.up.PRS.3SG
'Everything is allowed, but not everything is useful; everything is allowed, but not everything builds up.' [NA28]

The *Vulgate* follows the Greek source in this verse.

(140) Latin
omnia licent sed non omnia expediunt; omnia
all.NOM.PL.N be.allowed.PRS.3PL but NEG all.NOM.PL.N be.useful.PRS.3PL all.NOM.PL.N
licent sed non omnia aedificant
be.allowed.PRS.3PL but NEG all.NOM.PL.N build.up.PRS.3PL
'Everything is allowed, but not everything is useful; everything is allowed, but not everything builds up.' [*Vulgate*]

Tyndale and the *King James Version* employ Strategy A in both instances.

(141) All thynges are laufull unto me, but all thynges are not expedient. All thynges are lawfull, but all thinges edifye not. [Tyndale; 1526]

(142) All things are lawfull for me, but all things are not expedient: All things are lawfull for mee, but all things edifie not. [KJV; 1611]

Luther employs Strategy E.

(143) Early New High German
Jch hab-s zwar alles macht/ aber es ist nicht
I have.PRS.1SG-it indeed all.GEN.SG.N power.ACC.SG but it be.PRS.3SG NEG
alles nutzlich. Jch hab es alles macht/ aber es
all.NOM.SG.N useful I have.PRS.1SG it all.GEN.SG.N power.ACC.SG but it
bessert nicht alles.
improve.PRS.3SG NEG all.NOM.SG.N
'I have indeed power over everything, but it is not all useful. I have power over everything, but it does not all improve.' [Luther; 1522]

While the two parts of 1 Corinthians 10:23 are very similar to each other, some translations employ a different strategy in each case, as demonstrated in (144). Therefore, each sentence is considered a separate data point.

(144) Everything is allowable, but not everything is profitable. Everything is allowable, but everything does not build others up. [Weymouth; 1903]

Strategy	Text	Translation	Year
A	*All things are lawfull for me, but **all** things are **not** expedient: All things are lawfull for mee, but **all** things edifie **not**.*	KJV	1611
B	*All things are lawful; but **not all** things are expedient. All things are lawful; but **not all** things edify.*	ASV	1901
Paraphrase	*'We are free to do anything', you say. Yes, but is **everything** good for us? 'We are free to do anything', but does **everything** help the building of the community?*	NEB	1970

Table 5.11: Representative examples of English translations of 1 Corinthians 10:23

Table 5.11 provides representative examples of the various ways 1 Corinthians 10:23 has been translated into English.

1 Corinthians 15:39 This verse is part of Paul's extended discussion on the resurrection of the body. Paul clarifies that just as there are different types of bodies suited to different life forms on earth, so too will the resurrected bodies be different from our earthly bodies. He uses this analogy to prepare his readers for the idea that the resurrection body, while real and tangible, will be fundamentally transformed, adapted for eternal life, much like the distinctiveness between earthly creatures.

> 35 But someone will ask, "How are the dead raised? With what kind of body do they come?" 36 Fool! What you sow does not come to life unless it dies. 37 And as for what you sow, you do not sow the body that is to be but a bare seed, perhaps of wheat or of some other grain. 38 But God gives it a body as he has chosen and to each kind of seed its own body. 39 **Not all flesh is alike**, but there is one flesh for humans, another for animals, another for birds, and another for fish. 40 There are both heavenly bodies and earthly bodies, but the glory of the heavenly is one thing, and that of the earthly is another.
>
> [NRSVue, 1 Cor. 15:35-40]

The Greek text employs Strategy B in this verse (145).

(145) Koine Greek

Ou	pasa	sarx	hē	autē	sarx.
NEG	all.NOM.SG.F	flesh.NOM.SG	the.NOM.SG.F	same.NOM.SG.F	flesh.NOM.SG

'Not all flesh is the same flesh.' [NA28]

The *Vulgate* follows the Greek source.

Strategy	Text	Translation	Year
A	***All*** *flesh is* ***not*** *the same flesh*	KJV	1611
B	***Not all*** *flesh is the same*	NIV	2011
E	*Flesh is* ***not all*** *the same in kind*	Rutherford	1900
Paraphrase	*Similarly there are different kinds of flesh*	*New Living Bible*	1996

Table 5.12: Representative examples of English translations of 1 Corinthians 15:39

(146) Latin

non	omnis	caro	eadem	caro
NEG	all.NOM.SG.F	flesh.NOM.SG	same.NOM.SG.F	flesh.NOM.SG

‘Not all flesh is the same flesh.’ [*Vulgate*]

Tyndale and the *King James Version* employ Strategy A.

(147) All flesshe is not one manner of flesshe [Tyndale; 1526]

(148) All flesh is not the same flesh [KJV; 1611]

Notice that this verse involves a *same*-type predicate, so these are not considered genuine instances of Strategy A (see §2.3). Some languages, such as Catalan (149), employ Strategy A here, but not anywhere else in the text.

(149) Catalan

Totes	les	carns	no	són	iguals
all.PL	the.PL	flesh.PL	NEG	be.PRS.3PL	same.PL

‘All flesh is not the same’ [*Bíblia Evangèlica Catalana*; 2000]

Luther employs Strategy C in this verse, in which the negative particle is fronted to clause-initial position.

(150) Early New High German

Nicht	ist	alles	fleysch	eynerley	fleysch
NEG	be.PRS.3SG	all.NOM.SG.N	flesh.NOM.SG	same	flesh.NOM.SG

‘Not all flesh is the same flesh’ [Luther; 1522]

Table 5.12 provides representative examples of the various ways 1 Corinthians 15:39 has been translated into English.

1 Corinthians 15:51 This verse continues Paul’s theological discussion on the resurrection, introducing the idea that not all Christians will experience death before Christ’s return; instead, those who are alive will be transformed instantaneously. Paul refers to this as a “mystery”, indicating a divine revelation that was previously unknown but has now been disclosed through the gospel. This transformation involves a shift from mortal to immortal form, applicable to all believers, whether deceased or alive at the time of Jesus’ return.

> 50 What I am saying, brothers and sisters, is this: flesh and blood cannot inherit the kingdom of God, nor does the perishable inherit the imperishable. 51 Look, I will tell you a mystery! **We will not all die**, but we will all be changed, 52 in a moment, in the twinkling of an eye, at the last trumpet. For the trumpet will sound, and the dead will be raised imperishable, and we will be changed.
> [NRSVue, 1 Cor. 15:50-52]

The word order in the Greek text corresponds to Strategy A, assuming that inverse scope is the intended interpretation (151). The word *koimaō* 'sleep' is used figuratively in the sense of 'die'.

(151) Koine Greek

Pantes	ou	koimēthēsometha,	pantes	de	allagēsometha.
all.NOM.PL.M	NEG	sleep.PASS.FUT.1PL,	all.NOM.PL.M	but	change.PASS.FUT.1PL

'All of us will not sleep, but all of us will be changed.' [NA28]

1 Corinthians 15:51 is the most problematic verse among those considered here in terms of textual variants. Specifically, the translation of this verse in the *Vulgate* (153) is based on a variant where the negative marker appears in the second clause, thus completely changing the meaning of this sentence. In addition, instead of the verb *koimēthēsometha* 'we will sleep', this variant has *anastēsometha* 'we will rise' (152).

(152) Koine Greek

Pantes	men	anastēsometha,	ou	pantes	de	allagēsometha.
all.NOM.PL.M	indeed	rise.MID.FUT.1PL	NEG	all.NOM.PL.M	but	change.PASS.FUT.1PL

'All of us will rise, but not all of us will be changed.' [NA28 (variant)]

(153) Latin

omnes	quidem	resurgemus,	sed	non	omnes	inmutabimur.
all.NOM.PL.M	indeed	rise.FUT.1PL,	but	NEG	all.NOM.PL.M	change.PASS.FUT.1PL

'All of will indeed rise, but not all of will be changed.' [*Vulgate*]

Medieval translations and early catholic translations in the Reformation era followed the version that appears in the *Vulgate*. However, there is a consensus that the version that appears in (151) is the correct one. Already in the early 16th century, Erasmus modified the Latin translation in his *Novum Testamentum Omne*, a bilingual Latin-Greek New Testament.[6] This bilingual NT was later used by Martin Luther and William Tyndale for their translations.

[6] The same version appears in the first edition from 1516, known as *Novum Instrumentum Omne*, but with slightly different orthography.

(154) Latin

Non	omnes	quidẽ	dormiemus,	omnes	tñ	immutabimur
NEG	all.NOM.PL.M	indeed	sleep.FUT.1PL	all.NOM.PL.M	however	change.PASS.FUT.1PL

'Not all of us will sleep, but all of us will be changed.'
[Erasmus, *Novum Testamentum Omne*; 1519]

Notice that Erasmus employs Strategy B rather than adhering to the Greek word order (his edition features the Greek version in (151)). Importantly, he interprets (151) as an inverse-scope construction (Strategy A), even though a transparent-scope interpretation ('None of us will sleep') remains hypothetically possible. The distinction between these two interpretations hinges on whether Paul assumes that only some among them will be alive when Jesus returns or that all will be. Later translations usually agree with Erasmus that inverse scope is intended. Thus, Tyndale, the *King James Version*, and Luther employ Strategy E in this verse.

(155) Beholde I shewe a mistery unto you: we shall not all slepe: butt we shall all be chaunged [Tyndale; 1526]

(156) Behold, I shew you a mysterie: we shall not all sleepe, but wee shall all be changed [KJV; 1611]

(157) Early New High German

Wyr	werden	nicht	alle	entschlaffen/	wyr	werden	aber	alle	verwandelt	werdẽ
we	will.1PL	NEG	all.NOM.PL	fall.asleep.INF	we	will.1PL	but	all.NOM.PL	change.PST.PTCP	be.INF

'We will not all fall asleep, but we will all be changed.' [Luther; 1522]

However, some translators seem to have understood the Greek text based on the transparent-scope reading. For instance, the Swedish bible translator Paul Petter Waldenström preserves the Greek word order (158),and an explanatory footnote he added suggests that he did not interpret this verse as involving inverse scope: "All of us who are alive when the Lord comes again, we will not die. Paul thought that he and in general all the believers to whom he wrote would experience the day of the Lord's return." (Waldenström, *Nya Testamentet*; 1894: 187, originally in Swedish).

(158) Swedish

vi	alla	skola	icke	afsomna,	utan	vi	alla	skola	förvandlas
we	all.PL	shall.PL	NEG	fall.asleep.INF	but	we	all.PL	shall.PL	transform.PASS.INF

'We all shall not fall asleep, but we all shall be transformed.' [Waldenström; 1894]

Given this ambiguity, when translators retain the Greek word order, it is unclear whether they have a specific reading in mind or if they are intentionally preserving the ambiguity. Another example comes from Italian, a language that generally does not employ inverse scope: the 1974 edition of the CEI, the official version of the Italian Catholic Church,

Strategy	Text	Translation	Year
A(?)	***All** of us wo**n't** die*	CEB	2011
B	***not all** of us are to die*	Moffatt	1922
E	*we shall **not all** sleepe*	KJV	1611

Table 5.13: Representative examples of English translations of 1 Corinthians 15:51

employs Strategy B (159a), whereas the 2008 edition instead follows the Greek word order (159b). It is difficult to determine the motivation behind this choice and whether inverse scope was intended in the latter case. Consequently, such instances are not classified as Strategy A in this study.

(159) Italian

a. non tutti, certo, moriremo, ma tutti saremo trasformati
NEG all.PL.M certainly die.FUT.1PL but all.PL.M be.FUT.1PL transform.PTCP.PL.M
'Not all of us, certainly, will die, but all of us will be transformed.'
[CEI (1st edition); 1974]

b. noi tutti non moriremo, ma tutti saremo trasformati
we all.PL.M NEG die.FUT.1PL but all.PL.M be.FUT.1PL transform.PTCP.PL.M
'We all will not die, but all of us will be transformed.'
[CEI (2nd edition); 2008]

Table 5.13 provides representative examples of the various ways 1 Corinthians 15:39 has been translated into English.

2 Thessalonians 3:2 This verse addresses concerns about opposition to the gospel. This verse is part of Paul's closing exhortations to the Thessalonian church, where he emphasizes the importance of prayer and support for the apostolic mission. Paul's request highlights the reality that the spread of the gospel often met with skepticism and opposition from those who did not share the Christian faith.

> 1 Finally, brothers and sisters, pray for us, so that the word of the Lord may spread rapidly and be glorified everywhere, just as it is among you, 2 and that we may be rescued from wicked and evil people, **for not all have faith**. 3 But the Lord is faithful; he will strengthen you and guard you from the evil one.
> [NRSVue, 2 Thess. 3:1-3]

The Greek text employs Strategy B in this verse (160). Like in (110), the conjunction *gar* 'for, because' occurs in between negation and the quantifier. Notice that the quantifier is not the grammatical subject: *hē pistis* 'the faith' is the subject, and the quantifier is part of a non-verbal predicate. However, many translations employ a possessive construction instead, where the QP is the subject.

Strategy	Text	Translation	Year
A	*for* ***all*** *men haue* ***not*** *faith*	KJV	1611
B	*for* ***not everyone*** *has faith*	NIV	2011
D	*for it is* ***not everybody*** *who has faith*	Weymouth	1903
Paraphrase	*for the faith is* ***not*** *held by* ***all***	Moffatt	1922

Table 5.14: Representative examples of English translations of 2 Thessalonians 3:2

(160) Koine Greek

Ou	gar	pantōn	hē	pistis.
NEG	for	all.GEN.PL.M	the.NOM.SG.F	faith.NOM.SG

‘For not of all is the faith.’ [NA28]

The *Vulgate* follows the Greek source.

(161) Latin

non	enim	omnium	est	fides
NEG	for	all.GEN.PL.M	be.3SG	faith.NOM.SG

‘For not of all is the faith.’ [*Vulgate*]

Tyndale and the *King James Version* employ Strategy A in this verse.

(162) and thatt we maye be delivered from unresonable and evyll men. For all men have not fayth: [Tyndale; 1526]

(163) And that we may bee deliuered from vnreasonable and wicked men: for all men haue not faith. [KJV; 1611]

Luther, on the other hand, expresses possession through a genitive construction, like the Greek source, but reverses the word order. This construction is considered a paraphrase, and it is found in quite a few other translations.

(164) Early New High German

Denn	der	glawbe	ist	nicht	ydermans	ding
for	the.NOM.SG.M	faith.NOM.SG	be.PRS.3SG	NEG	everyone.GEN	thing.NOM.SG

‘For faith is not everyone’s thing.’ [Luther; 1522]

Table 5.14 provides representative examples of the various ways 2 Thessalonians 3:2 has been translated into English.

Chapter 6

Corpus evaluation: Case studies

This chapter evaluates the reliability and representativeness of the NT corpus as evidence for the distribution of strategies employed by a given language. The objective is to detect and analyze any false positives, false negatives, or other potential biases in this corpus. To this end, I compare the distribution of strategies in the NT corpus with that of a larger corpus, specifically the online news section of the Leipzig Corpora Collection (Goldhahn et al. 2012). This collection contains randomly sampled sentences from online news websites, covering a variety of topics, such as politics, sports, and culture. I focus on news rather than general web data, which are also found in this corpora collection, as news data are likely to be more comparable across languages in terms of genre and register.

Collecting the relevant data from non-parallel corpora is considerably more time consuming, as it is not easy to query for the relevant constructions, and a fair amount of time is needed to manually sift through the hits. Therefore, I restrict this procedure to a selected group of languages as a proof of concept. Specifically, I focus on the standard Germanic languages, as these languages share structural similarities, and yet, according to the results of the NT corpus, there is considerable variation between them in the expression of 'Not all X are Y' propositions.

Given the low frequency of 'Not all X are Y' propositions, I use corpora of 1M sentences when available, so as not to miss rare strategies. These corpora contain raw text without part-of-speech tagging or syntactic parsing. The first step in the analysis was narrowing down the list of 1M sentences to those containing a negative marker and a universal quantifier within a distance of up to seven words between each other. The relevant sentences were then tokenized, lemmatized and annotated for parts of speech and dependency relations with the help of the R package UDPipe (Wijffels et al. 2023). This package provides pre-trained models based on the Universal Dependencies corpora and uses these models to automatically annotate large volumes of text. The resulting annotation is not perfect (see Straka and Straková 2017 for details), but it is possible to adjust the queries so as to make room for inconsistencies in the annotation. Once the relevant sentences were annotated, I designed language-specific queries to search for the various strategies.

To ensure comparability between the two corpora, I restricted the analysis of the news data to declarative sentences containing ‘Not all X are Y’ constructions, in which the quantifier occurs in subject position. For example, instances like *It’s not every day a healthcare company buys sapphires* are not included in the count. However, there still remain inherent differences in the structure of each corpus. First, the news corpus contains formulaic expressions such as *All is not lost*, which in some languages constitute a significant portion of the total instances of Strategy A and are absent from the NT corpus. Second, some Germanic languages sometimes employ XVS word order, usually (but not always) corresponding to Strategy C, where X is a clause-initial constituent other than the subject or the negative marker. This word order pattern tends to be more frequent in the news corpus, presumably because there happen to be more opportunities to use it, given that this corpus seems to have more syntactic complexity, on average, in terms of adverbial clauses and so forth. Third, Strategy E is extremely rare in some languages in the news corpus, presumably because this strategy is most often used with a referential personal pronoun in subject position, such as *We’re not all happy about it*, and this particular context tends to be underrepresented in the news corpus compared to the NT corpus. Finally, the news corpus contains contexts that are favorable toward Strategy A, such as cases where a ‘Not all X are Y’ construction is embedded in a Strawson-downward-entailing environment (see §2.3), and these are not represented in the NT corpus. Consequently, part of the observed variations between the corpora might be attributed to the above-mentioned factors.

The statistical analysis involved a Chi-squared test, with the null hypothesis being that the distribution of strategies is independent of the corpus. Given that some cells in the contingency tables had low counts, potentially compromising the validity of the Chi-squared test, Fisher’s exact test was also applied. This test was used to compare the relative frequency of Strategy A, which is the primary interest of this study, between the corpora. In this test, the combined frequencies of all remaining strategies (i.e., Strategies B to E) were collapsed together to produce a 2×2 matrix. Overall, the distributions of strategies tended to be similar across the two corpora, except for Faroese, where the most recent translations of the NT available to me were about 100 years old.

The remainder of this chapter is structured as follows: Section 6.1 presents the results for the West Germanic languages (English, German, Dutch, and Afrikaans), where Strategy B is the dominant strategy. Section 6.2 examines the North Germanic languages (Swedish, Norwegian, Danish, Icelandic, and Faroese), where the dominant strategies vary: Swedish and Faroese primarily employ Strategy A, while Danish and Norwegian prefer Strategy B, and Icelandic is unique in favoring Strategy C. Finally, Section 6.3 summarizes and discusses the results, comparing trends across languages and corpora.

6.1 West Germanic

6.1.1 English

The online news corpus consists of randomly sampled sentences from online news websites across the broad English-speaking world. Consequently, the results do not represent a specific dialect of English but rather an aggregate of various dialects.

A sample of 1M sentences yielded a total of 679 'Not all X are Y' constructions with a universal quantifier in subject position. Strategy B is by far the dominant strategy (85.8%), followed by Strategy E (9.5%). Strategy A is relatively rare (4.3%), whereas Strategy D is extremely rare (0.4%).

These results were compared to a collection of four translations published in the last 40 years: the *New Revised Standard Version* (1989), the *New Living Translation* (1996), the *New International Version* (2011 revision), and the *Common English Bible* (2011). Strategy B is the dominant strategy in all four translations.

Figure 6.1 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for these strategies.

A Chi-squared test comparing the distribution of strategies between the two corpora yielded a χ^2 value of 5.41 with 3 degrees of freedom and a p-value of 0.144. This p-value suggests that there is no statistically significant difference in the distribution of strategies between the two corpora, as it exceeds the conventional significance threshold of 0.05.

Due to small expected counts in some cells of the matrix, which might affect the reliability of the Chi-squared test results, Fisher's exact test was also applied for a more accurate measure. For the purpose of this test, the contingency table was converted to a 2×2 matrix, comparing the relative frequency of Strategy A across corpora. The Fisher's test did not show a significant association either, yielding a p-value of 0.081. The odds ratio from this test was 0.41, with a 95% confidence interval ranging from 0.15 to 1.42, further indicating that the difference between the news and NT corpora in the relative frequency of Strategy A could not be considered statistically significant. Overall, both tests support the conclusion that there is no substantial evidence of dependency between the corpus and the distribution of strategies.

Strategy A The online news corpus contains 29 instances of Strategy A, 10 of which are formulaic, as exemplified in (165). Of the remaining 19, seven occur in Strawson-downward-entailing environments, such as the complement of an emotive factive predicate (166a) or a counterfactual conditional (166b). These examples are not considered as independent evidence for the use of Strategy A (see §2.3). Similarly, one instance involves a *same*-type predicate (167), and another features a possibility modal scoping between negation and the quantifier (168). These, too, are not treated as genuine occurrences of Strategy A (see §2.3).

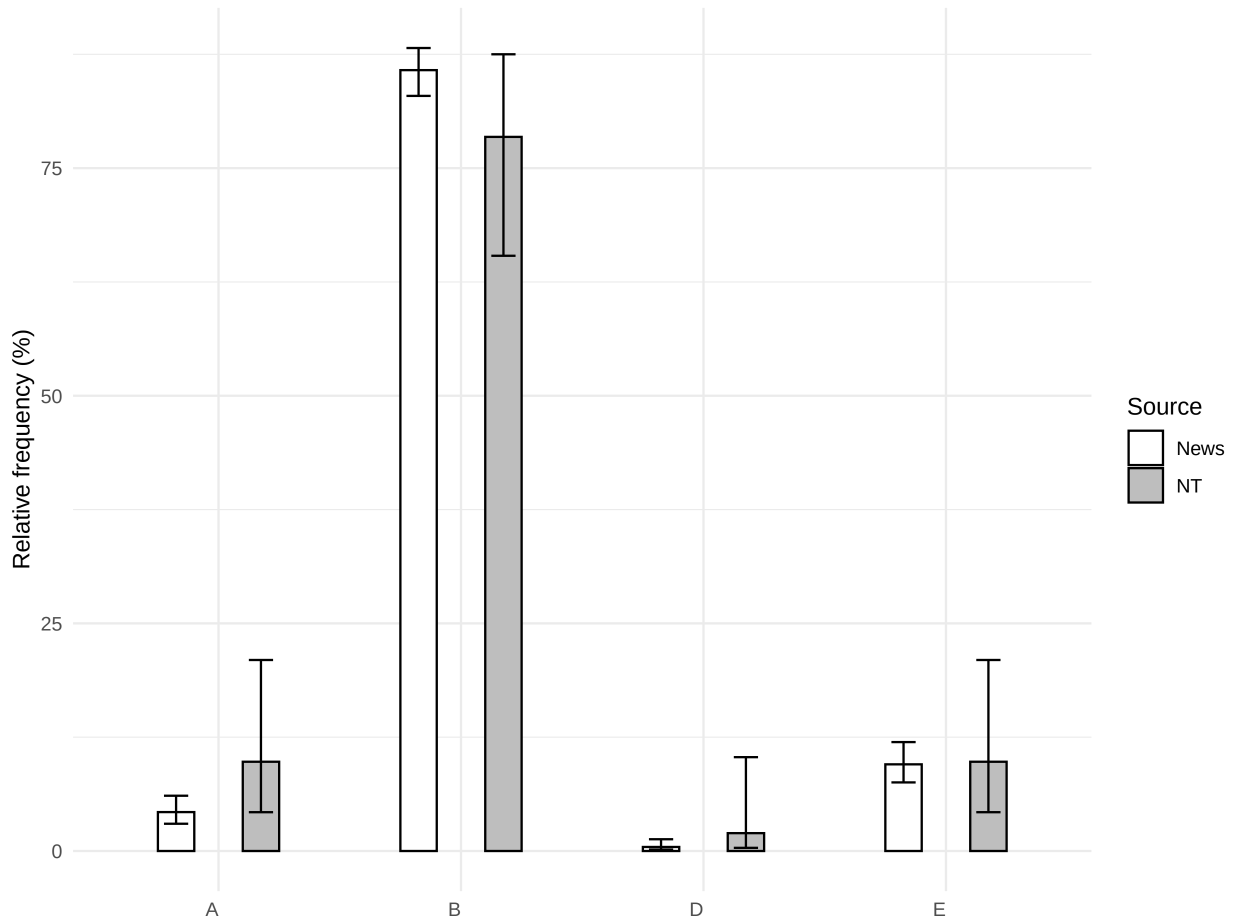


Figure 6.1: The distribution of strategies in English in the online news corpus compared to the NT corpus

(165) a. All is not lost for Mercedes, as Russell qualified in third, for his second-best qualifying run of the season and his second-best performance since back at the Australian Grand Prix.
b. But all is not well among the baskets, birdhouses and earthenware bowls.
c. So, according to Talmage, all is not gloom and doom, but a future for the region that shines brighter than ever for growth and prosperity.

(166) a. But the Union Army was so well equipped it is a wonder that every soldier did not gain at least thirty pounds.
b. Harsh Vardhan said he was "overjoyed" to have been elected as a Member of Parliament (MP) from the "prestigious" constituency of Chandni Chowk and this could never have happened if all communities had not supported him.

(167) By beginning to change the mowing regimes through this pilot, we can look to make a start - but there has to be a recognition that all verges are not the same.

(168) The need to cool down the temperature has developed a thirst for cold drinks but in a country where power outage is a phenomenon, everybody cannot cool his or her drinks.

This leaves us with merely 10 instances of Strategy A that meet the criteria of the present study for genuine evidence for the use of Strategy A, as demonstrated in (169). It is clear, then, that Strategy A is extremely rare in regular cases of 'Not all X are Y' sentences.

(169) a. We have responded to a complaint from Dawn Deaville and apologised that all options for pain relief were not discussed at the time.
b. Despite this ability looking good on paper, every squad is not taking advantage of the class and gameplay perks offered by Support legends.
c. Everybody is not that hardened criminal.
d. It's always cool to have a good response at the end when you spend so much effort, it's just a bit sad when everybody's not liking it but, I don't take it too personally.

In the NT corpus, Strategy A appears only in the *Common English Bible*, but it occurs five times in this version (170). Presumably, the disproportionately high frequency of Strategy A in this particular translation compared to the other versions is not a mere coincidence but rather indicative of dialectal variation (see §5.3). The absence of Strategy A in the remaining three translations is consistent with the low frequency of this strategy in the news corpus.

(170) a. But everyone hasn't obeyed the good news. [CEB, Rom. 10:16; 2011]
b. Everything is permitted, but everything isn't beneficial. Everything is permitted, but everything doesn't build others up. [ibid, 1 Cor. 10:23]
c. All flesh isn't alike. [ibid, 1 Cor. 15:39]
d. Pray too that we will be rescued from inappropriate and evil people since everyone that we meet won't respond with faith. [ibid, 2 Thess. 3:2]

Strategy B Strategy B is the dominant strategy in both corpora. Moreover, this strategy is rarely used in formulaic expressions, a fact already noted by Tottie and Neukom-Hermann (2010, 162-163), who write: "[The British National Corpus] does not contain a single instance of *not all is/was lost* [...] [s]imilarly, there are no instances of *not all is/was well*". Thus, if we only consider non-formulaic instances, the dominance of Strategy B is even greater. Representative examples are given in (171).

(171) a. When baking with apples, it's important to remember that not all apple varieties will work.
b. I know not everyone believes in this kind of thing.

c. The Elite have enlisted the help of the inaugural IWGP World Heavyweight Champion for this match, a man that every member of BCC is aware of but not all have stepped in the ring with.

Strategy B is the dominant strategy in all four translations (172).

(172) a. Not everyone can accept this teaching [NRSV, Matt. 19:11; 1989]
b. But not everyone welcomes the Good News [*New Living Translation*, Rom. 10:16; 1996]
c. But not everyone possesses this knowledge. [NIV, 1 Cor. 8:7; 1978 (2011)]
d. Not all who are descended from Israel are part of Israel. [CEB, Rom. 9:6; 2011]

Strategy D Strategy D is extremely rare in the news corpus, except in the formulaic expression *It's not every day that*. These instances are not included in the count since the quantifier is not the subject in these constructions but part of an adjunct. There are 11 instances in the corpus of *It's not every day that* and its variant *It's not all the time that* (173), but only three instances of Strategy D where the quantifier is the grammatical subject (174). These are all clefts, and there are no instances of Strategy D involving superordinate negation, such as *It is not (the case) that everyone came.*

(173) a. After all, it's not every day a member of the Royal Family comes into the nursing home!
b. It's not every day a healthcare company buys sapphires.
c. Sani, however, reacting to the development, in a post shared via Twitter, averred that it is not all the time when people close their eyes that they are sleeping or taking a nap.

(174) a. Because it is not everyone that is a politician.
b. But it's not everyone who is subject to these extreme limitations.
c. And I often say that it's not all pregnancies where (babies) are exposed to alcohol, cigarettes or cannabis that they're going to have negative outcomes.

Similarly, Strategy D appears only once in one of the NT version (175).

(175) It is not everyone, however, who has this knowledge. [NRSV, 1 Cor. 8:7; 1989]

Strategy E About half of the instances of Strategy E are with *it* as a subject pronoun. Usually, *it* functions as an expletive pronoun (176a), but it is sometimes used anaphorically (176b). The remaining instances are with referential pronouns (177) and NPs (178).

(176) a. However, it is not all smooth sailing for bookmakers.
b. So if your goal is to shrink government, and lower taxes, and do an IRA, it doesn't all fit together.

(177) We aren't all carrying shiny new phones right now.

(178) Rome's influences are not all positive.

Strategy E is often used in formulaic expressions, as demonstrated in (179). There are also many semi-formulaic expressions, such as *It's not all good/bad news* (180) or *It's not all about* (181).

(179) Bear market conditions have certainly trimmed dialogue around the most crypto-friendly markets in the US, but one new report suggests that it's not all doom and gloom.

(180) But it's not all bad news for clean energy.

(181) Carol has always told me that it's not all about the money but it's about making others in need full of both heart and respect.

Strategy E is also found in all NT translations (182), but no more than twice out of 14 times in each version.

(182) a. For just as each of us has one body with many members, and these members do not all have the same function [NIV, Rom. 12:4; 1978 (2011)]
b. Listen, I will tell you a mystery! We will not all die, but we will all be changed [NRSV, 1 Cor. 15:51; 1989]

6.1.2 German

A sample of 1M sentences from online news yielded a total of 738 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is B (65.8%), followed by Strategies C (29.5%) and E (4.2%). Strategy A is extremely rare (0.5%), appearing only four times in this corpus.

The online news corpus was compared to a collection of three translations of the NT from the 21st century: *BasisBibel* (2010 [2021]), *Neue Genfer Übersetzung* (2011), and *Hoffnung für alle* (2015).

Figure 6.2 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. The confidence intervals of Strategies A and B a similar across corpora, but there is a significant difference in the distribution of Strategies C and E. This discrepancy is also found in other Germanic languages with a V2 word order, but it happens to be more pronounced in German. It probably arises from underlying differences between these two corpora in the distribution of linguistic contexts: on the one hand, the NT corpus has a higher proportion of clauses where the subject is a personal pronoun, a context that is strongly associated with Strategy E; on the other hand, the news corpus has a higher proportion of XVS clauses, corresponding to Strategy C, where some constituent other than the subject occurs in clause-initial position.

A Chi-squared test was conducted without the counts for Strategy A, since this test is not designed for such low counts. The test produced a χ^2 value of 25.28 with 2 degrees of

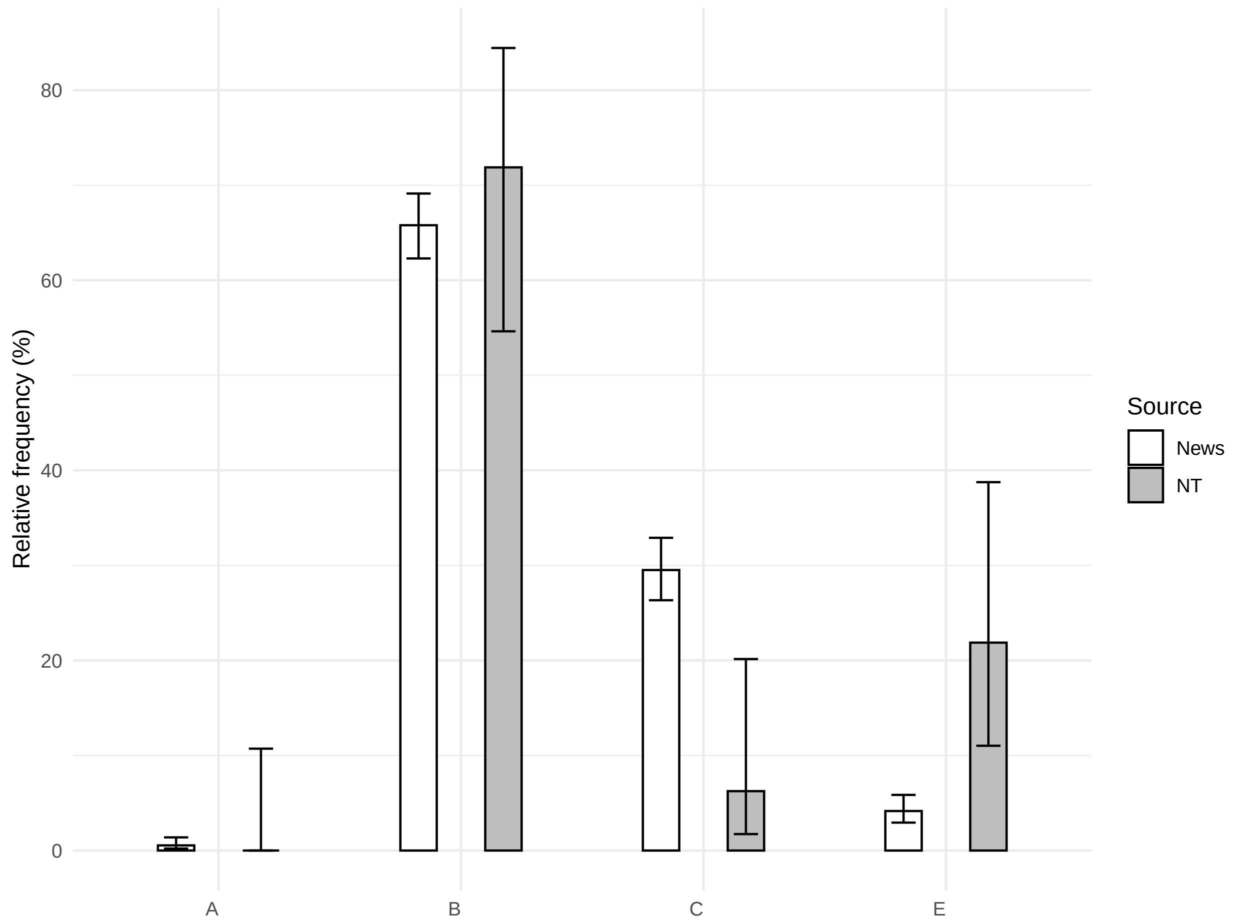


Figure 6.2: The distribution of strategies in German in the online news corpus compared to the NT corpus

freedom and a p-value of less than 0.001. This result suggests that there is a significant difference in the distribution of the remaining strategies between the two corpora, which is probably due to the variation in the frequencies of Strategies C and E, as the relative frequency of Strategy B is similar across corpora.

Strategy A Strategy A is extremely rare in the news corpus, appearing only four times. In three of these instances, the quantifier occurs in the canonical subject position, as exemplified in (183), while the remaining instance involves XVS word order (184). On the other hand, the NT corpus does not contain instances of Strategy A.

(183) Alle iPhones werden allerdings nicht mit dem OS-Update bedacht
all iPhones are however NEG with the.DAT.SG.N OS-Update considered
'However, not all iPhones are considered for the OS update [– Apple usually provides operating system updates for five years.].'

(184) Doch von heute auf morgen nutzbar sind alle nicht.
but from today to tomorrow usable are all NEG
'[There are almost 600 bunkers between Flensburg and Freilassing.] But not all are usable overnight. [… Reopening these bunkers could take two days, two weeks or two years, depending on their condition.]'

Strategy B Strategy C is used in XVS clauses, a construction found both in the news corpus (185) and the NT corpus (186).

(185) Am Zebrastreifen ist nicht alles Schwarz und Weiß, aber doch vieles.
at the.crosswalk is NEG everything black and white, but yet much
'Not everything at the crosswalk is black and white, but a lot is.'

(186) Allerdings gehören nicht alle Israeliten auch wirklich zu Israel.
however belong.PRS.3PL NEG all Israelites also really to Israel
'However, not all Israelites truly belong to Israel.'
[*BasisBibel*, Rom. 9:6; 2010 (2021)]

Strategy E Strategy E is primarily used with personal and expletive pronouns, including expletive inversion constructions such as (187). Although relatively rare, this construction also appears in the NT corpus (188).

(187) Es wohnen ja nicht alle Grünenwähler in den reichen Speckgürteln
it live.PRS.3PL indeed NEG all Green.voters in the rich commuter.belts
der Großstädte.
of.the big.cities
'Not all Green party voters live in the wealthy suburbs of the big cities.'

(188) Aber es gehören eben nicht alle Israeliten zum 'wahren' Israel.
but it belong.PRS.3PL just NEG all Israelites to.the true Israel
'But not all Israelites belong to the 'true' Israel.'
[*Neue Genfer Übersetzung*, Rom. 9:6; 2011]

6.1.3 Dutch

A corpus of 1M sentences of online news yielded a total of 1373 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is Strategy B (69%), followed by Strategies C (17.7%) and E (10.8%). Strategy A is rare (2.5%).

The online news corpus was compared to a collection of three translations of the NT from the 21st century: *Nieuwe Bijbelvertaling* (2004 [2011]), *Het Boek* (1988 [2008]), and *BasisBijbel* (2013).

Figure 6.3 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for these strategies. The discrepancy in the relative frequencies of Strategies C and E is also observed in other Germanic languages with V2 word order.

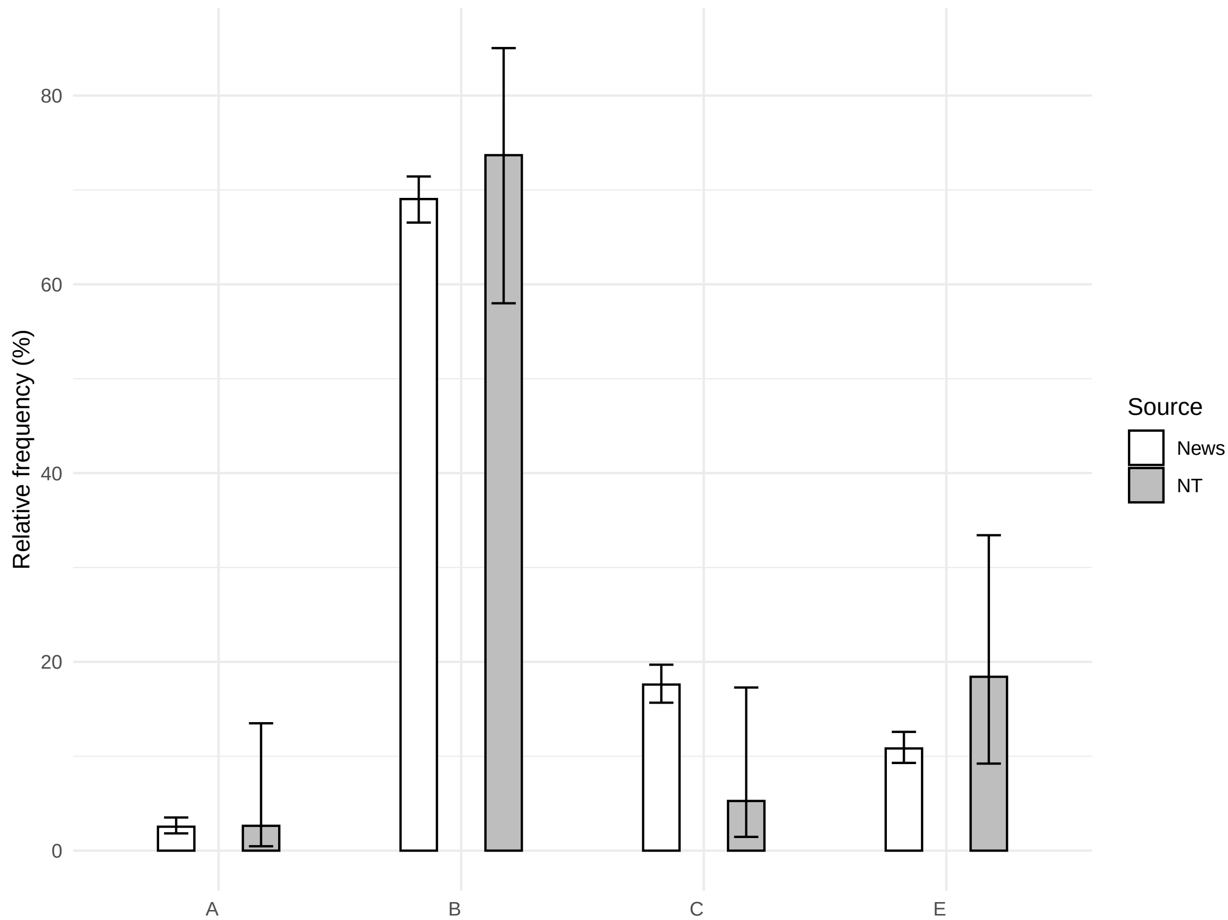


Figure 6.3: The distribution of strategies in Dutch in the online news corpus compared to the NT corpus

A Chi-squared test comparing the distribution of strategies between the two corpora yielded a χ^2 value of 5.3 with 3 degrees of freedom and a p-value of 0.151, suggesting that there is no statistically significant difference in the distribution of strategies between the two corpora.

Fisher's exact test was also conducted to compare the frequency of Strategy A between the two corpora. As the proportion of Strategy A happens to be extremely similar between these two corpora, this test yielded a p-value of 1. The odds ratio is 0.97 with a 95% confidence interval ranging from 0.15 to 40.32, further indicating that there is no significant difference between the corpora in the relative frequency of Strategy A.

Strategy A is rare in the news corpus. Most instances involve a contrast with a salient affirmative universal statement, which is either explicitly or implicitly present in the context, as demonstrated by the first sentence in (189). There are also formulaic instances like (190). Overall, Strategy A is by no means a common strategy in the news corpus.

(189) Net zoals dat elke niet-gevaccineerde niet meteen een antivaxx-wappie is is
just like that every not-vaccinated NEG immediately a anti.vax-nut is is
niet elke alcoholdrinker een alcoholist.
not every alcohol.drinker a alcoholic
'Just like not every unvaccinated person is immediately an anti-vax nut, not every alcohol drinker is an alcoholic.'

(190) Alle hoop is nog niet verloren voor Europa.
all hope is still NEG lost for Europe
'All hope is not yet lost for Europe.'

Similarly, the NT corpus contains only one instance of Strategy A, and it involves a *same*-type predicate (191).

(191) Elk vlees is niet gelijk
every flesh is NEG alike
'Not all flesh is alike.' [*Het Boek*, 1 Cor. 15:39, 1988 (2008)]

The distribution of Strategies C and E is fairly similar to German, although there are no instances of an expletive inversion construction.

6.1.4 Afrikaans

Although this dataset is intended to consist of online news, similar to the corpora used for other languages in this sample, approximately half of the instances seem to originate from non-news websites, particularly those related to Christian discourse. Consequently, the genre of this corpus differs from that of the other news corpora. Nevertheless, this dataset still provides valuable independent evidence concerning the distribution of strategies in Afrikaans, and the substantial number of online news examples allows for reliable generalizations.

A corpus of 300K sentences yielded a total of 188 'Not all X are Y' constructions with a universal quantifier in subject position. Considering both news and non-news sources, the dominant strategy is Strategy B (61.3%), followed by Strategies A (18.3%), E (14.3%), C (4.1%), and D (2%).

These results were compared to a collection of three translations of the NT from the 21st century: *Nuwe Lewende Vertaling* (2006), *Bybel vir Almal* (2008), and the official translation by the South African Bible Society (2020).

Figure 6.4 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for these strategies.

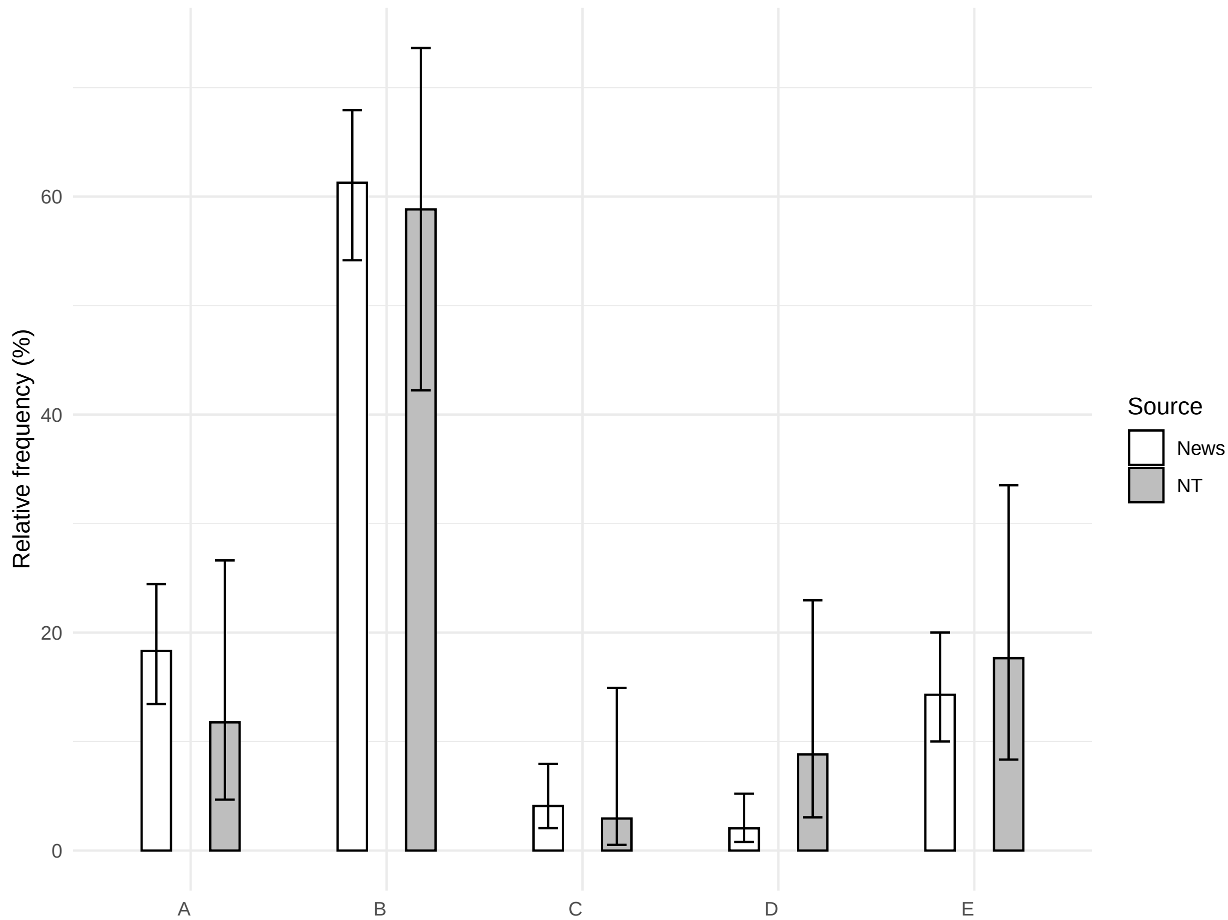


Figure 6.4: The distribution of strategies in Afrikaans in the online news corpus compared to the NT corpus

A Chi-squared test comparing the distribution of strategies between the two corpora yielded a χ^2 value of 5.27 with 4 degrees of freedom and p-value of 0.261, suggesting that there is no statistically significant difference in the distribution of strategies between the two corpora.

Fisher's exact test was also conducted to compare the frequency of Strategy A between the two corpora. This test yielded a p-value of 0.464. The odds ratio is 1.69 with a 95% confidence interval ranging from 0.54 to 7.03, further indicating that there is no significant difference between the corpora in the relative frequency of Strategy A.

Strategy A Strategy A is fairly common in the news corpus, and it often appears in non-contrastive contexts, such as in (192). Nevertheless, about a third of the instances of Strategy A are formulaic, most commonly the expression in (193).

(192) Elke rugbybal gaan nie oor die dwarslat seil nie, maar moenie moed verloor nie.
every rugby.ball go.PRS NEG over the crossbar sail.PRS NEG, but PROH courage lose.INF NEG
'Not every rugby ball sails over the crossbar, but do not lose courage.'

(193) Maar alles is nie pluis nie.
but everything is NEG right NEG
'But all is not well.'

Among the NT translations considered here, Strategy A only appears in the *Bybel vir Almal* (194), although it is also attested in earlier translations into Afrikaans, such as the 1983 edition by the South African Bible Society.

(194) Maar al die gelowiges verstaan nie goed dat daar net een God is nie
but all the believers understand NEG well that there only one god is NEG
'But not all the believers understand well that there is only one god'
[*Bybel vir Almal*, 1 Cor. 8:7; 2008]

Strategy B Strategy B involves negative concord (195)-(196).[1]

(195) Maar nie almal sal van HAIRSPRAY hou nie.
but NEG everyone will of Hairspray like NEG
'But not everyone will like Hairspray.'

(196) Maar nie alle gelowiges besef dit nie.
but not all believers realize this not
'But not all believers realize this.' [*Nuwe Lewende Vertaling*, 1 Cor. 8:7; 2006]

Strategy D Strategy D is relatively rare, but it appears in non-formulaic contexts (197)-(198).

(197) Dis nie alle afgestorwenes wat na hul dood spook nie.
it.is NEG all deceased who after their death haunt NEG
'Not all the deceased haunt after their death.'

(198) Dit is nie al die mense wat kan verstaan wat Ek gesê het nie
it is NEG all the people who can understand what I said have NEG
'Not all people can understand what I said' [*Bybel vir Almal*, Mt 19:11, 2008]

[1] There is one example without negative concord, but this might be a typo.

(i) Nie almal was bedoel om daai trui te dra.
not everyone was intend.PTCP to that sweater to wear
'Not everyone was meant to wear that sweater.'

6.2 North Germanic

6.2.1 Swedish

A corpus of 1M sentences of online news yielded a total of 717 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is Strategy A (72.3%), followed by Strategy B (11.3%), Strategy C (9.3%), and Strategy D (6.7%). Strategy E is extremely rare in this corpus (0.4%).

These results were compared to a collection of three translations of the NT published in the last 30 years: the *Svenska Folkbibeln* (1998), the official *Bibel 2000* (2000), and the *nuBibeln* (2015). Strategy A is the dominant strategy in all three translations.

Figure 6.5 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for all strategies, with the notable exception of Strategy E, which is extremely rare in the news corpus.

A Chi-squared test comparing the distribution of strategies between the two corpora yielded a X^2 value of 51.41 with 4 degrees of freedom and a p-value of less than 0.001, indicating substantial differences between the two corpora. These results raise the suspicion that the disproportionately low frequency of Strategy E in the news corpus biased the results of the Chi-squared test. Therefore, Strategy E was excluded from an additional analysis to isolate its impact on the statistical results. The revised Chi-squared test produced a X^2 value of 1.64 with 3 degrees of freedom and a p-value of 0.651. As expected, this outcome suggests that, in the absence of Strategy E, there is no statistically significant difference in the distribution of the remaining strategies between the two corpora.

Additionally, Fisher's exact test was applied to compare the relative frequencies of Strategy A across corpora. This test produced a non-significant p-value of 0.22. The odds ratio is 1.64 with a 95% confidence interval ranging from 0.71 to 3.64, further indicating that the difference in the frequency of Strategy A between the two corpora is not statistically significant.

Overall, both tests suggest that there is no substantial evidence for a systematic difference between the two corpora in the distribution of the various strategies, except for Strategy E, which is unexpectedly more frequent in the NT corpus, even in absolute numbers.

Strategy A In the news corpus, Strategy A appears in a variety of contexts, spanning both main clauses (199) and dependent clauses (200). Interestingly, it is also sometimes employed in XVS clauses (201).

(199) Alla har inte ens en mobil eller dator.
everyone have.PRS NEG even a cell.phone or computer
'Not everyone even has a cell phone or computer.'

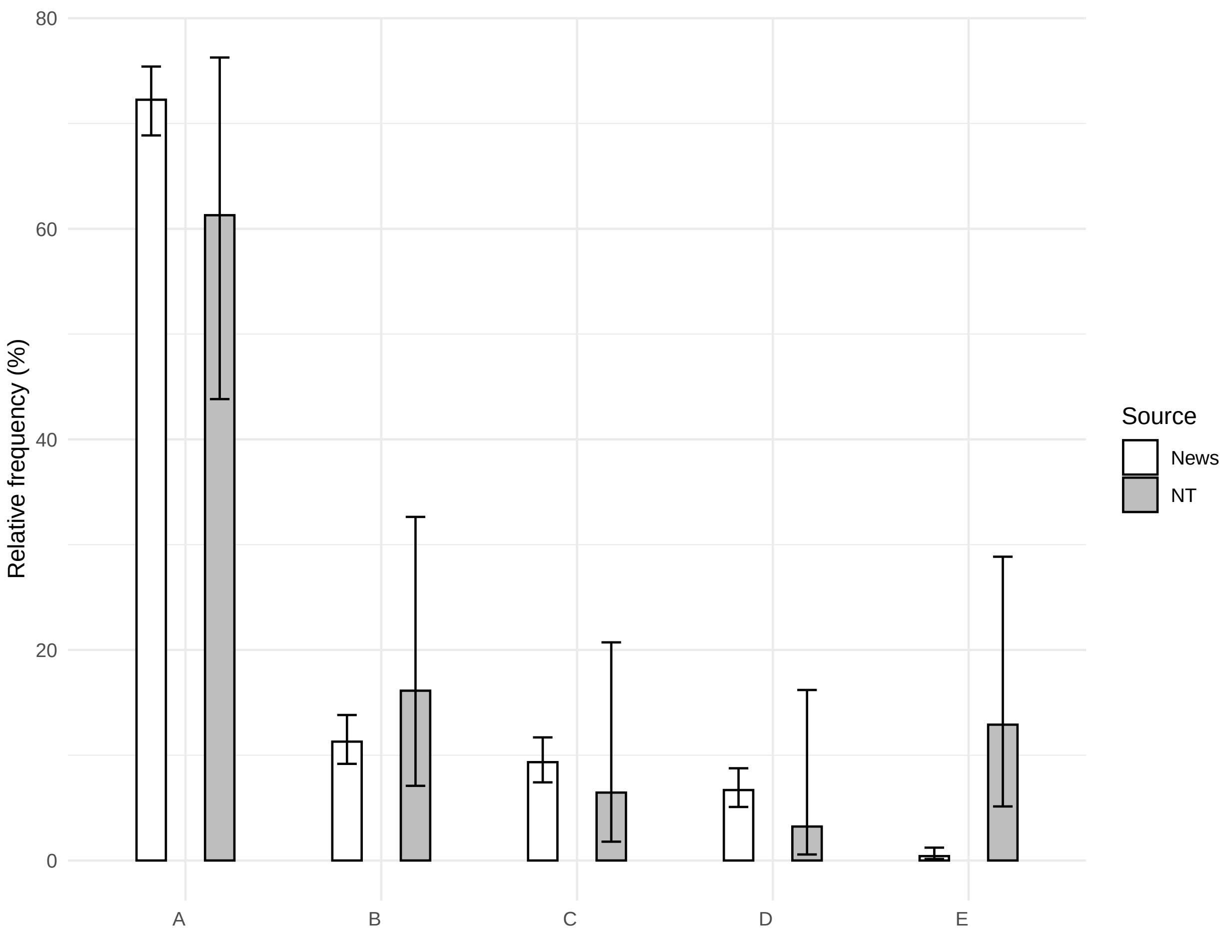


Figure 6.5: The distribution of strategies in Swedish in the online news corpus compared to the NT corpus

(200) En osäkerhet är att alla inte väljer att svara på undersökningarna.
one uncertainty be.PRS that everyone NEG choose.PRS to answer.INF on surveys.DEF
'One uncertainty is that not everyone chooses to answer the surveys.'

(201) Dessutom är allt inte helt klart ännu
besides be.PRS everything NEG completely ready.SG.N yet
'Besides, not everything is completely ready yet'

The prevalence of Strategy A is also manifested in the NT corpus.

(202) Men alla ville inte lyda evangeliet.
but all want.PST NEG obey.INF the.gospel
'But not everyone wanted to obey the gospel' [*Svenska Folkbibeln*, Rom. 10:16; 1998]

(203) Men alla israeliter trodde inte på evangeliet
but all Israelites believe.PST NEG on the.gospel
'But not all the Israelites believed the gospel' [*nuBibeln*, Rom. 10:16; 2015]

(204) Alla kan inte tillägna sig detta
all can NEG acquire.PRS REFL this
'Not everyone can accept this' [*Bibel 2000*, Matt. 19:11; 2000]

Strategy B Strategy B is used less frequently than Strategy A in most contexts, including main clauses (205). However, it appears that one context where Strategy B is preferred over Strategy A is within relative clauses, such as (206): 11 out of 81 (13.6%) instances of Strategy B are in relative clauses, compared to merely 2 out of 518 (0.39%) instances of Strategy A.

(205) Inte alla populister är blivande tyranner.
NEG all populists be.PRS would.be tyrants
'Not all populists are would-be tyrants.'

(206) Det är en ny sport som inte alla hört om.
it be.PRS a new sport which NEG everyone hear.SUP about
'It is a new sport that not everyone has heard of.'

Strategy B is also fairly common in the NT corpus.

(207) Inte alla som säger 'Herre, herre' till mig skall komma in i himmelriket
NEG all who say Lord Lorde to me will come into kingdom.of.heaven.DEF
'Not everyone who says "Lord, Lord," to me will enter the kingdom of heaven'
[*Bibel 2000*, Matt. 7:21; 2000]

(208) Inte alla förstår det ordet
NEG everyone understand.PRS that word.DEF
'Not everyone understands that word' [*Svenska Folkbibeln*, Matt. 19:11; 1998]

(209) Det var därför han sa att inte alla var rena.
it be.PST therefore he say.PST that NEG everyone be.PST clean
'That is why he said that not everyone was clean.' [*nuBibeln*, John 13:11; 2015]

Strategy C Strategy C is mostly used in XVS clauses, such as (210).

(210) Dessutom har inte alla alternativ utretts.
besides have.PRS NEG all alternatives investigate.PASS.SUP
'Besides, not all alternatives have been investigated.'

In the NT corpus, Strategy C appears not only in XVS clauses (211), but also with negative fronting (212). A speaker I consulted considered this construction archaic (Alice Bondarenko, p.c.).

(211) På samma sätt är inte alla Abrahams efterkommande hans barn.
on same way be.PRS NEG all Abraham's descendants his children
'Likewise, Abraham's descendants are not all his children.'
[*nuBibeln*, Rom. 9:7, 2015]

(212) Inte skall var och en som säger 'Herre, Herre' till mig komma in i himmelriket
NEG will everyone who say Lord Lord to me come into kingdom.of.heaven.DEF
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven.'
[*Svenska Folkbibeln*, Matt. 7:21; 1998]

Strategy D Strategy D is mostly used in main clauses, as exemplified in (213). The copula can precede the expletive pronoun in XVS clauses, as in (214).

(213) Det är inte alla som är så bekanta med den svenska politiken.
it be.PRS NEG everyone who be.PRS so familiar with the Swedish politics
'It is not everyone who is that familiar with Swedish politics.'

(214) Enligt vittnesmålen är det inte alla som återvänder från processen.
according.to testimonies.DEF be.PRS it NEG everyone who return.PRS from process.DEF
'According to the testimonies, it is not everyone who returns from the process.'

In the NT corpus, Strategy D appears only once (215). Overall, Strategy D appears to be less common in Swedish than in Norwegian and Danish (see §6.2.2, §6.2.3).

(215) för det är inte alla som tror.
for it be.PRS NEG everyone who believe.PRS
'For it is not everyone who believes.' [*nuBibeln*, 2 Thess. 3:2; 2015]

Strategy E Strategy E appears only three times in the news corpus (216).

(216) Vi kan inte alla vara Walter Munk
we can NEG all be.INF Walter Munk
'We cannot all be Walter Munk'

In the NT corpus, Strategy E appears four times, but only when the subject is a personal pronoun.

(217) Därför sade han att de inte alla var rena.
therefore say.PST he that they NEG all be.PST clean
'That is why he said that they were not all clean.'
[*Svenska Folkbibeln*, John 13:11; 1998]

(218) vi skall inte alla dö, men vi skall alla förvandlas
we shall NEG all die but we shall all transform.PASS.INF
'We will not all die, but we will all be transformed' [*Bibel 2000*, 1 Cor. 15:51; 2000]

6.2.2 Norwegian

A corpus of 1M sentences of online news in the Bokmål standard yielded a total of 950 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is Strategy B (61.7%), followed by Strategy D (23%), Strategy A (12.2%), and Strategy C (3.1%). Strategy E is not attested in this corpus.

The online news corpus was compared to a collection of four translations of the NT published in the last 40 years: the 1985 revision of the 1978 translation by the Bible Society, *En Levende Bok* (2005), *Guds Ord* (2017), and *Guds Ord Hverdagsbibelen* (2018).

Figure 6.6 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for all strategies, except for Strategy E, which is only attested in the NT corpus.

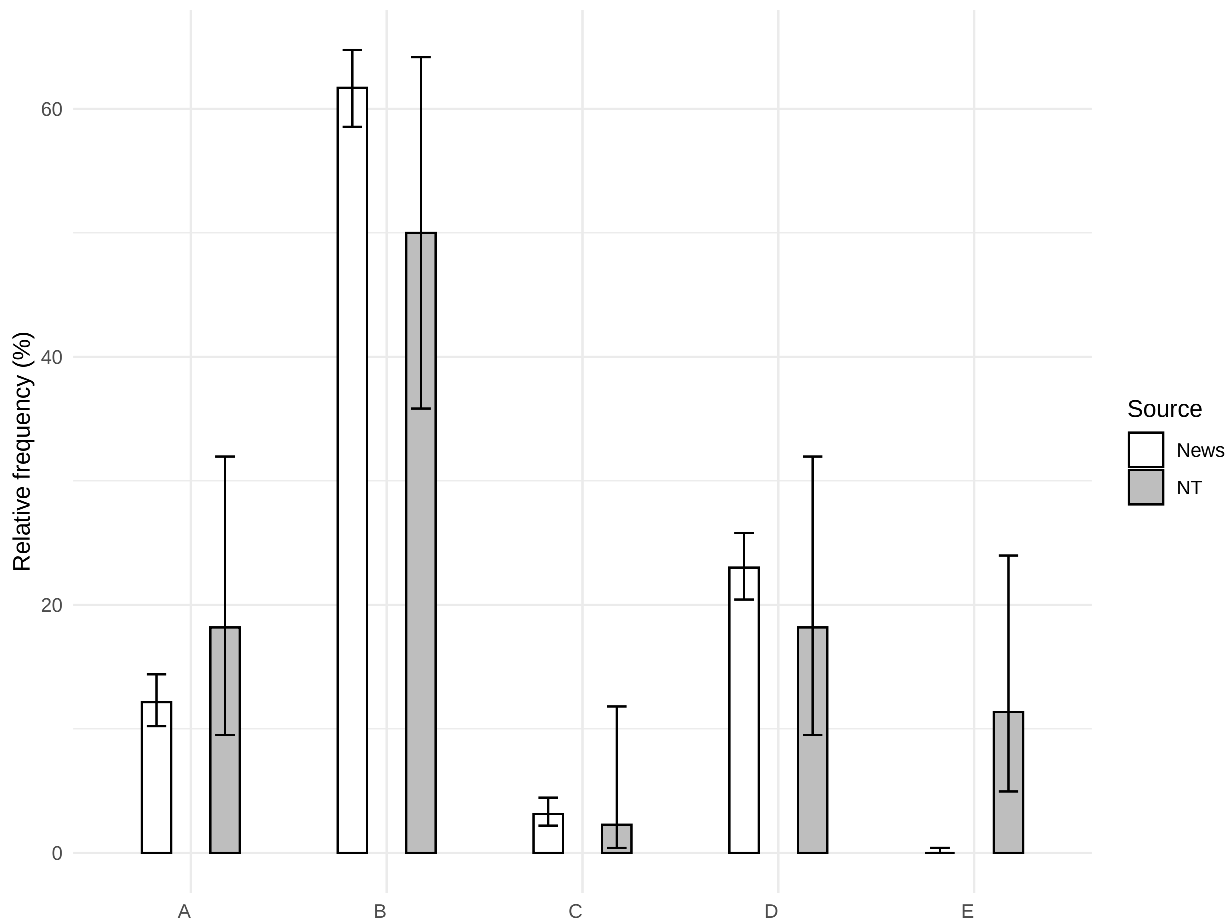


Figure 6.6: The distribution of strategies in Norwegian (Bokmål) in the online news corpus compared to the NT corpus

Strategy E was excluded from the Chi-squared test, as its absence in the news corpus would bias the results. This test yielded a X^2 value of 2.43 with 3 degrees of freedom and a p-value of 0.488, suggesting that, in the absence of Strategy E, there is no statistically significant difference in the distribution of the remaining strategies between the two corpora.

Fisher's exact test was also applied to compare the relative frequency of Strategy A between the two corpora. This test produced a non-significant p-value of 0.24. The odds ratio is 0.62 with a 95% confidence interval ranging from 0.28 to 1.59, further indicating that the difference in the frequency of Strategy A between the two corpora is not statistically significant. Overall, both tests suggest that there is no substantial evidence for a systematic difference the two corpora in the distribution of the various strategies, except for Strategy E, which is unexpectedly more frequent in the NT corpus.

Strategy A Strategy A is occasionally used in Norwegian, often appearing in non-contrastive and non-formulaic contexts, as exemplified in (219)-(220).

(219) Alt er ikke tipp topp på nyåpnede Onda Bar & Grill.
everything be.PRS NEG tip top at newly-opened Onda Bar & Grill
'Not everything is tip-top at the newly-opened Onda Bar & Grill.'

(220) Vi har en god velferd i Norge, men alle får ikke den hjelpen de trenger.
we have.PRS a good welfare in Norway but all get.PRS NEG the the.help they need.PRS
'We have good welfare in Norway, but not everyone gets the help they need.'

In the NT corpus, the use of Strategy A varies across translations. It is frequently found in *En Levende Bok* (221) and *Guds Ord Hverdagsbibelen* (222), both of which employ dynamic equivalence and paraphrases. In contrast, this strategy is uncommon in the other two translations, which predominantly employ formal equivalence: the 1978/1985 translation does not use Strategy A at all, while *Guds Ord* employs it only once with a *same*-type predicate (223). It remains unclear whether this discrepancy is due to chance, dialectal variation, or a deliberate stylistic choice.

(221) Alle israelittene trodde ikke på de gode nyhetene.
all Israelites.DEF believe.PST NEG on the good news.DEF
'Not all the Israelites believed the good news.' [*En Levende Bok*, Rom. 10:16; 2005]

(222) Alle ga likevel ikke respons på disse gode nyhetene.
everyone give.PST however NEG response on these good the.news
'Not everyone, however, responded to these good news.'
[*Guds Ord Hverdagsbibelen*, Rom. 10:16; 2018]

(223) Alt kjøtt er ikke av samme slag
all flesh be.PRS NEG of same kind
'Not all flesh is of the same kind.' [*Guds Ord*, 1 Cor. 15:39, 2017]

Strategy B Strategy B is the most common strategy in both the news corpus (224) and the NT corpus (225).

(224) Ikke alle bryr seg.
NEG everyone bother.PRS REFL
'Not everyone cares.'

(225) Jeg har lov til alt, men ikke alt er like sunt for meg.
I have.PRS law to everything but NEG everything be.PRS equally healthy for me
'I am allowed everything, but not everything is equally good for me.'
[*Guds Ord Hverdagsbibelen*, 1 Cor. 6:12, 2018]

Strategy C Strategy C is used in XVS clauses (226)-(227).

(226) Men selv om det blir varmere i hele landet får ikke alle sol fra blå himmel hele dagen.
but although it become.PRS warmer in entire the.country get.PRS NEG everyone sun from blue sky whole the.day
'But even though it is getting warmer throughout the country, not everyone gets sun from blue skies all day.'

(227) På samme måten er ikke alle som i fysisk betydning er Abrahams etterkommere, blitt de barna som Gud lovet Abraham.
on same way,DEF be.PRS NEG all who in physical meaning be.PRS Abraham's descendants become.PTCP the children.DEF who God promise.PST Abraham
'In the same way, not all who are descendants of Abraham in the physical sense have become the children that God promised to Abraham.'
[*En Levende Bok*, Rom. 9:6; 2005]

Strategy D Strategy D is a common strategy, including in XVS clauses (228). It is found in all four NT translations (229).

(228) Den forklaringen er det ikke alle som kjøper.
that explanation,DEF be.PRS it NEG everyone who buy.PRS
'Not everyone buys that explanation.'

(229) Men det er ikke alle som har denne kunnskapen.
but it is NEG all who have this knowledge
'But not everyone has this knowledge.' [*Bokmål 1978/1985*, 1 Cor. 8:7; 1978 (1985)]

Strategy E By contrast, Strategy E is not attested in the news corpus. As previously discussed, this absence is likely due to a genre-related bias, as Strategy E is primarily used with a referential personal pronoun in subject position. Such contexts are better represented in the NT corpus, where this strategy appears more frequently (230).

(230) Dere er ikke alle rene.
you be.PRS NEG all clean
'You are not all clean.' [*Bokmål 1978/1985*, John 13:11; 1978 (1985)]

6.2.3 Danish

A corpus of 1M sentences of online news yielded a total of 786 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategies are B (42.5%) and D (40.3%), followed by Strategies A (9.5%), E (6.6%), and C (1.1%).

The online news corpus was compared to a collection of three translations of the NT published in the last 40 years: the 1992 Authorized Version, the *Bibelen på hverdagsdansk* (1992), and a translation of the NT by Ole Wierøds (1997).

Figure 6.7 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Overall, the distributions are similar across the two corpora, with considerable overlap in the confidence intervals for all strategies. However, Strategy A is practically absent from the NT corpus if instances with *same*-type predicates are excluded, as discussed below.

A Chi-squared test produced a X^2 value of 11.43 with 4 degrees of freedom and a p-value of 0.022. This result suggests that there is a significant difference between the two corpora in the distribution of strategies. However, some of the cells in the matrix have low counts, compromising the reliability of these results.

Fisher's exact test was also applied to compare the relative frequency of Strategy A between the two corpora. This test yielded a non-significant p-value of 0.249. The odds ratio is 4 with a 95% confidence interval ranging from 0.66 to 164.25, further indicating that the difference in the frequency of Strategy A between the two corpora is not statistically significant.

Strategy A Strategy A is occasionally used in the news corpus. However, about half of the instances of Strategy A are either formulaic, as exemplified in (231), or contain a *same*-type predicate, or are otherwise not considered true instances of Strategy A according to the criteria of this study (see §2.3). Nevertheless, the remaining examples suggest that Strategy A is also used outside these contexts, as demonstrated in (232). Like in other Germanic language with V2 word order, Strategy A sometimes appears in XVS clauses (233) .

(231) Alt er dog ikke fryd og gammen
everything be.PRS however NEG joy and merriment
'However, it's not all joy and merriment.'

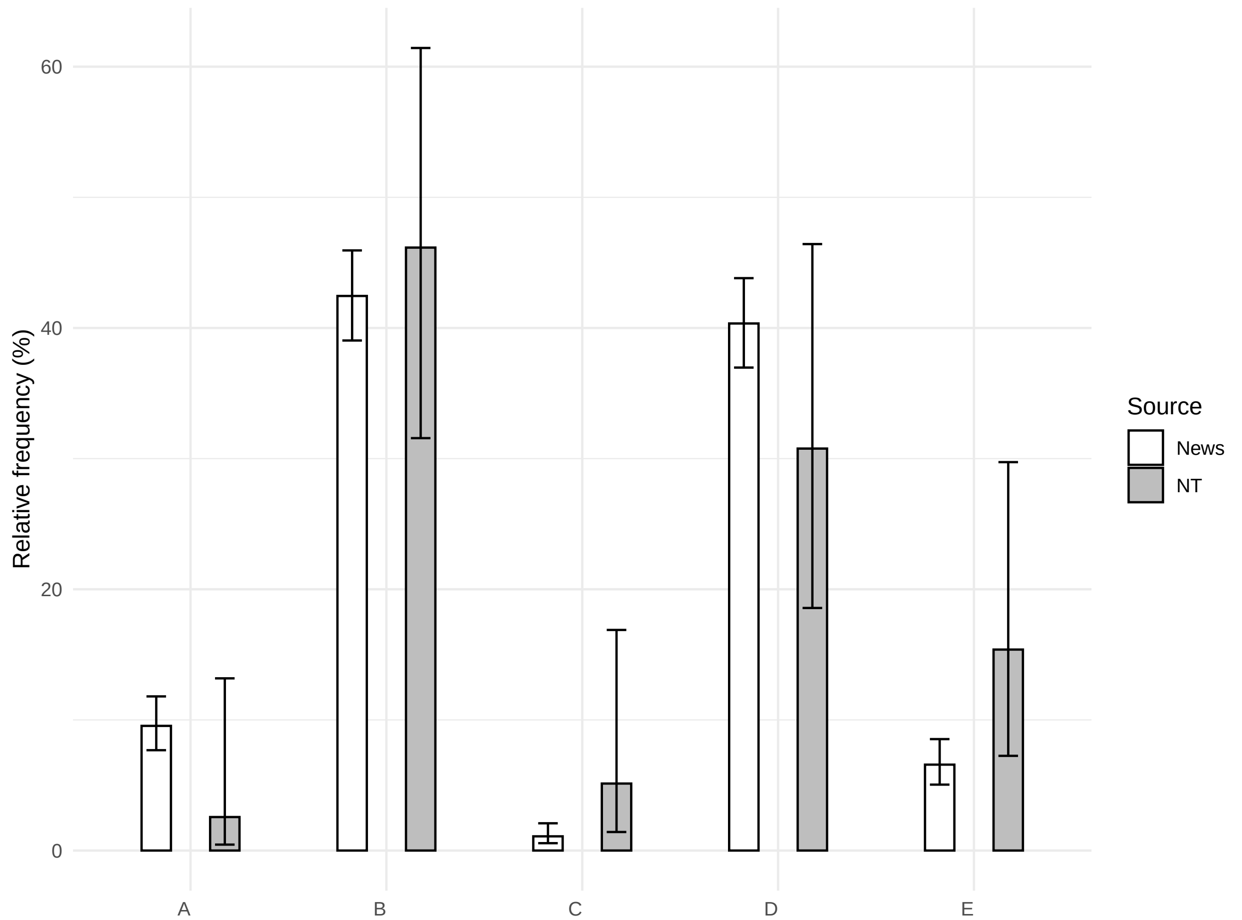


Figure 6.7: The distribution of strategies in Danish in the online news corpus compared to the NT corpus

(232) Alle kan ikke gøre brug af de nye billetter, som kun er til unge
everyone can NEG make.INF use of the new tickets which only be.PRS for young
under 25 år.
under 25 years
'Not everyone can use the new tickets, which are only for young people under the age of 25.'

(233) Det var alle langs ruten ikke lige begejstrede for.
it be.PST all along the.route NEG just excited for
'Not everyone along the route was thrilled about it.'

On the other hand, Strategy A is not attested in the NT corpus, except for a single instance with a *same*-type predicate (234). Setting aside this instance, the absence of Strategy A in the NT corpus instantiates a false negative.

(234) men alle lemmer ikke har den samme opgave
but all limbs NEG have.PRS the same task
'but not all limbs have the same task' [Wierøds, Rom. 12:4; 1997]

Strategy B Strategy B is common in both corpora (235)-(236).

(235) Ikke alle edderkopper bruger spind til at jage.
NEG all spiders use web for to hunt
'Not all spiders use webs to hunt.'

(236) Men ikke alle ønskede at tage imod det glædelige budskab.
but NEG everyone wish.PST to take.INF against the glad news
'But not everyone wanted to accept the good news.'
[*Bibelen på hverdagsdansk*, Rom. 10:16; 1992]

Strategy D Strategy D is very frequent, too (237)-(238).

(237) Men det er ikke alle stenene, der er fjernet.
but it is not all stones.DEF that are removed
'But not all the stones have been removed.'

(238) Dog var det ikke alle, der adlød budskabet
however be.PST it NEG everyone who obey.PST news.DEF
'However, it was not everyone who obeyed the news.'
[*Den autoriserede oversættelse*, Rom. 10:16; 1992]

Strategy E Strategy E is more frequently used in Danish than in Swedish and Norwegian, sometimes even appearing with a common NP (239). However, it remains unclear whether this reflects genuine cross-linguistic variation among the mainland Scandinavian languages, as the frequency of Strategy E in the NT corpus is similar across these languages.

(239) Men ø-beboerne er ikke alle lutter begejstrede.
but island-inhabitants.DEF are not all just excited
'But the islanders are not all enthusiastic.'

6.2.4 Icelandic

The UDPipe package does not offer a pre-trained model for Icelandic, making it impractical to process large datasets. For this reason, I analyzed a sample of 30K sentences from online news, manually examining instances where negation and a universal quantifier appear within a distance of up to seven words from each other.

The online news corpus yielded a total of 23 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is Strategy C (73.9%, n = 17), followed by Strategy E (26.1%, n = 6). The remaining strategies are unattested in this dataset. These results were compared to a single translation of the NT, published by the Icelandic Bible Society in 2007.

Figure 6.8 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. The distributions are similar across the two corpora, with considerable overlap in the confidence intervals for these strategies.

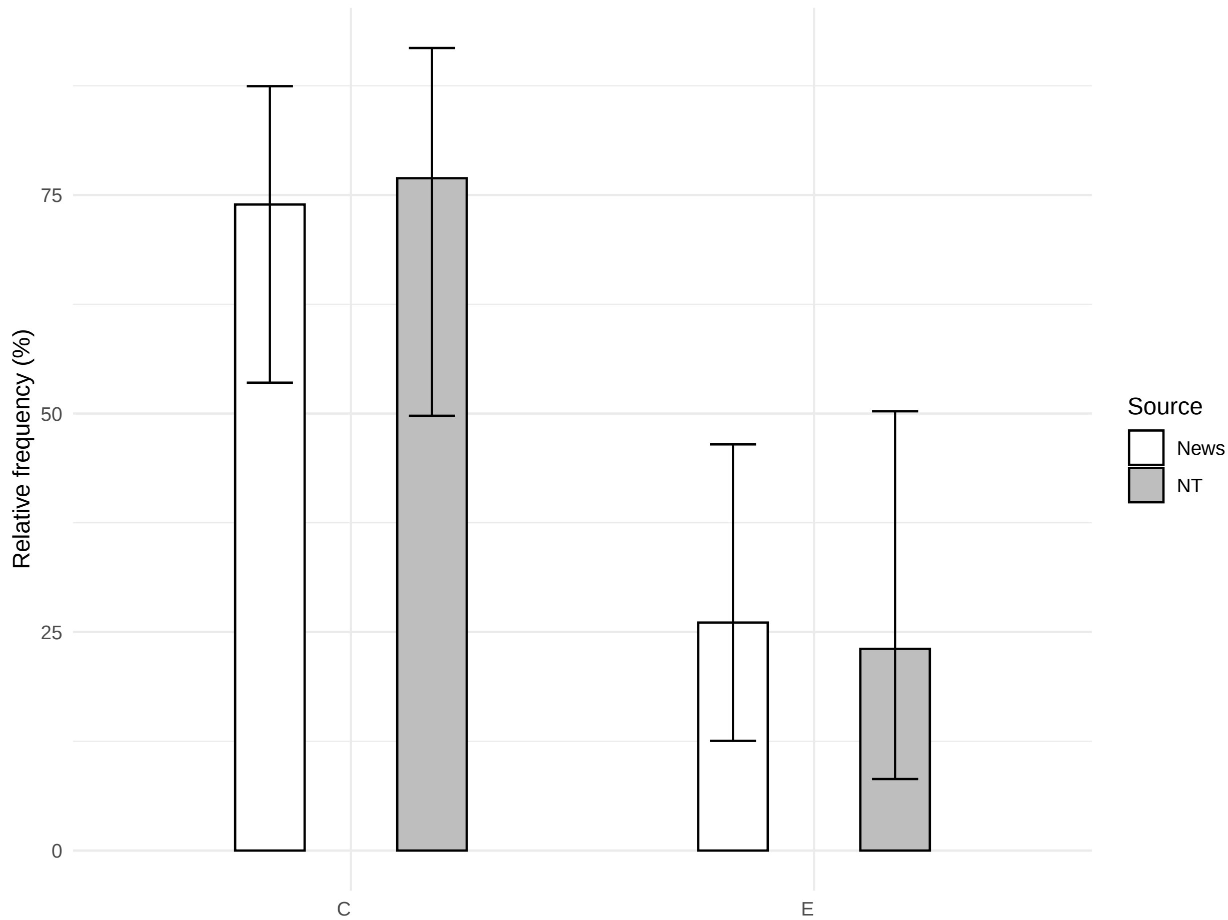


Figure 6.8: The distribution of strategies in Icelandic in the online news corpus compared to the NT corpus

Given that only two strategies are attested in these corpora, a Chi-squared test was not required, and Fisher's exact test was applied instead to compare the two corpora. This test yielded a p-value of 1. The odds ratio is 0.85 with a 95% confidence interval ranging from 0.11 to 5.17, further indicating that there is no significant difference between the corpora in the distribution of strategies.

Strategy C Although Strategy C is also found in other Germanic languages, its distribution in Icelandic is very different. In the online news corpus, only 5 out of 17 instances of

Strategy C involved XVS word order, where X is a clause-initial constituent other than the subject or the negative particle, as exemplified in (240). On the other hand, 11 instances involved NegVS word order, where the negative particle is fronted to clause-initial position, as in (241). The remaining instance involved V1 word order (242).

(240) Sem betur fer eru ekki allir syngjandi úti á svölum
fortunately are NEG everyone singing out on balconies
'Fortunately, not everyone is singing out on the balconies'

(241) Ekki var þó allt neikvætt
NEG was though everything negative
'However, not everything was negative'

(242) Er þó ekki allt upp talið af þeirri grósku sem í gangi var.
is however NEG everything up counted from that growth which in progress was
'However, not everything is accounted for by the growth that was going on.'

NegVS word order is also common in the NT translation (243).

(243) Ekki hafa allir þessa þekkingu.
NEG have everyone this knowledge
'But not everyone have this knowledge.' [Icelandic Bible Society, 1 Cor. 8:7; 2007]

Strategy D Strategy D is not attested in either of the corpora, but it is found in a larger sample of online news data that was not thoroughly analyzed (244). Apparently, this strategy is not as frequently used as Strategies C and E.

(244) en það voru ekki allir sem hlógu.
but it were NEG everyone who laugh.PST.3PL
'But not everyone laughed.'

Strategy E Similarly to German (§6.1.2), Icelandic allows expletive inversion constructions as Strategy E.

(245) Það eru þó ekki allir sammála Brown.
it are however NEG everyone in.agreement Brown.DAT
'However, not everyone agrees with Brown.'

6.2.5 Faroese

Similarly to Icelandic, the UDPipe package does not offer a pre-trained model for Faroese, making it impractical to process large datasets. For this reason, I analyzed a sample of 30K sentences from online news, manually examining instances where negation and a universal quantifier appear within a distance of up to seven words from each other.

The online news corpus yielded a total of 29 'Not all X are Y' constructions with a universal quantifier in subject position. The dominant strategy is Strategy A (48.3%, n = 14), followed by Strategies D (24.1%, n = 7), B (17.2%, n = 5), E (7%, n = 2), and C (3.4%, n = 1).

These results were compared to two old translations of the NT: a 1937 translation by Victor Danielsen and a 1923 translation by Jákup Dahl, published by the Danish Bible Society in 1961. Newer translations were not available.

Figure 6.9 compares the relative frequency of each strategy within the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies. Unlike other Germanic languages in this sample, the distribution of strategies varies significantly across corpora. First, Strategies A and D are not attested in the NT translations, with the exception of a single instance of Strategy A involving a *same*-type predicate; yet, they are the most frequent strategies in the news corpus. Second, Strategy C is commonly used in the NT translations, but it is rare in the news corpus.

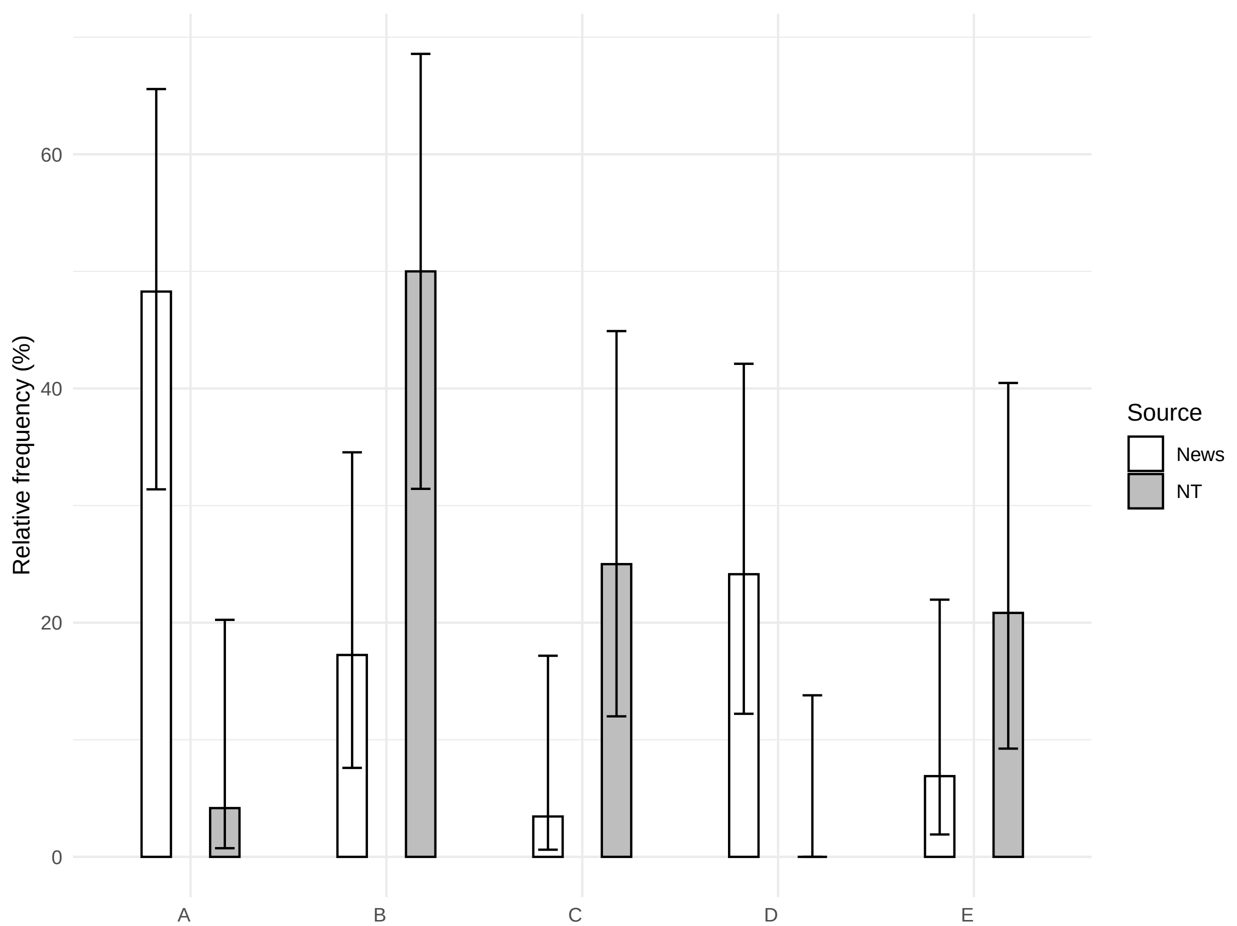


Figure 6.9: The distribution of strategies in Faroese in the online news corpus compared to the NT corpus

A Chi-squared test comparing the distribution of strategies between the two corpora yielded a X^2 value of 25.76 with 4 degrees of freedom and a p-value of less than 0.001. These results suggest that there is a significant difference between the two corpora.

Fisher's exact test was also applied to compare the relative frequency of Strategy A across the two corpora. This test, too, produced a p-value of less than 0.001. The odds ratio is 20.36 with a 95% confidence interval ranging from 2.6 to 941.12, further indicating that there is a significant difference between the corpora in the relative frequency of Strategy A.

The discrepancies observed between these corpora could be attributed to the periods in which they were composed. Furthermore, notable differences are evident between the two NT translations: Dahl utilizes Strategy C with NegVS word order on five occasions, while Danielsen employs this strategy just once, and even that in an XVS clause. This difference might be due to dialectal variation, as Dahl was a speaker of the Southern Dialect, whereas Danielsen's native dialect was the Northwestern one, based on their respective places of birth.[2] It is likely that the online news corpus mostly represents the Central Dialect, which is spoken in Tórshavn, the capital and largest city of the Faroe Islands. However, it is hard to tell whether the variation is mostly explained by period of composition, dialect, or genre.

Strategy A In the news corpus, Strategy A is used in both SV and XVS clauses, as demonstrated in (246) and (247), respectively. This strategy only appears once in the NT corpus, in a verse with a *same*-type predicate (248).

(246) Men øll fingu ikki pengar.
but everyone get.PST.PL NEG money.ACC
'But not everyone received money.'

(247) Enn hava øll ikki fingið streymin aftur í Hvalba og Sandvík.
still have.PRS.PL everyone NEG get.SUP power.ACC back in Hvalba and Sandvík
'Not everyone has gotten the power back yet in Hvalba and Sandvík.'

(248) Men allir limirnir hava ikki sama starv.
but all.NOM.PL.M members.DEF have.PRS NEG same.NOM.SG.N job.NOM
'But not all the members have the same job.' [Danielsen, Rom. 12:4; 1937]

Strategy B Strategy B is also relatively common in the news corpus.

(249) Men ikki øll lond, eru so fegin um tilráðingarnar frá ST-stovninum.
but NEG all.NOM.PL.N countries.NOM be.PRS.PL so pleased about recommendations.DEF from UN-agency.DEF
'But not all countries are so pleased with the recommendations from the UN agency.'

Strategy C Strategy C appears only once in the news corpus, in an XVS clause (250).[3]

[2] The dialect classification is based on Petersen (2022).

[3] The pronoun *tað* 'it' is an argument of the verb *dáma* 'like' in (250), not an expletive subject pronoun.

(250) Tað dámar ikki øllum í hesum koronutíðum við tilmælum
it pleases NEG everyone.DAT in these.DAT.PL corona.times with recommendations.DAT
um fráststøðu og øðrum.
about distancing and other.DAT.PL
'Not everyone likes it in these corona times with recommendations about distancing and other [measures].'

By contrast, Dahl employs Strategy C five times with NegVS word order, as exemplified in (251), a construction that is frequently used in Icelandic but rare in other Scandinavian languages. This construction seems to have been lost or possibly remained a dialectal feature.

(251) Alt er loyvt, men ikki er alt gagnligt. Alt er loyvt, men
everything is allowed but NEG is everything useful everything is allowed but
ikki alt byggir upp.
NEG everything builds up
'Everything is allowed, but not everything is useful. Everything is allowed, but not everything builds up.' [Dahl, 1 Cor. 10:23; 1937 (1961)]

Strategy D Strategy D is also a common strategy in the news corpus (252), similarly to other Scandinavian languages. It is absent from the NT translations.

(252) Men tað eru ikki allar bygdirnar sum hava sæð
but it be.PRS.PL NEG all.NOM.PL.F villages.NOM.PL.DEF which have.PRS.PL see.SUP
hendan vøksturin.
this.ACC.SG.M growth.ACC
'But not all the villages have seen this growth.'

Strategy E Strategy E appears with subject pronouns and NPs (253), while there are no examples of an expletive inversion construction.

(253) Men tey eru ikki øll gomul.
but they.N be.PRS NEG all.NOM.PL.N old.NOM.PL.N
'But they are not all old.'

6.3 Discussion

Figure 6.10 presents the relative frequencies of Strategy A across the languages in the sample, calculated from the total instances of 'Not all X are Y' constructions in each language. For each language, a comparison is made between the relative frequency of Strategy A in the online news corpus and the NT corpus, with error bars representing 95% Wilson confidence intervals for these frequencies.

Overall, the relative frequencies of Strategy A align well across the two corpora in all languages except Faroese. In most cases, the confidence intervals from each corpus overlap

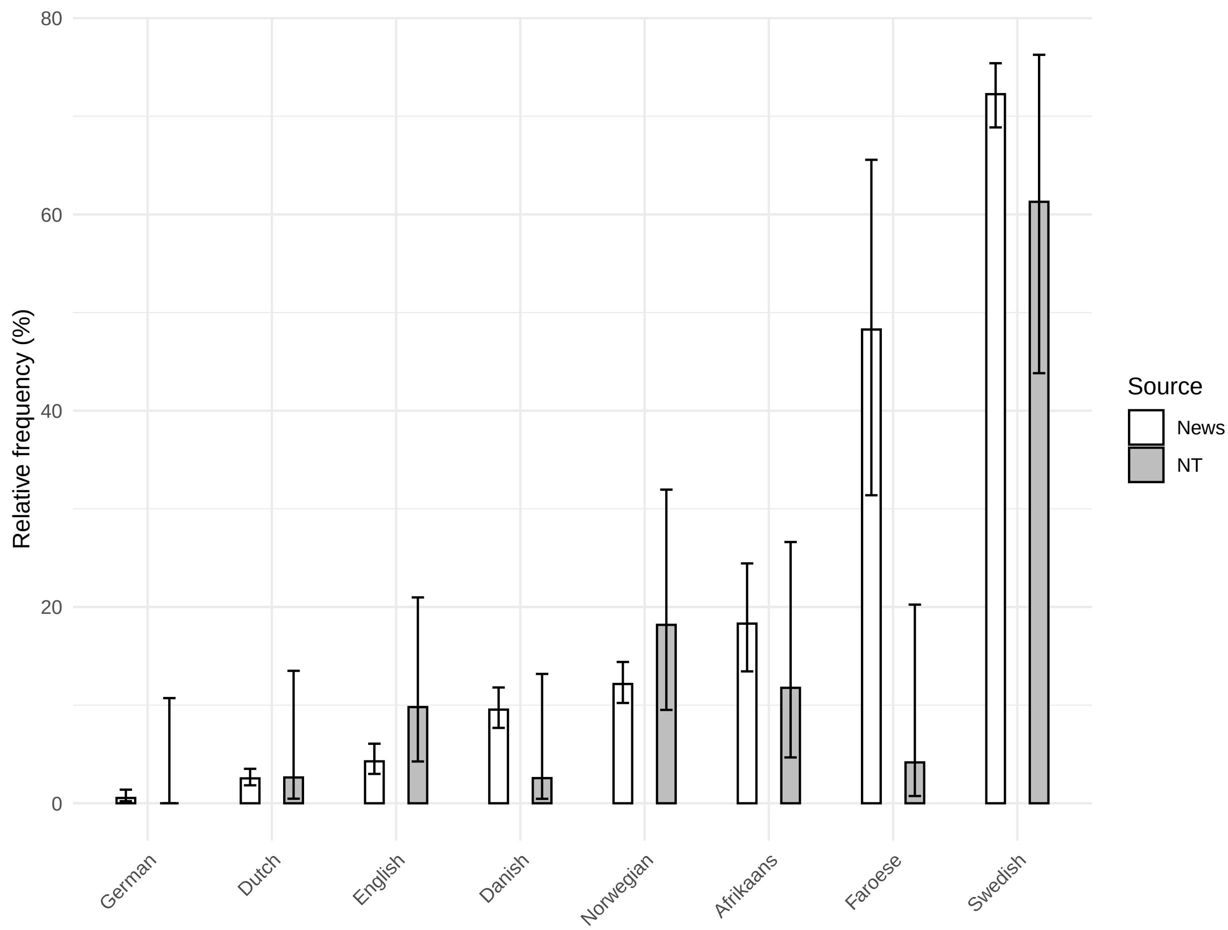


Figure 6.10: The relative frequency of Strategy A in the online news corpus compared to the NT corpus

considerably, suggesting that the NT corpus provides a reasonable approximation of the usage frequency of Strategy A despite its smaller sample size. This indicates that the NT corpus can be used as a proxy for formal written language, yielding results that are generally comparable to those obtained from a news corpus. Moreover, there is little evidence to suggest that the NT corpus systematically favors older or more conservative structures. The only instance that might suggest an archaic pattern is a single occurrence of Strategy C in Swedish, but beyond this, the NT corpus does not exhibit clear signs of deviation from contemporary usage.

One important limitation in comparing the two corpora is that the NT corpus contains only 14 instances of 'Not all X are Y' propositions, meaning that the range of available syntactic environments is much more constrained than in the larger news corpus. The analysis compares choice rates between the NT corpus and the news corpus, meaning that it examines how frequently each strategy is selected when the opportunity arises to use it (Wallis 2020, 7). In this case, "opportunity" is defined as any instance of a 'Not all X are Y' proposition. However, this definition treats all such instances as equivalent, when in reality, factors such as subject type, clause type, and information structure influence which strategies are actually chosen. This means that the observed choice rates are aggregated across different syntactic and discourse environments, even though the distribution of these environments differs between the two corpora. As a result, differences in strategy frequency between the corpora may reflect not just variation in linguistic preferences but also differences in the types of contexts that each corpus contains.

This distinction is particularly relevant for phenomena such as V2 word order in Germanic languages. In the news corpus, fronted adverbs and adverbial clauses—both of which favor V2 structures—happen to occur at a higher rate than in the relevant NT verses. As a result, Strategy C, which involves subject-verb inversion, is more frequent in news texts not necessarily because these writers prefer it in general, but because news texts contain more contexts that make it available. Similarly, Strategy E, which is commonly used with referential pronouns, is more frequent in the NT corpus because personal pronouns occur more frequently in a text with narrative and dialogue elements than in news reporting.

Thus, the apparent discrepancies in strategy frequencies between corpora do not necessarily indicate a fundamental difference in language use. Instead, they largely reflect differences in the distribution of the contexts in which a given strategy is possible. The key takeaway is that the NT corpus, despite its small sample size, generally aligns with the news corpus in its representation of choice rates, provided that we account for contextual influences on strategy selection.

False negatives present a more serious challenge. The clearest example is Danish, where Strategy A is missing from the NT corpus except for a single instance with a *same*-type predicate, yet it occurs in a wider range of contexts in the news corpus. While the overlapping confidence intervals suggest that the NT corpus does not drastically misrepresent the distribution of Strategy A, this case underscores an important limitation: the absence of a strategy in the NT corpus does not provide strong evidence that the strategy is absent from the language. Instead, such an absence should be taken as an indication that the strategy is likely rare, though the corpus size may make it difficult to distinguish between rarity and complete absence. A similar issue arises in Icelandic, where Strategy D is absent from both the NT corpus and the news sample, yet it is attested elsewhere. Given the small sample size for Icelandic, this further indicates that some rare strategies may be underrepresented.

Beyond the issue of corpus size, the greatest deviation between the two corpora is found in Faroese. As discussed in §6.2.5, the most recent translations of the NT into Faroese are nearly 100 years old, whereas for other languages in this sample, the translations are at most 40 years old. This difference likely accounts for the greater discrepancy in Strategy A frequencies in Faroese compared to other languages, suggesting that older translations may not reliably reflect the current state of a language, not even as a rough approximation. This case highlights an important consideration when using NT corpora for linguistic analysis: while they may serve as a reasonable proxy for formal written language, older translations should be treated with more caution, as they may reflect an earlier stage of the language.

Taken together, these findings indicate that the NT corpus is largely representative of formal written language, with its primary limitation being sample size rather than any fundamental bias in strategy distribution. This limitation can be mitigated when multiple translations per language are available, as they allow for a more robust assessment of variation. By contrast, for languages with only a single contemporary translation, such as Icelandic, the results should be interpreted with greater caution, as individual translation choices may have a disproportionate impact. Nevertheless, for most languages in the sample, the overall trends appear stable, supporting the use of NT corpora as a valuable resource for studying cross-linguistic variation in 'Not all X are Y' strategies.

Part III

How languages avoid inverse scope

Chapter 7

Strategies for expressing 'Not all X are Y' propositions

A central question in this study is how languages express propositions of the form 'Not all X are Y' and whether they favor scope-transparent constructions over inverse-scope alternatives. This chapter serves as the exploratory component of the study, providing a detailed survey of the syntactic strategies used to encode such meanings across languages. By first mapping the empirical landscape, this chapter lays the groundwork for the theoretical and quantitative analyses that follow.

The following two chapters build upon these findings by analyzing them through the lens of the Scope Transparency Hypothesis (11): Chapter 8 examines the extent to which the observed cross-linguistic distributions align with the predictions regarding the bias for scope transparency, while Chapter 9 considers the linguistic means recruited for establishing transparent-scope strategies.

Languages employ a range of syntactic constructions to convey 'Not all X are Y' propositions, which can be broadly classified into five strategies. Table 7.1 provides a schematic overview of these strategies, distinguishing between inverse-scope constructions and various scope-transparent alternatives.

Strategy	Schematic example
A	*All the cookies didn't burn.*
B	*Not all the cookies burned.*
C	*Not burned all the cookies.*
D	*It's not all the cookies that burned.*
	It's not that all the cookies burned.
E	*The cookies did not all burn.*

Table 7.1: A schematic representation of five strategies for expressing 'Not all X are Y' propositions

While Strategy A involves an inverse-scope construction, the remaining strategies employ distinct syntactic mechanisms to maintain scope transparency. Strategy B places the negative marker before the QP. Strategy C relies on clausal negation while positioning the negated verb or auxiliary before the subject QP. Strategy D encompasses focus constructions, such as negated clefts, which allow the negative marker to be higher than the QP. Finally, in Strategy E, the quantifier appears lower in the clause, while another constituent occupies the canonical subject position.

Each of these broad strategies consists of multiple distinct constructions, which share certain fundamental properties but differ in their specific syntactic realization. To illustrate these patterns, each construction will be demonstrated using representative examples from a selection of languages, highlighting both the similarities and differences across languages.

The classification of these constructions emerged through an iterative process typical of typological research. It was initially based on languages where the relevant constructions were well-documented, and as the study expanded to a broader and more diverse sample, additional construction types were identified.

While each construction exhibits distinct syntactic properties, they are grouped into broader strategies based on shared mechanisms for maintaining scope transparency (except for Strategy A, which violates it). Strategies B, C, and E leverage word order flexibility—whether through syntactic movement or different attachment sites of the negative marker—to avoid inverse scope. Strategy D, by contrast, relies on specialized negative constructions rather than reordering elements within the clause. Thus, this classification captures structural similarities across different constructions, even when their specific syntactic implementations vary.

The remainder of this chapter is structured as follows. Sections §7.1–§7.5 each examine one of the five primary strategies in detail. Beyond these core strategies, Section 7.6 considers borderline cases—constructions that do not fit neatly into a single category and may exhibit properties of multiple strategies. Lastly, Section 7.7 examines rare constructions that deviate from the five main strategies, providing insight into less common ways of expressing 'Not all X are Y' propositions.

7.1 Strategy A

Strategy A is an inverse-scope construction, as defined in (254).

(254) *Strategy A*
A construction in which a universal QP in subject position precedes clausal negation within the same clause.

As discussed in §2.4, Strategy A is defined in terms of precedence rather than hierarchical relations. While many instances of Strategy A involve scope reconstruction at LF, this is not a defining property of the strategy. What matters is that the subject QP precedes clausal negation within the same clause, regardless of whether scope reconstruction is required. For a detailed justification of this approach, see §2.4.

In English, subjects typically move from their base position to the specifier of IP. For instance, in (255) from Early Modern English, the subject is initially generated in [Spec, vP] and then raises to the canonical subject position. Importantly, negation is lower than the surface position of the subject but higher than its base position. As a result, for negation to take wide scope over the quantifier, the subject must reconstruct at LF.

(255) for [$_{IP}$ all men$_1$ have$_2$ not [$_{vP}$ t_1 t_2 faith]] [KJV, 2 Thess. 3:2; 1611]

From a constraint-based perspective (see §3.1.1), the violation of scope transparency in (255) can be seen as a last resort given the LF from which it is derived. The subject must raise to [Spec, IP] to satisfy EPP, meaning that it cannot remain lower than negation at PF. In other words, there is no alternative transparent-scope PF that could be derived from the same LF (assuming that the negated QP construction *Not all men have faith* is derived from a distinct LF, see §3.1.1).

However, other languages allow inverse scope even in cases where violating scope transparency is not mandated by hard syntactic constraints such as the EPP in English. Consider, for instance, the structure in (256) in Swedish, where the subject raises to [Spec, IP] but reconstructs at LF, allowing for an inverse-scope reading.[1]

(256) Swedish

[$_{CP}$ Tyvärr	är$_2$	[$_{IP}$ alla	hundar$_1$	inte	[$_{PredP}$ t_1	t_2	vatten	tokiga	Golden	retrievers]]]
unfortunately	are	all	dogs	NEG			water	crazy	Golden	Retrievers

'Unfortunately, not all dogs are water-crazy Golden Retrievers.' [LCC: News; 2022]

Unlike in (255), subject raising in this construction is optional, as a transparent-scope PF can be derived from the same LF, as in: *Tyvärr är inte alla hundar vatten tokiga Golden retrievers*. In this alternative structure, the subject remains below negation, making scope reconstruction unnecessary. This crucially differs from (255), where no such alternative PF exists. This suggests that violations of scope transparency may sometimes reflect language-specific syntactic options rather than an unavoidable consequence of conflicting constraints.[2]

Strategy A can also arise in raising constructions, where the subject QP originates in an embedded clause but moves to the subject position of the matrix clause. In (257) from Kinyarwanda, the QP raises from the embedded clause into the matrix subject position.

[1] In Swedish, the verb-second constraint requires the finite verb to move to C, resulting in XVS word order in (256).

[2] The availability of inverse scope in (256) can potentially be analyzed within the framework Bobaljik and Wurmbrand (2012), where conflicting soft constraints allow both (256) and its transparent-scope alternative to emerge as permissible candidates. Crucially, the key observation here is that the violation of scope transparency is not a last resort in the strict sense, as an alternative PF exists that expresses the same LF without requiring reconstruction. The same applies to example (257) from Kinyarwanda below.

This results in an inverse-scope configuration, where the negation appears lower than the subject, but the intended reading is derived through reconstruction.[3]

(257) Kinyarwanda
Ariko [$_{IP}$ bose$_1$ si [$_{CP}$ ko t_1 ba-bi-fit-emo ubujijuke]]
but all.2 NEG.COP COMP SBJ2-OBJ-have-LOC knowledge
‘But not everyone has knowledge of it.’ [*Bibiliya Ntagatifu*, 1 Cor. 8:7; 2012]

Crucially, however, this is not the only available PF given this LF. Kinyarwanda also allows an alternative structure in which the QP remains in the embedded clause, and the negative copula *si* negates an embedded clause headed by the complementizer *ko* (‘It is not the case that…’). In this structure, scope transparency is maintained, and no reconstruction is required to derive the LF where negation outscopes the universal quantifier. Thus, as in the Swedish case above, violating scope transparency is not forced by syntactic constraints.

Finally, Strategy A also includes constructions where the subject QP precedes clausal negation within the same clause, but scope reconstruction is not required. This applies to languages where negation is structurally higher than the canonical subject position in the clausal spine. For instance, Öztürk (2005, 131) argues that in Turkish, a non-scrambled subject remains low in its base position, where it is c-commanded by negation, as shown in (258). This construction is considered Strategy A because the QP precedes clausal negation within the same clause (see §2.4 for a detailed discussion).

(258) Turkish (Öztürk 2005, 131)
[$_{IP}$ [$_{vP}$ bütün çocuklar o test-e t_1] gir-me-di$_1$].
all children that test-DAT take-NEG-PST
‘Not all of the children took that test.’

7.2 Strategy B

Strategy B is defined in (259).

[3] The raising construction in (257) differs from the familiar English pattern. In English, raising is restricted to non-finite clauses, as illustrated in (i).

(i) a. John seems to have left.
b. *John seems (that) has left.

Other languages, however, including many Bantu languages like Kinyarwanda, allow raising out of a finite clause—a construction known as hyper-raising (Ura 1994; Carstens and Diercks 2009; Halpert 2016). An example from Zulu is provided in (ii).

(ii) Zulu (Halpert 2016)
uZinhle$_1$ u-bonakala [$_{CP}$ ukuthi t_1 u-zo-xova ujeqe]
AUG.1.Zinhle 1SBJ-seem that 1SBJ.FUT.make AUG.1.steamed.bread
‘It seems that Zinhle will make bread.’

(259) *Strategy B*
A monoclausal construction in which a negative marker modifies a subject QP or appears in clause-initial position before a subject QP.

This strategy comprises two types of constructions: negated QP constructions (§7.2.1-§7.2.2) and clause-initial negative markers (§7.2.3). In both types, the negative marker appears before the subject QP, but its function is different: in negated QP constructions, negation directly modifies the QP, whereas in clause-initial negation, it applies to the entire clause.

7.2.1 Negated quantifier phrase constructions

A negated QP construction is a negative construction distinct from standard negation in which a negative marker modifies a QP. For example, in English, standard negation is expressed by a construction where the negative particle *not* follows the auxiliary (260). However, when the subject is a universal quantifier, *not* can appear before the QP, as shown in (261).[4]

(260) The kids didn’t like the movie.

(261) [$_{IP}$ [Not [$_{QP}$ all the kids]]$_1$ [$_{vP}$ t_1 liked the movie]]

[4] Broadly speaking, three types of structural analyses have been proposed for negated QP constructions, with some rejecting the idea that negation directly modifies the QP. The analysis assumed here, as reflected in (261), treats negation as directly modifying the QP, which is also the motivation behind the term “negated QP construction”. However, adjudicating between these competing analyses falls outside the scope of this study, and I adopt this approach without arguing for it over the alternatives.

Consider, for instance, the sentence *Not all the cookies burned*. Under the analysis in (ia), proposed by Hoeksema (1987), negation combines directly with the universal quantifier, forming the complex negative quantifier *not all*, which then takes the DP as its complement. On this view, *not all* is structurally similar to negative quantifiers such as *no* and *none*, as in *None of the cookies burned*, differing only in its morphological composition.

By contrast, Collins (2020) argues that negation applies to the entire QP, forming a negated QP constituent, as shown in (ib). A related analysis is proposed by Cirillo (2009, 21), who places the negative marker in the specifier of the QP, rather than treating it as a head of NegP that takes the QP as its complement. This approach has been suggested particularly for Germanic languages, where, according to Zeijlstra (2004), the negative marker is not a syntactic head projecting NegP but rather a phrasal element occupying a specifier position. Crucially, under these approaches, *not all* is neither a constituent nor a complex head, meaning that it does not share the same syntactic structure as negative quantifiers like *no* and *none*.

Finally, Etxepare and Uribe-Etxebarria (forthcoming) propose an alternative analysis, shown in (ic), in which negation is structurally high in the left periphery. Under this view, the QP appears adjacent to negation due to focus movement, rather than because it forms a constituent with negation. As a result, this construction is described as “illusory constituent negation”, since negation does not modify the quantifier or the QP itself.

(i) a. [$_{IP}$ [$_{QP}$ [$_{Q}$ Not all [$_{DP}$ the cookies]]] burned].
b. [$_{IP}$ [Not [$_{QP}$ all the cookies]] burned].
c. [$_{CP}$ [$_{NegP}$ Not [$_{FocP}$ [$_{QP}$ all the cookies]$_1$ Foc^o] [$_{IP}$ t_1 burned]]].

Crucially, this constituent negation construction is restricted to QPs, and it cannot be employed as a strategy for negating regular clauses, as shown in (16b).

(262) *Not the kids liked the movie.

The remainder of this subsection outlines several key properties of negated QP constructions.

Similarity to constituent negation of a focused DP In many languages, non-quantificational subjects, such as definite DPs, are incompatible with constituent negation in a free-standing clause, as demonstrated by English in (263).

(263) *Not the kids liked the movie. (Intended: ‘It wasn’t the kids who liked to movie.’)

In other languages, however, negated QP constructions are structurally similar to constituent negation constructions where the negated constituent is a focused DP, exemplified by Hebrew in (264).

(264) Hebrew

a. lo kulam hizminu bira.
NEG everyone order.PST.3PL beer.
‘Not everyone ordered beer.’

b. lo [ani]$_F$ hizmanti bira.
NEG 1SG order.PST.1SG beer.
‘It wasn’t me who ordered a beer.’ (lit. ‘Not me ordered a beer.’)

Pre-verbal position In many languages, negated QPs cannot appear post-verbally. For instance, in English negated QPs may not appear as post-verbal objects or adjuncts (Lasnik 1972, 9-10), as demonstrated by the minimal pairs in (265) and (266), respectively.

(265) Lasnik (1972, 9)

a. *The students solved not all of the problems.
b. Not all of the problems were solved by the students.

(266) Lasnik (1972, 10)

a. Not everyone saw the play.
b. *The play was seen by not everyone.

Similarly, in Hebrew, subjects of unaccusative verbs may occur post-verbally (267a), but negated QPs may not (267b). Thus, either the negated QP occurs pre-verbally (267c), or clausal negation is employed (267d).

(267) Hebrew

a. nisrefu l-i kol ha-ugiy-ot.
burn.PST.3PL DAT-1SG all the-cookie-PL
‘All of my cookies burned.’

b. *nisrefu l-i lo kol ha-ugiy-ot.
burn.PST.3PL DAT-1SG NEG all the-cookie-PL
(Intended: 'Not all of my cookies burned.')

c. lo kol ha-ugiy-ot nisrefu l-i.
NEG all the-cookie-PL burn.PST.3PL DAT-1SG
'Not all of my cookies burned.'

d. lo nisrefu l-i kol ha-ugiy-ot.
NEG burn.PST.3PL DAT-1SG all the-cookie-PL
'Not all of my cookies burned.'

However, there are also languages where negated QPs may appear as post-verbal subjects or objects. This is demonstrated in Russian (268a)-(268b) and Ukrainian (269). I do not have quantitative data regarding the cross-linguistic prevalence of this phenomenon, but it seems to be relatively rare.

(268) Russian

a. Odnako v eto poverili ne vse.
however in this believe.PST.PFV.PL NEG everyone
'However, not everyone believed this.' [LCC: News; 2022]

b. My rešili ne vse zadači.
we solve.PST.PFV.PL NEG all problems-ACC
'We didn't solve all the problems.' (lit. 'We solved not all the problems.')
(Borschev et al. 2006)

(269) Ukrainian
Use meni mozhna, ta buduye ne vse!
everything 1SG.DAT is.possible but build.PRS.3SG NEG everything
'Everything is allowed to me, but not everything is constructive!'
[Ohienko, 1 Cor. 10:23; 1939 [1962]]

Separation of negation from the QP In some languages, the negative marker can modify not only QPs but also prepositional phrases (PPs) containing a QP. When the QP functions as the complement of a preposition, the preferred word order often places the negation before the entire PP, yielding a [Neg + Prep + QP] structure, as illustrated in (270a) for Spanish. By contrast, the alternative word order, where the preposition precedes the negated QP [Prep + Neg + QP], as shown in (270b), is dispreferred in Spanish. Some speakers reject it outright, and it is entirely unavailable with certain quantifiers (Etxepare and Uribe-Etxebarria, forthcoming).

(270) Spanish (Etxepare and Uribe-Etxebarria, forthcoming)

a. No en todos los hoteles se duerme bien.
NEG in all the hotels IMPERS sleep.PRS.3SG well
'Not in all hotels can you sleep well'

b. En no todos los hoteles se duerme bien.
in NEG all the hotels IMPERS sleep.PRS.3SG well
‘Not in all hotels can you sleep well’

The same preference is also found in English (271) from the Corpus of News on the Web (NOW, Davies 2016), and in Hebrew (272), where the alternative word order is not possible.

(271) a. Not in every place do they react that way, but in Vegas they certainly seem to, so, we’re excited to spend more time with them.
[NOW: US; 2022]
b. ?In not every place (do) they react that way, ….

(272) Hebrew
a. lo al kol mucar mecuyan ha-mexir.
NEG on every product mark.PASS.PRS.3SG.M the-price
‘Not every product has a price tag on it.’
b. ?al lo kol mucar mecuyan ha-mexir.
on NEG every product mark.PASS.PRS.3SG.M the-price
(Intended: ‘Not every product has a price tag on it.’)

In some languages, even conjunctions can appear between negation and the QP, as demonstrated by Koine Greek (273), where the conjunction *gar* ‘for, since’, which typically appears in the second position in the clause, follows the negative particle. A similar structure is observed in Latin (274).

(273) Koine Greek
ou gar pantes hoi ex Israēl houtoi Israēl.
NEG because all.NOM.PL.M the.NOM.PL.M from Israel these.NOM.PL.M Israel
‘For not all who are from Israel, these are Israel.’ [NA28]

(274) Late Latin
non enim omnes in priore uita geometrae fuerunt
NEG because all.PL.NOM in prior.SG.ABL life geometrician.PL.NOM be.PERF.3PL
‘For not all have been geometricians in their previous life’
[Augustine, *De Trinitate*, bk. 12, chap. 15; about 400-428 CE]

Type of negative marker In the vast majority of languages, the negative marker used in negated QP constructions is the same one that appears in NEG-stripping—an ellipsis construction exemplified in (275) (see §9.2 for further details).

(275) Some of the students passed, but not all of them.

Notably, the negative marker employed in NEG-stripping and negated QP constructions is not necessarily the one that is employed in standard negation. For instance, in Albanian, standard negation is expressed by a construction where the negative particle *nuk* or *s'* immediately precedes the finite verb or auxiliary (Turano 2000), as demonstrated in (276).

(276) Albanian (Turano 2000, 83)
Nuk kam ngrënë.
NEG have.1SG eaten
'I haven't eaten.'

By contrast, the negative marker *jo* is used both in NEG-stripping (277a)-(277b) and in a negated QP construction (277c). The main function of *jo* is as a negative response particle.

(277) Albanian
a. Lexo librin e historis jo atë të gjografis.
read.IMP the.book of history NEG that of geography
'Read the history book, not the geography book.' (Turano 2000, 86)
b. Ju jeni të pastër, por jo të gjithë.
you are ART clean but NEG ART all
'You are clean, but not all [of you].'
[*Interconfessional Version*, John 13:10; 2007]
c. Por jo të gjithë iu bindën ungjillit.
but NEG ART all AUX obeyed gospel
'But not everyone has obeyed the gospel.' [ibid, Rom. 10:16]

Two exceptions to this generalization are found in the sample: Portuguese and Standard Malay, both of which are discussed in §9.2.

Modification by adverbs In some languages, negated QP constructions can be modified by intensifying adverbs, as illustrated in German (278). Evidence that *längst nicht* does not modify the entire clause comes from the lack of subject-verb inversion, which would otherwise be required by the verb-second constraint. However, it remains unclear whether *längst* directly combines with *nicht* as a single unit or whether it modifies the entire negated QP.

(278) German
Längst nicht alle Menschen haben ein Smartphone
by.far NEG all people have.PRS.3PL a smartphone
'Not everyone, by far, has a smartphone' [LCC: News; 2023]

7.2.2 Negated quantifier phrase constructions with negative concord

In some languages, negated QPs co-occur with clausal negation. For instance, in Estonian, standard negation is expressed by a non-inflected negative auxiliary *ei*, which immediately precedes the connegative lexical verb (Tamm 2015).

(279) Estonian (Tamm 2015)

Me	ei	karda	raamatu-id
we.NOM	NEG	fear.CNG	book-PART.PL

'We are not afraid of books [that describe crimes committed against Finno-Ugric peoples.]'

In a negated QP construction, the negative particle *mitte* modifies the QP, co-occurring with the negative auxiliary *ei*, as demonstrated in (280).[5]

(280) Estonian

Siiski	mitte	kõik	ei	ole	saanud	kuulekaks	evangeeliumile.
however	NEG	all.NOM.PL	NEG	be.PRS.CNG	become.PTCP.PERF	obedient.TRANSL.SG	gospel.ALL.SG

'However, not all have become obedient to the gospel.'

[Bible Society, Rom. 10:16; 1997]

A relatively rare variant of this construction appears in 16th-century French, where both negative markers are stacked in the pre-QP position, as demonstrated in (281).[6]

(281) French

Non	point	ung	chascũ	qui	me	dit	Sire	Sire	entrera	au	royaulme	des	cieulx
NEG	NEG	one	everyone	who	to.me	say.PRS.3SG	Lord	Lord	enter.FUT.3SG	to.the.SG.M	kingdom	of.the.PL	heaven

'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'

[Lefèvre, Matt. 7:21; 1523 (1525)]

Negated QP constructions with negative concord are typically found in languages with bipartite negative markers. Their cross-linguistic distribution is independent of the distribution of negative concord involving negative indefinites. For further discussion, see §9.2.

[5] In Strategy A, only *ei* is used, as this construction does not involve constituent negation.

(i) Estonian (Liina Lindström, p.c.)

Kõik	ei	tea	sellest.
everyone	NEG	know.CNG	it.ELA

'Not everyone knows about this.'

[6] See §10.2.5 for additional examples of this short-lived construction in French.

7.2.3 Clause-initial negation

In some languages, the negative marker appears in clause-initial position but does not directly modify the QP. For example, in Ojibwe (Algonquian), standard negation is expressed by a construction where the the negative particle *gaawiin* can appear either before or after the subject, while the negative suffix *-siin* attaches to the verb (Tilleson 2019, 243, 253), as demonstrated in (282).

(282) Ojibwe (Tilleson 2019, 253)

{Inini	gaawiin /	Gaawiin	inini}	o-gii-waabam-aa-siin	ikwe-wan.
man	NEG	NEG	man	3SG-PST-see-DIR-NEG	woman-OBV

'The man didn't see the woman.'

Similarly, the negative particle can also precede a subject QP, as demonstrated in (283). However, since this construction is structurally identical to standard negation, it is not classified as a negated QP construction.

(283) Northwestern Ojibwa

[$_{CP}$ Kaawiin	tahsh	[$_{IP}$ kahkina	o-kii-'otaahpin-an-sii-naawaa	minwaacimowin]
NEG	but	all	3-PST-accept-DIR-NEG-3PL	good.news

'But not all have accepted the good news.'

[*Ojibwe Kihcimasina'ikan*, Rom. 10:16; 1988]

While these constructions are most commonly found in languages where standard negation is expressed by clause-initial negative markers, they also occur to a limited extent in other languages. In English, the adverb *by no means* typically triggers subject-auxiliary inversion when it appears in clause-initial position, as illustrated in (284). However, when the subject is a QP, inversion becomes optional, though it remains the preferred pattern. Thus, while the examples in (285)—which involve subject-auxiliary inversion—are classified as Strategy C (specifically, see §7.3.2), those without inversion (286) fall under Strategy B.

(284) By no means do I think that this is the end of me playing good golf.
[COCA: Newspaper; 2019]

(285)
a. By no means do all parents do that, but in my own experience, a number of them do. [COCA: Web; 2012]
b. By no means do all liberals adhere to them, but they are mainstream in left-of-center thinking. [COCA: Blog; 2012]
c. By no means are all of McCarthy's customers millionaires
[COCA: Magazine; 1998]
d. Texas City's unions have been crippled, and the plants are filled with contract workers from out of town. **By no means are all of them reckless**, but without job security or the union-guaranteed right to refuse unsafe work, some no doubt take unwise chances for fear of being fired. [COCA: Magazine; 1992]

(286) a. The danger increased when converted pagans began to outnumber Jews in the Church. Many were attracted to it by the same attitudes that explain the success of Gnosticism. **By no means all of them could immediately appreciate the vital distinction between Christianity and the Eastern Hellenistic mystery religions**; they saw in the Church what they wanted to see, the crowning of their own religious experience. [COCA: Blog; 2012]

b. Knoedler's entire commerce in the paintings of Trayer fell within a five-year period, 1863 to 1867, and comprised some twelve pictures, including the Beaumont painting. **By no means all stemmed from Goupil**. [COCA: Magazine; 2006]

It is unlikely that the negative adverb *by no means* directly modifies the QP in (286), as this adverb does not function as constituent negation in other contexts. However, the presence of a QP, rather than a regular DP, appears to be crucial for the acceptability of this word order pattern. This may be due to an analogy with negated QP constructions involving *not*, which also do not require inversion.

7.3 Strategy C

Strategy C is defined in (287).

(287) *Strategy C*
A monoclausal construction in which a QP occurs in post-verbal subject position, while clausal negation appears earlier in the clause.

This strategy encompasses several distinct construction types. In NegVS word order, the negated verb appears in clause-initial position, followed by the subject (§7.3.1). In NegVS with fronting, the verb appears in the second position, preceded by a fronted negative particle (§7.3.2). Finally, in XVS word order, the verb also appears in second position, but it is preceded by a constituent other than the subject or the negative marker (§7.3.3).

7.3.1 NegVS word order

In some languages, the basic word order is VS, and clausal negation precedes non-topicalized/scrambled subjects. For example, in Irish, standard negation is expressed by the pre-verbal negative particle *ní* (Stenson 2019, 42), exemplified in (288).

(288) Irish (Stenson 2019, 42; 78)

a. Ní labhraíonn Brian Gaeilge.
NEG speaks Brian Irish
'Brian doesn't speak Irish.'

b. Níl sé tinn.
NEG.COP.PRS he sick
‘He isn’t sick.’

Strategy C looks the same as standard negation and negative copular clauses, as shown in (289).[7]

(289) Irish
ach [$_{IP}$ Níl$_1$ [$_{PredP}$ gach ní t_1 fóinteach]].
but NEG.COP.PRS every thing helpful
‘but not everything is helpful.’ [*An Bíobla Naofa*, 1 Cor. 10:23; 1981]

In some languages, VS word order is not the default, but it commonly occurs in negative clauses. For example, in Old English, the basic word order in main declarative clauses follows the V2 constraint, typically resulting in SV word order. However, in negative clauses, the negative clitic *ne* usually occurs clause-initially immediately before the finite verb, producing NegVS word order, as shown in (290). Crucially, this NegVS pattern does not qualify as V2, since *ne* is a preverbal clitic, not a phrasal constituent (Cichosz 2020).

(290) Old English (van Kemenade 1997, 96)
Ne mihte se deað him genealæcan
NEG can.PST.3SG the death 3SG.DAT approach.PTCP.PST
‘Death could not have approached him’

This negation pattern also appears in Strategy C, as illustrated in (291).

(291) Old English
Ne under-foð ealle men þis word
NEG receive.PRS.3PL all men this word
‘Not everyone can accept this teaching’ [*Wessex Gospels*, Matt. 19:11; 990]

Finally, in some languages, VS word order is pragmatically marked in both affirmative and negative clauses, often associated with focus and other information-structural factors. However, in ‘Not all X are Y’ sentences, VS order emerges as the neutral option, despite its marked status elsewhere.

This pattern is particularly common in the Balkan linguistic area, where Strategy C is the dominant strategy in Greek, Serbian-Croatian-Bosnian, and Slovenian—languages that lack Strategy B. For instance, in Serbian-Croatian-Bosnian, (292) represents the most natural way to convey the intended meaning.

[7]For simplicity, the VSO word order in (289) is represented as head movement of the verb to I^0. However, alternative analyses exist, including phrasal movement of the VP following the extraction of verbal arguments (Carnie and Guilfoyle 2000). Since the derivation of verb-initial word order is not directly relevant to the phenomenon under investigation, I set this issue aside here.

(292) Serbian-Croatian-Bosnian (Milica Denić, p.c.)
Ne puše svi moji prijatelji.
NEG smoke-PRS.3PL all.NOM.PL my.NOM.PL.M friend.NOM.PL
'Not all my friends smoke.'

7.3.2 NegVS word order with negative fronting

In some languages, VS word order is triggered by the fronting of the negative particle, resulting in NegVS word order. In Icelandic, standard negation is expressed with the negative particle *ekki*, which typically follows the auxiliary or finite verb, as shown in (293a). However, in certain contexts, *ekki* can be fronted to clause-initial position, preceding the auxiliary or finite verb, as illustrated in (293b) (Thrainsson 2007, 82).

(293) Icelandic (Thrainsson 2007, 66, 82)

a. Nemandinn las hana ekki.
student.DEF read her NEG
'The student didn't read it.'

b. Ekki hafa þeir lokið verkinu í dag.
NEG have they finished work.DEF in day
'They have not finished the work today.'

This same negative fronting construction is employed in Strategy C, as shown in (294).

(294) Icelandic
[$_{CP}$ Ekki$_2$ hafa$_1$ [$_{IP}$ allir t_1 t_2 þessa þekkingu]
NEG have.PRS.3PL all.NOM.PL.M this.ACC.SG.F knowledge.ACC
'Not everyone has this knowledge.' [Icelandic Bible Society, 1 Cor. 8:7; 2007]

A similar construction is found in Finnish, but where the negative marker is an auxiliary rather than a particle. In Finnish, standard negation is expressed by an inflecting negative auxiliary, which precedes a non-finite lexical verb. The negative auxiliary usually follows the subject, but it can also precede it in certain contexts (Vilkuna 2015, 458, 472-473), as demonstrated in (295).

(295) Finnish (Vilkuna 2015, 472-473)

a. Anna-lla ei ole koira-a.
Anna-ADE NEG-3SG be.CNG dog-PART
'Anna does not have a dog.'

b. Ei Anna-lla ole koira-a.
NEG-3SG Anna-ADE be.CNG dog-PART
'Anna does not have a dog.'

When the subject is a universal quantifier, the negative auxiliary can, but need not, precede the subject QP (296).

(296) Finnish (Maria Vilkuna, p.c.)

[$_{CP}$ Ei$_1$ [$_{IP}$ kaikki-lla t_1 ole koira-a]].
NEG-3SG everyone-ADE be.CNG dog-PART
'Not everyone has a dog.'

7.3.3 XVS word order

In some languages, Strategy C is primarily employed when a constituent other than the subject or the negative marker appears in clause-initial position. This is characteristic of V2 languages, where the verb normally cannot precede the subject, unless the verb itself is preceded by another constituent (XVS word order). This is demonstrated by example (297) in German, where the verb appears in the second position, preceding the subject QP.

(297) German

[$_{CP}$ Am Zebrastreifen$_3$ ist$_2$ nicht alles$_1$ [$_{PredP}$ t_1 t_2 Schwarz und Weiß t_3]]
at the.crosswalk is NEG everything black and white
'Not everything at the crosswalk is black and white.' [LCC: News; 2022]

7.4 Strategy D

Strategy D is defined in (298).

(298) *Strategy D*
A monoclausal or biclausal construction in which the QP functions as a cleft or focus constituent and/or negation is superordinate.

This strategy includes various constructions that contain at least one additional structural element beyond the negative marker and the QP. This additional element may take the form of a copula, a focus marker, a relativizer, or a complementizer.

7.4.1 Negated clefts

A cleft is a focus construction that structurally separates the focused constituent from the rest of the sentence, which is typically given information. As a terminological choice, I use the term cleft exclusively for biclausal structures; monoclausal focus constructions are discussed in the next subsection.

A constituent cleft, also known as an *it*-cleft, is a copulative construction consisting of some or all of the following ingredients: an impersonal subject pronoun (the cleft pronoun), a copula, a focused constituent (the cleft phrase, also known as the pivot), and a relative clause (the cleft clause), sometimes headed by a relativizer or a relative pronoun (Hartmann and Veenstra 2013, 1).

Many languages employ negated clefts where a QP functions as the cleft phrase, while the scope of the quantifier is determined by the cleft clause, as demonstrated in (299).

(299) It is not everybody [$_{CP}$ who can live up to this] [Phillips, Matt. 19:11; 1958]

In present-day English, negated clefts with QPs are relatively rare, and they are apparently most commonly used to express generic statements, as in (299) and (300).

(300) Look on the bright side. It's not every student who has access to their teacher 24 hours a day. [COCA: TV; 2001]

However, in other languages, negated clefts occur more frequently and can even function as the dominant strategy for expressing 'Not all X are Y' propositions. In Danish, for example, negated clefts are about as common as negated QP constructions (Strategy B) and can be used in episodic contexts, as in (301) (see §6.2.3). This suggests that negated clefts are not inherently tied to a specific discourse function across languages.

(301) Danish
Der var i alt 80 tilmeldte, og faktisk var det ikke alle, der troppede op.
there be.PST in total 80 registered.PL and actually be.PST it NEG all who show.PST up
'There were a total of 80 registrants, and in fact not everyone turned up.'
[LCC: News; 2023]

In some languages, clefts are recognized by only one or two of the ingredients associated with clefts in English. For instance, in Belize Kriol English, negated clefts (302a) differ from Strategy B (302b) by the presence of the copula *da*.

(302) Belize Kriol English
a. bikaaz da noh evribadi bileev eena di Laad.
because COP NEG everybody believe in the Lord
'because not everybody believes in the Lord.'
[Wycliffe Bible Translators, 2 Thess. 3:2; 2012]
b. Bot noh evribadi noa dis.
but NEG everybody know this
'But not everybody knows this.' [ibid, 1 Cor. 8:7; 2012]

In Morisyen, negated clefts are identified by the presence of the relativizer *ki*, as shown in (303).

(303) Morisyen
Me pa tou dimoun ki konn sa laverite la.
but NEG all people who know this truth the
'But it is not everyone who knows this truth.'
[Bible Society of Mauritius, 1 Cor. 8:7; 2009]

It is important to note that not all languages that have cleft constructions allow QPs to serve as the cleft phrase. Some languages permit QPs in cleft-like structures involving ellipsis, but not in fully realized clefts.[8]

For example, the Hebrew construction in (304a) resembles a cleft construction where the cleft clause has been elided. However, a full cleft in the same context, as shown in (304b), is relatively marked or unnatural, though not necessarily ungrammatical. This suggests that while some languages have extended the function of clefts to accommodate QPs, this requires conventionalization and does not automatically follow from the availability of cleft constructions.

(304) Hebrew

a. anašim tofs-im homo-im k-a-benadam ha-ze b-a-televizia
people conceive.PRS-PL.M gay-PL.M as-the-person the-this in-the-television
še-ose kaxa ve-mitlabeš kaxa, še-gam ze beseder,
that-do.PRS.SG.M like.this and-dress.PRS.SG.M like.this that-also this okay
aval ze lo kulam, kmo še-lo kol ha-streyt-im oto davar.
but this NEG everyone like that-NEG all the-straight-PL.M same thing
'People tend to think of gay people as that person on TV who acts this way and dresses that way — and that's fine, too, but it's not everyone (who is like that), just as not all straight people are the same.' [web example[9]]

b. ?ze lo kulam še-mitlabš-im kaxa.
this NEG everyone that-dress.PRS-PL.M like.this
(Intended: 'It's not everyone who dresses this way.')

Even in English, where some QPs can appear in negated clefts, certain QP types remain marked in this construction. As previously noted, negated clefts with QPs are relatively rare in present-day English, with most occurrences found in the formulaic expression *It is not every day that…* (see §6.1.1).

Beyond this fixed expression, most naturally occurring instances involve QPs headed by *every*, as in (300), or pronouns like *everybody*, as in (299). By contrast, QPs headed by *all* are much less frequent in this construction, although naturally occurring examples, such as (305), can still be found. This distribution suggests that negated clefts with QPs may be partially formulaic, rather than a fully productive pattern in English.

[8]This parallels restrictions on constituent negation in ellipsis constructions compared to negated QP constructions, as discussed in §9.2.

[9]Katzir, Itamar. *Haaretz*, 15 August 2023, `https://www.haaretz.co.il/gallery/music/2023-08-15/ty-article-magazine/.premium/00000189-f3b2-d8b0-a1bd-fbf6f74c0000`. Accessed on 08 October 2024.

(305) It upsets me when people have a go at students [...] It's unfair because it's not all students who want to ban things; campaign groups and students' unions often represent a tiny minority. [web example[10]]

7.4.2 Monoclausal focus constructions

Distinguishing clefts from monoclausal focus constructions that do not involve subordination can often be challenging. In many languages, copulas frequently develop into focus markers, a process typically accompanied by a structural reanalysis that results in a monoclausal construction (Heine and Reh 1984; Hopper and Traugott 1993).

For instance, in Jamaican Creole English, the focus marker *a* is homophonous with the equative copula *a*, and was likely historically derived from it (Durrleman 2008, 106). This focus marker precedes the focused constituent, as illustrated in (306).

(306) Jamaican Creole English (Durrleman 2008, 74, 81)

a. A di bami Piita nyam.
FOC the bammy Peter eat.
'What Peter ate was the bammy.'

b. A no aadineri wok im a wok.
FOC NEG ordinary work 3SG PROG work.
'It's not an ordinary work that he's doing.'

This monoclausal focus construction is also employed in Strategy D, as shown in (307). According to Durrleman (2008, 114–119), this structure is monoclausal, with the focused constituent fronted to a focus position in the left periphery.

(307) Jamaican Creole English
Bot [$_{CP}$ a$_2$ no [$_{FP}$ evribadi$_1$ t_2 [$_{IP}$ t_1 nuo dis]]].
but FOC NEG everybody know this
'But not everybody knows this.' [Bible Society of the West Indies, 1 Cor. 8:7; 2012]

Notably, the construction in (307) is superficially similar to the one found in Belize Kriol English (302a), which has been analyzed as a cleft. However, Durrleman (2008, 106–114) argues that *a* in (306) cannot be analyzed as an equative copula, as Jamaican Creole English generally does not allow null subjects, and *a* in this construction cannot be modified by tense.

By contrast, in Belize Kriol English, similar structures sometimes contain a relativizer, as in (308a), though they can also occur without one, as in (308b). The presence of a relativizer in some cases suggests that these are biclausal clefts. In contrast, the focus construction in

[10] Lock, Helen. *The Guardian*, 1 March 2016, `https://www.theguardian.com/higher-education-network/2016/mar/10/student-protests-are-young-people-too-sensitive-these-days`. Accessed on 07 October 2024.

Jamaican Creole English never involves a relativizer, supporting the analysis that it is not a cleft (Durrleman 2008, 110–111). Instead, in Jamaican Creole English, the F head takes an IP as its complement, rather than an embedded CP.

(308) Belize Kriol English (Decker 2005, 98; 85)

a. Da Jan weh gwain da Jamayka.
COP John who go.PRS to Jamaica
'It is John who is going to Jamaica.'

b. Da shee du it, noh mee!
COP she do it NEG me
'It was her who did it, not me!'

It is also noteworthy that in (307), the focus marker is further away from the QP compared to the negative marker, even though negation is not within the scope of the focus operator.

More generally, negated clefts and other negative focus constructions can only address a positive question under discussion (Amiraz 2022). For instance, the negated cleft *It wasn't my dog that bit the mailman* serves as a response to the positive question "Whose dog bit the mailman?", but not to the negative question "Whose dog didn't bite the mailman?". This suggests that negation does not fall within the scope of the cleft or focus operator, which is responsible for generating the alternative set.

From this perspective, the structure of (307) is unexpected: the focus marker precedes the negative marker, yet negation apparently takes wider scope.

A similar hierarchical structure is found in Lingala (Central-Western Bantu), but in the reverse order. Standard negation in Lingala is expressed by the clause-final negative particle *tɛ́* (Redden 1963, 11), as shown in (309a). The focus marker *ndé* follows the focused constituent (van der Wal and Maniacky 2015, 59), as exemplified in (309b).

(309) Lingala

a. na-yéb-í Anglais tɛ́.
1SG-know-PRS English NEG
'I don't speak English.' (Redden 1963, 11)

b. ngáí ndé nazalí koloba.
1SG FOC 1SG.be.PFV 15.talk
'It is me who is talking (rather than someone else).'
(van der Wal and Maniacky 2015, 59)

Similarly to Jamaican Creole English, this focus construction is also employed as Strategy D, as shown in (310a) and (310b).[11]

[11] This construction is distinct from the cleft construction in (i), where there is an overt copula.

(i) Lingala
Pamba te, e-zal-i bakitani nyonso te ya Isalaele nde ba-zal-i bato ya
because 3SG.INAN-be-PRS descendants all NEG of Israel FOC 3PL-AN-be-PRS people of

(310) Lingala

a. Pamba te bango nyonso te nde ba-zal-i na ko-ndim-a
because 3PL.AN all NEG FOC 3PL.AN-be-PRS with INF-accept-FV
'For not all have faith.' [*Mokanda na Bomoi*, 2 Thess. 3:2; 2002]

b. Kasi, bato nyonso te ya Isalaele nde ba-ndim-aki Sango Malamu.
but people all NEG of Israel FOC 3PL.AN-accept.PST message good
'But not all the Israelites accepted the good news.' [ibid, Rom. 10:16]

Notably, negation appears structurally closer to the QP, even though it does not fall within the scope of the focus operator.[12]

In other languages, the negative marker and the focus marker appear on opposite sides of the QP. In Malagasy (Austronesian), for instance, the focused constituent or QP is preceded by the negative marker *tsy* and followed by the focus marker *no*, as demonstrated in (311a) and (311b).[13]

Isalaele
Israel
'For it is not all the descendants of Israel who are the people of Israel.' [ibid, Rom 9:6]

[12] I assume that in this construction, negation is structurally higher than the QP, as in (307). Importantly, I do not classify this as inverse scope, since the hierarchical relations can be read off the surface structure—there is no possible syntactic position for negation below the QP in this construction.

[13] According to Paul (2000), the focused constituent in (311a) is not fronted to a dedicated focus position. Instead, it functions as a non-verbal predicate, with the pseudocleft subject headed by *no*, which is analyzed as a determiner. Since Malagasy has basic VS word order, the structure in (311a) is at least superficially similar to non-verbal predicate clauses, such as (i). Under this analysis, (311a) can be paraphrased as 'The one(s) who kissed Rakoto is/are not Rasoa'.

(i) Malagasy (Paul 2000, 167)
Tsy mpianatra Rasoa.
NEG student Rasoa
'Rasoa is not a student.'

The key difference between focus constructions and regular predicate-initial clauses lies in the final position of the predicate: in regular predicate-initial clauses, the predicate moves to [Spec, TP] to satisfy EPP, whereas in focus constructions, the predicate undergoes further movement to [Spec, FocP] to obtain a focus interpretation (Paul 2000, 164).

However, the pseudocleft analysis cannot be straightforwardly extended to languages where non-verbal predicates require an overt copula. For example, in Lingala, (310a) and (310b) lack a copula, whereas nominal predicates in Lingala do co-occur with one, as shown in (ii). This suggests that (310a) and (310b) do not have the same structure as (i), where an overt copula is present.

(ii) Lingala (Meeuwis 2020, 263)
Bo-yók-í éte a-zál-í mw-ána na bísó té
2PL-hear-PRS COMP 3SG.AN-be-PRS 1-child CONN 1PL NEG
'You've heard that he is not our child.'

Given the cross-linguistic similarities between Jamaican Creole English, Lingala, and Malagasy focus constructions, a unified analysis may be preferable—one that does not rely on the possibility of a zero copula. From this perspective, an analysis in which the focused constituent is an argument (rather than a predicate)

(311) Malagasy

a. Tsy Rasoa no nanoroka an-dRakoto.
NEG Rasoa FOC PST.AT.kiss ACC-Rakoto
'It's not Rasoa who kissed Rakoto.' (Paul 2000, 166)

b. Tsy ny ankizy rehetra no be tomany.
NEG DET child all FOC a.lot cry
'Not all children cry a lot.' (Hanitramalala and Paul 2012, 629)

7.4.3 Wide-scope negation constructions

Another class of Strategy D constructions involves cases where negation appears either in a superordinate clause or in a high position in the left periphery, while the QP remains within the same clause as the predicate, without being separated by a clause boundary or a focus marker. These constructions can be roughly translated into English as 'it is not the case that…'.

In Mandarin Chinese, standard negation is expressed by a construction where the negative particle *bù* or *méiyǒu* precedes the verb, with the choice depending on aspect (Miestamo 2005, 90-91). This is demonstrated in (312).

(312) Mandarin (Wiedenhof 2015, 201)
Wǒ bú gàosu nǐ.
I NEG tell you
'I won't tell you.'

The most frequent strategy for expressing 'Not all X are Y' propositions is to negate the clause with *bùshì*, which can be decomposed into the negative particle *bù* and the copula *shì*, as shown in (313).

(313) Mandarin (Fan 2017, 46)
Bu-shi mei-pi ma dou tiaoguo-le liba.
NEG-COP every-CLF horse all jump-over-ASP fence
'Not every horse jumped over the fence.'

This construction is closely related to a focus structure, sometimes referred to as a bare-*shì* cleft, which can occur in both affirmative and negative contexts, as illustrated in (314).[14]

that moves to a dedicated focus position is perhaps more theoretically desirable. However, the extent to which this analysis applies across languages remains an open question.

[14] Mandarin also has another focus construction, known as a *shì…de* cleft (Cheng 2008), as exemplified in (i). However, this construction is not regularly used in 'Not all X are Y' sentences where the universal QP serves as the subject.

(i) Mandarin (Paul and Whitman 2008, 440)
Wo bu shi zuotian xie de shi.
1SG NEG COP yesterday write DE poem
'It was not yesterday that I wrote a poem.'

(314) Mandarin (Jin and Chen 2022, 303, 319)

a. Shì zhāngsān yào lái.
COP Zhangsan will come
'It's Zhangsan that will come.'

b. Bú shì zhāngsān yào lái.
NEG COP Zhangsan will come
'It's not Zhangsan that will come.'

Although the bare-*shì* construction may resemble constituent clefts (i.e., *it*-clefts), it differs in that it can also associate with broad focus over the entire clause, as shown in (315)—a possibility not available in constituent clefts.

(315) Mandarin (Paul and Whitman 2008, 424)

a. Shi xia yu le, bu pian ni.
COP fall rain PART NEG trick 2SG
'It really is that it's raining, I kid you not.'

b. Bu shi [yifu tai shou]$_F$, shi [ni tai pang le]$_F$.
NEG COP clothing too slim COP 2SG too fat PART
'It's not that the clothing is too small, it's just that you're too big.'

Similar constructions are also found in verb-final languages. In Japanese, the complex negation construction *wake de wa nai*—which roughly translates to 'it is not the case that…'—is commonly used for wide-scope negation. As shown in (316), the QP appears within a nominalized embedded clause, which serves as the complement of the nominalizing noun *wake* 'conclusion, reason, cause', while negation targets the main clause copula.[15]

(316) Japanese (Akitaka Yamada, p.c.)
[Seito-tati-no zenin-ga siken-ni uka-tta $_{CP}$] wake-de-wa na-i.
student-ASS-GEN all-NOM exam-DAT pass-PST conclusion-COP-CNTR NEG-PRS
'Not all of the students passed to exam.'
(lit. 'It is not the conclusion that all of the students passed to exam.')

In Korean, standard negation can be expressed by a construction in which the negative prefix *ani-* attaches to the auxiliary verb *ha* 'do', while the lexical verb is followed by the nominalizing suffix *-ci*, as shown in (317a). This construction is referred to as long-form negation in Korean linguistics.

A superficially similar construction involves the contrastive focus marker *-nun*, which is affixed to *-ci*, as seen in (317b). In this case, negation takes wide scope over preceding scope-bearing elements (Sells and Kim 2006; Park and Dubinsky 2020). However, unlike the corresponding constructions in Mandarin and Japanese, the Korean construction requires

[15] This construction is not considered an instance of inverse scope, even though negation linearly follows the quantifier. Since there is no syntactically possible position for negation below the QP, there is no potential for structural ambiguity.

narrow focus on a subclausal constituent and cannot be used to express broad focus over the entire clause.

(317) Korean (Eunsun Jou, p.c.)

a. na-uy kay-ka cippaywon-ul mwul-ci anh-ass-ta.
1SG-POSS dog-NOM mailman-ACC bite-NMLZ NEG-PST-DECL
‘My dog didn’t bite the mailman.’

b. [na-uy]$_F$ kay-ka cippaywon-ul mwul-ci-nun anh-ass-ta.
1SG-POSS dog-NOM mailman-ACC bite-NMLZ-CNTR NEG-PST-DECL
‘It wasn’t my dog that bit the mailman.’

The same focus construction is employed in ‘Not all X are Y’ sentences, as illustrated in (318). According to Park and Dubinsky (2020), the focus-marked nominalized verb *phwul-ci-nun* undergoes head raising to check the focus feature of the affix *nun*. This movement proceeds from V^0 through T^0 and C^0 to Foc^0, where its feature is checked. As a result of this movement, the complex negated verb appears above TP, taking scope over the subject QP. Crucially, while the verb moves to a focus position, the focused constituent itself (the QP) remains in situ and does not undergo movement to a dedicated focus position.

(318) Korean (Park and Dubinsky 2020)
motun haksayng-i ku mwuncey-lul phwul-ci-nun anh-ss-ta.
every student-NOM the question-ACC answer-NMLZ-CNTR NEG-PST-DECL
‘Not every student answered the question.’

In rare cases, the standard negation construction itself is biclausal. In Fijian, standard negation is formed using a semi-auxiliary verb, *sega*, which negates a complement clause containing the lexical predicate (Dixon 1988, 40). The subject pronoun of the complement clause can optionally raise to the main clause, as illustrated in (319) (Dixon 1988, 280).

(319) Fijian (Dixon 1988, 280)

a. au sega ni la’o.
1SG NEG COMP go
‘I’m not going.’

b. e sega ni=u la’o.
3SG NEG COMP=1SG go
‘I’m not going.’ (lit. ‘It is not that I’m going.’)

The same biclausal structure is employed in ‘Not all X are Y’ sentences, as shown in (320).

(320) Fijian
Dou sa sega ni savasava kece.
2PL ASP NEG COMP clean all
‘You are not all clean.’ [*Fijian New Translation*, John 13:11; 2010]

7.5 Strategy E

Strategy E is defined in (321).

(321) *Strategy E*
A monoclausal construction in which the subject position is occupied by a lexical noun phrase or a pronoun, while clausal negation precedes a floating, adverbial, or right-dislocated quantifier.

This strategy includes several distinct construction types, all of which share the characteristic that the quantifier does not function as the head of the subject argument. Instead, the subject position is filled by another constituent, which either serves as the restrictor of the quantifier or functions as an expletive.

7.5.1 Floating quantifier constructions

In some languages, ‘Not all X are Y’ propositions can be expressed using a floating quantifier construction, in which a quantifier that typically functions as a determiner is separated from the noun phrase or pronoun that serves as its restrictor, with the latter occupying the subject position.

Floating quantifiers are often analyzed as base-generated as the head of a QP in argument position, with their complement moving to a higher position, thereby stranding the quantifier in its original position (Sportiche 1988, and subsequent work).[16] This analysis is illustrated in (322).

(322) a. [$_{IP}$ The Tea Party people$_1$ are$_2$ not [$_{PredP}$ all t_1 t_2 Republicans]]
[COCA: Spoken; 2010]
b. [$_{IP}$ We$_1$ are$_2$ not [$_{PredP}$ all t_1 t_2 vampires]]
[COCA: Fiction]

In some cases, it is difficult to determine whether the subject pronoun functions as the restrictor of the quantifier or serves as an expletive. For instance, in (323a), *it* lacks a clear referent, and the paraphrase in (323b) sounds unnatural.

(323) a. It’s all fun and games until someone loses an eye.
b. #All of it is fun and games until someone loses an eye.

A similar issue arises with floating quantifiers under negation, as in (324), which can be paraphrased as ‘Not everything is fun and games with Gandalf’. Since *it* in these cases does not straightforwardly restrict the domain of the quantifier—except perhaps in the broad

[16] Alternatively, floating quantifiers have been analyzed as adverbial, in which case they are not base-generated as the head of a QP (Kayne 1975, and following work). Under this analysis, floating quantifiers belong to the same category as adverbial quantifiers that cannot function as determiners (see §7.5.4).

sense of limiting it to non-humans—analyzing *it* as an expletive in (323a) and (324) seems more appropriate.

(324) From the first time audiences catch an eye of Gandalf sitting atop a wagon of fireworks, it was easy to tell that there was a lot to love with this character. **But it’s not all fun and games with Gandalf**, as the wizard proves to be powerful and stern throughout the journey to destroy the Ring once and for all. [NOW: US; 2005]

By contrast, when the domain of quantification is restricted to humans, it is typically not possible to use a floating quantifier construction unless the restrictor is explicitly specified by a DP, pronoun, or relative clause. For example, (325a) cannot be paraphrased as (325b), suggesting that floating quantifier constructions generally require a more specific restrictor when referring to humans.

(325) a. Not everybody is going to be a musician. [COCA: Newspaper]
b. People/They are not all going to be musicians.

This contrasts with cases where the domain of quantification is restricted to non-humans, as in (326), where the pronoun and the floating quantifier construction are largely interchangeable.

(326) a. Everything is going to be alright.
b. It’s all going to be alright.

Finally, in some languages, the subject pronoun can be modified by a relative clause. In (327), from Middle English, the relative clause is extraposed, yielding a structure that can be paraphrased as ‘That which glitters is not all gold’. Similarly, in (328), from Early New High German, a comparable construction is used when the quantifier’s domain is restricted to humans.

(327) Middle English
Hit is not al gold, that glareth.
‘Not all that glitters is gold.’ [Chaucer, *The House of Fame*, line 272; 1374-1385]

(328) Early New High German

denn	es	sind	nicht	alle	Jsraeliter/	die	von	Jsrael	sind/
for	it	be.PRS.3PL	NEG	all.NOM.PL	Israelites	who.NOM.PL	from	Israel	be.PRS.3PL

‘For they (lit. ‘it’) are not all Israelites who are from Israel.’
[*Luther Bible*, Rom. 9:6; 1522]

7.5.2 Expletive inversion constructions

Another type of Strategy E construction features an expletive pronoun in subject position, while the quantifier functions as a determiner of a QP in post-verbal position. This pattern is structurally similar to Strategy C constructions involving NegVS word order (§7.3.1).

However, here an expletive pronoun is required to satisfy syntactic subject requirements. This construction is illustrated in (329) from German.

(329) German

Es	wohnen	ja	nicht	alle	Grünenwähler	in	den	reichen	Speckgürteln
it	live.PRS.3PL	indeed	NEG	all	Green.voters	in	the	rich	commuter.belts

der	Großstädte.
of.the	big.cities

'Not all Green party voters live in the wealthy suburbs of the big cities.'
[LCC: News; 2023]

Crucially, in (329), the negative marker modifies the verb (clausal negation) rather than the QP (constituent negation). This is supported by two key observations. First, negated QPs are disallowed in post-verbal position in most languages (see §7.2.1). Second, this expletive inversion structure is also found in Icelandic, which does not have negated QP constructions, further supporting the analysis of (329) as a clausal negation structure rather than a negated QP construction.

(330) Icelandic

Það	komast	ekki	allir	í	eitt	mót.
it	get.PRS.3PL	NEG	all.NOM.PL.M	into	one.ACC.SG.N	competition.ACC

'Not everyone can get into one competition.' [LCC: News; 2020]

7.5.3 Right-dislocated quantifiers

In some languages, the quantifier is separated from its NP complement, similar to floating quantifiers, but it does not remain in its base position. Instead, the quantifier appears in a right-dislocated position, occurring at the right edge of the clause rather than in argument position.

In Zulu, an SVO language, a universal quantifier in subject position can appear either inside or outside the *v*P. In (331a), the object is right-dislocated, while the quantifier remains inside *v*P. By contrast, in (331b), the quantifier is right-dislocated, appearing at the end of the clause (Buell 2008, 44).

(331) Zulu (Buell 2008, 46)

a.

[$_{IP}$ Izingane$_1$	zi-wa-thanda$_2$	[$_{vP}$ zonke	t_1 t_2 t_3]]	amaswidi$_3$.
10.children	10.SBJ-6.OBJ-love	10.all		6.sweets

'All (the) children like sweets.'

b.

[$_{IP}$ Izingane$_2$	zi-wa-thanda$_3$	[$_{vP}$ t_1 t_2 t_3	amaswidi]]	zonke$_1$.
10.children	10.SBJ-6.OBJ-love		6.sweets	10.all

'All (the) children like sweets.'

In (332), the quantifier follows the predicate, occurring at the right edge of the clause, similarly to (331b).

(332) Zulu (Buell 2008, 46)
[$_{IP}$ Izingane$_2$ a-zi-thand-i$_3$ [$_{vP}$ t_1 t_2 t_3 amaswidi]] zonke$_1$.
10.children NEG-10.SBJ-love-NEG 6.sweets 10.all
'Not all children like sweets.'

7.5.4 Adverbial quantifiers

Unlike floating quantifiers, which are often analyzed as base-generated as determiners that are then stranded by their NP complement, adverbial quantifiers are not derived from or structurally related to determiners. Instead, they function as independent quantificational adverbs that modify the predicate.

In Mandarin, the adverbial quantifier *dōu* always precedes the verb and cannot function as a determiner (Cheng 1995, 198-199). In (333), the negative particle *méiyǒu* precedes *dōu*, while the subject position is occupied by a DP or pronoun that serves as the restrictor of the quantifier.

(333) Mandarin

a. [$_{IP}$ neixie ren$_1$ meiyou dou [$_{vP}$ t_1 kan-guo neiben shu]]
those person NEG all read-ASP that book
'Not all of these people read that book.' (Cheng 1995, 199)

b. [$_{IP}$ wǒmen$_1$ méiyǒu dōu [$_{vP}$ t_1 qù]]
1PL NEG all go
'Not all of us have gone.' (Zhan and Sun 2013, 779)

7.6 Borderline cases

This section examines borderline cases where a construction could potentially be classified under more than one strategy type.

7.6.1 Strategy B vs. Strategy D

The distinction between Strategy B (negated QP constructions) and Strategy D (negated clefts and focus constructions) is generally straightforward when a construction is biclausal, as Strategy B is strictly monoclausal by definition. However, the classification becomes more complex when Strategy D constructions are also monoclausal. In such cases, the presence of a focus marker or a copula is typically a key diagnostic for Strategy D. Yet, it is not always clear whether a given focus marker or copula functions as an independent syntactic element or has been reanalyzed as part of the negative marker itself.

An example of this ambiguity is found in Jewish Babylonian Aramaic (JBA) and Syriac, two closely related dialects of Late Aramaic, both of which developed a clause-initial negative particle through the grammaticalization of an earlier negated cleft construction.

In the first stage, the form *lāw* emerged through the univerbation of the regular negator *lā* with the third-person pronominal copula *=hu* (Bar-Asher Siegal and De Clercq 2019), as demonstrated in (334a). However, this contracted form does not appear when a verbal copula is present, which occurs in non-present tenses, as shown in (334b).

(334) Syriac (Pat-El 2006, 339-340)

a. lā=w Ḥīm ’ītaw=y wa
NEG=3SG.M Ham exist=SG.F be.PST.3SG.M
‘[It is clear that] it was not Ham, [since Ham is the middle [son], not the youngest].

b. lā=wā men ’ūlṣānā=hu
NEG=be.PST.3SG.M from coercive=3SG.M
‘It was not out of coersion’

In the second stage, *lāw* was reanalyzed as a negative marker, typically appearing in clause-initial position, as illustrated in (335a) and (335b). The same syntactic structure was employed in ‘Not all X are Y’ sentences, as shown in (335c).

(335) Jewish Babylonian Aramaic (Bar-Asher Siegal and De Clercq 2019, 232, 234, 243)

a. lāw gazlān-e ninhu
NEG thief-PL COP.3PL.M
‘They are not thieves.’

b. lāw ’akbrā gnab ’ella ḥora gnab
NEG mouse steal.PST.3SG.M but hole steal.PST.3SG.M
‘It is not the case that the mouse stole, the hole stole.’

c. lāw kulle-h ’ālmā ḥazu lsahdūtā
NEG all-3SG.M world see-PASS.PTCP.3PL.M to-testemony
‘It is not the case that everyone is eligible (to give) testimony.’

The construction in (335c) is classified as Strategy B, as *lāw* had already been reanalyzed as a single negative marker at this stage in Late Aramaic. However, the borderline case arises when analyzing earlier Syriac examples, such as (336), where *lāw* could still be interpreted as a combination of the negator *lā* and a copular element.

(336) Syriac
lā=w kull-an neḏmaḵ
NEG=3SG.M all-1PL sleep.IMPERF.1PL
‘Not all of us will sleep.’ [*Peshitta*, 1 Cor. 15:51; 3rd-5th century]

Importantly, syntactic reanalysis is only observable retrospectively, once a construction has already been extended to new syntactic contexts (Harris and Campbell 1995; Bar-Asher Siegal 2024). Thus, it is impossible to pinpoint the exact moment when *lāw* became a fully grammaticalized negative marker. This makes (336) a borderline case, highlighting how the distinction between Strategies B and D is not always clear-cut.

7.6.2 Strategy C vs. Strategy E

A key distinction between Strategy C and Strategy E lies in the grammatical role of the QP. In Strategy C, the QP functions as the grammatical subject of the clause, whereas in Strategy E, the subject position is occupied by a pronoun or a lexical noun phrase, with the QP appearing as a floating or right-dislocated element.

This distinction becomes more challenging to determine in null-subject languages, where the absence of an overt pre-verbal subject does not necessarily indicate that the QP itself is a post-verbal subject. In such cases, two competing analyses must be considered: either the QP serves as a post-verbal subject (Strategy C), or the subject is a null pronoun (*pro*), with the QP functioning as a floating quantifier or a right-dislocated element (Strategy E).

For example, in the first clause of (337), classification depends on whether *todos* is the grammatical subject, which would indicate Strategy C, or whether the subject is an unexpressed *pro*, with *todos* as a floating quantifier, aligning with Strategy E. Crucially, subject agreement does not resolve this ambiguity, since first-person plural agreement is also possible when *todos* is the subject, as seen in the second clause of the example. Given this structural uncertainty, cases of this type are labeled as ‘C/E’ and excluded from the statistical analysis.

(337) Spanish
No moriremos todos, mas todos seremos transformados.
NEG die.FUT.1PL all.PL.M but all.PL.M be.FUT.1PL transform.PST.PTCP.PL.M
‘We will not all die, but we will all be transformed.’
[*Biblia de Jerusalén*, 1 Cor. 15:51; 1973]

7.6.3 Strategy A vs. Strategy E

Some languages allow, or even require, the co-occurrence of a subject QP and an adverbial quantifier, a phenomenon known as distributive concord (Oh 2006). When the negative marker appears between these two quantifiers, the distinction between transparent scope and inverse scope is blurred.

For example, in (338), from Somali, the pronominal quantifier—derived from *dhán* ‘all’ (Saeed 1982, 535)—appears alongside the adverbial quantifier *wada* ‘together, completely, all’, with the negative particle *ma* intervening between them. Given that this structural configuration may facilitate the processing of scope relations, such cases are not classified as instances of inverse scope. Instead, they are labeled as ‘A/E’ and excluded from the statistical analysis.

(338) Somali
Dhammaan-tiin ma wada nadiifsan-idin.
all-2PL NEG all clean-2PL
‘Not all of you are clean.’ [*Kitaabka Quduuska Ah*, John 13:11; 2008]

7.7 Rarities

This section examines rare constructions that deviate from the five main strategies, providing insight into less common yet attested ways of expressing 'Not all X are Y' propositions.

7.7.1 Negated quantifier verbs

Some languages have verbal quantifiers that can be directly negated. In Garifuna (Arawakan), the stative verb *gíbe-* 'be many, be much' (339a) can be negated like a verb (339b).[17]

(339) Garifuna (Barchas-Lichtenstein 2012, 176; 214)

a. G-íbe-tu féin.
AFF-be.much-T3F bread
'There is a lot of bread.'

b. M-íbe-tiyan óunli éibgua-tiyan
NEG-be.much-T3PL dog run-T3PL
'Not many dogs run.'

Although Garifuna is predominantly VSO (Munro and Gallagher 2014, 14), the quantifier verb is not the main predicate in (339b) but rather a prenominal modifier. This is evident in constructions where the quantifier verb modifies the object, as in (340a). Structurally, this parallels indefinite attributive adjectives (Barchas-Lichtenstein 2012, 177-178), such as (340b).[18]

[17] Garifuna has several series of verbal person markers. In the gloss, T corresponds to the T-series (see Munro and Gallagher 2014, and references therein).

[18] Negative indefinites in Garifuna are derived from a negative existential construction (Barchas-Lichtenstein 2012, 179), as demonstrated in (i). The resulting form can function as either a pronoun (ia) or a nominal modifier (ib), similarly to other quantifier verbs in the language.

(i) Garifuna (Barchas-Lichtenstein 2012, 179; 214)

a. Úwa-ti erémuha-ti brídu.
NEG.EX-T3M sing-T3M well
'Nobody sings well.'

b. Úwa-ti bálu áfaru-ti budéin.
NEG.EX-T3M bullet hit-T3M bottle
'No bullets hit the bottle.'

(340) Garifuna

a. Agányeha-tìna gíbe-tu fáluma
buy-T1SG many-T3F coconut
'I bought a lot of coconuts' (Haurholm-Larsen 2016, 195)

b. Busíye-tina ába harú-ti óunli.
want-T1SG one be.white-T3M dog
'I need a white dog.' (Barchas-Lichtenstein 2012, 178)

The primary universal quantifier in Garifuna is the determiner *sún* 'all' (341a). In negative contexts, Strategy B is the dominant strategy, as exemplified in (341b).

(341) Garifuna (Barchas-Lichtenstein 2012, 181; 188)

a. Sún mútu busíye-tiyan l-ún t-amáredu h-aráü l-úma ába surúsiya.
all person want-T3PL P3M-to P3F-marry P3PL-child P3M-with one doctor
'All people want their daughter to marry a doctor.'

b. Máma sún óunli éibgua-tiyan
NEG all dog run-T3PL
'Not all dogs run.'

However, another alternative strategy involves negating a quantifier verb, similar to (339b), as demonstrated in (342).[19]

(342) Garifuna
Má-su-tiña gayára ja-bágari-dun má-mari-ga
NEG-be.all-T3PL be.able VBLZ-life-VBLZ NEG-spouse-GA
'Not everyone can remain single.' [Wycliffe Bible Translators, Matt. 19:11; 2012]

Interestingly, *másutiña* 'NEG-be.all-T3PL' appears to lack an affirmative counterpart: the expected form *gásutiña* 'AFF-be.all-T3PL' (parallel to the attested form *gíbetiña* 'AFF-be.much-T3F') does not appear in the New Testament translation or in any other online sources, as far as I can determine. Given that affirmative 'All X are Y' statements are generally far more frequent than their negative counterparts ('Not all X are Y'), this absence is unlikely to be accidental. Instead, it suggests that *másutiña* originated as a stative verb meaning 'be all', which later fell out of use in affirmative contexts but was retained as a fixed form in negation.

Even if *másutiña* is analyzed as a negative verbal quantifier, rather than as a lexicalized fixed form, it still does not fit neatly into any of the proposed strategies. On the one hand, negation modifies a quantificational expression, similar to a negated QP construction (Strategy B). On the other hand, this construction clearly involves clausal negation rather than constituent negation, setting it apart from Strategy B. As a result, the construction in (342) falls outside the scope of the five main strategies.

[19] The suffix *-tiña* is a spelling variant of the third-person plural suffix suffix *-tiyan*. Additionally, see Haurholm-Larsen (2016, 190-191) for a discussion of the function of *-ga*.

Moreover, Garifuna presents a potential counterexample to the long-standing claim that no language has a lexicalized expression meaning ‘not all’——a generalization first proposed as an absolute linguistic universal by Horn (1972). The form *másutiña* is not transparently decomposable into a negative marker plus a universal quantifier, since the universal quantifier verb appears to be restricted to this particular negative form, although this appears to have been the lexical source of this form.[20] For a broader discussion of the lexicalization of ‘not all’, see §9.3.

While Garifuna provides an example of a negated quantifier verb, it is worth noting that some languages have quantifier verbs functioning as nominal modifiers, but these verbs are not the ones that are negated in ‘Not all X are Y’ sentences.

A case in point is Chickasaw (Muskogean), where the primary universal quantifier is the verb *mó̱ma* ‘be all’, which bears switch-reference marking (Munro 2017, 151). Similar to Garifuna, this quantifier verb functions as a nominal modifier, as seen in (343a), akin to the attributive adjective verb in (343b).

(343) Chickasaw (Munro 2017, 151; 131)

a. Ofi’ mó̱ma-kat woochi.
dog be.all-COMP.SS bark
‘All dogs bark.’

b. Ofi’ tohbi-hoot sa-kisi-tok.
dog be.white-FOC.SS 1SG.II-bite-PST
‘The/A white dog bit me.’

This construction is also employed in ‘Not all X are Y’ sentences, but crucially, negation does not target the quantifier verb itself. Instead, negation modifies the main predicate, either periphrastically or morphologically, as demonstrated in (344a) and (344b), respectively.

(344) Chickasaw (Munro 2017, 174)

a. Kowi’-at mó̱ma-kat tohbi ki’yo.
cat-NOM be.all-COMP.SS be.white NEG
‘Not all cats are white.’

b. Kowi’-at mó̱ma-kat ik-tohb-o.
cat-NOM be.all-COMP.SS HYP-be.white-NEG
‘Not all cats are white.’

Importantly, negation modifies the main predicate, not the quantifier verb, in these constructions. This makes these cases instances of Strategy A rather than a novel strategy, since clausal negation follows the constituent containing the quantifier.

[20] I am grateful to Steffen Haurholm-Larsen (p.c.) for insightful discussions on the Garifuna data.

7.7.2 Affixal quantifiers

In Paraguayan Guaraní, the completive aspect suffix *-pa* (and its allomorph *-mba*) typically express completion or termination, as demonstrated in (345a), but it can also express universal quantification (Tonhauser 2006, 267-268), as show in (345b).

(345) Paraguayan Guaraní (Tonhauser 2006, 267-268)

a. A-ñani-mba.
A1SG-run-COMPL
‘I finished / stopped running.’

b. O-ñe-mondý̃i-mba
A3-JE-scare-COMPL
‘Everybody was scared.’

The same suffix appears in (346), where it again conveys universal quantification. Importantly, this construction does not fit into any of the proposed strategies, as the universal quantifier is neither the subject nor an adverb.

(346) Paraguayan Guaraní
Na-ña-mano-mba mo’ã-i-ko
NEG-1PL-die-COMPL FUT.NEG-NEG-EMPH
‘We will not all die.’ [Paraguayan Bible Society, 1 Cor. 15:51; 2006]

Chapter 8

Quantitative results

This chapter presents a quantitative analysis of the cross-linguistic distribution of strategies used to express 'Not all X are Y' propositions. The goal is to assess how frequently each strategy is employed across languages and whether certain strategies are more prevalent in particular linguistic macro-areas. Additionally, the chapter examines whether the observed distribution aligns with theoretical expectations regarding a bias for scope transparency.

The chapter is structured as follows. Section 8.1 provides an overview of the cross-linguistic frequencies of the strategies, based on a stratified sampling approach that accounts for genealogical and areal biases. The results reveal that Strategy A is the most widespread across macro-areas, though its dominance varies by region. Strategy B and Strategy D also show broad but uneven distributions, while Strategies C and E are much less frequent as dominant strategies but appear more regularly as secondary strategies.

Following this, Section 8.2 tests the Synchronic Hypothesis, which posits that inverse-scope constructions are less common than what would be expected if languages had no preference for either transparent or inverse scope. To evaluate this hypothesis, the analysis compares the observed frequency of Strategy A to a baseline expectation derived from the position of negation relative to subjects in standard negation constructions. The results indicate statistically significant deviations from this baseline in Eurasia and Africa, suggesting that these macro-areas exhibit a systematic preference for transparent-scope strategies over inverse-scope constructions. While additional effects were found in Papunesia and North America when all available languages were considered, these results were not robust under more conservative sampling methods, possibly due to limited genealogical diversity in the sample.

8.1 Cross-linguistic frequencies

To estimate the distribution of strategies for expressing 'Not all X are Y' propositions, this study employs a non-proportional stratified sampling approach, as outlined in §4.2. This method reduces genealogical biases by using genera as the primary unit of analysis, rather than individual languages, ensuring that no single language family disproportionately influences the results. Additionally, macro-areas are used to account for broader areal effects.

Each genus was assigned equal weight in the analysis. For genera containing only one language, the dominant strategy of that language directly represented the genus. However, for genera containing multiple languages, the relative frequencies of each strategy within the genus were first calculated based on the dominant strategies of its languages. These within-genus frequencies were then used to contribute proportionally to the overall frequency calculations.

In each iteration, one language was randomly selected from each genus, ensuring equal representation of genera in the sample. This process was repeated 1,000 times to account for variability in language selection within multi-language genera. This iterative sampling approach allowed for robust estimates of the overall relative frequencies by incorporating intra-group variability. The resulting confidence intervals reflect the combined uncertainty from both the random selection process and the diversity within genera.

This process is an example of subsampling, where a subset of items (in this case, one language per genus) is randomly selected from a larger group. By repeating the process across multiple iterations, subsampling captures the variability within genera that contain multiple languages, ensuring that the results are not overly dependent on a specific selection of languages. This method enhances the reliability of the estimates and provides a more accurate reflection of the diversity within the sample.[1]

Figure 8.1 presents the relative frequencies of dominant strategies across macro-areas. The mean relative frequency is calculated as the average proportion of genera in which each strategy is dominant across 1,000 iterations, with one language randomly sampled per genus in each iteration.

Figure 8.2 provides a visual representation of the dominant strategy for each language in the sample. The data are based on translations of the New Testament published after 1950, which may not fully reflect the current state of these languages. The map includes all the languages in the sample and does not account for the overrepresentation of certain language families and regions, unlike the adjusted analysis of cross-linguistic frequencies. The maps shows certain areal patterns but also a fair amount of variation within regions.

Figure 8.3 shows the proportion of languages in each macro-area where a strategy is attested, based on 1,000 iterations of random sampling. Unlike dominant strategies, which reflect the most frequently used strategy in each language, this figure considers all strategies that are attested at least once in the corpus, regardless of their relative frequencies. This provides a broader view of the diversity of available strategies in each macro-area, rather than highlighting a single preferred option per language. However, these frequencies may be

[1] An alternative to subsampling is to use statistical modeling techniques, such as mixed-effects models, which can explicitly account for genealogical and areal biases by incorporating group-level effects (e.g., genera or macro-areas) as random factors. These models allow for unbalanced datasets and estimate the contribution of group-level variability to the overall results, making them particularly effective for large-scale studies where full balance across groups is difficult to achieve. For a detailed discussion and implementation of such methods, as well as a review of other methods of bias control in typological research, see Guzmán Naranjo and Becker (2022).

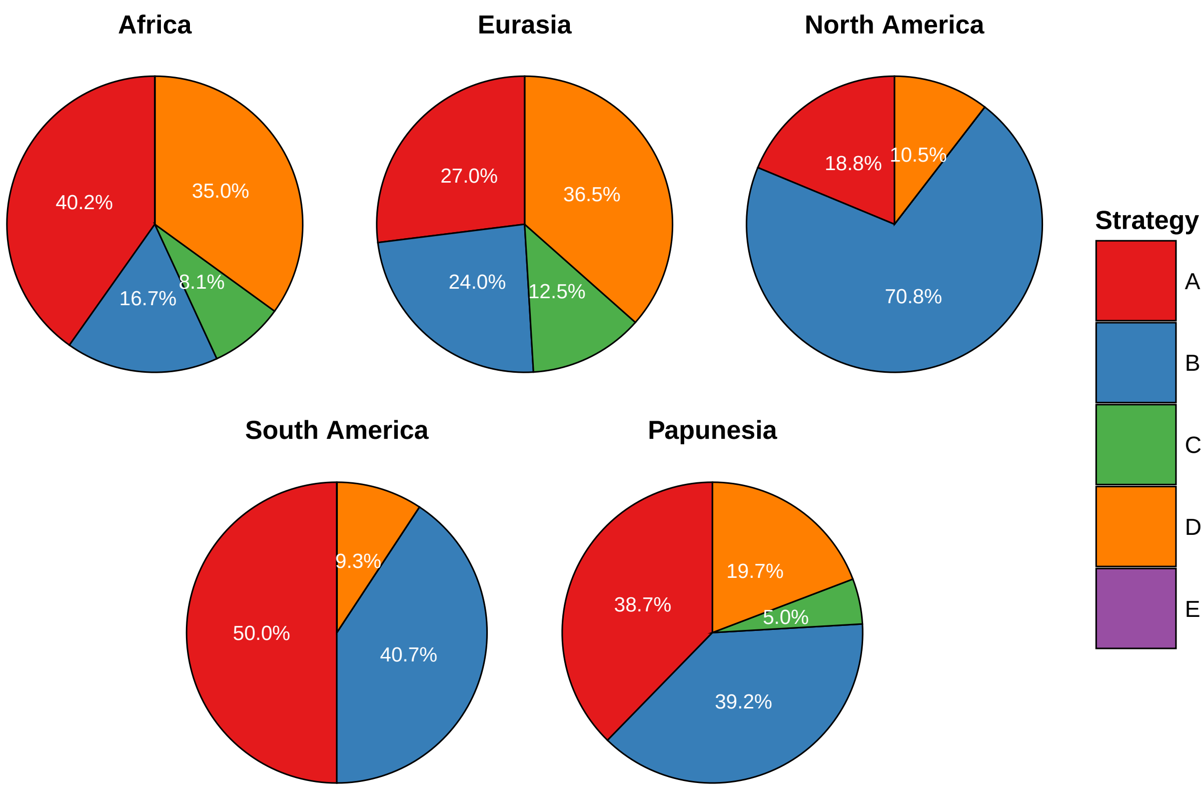


Figure 8.1: Distribution of dominant strategies across macro-areas

slightly underestimated due to potential false negatives—cases where a strategy is available in a given languages but rarely used and therefore not captured in the data (see §5.2).

The 95% confidence intervals reflect the uncertainty in the sampling of genera, assuming that each genus is an independent observation and that the genera were randomly sampled. However, the dataset is based on a non-random sample of genera, reflecting the available data rather than a true random sampling process. As a result, these confidence intervals may underestimate the true uncertainty, serving instead as a best-case approximation for variability in the dataset.[2]

As shown in Figure 8.3, Strategy A is relatively widespread across all macro-areas and exhibits the least variation in frequency between them. This supports the view that its distribution is primarily shaped by language-internal factors, with language contact playing a secondary role. The only macro-area where Strategy A is notably less common is North America. One possible explanation is that Mesoamerican languages are overrepresented in

[2]Note that the uncertainty captured by the confidence intervals pertains to the sampling of genera, not the variability within each genus. The latter is accounted for only in the calculation of the mean frequencies, as one language is randomly sampled from each genus in every iteration.

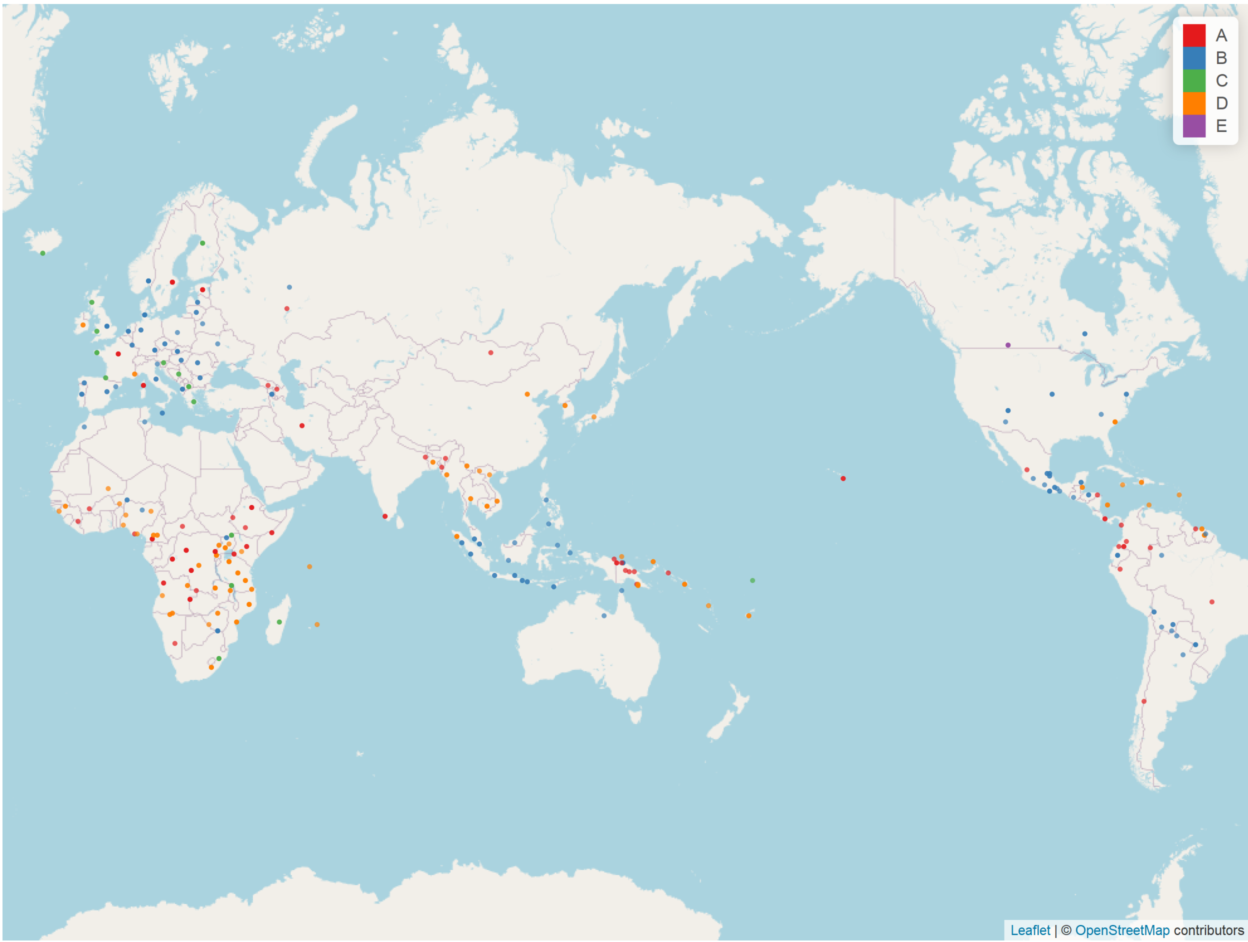


Figure 8.2: Geographic distribution of dominant strategies across languages in the sample

the sample due to data availability, whereas other macro-areas allow for a more balanced geographic distribution.

Strategies B and D are also widely attested but show a more uneven distribution. Strategy D is more frequent than Strategy B in Africa and, to a lesser extent, in Eurasia—particularly in East Asia and Mainland Southeast Asia, as will be discussed shortly. In contrast, the remaining macro-areas exhibit a stronger tendency toward Strategy B.

Strategy C is rarely the dominant strategy, while Strategy E is dominant in only one language in the sample—Plautdietsch, an East Low German variety spoken in North America. However, both Strategies C and E are moderately common as secondary strategies in certain macro-areas, indicating that while they are rarely the primary means of expressing 'Not all X are Y' propositions, they are still employed in specific contexts in many languages.

Furthermore, the differences between Figure 8.1 and Figure 8.3 highlight the extent to which each strategy is employed as a secondary rather than a dominant strategy. Strategy A is widely attested in Eurasia, but it serves as the dominant strategy in only about half of the languages that use it; in the remaining cases, it is employed as a secondary strategy. In other macro-areas, however, it is more often the dominant strategy, though it is not uncommon for it to be employed as a secondary strategy as well. In contrast, Strategy B is overwhelmingly used as the dominant strategy in the languages where it is attested. Strategy D, on the other hand, patterns more like Strategy A, being fairly common as a secondary strategy while still serving as the dominant one in most languages where it is available. Finally, as previously noted, Strategies C and E are more frequently attested as secondary strategies rather than dominant ones.

Figure 8.4 illustrates the co-occurrence patterns between Strategy A and each of the other attested strategies (B, C, D, and E). The underlying data were averaged across macro-areas, assigning equal weight to each macro-area. Each Venn diagram represents the proportion of languages where Strategy A and the given strategy co-occur, where only Strategy A is attested, where only the other strategy is attested, and where neither strategy appears. The relative sizes of the overlapping and non-overlapping areas are proportional to the observed frequencies in the dataset. This visualization highlights the degree to which Strategy A tends to co-occur with other strategies or, conversely, whether certain strategies tend to be mutually exclusive.

As shown in the figure, languages are less likely to have both Strategy A and Strategy B or C, whereas co-occurrence with Strategy D or E is more common. Among languages that use Strategy B, approximately 28% also use Strategy A, while the proportion is slightly lower for Strategy C at 25%. In contrast, 45% of languages that use Strategy D also have Strategy A, and this proportion is 41% for languages that use Strategy E.

Setting aside Strategy E for the moment, the differences between Strategies B and C, on the one hand, and Strategy D, on the other, likely reflect the extent to which these strategies function as pragmatically marked constructions rather than as neutral means of expressing 'Not all X are Y' propositions. Strategy B is rarely marked in this respect, though there are exceptions, such as in Swedish (see §6.2.1). Strategy C is more likely to serve as a marked alternative to Strategy A, primarily in verb-medial languages. However, in verb-initial languages where Strategy C is attested, it is often preferred over Strategy A, since the latter involves a deviation from the default word order.

By contrast, while Strategy D frequently serves as the dominant, neutral strategy, it also remains pragmatically marked in many cases, retaining the discourse function typically associated with clefts and other focus constructions (see §9.1.3). Finally, Strategy E is almost never the dominant strategy, indicating that its presence or absence alongside Strategy A is largely shaped by the availability of other competing strategies.

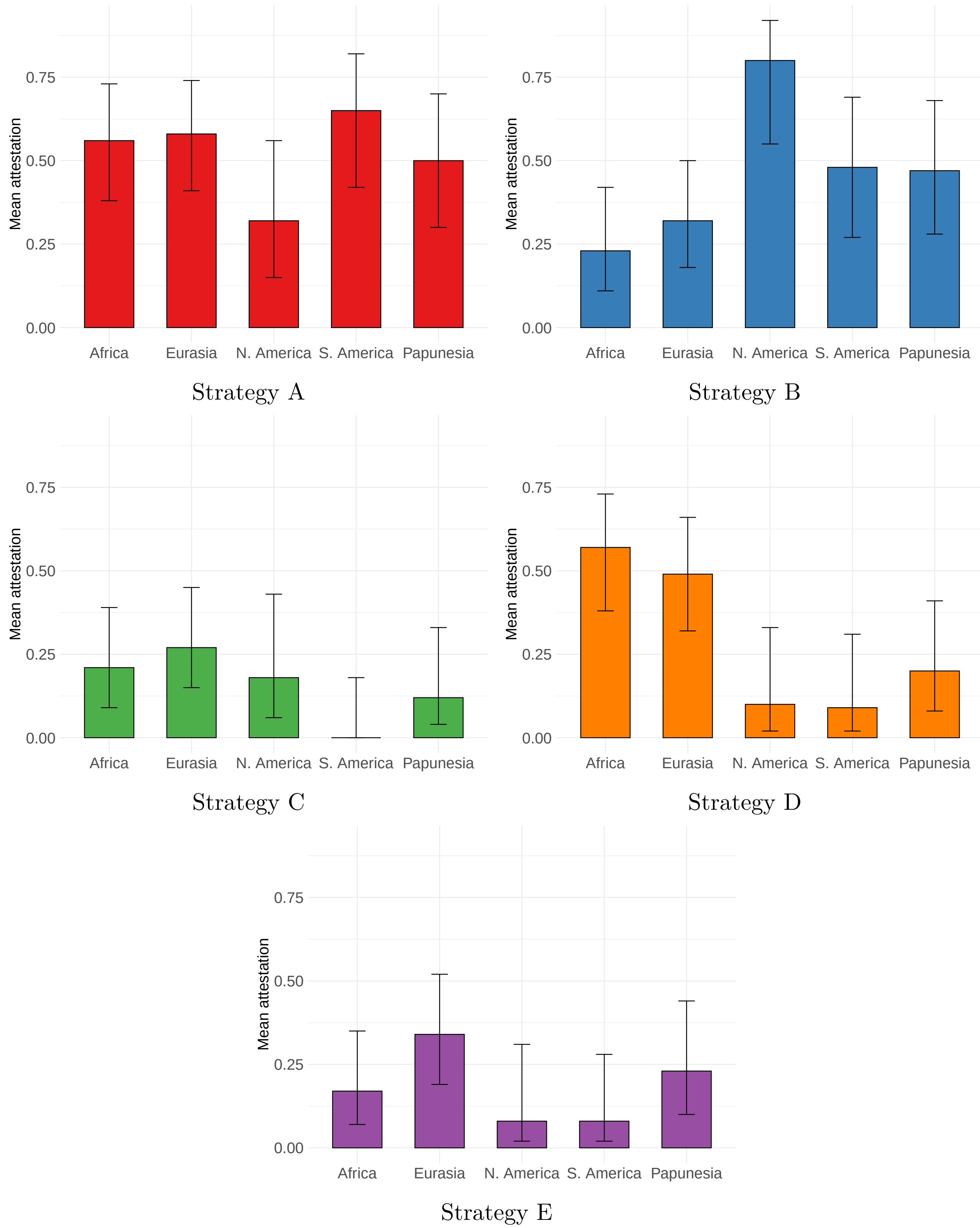
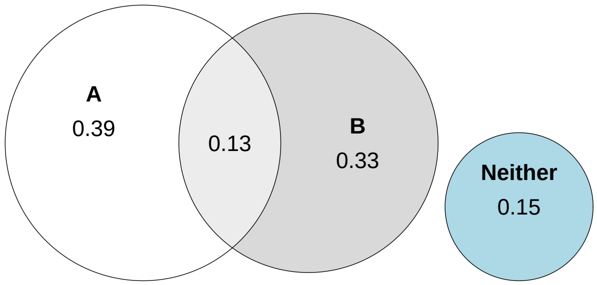
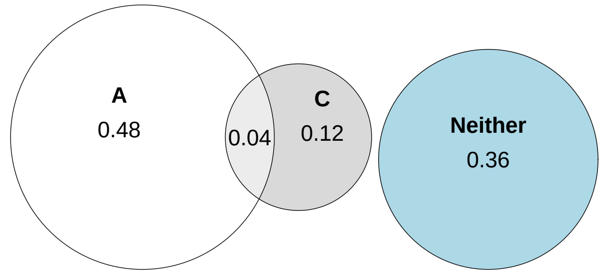


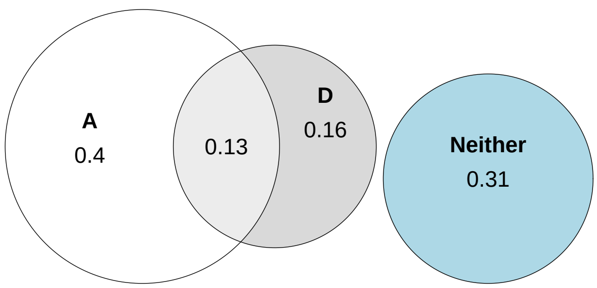
Figure 8.3: Proportion of languages in each macro-area where each strategy is attested, with 95% confidence intervals



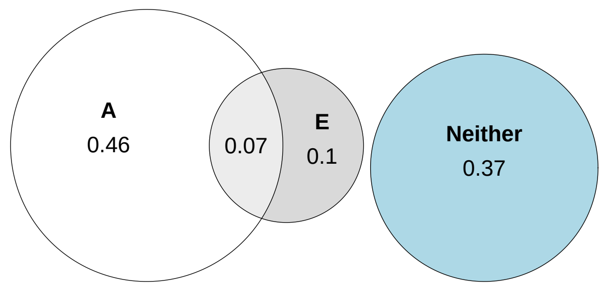
Strategies A and B



Strategies A and C



Strategies A and D



Strategies A and E

Figure 8.4: Cross-linguistic co-occurrence patterns of Strategy A with other strategies

Figure 8.5 presents the geographic distribution of Strategy A across languages in the sample, representing its relative frequency within each language. Strategy A is prevalent across all macro-areas but is notably rare in Southeast Asia, where Strategies D (Figure 8.6b) and B (Figure 8.6a) dominate. Strategy B, in contrast, is more geographically concentrated, with Strategy D serving as its main competitor in areas where B is less frequent.

Strategy C (Figure 8.7a) is even more geographically restricted, appearing commonly in the Balkans, certain Bantu languages, and some verb-initial languages such as Celtic and Austronesian. However, other verb-initial languages, such as those in Mesoamerica, tend to favor Strategy B. Strategy D is almost in complementary distribution with Strategy B, being more frequent in Africa, East Asia, Mainland Southeast Asia, and among creole languages. In Europe, Strategy D is less common and usually serves as a secondary strategy. Lastly, Strategy E (Figure 8.7b) is primarily found in European languages, where it is typically employed as a secondary strategy.

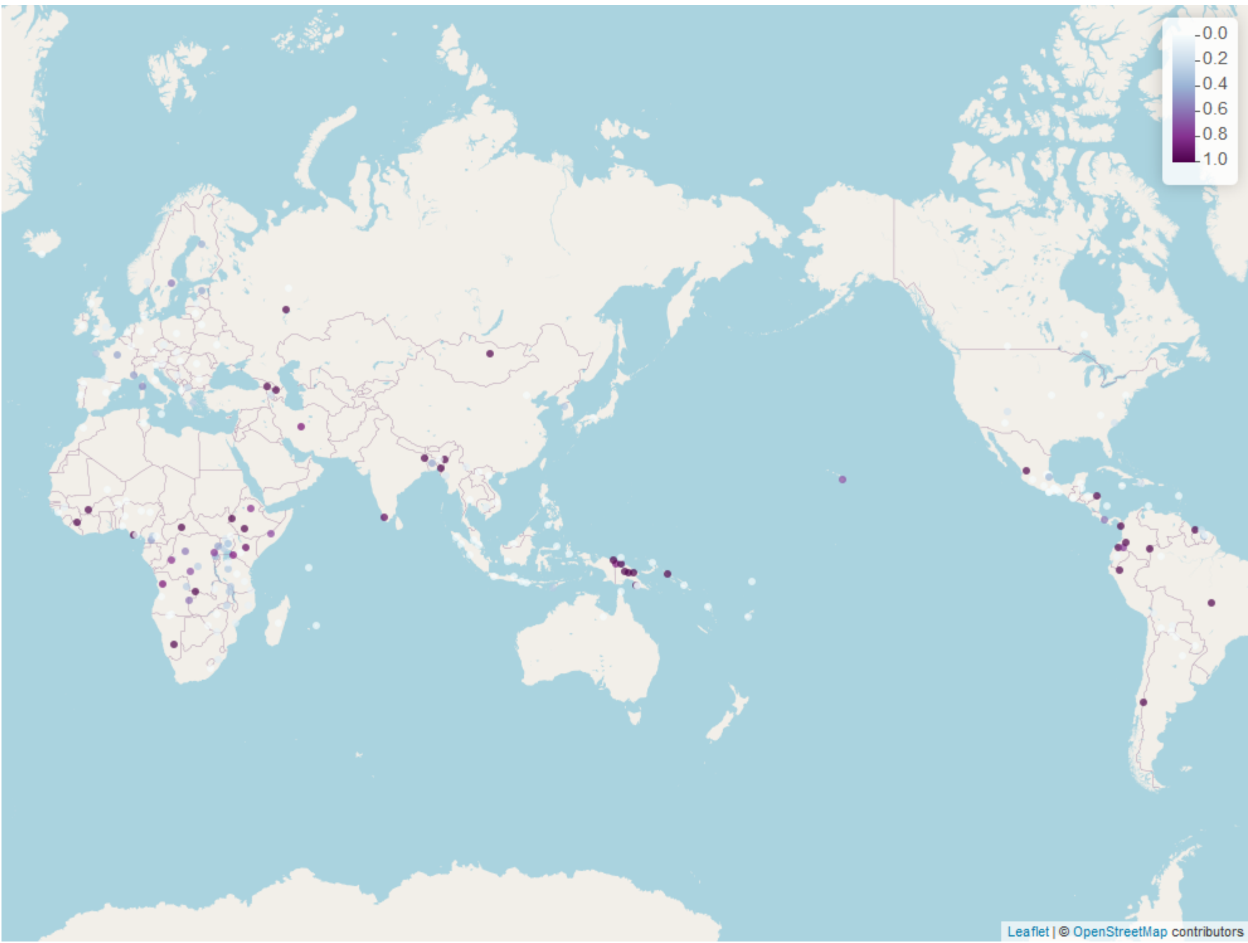
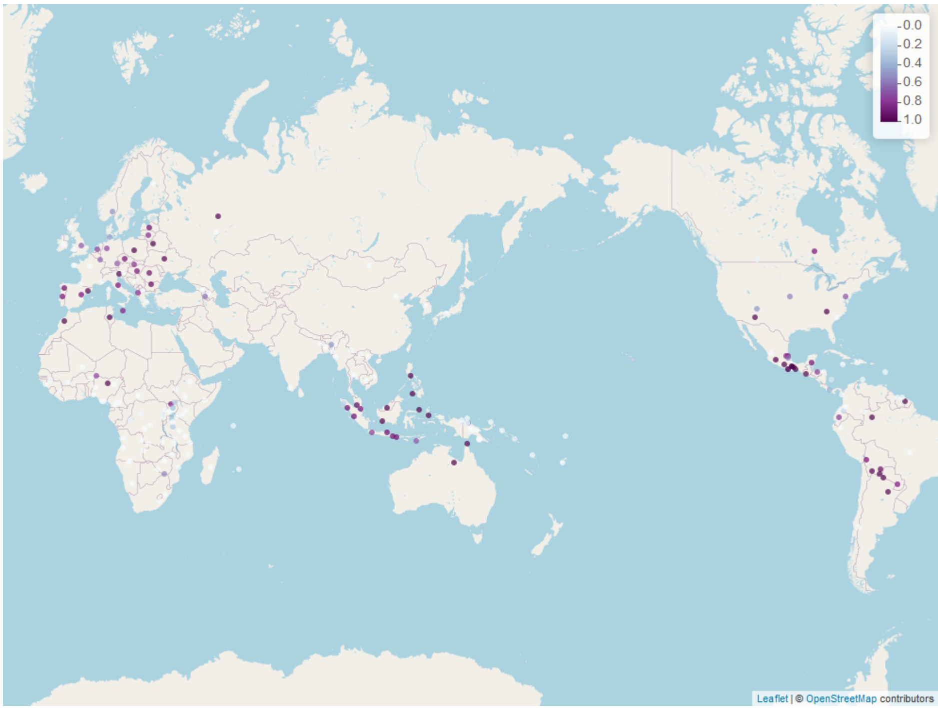
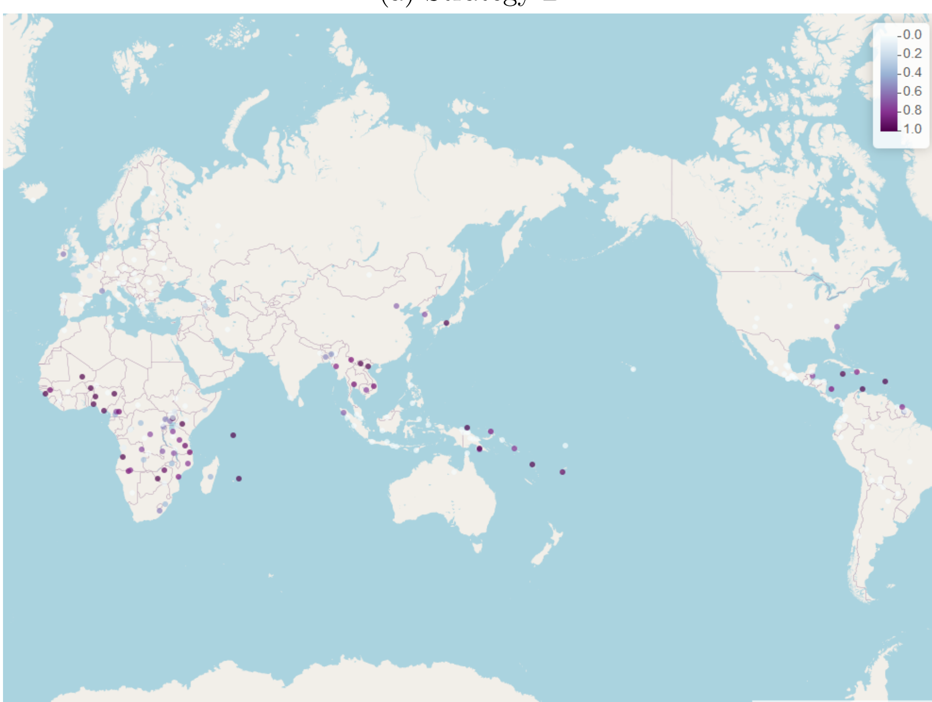


Figure 8.5: Geographic distribution of Strategy A across languages in the sample, representing its relative frequency within each language.

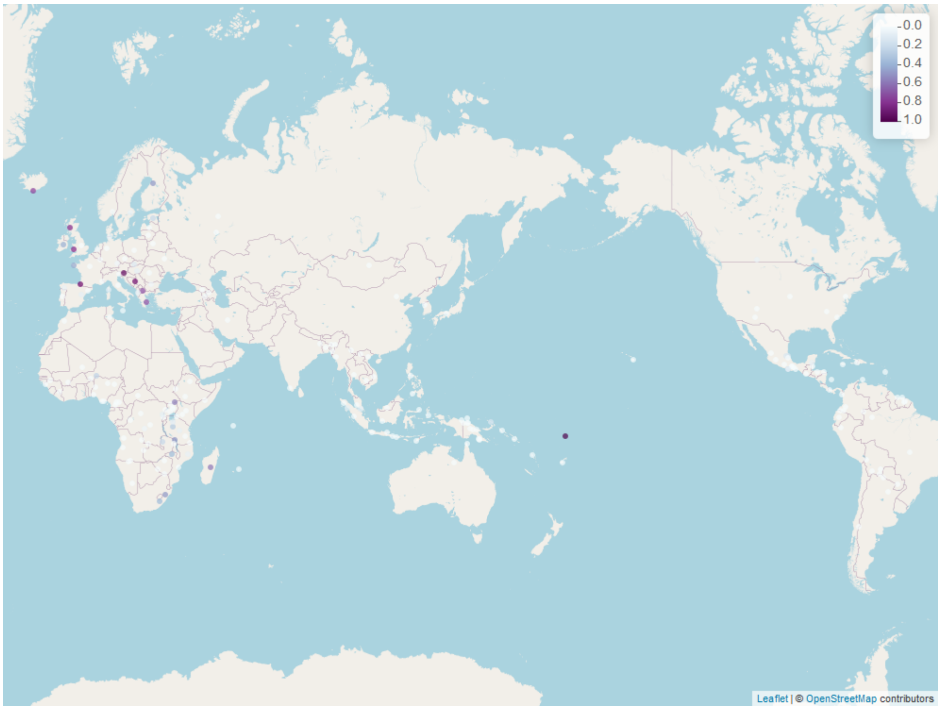


(a) Strategy B

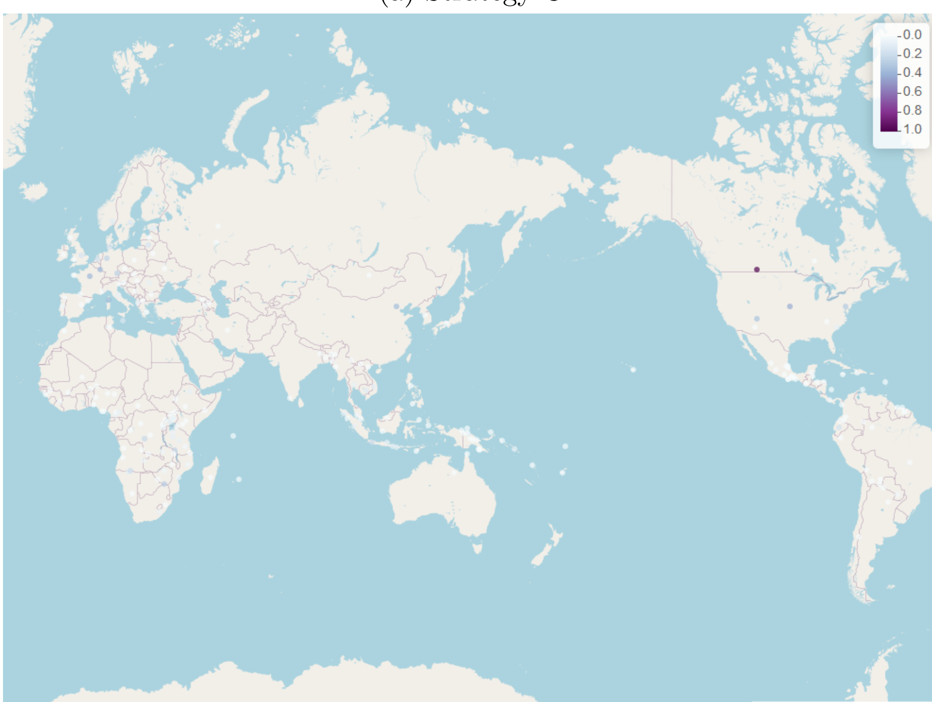


(b) Strategy D

Figure 8.6: Geographic distribution of Strategies B and D across languages in the sample. Each subfigure represents the relative frequency of a specific strategy.



(a) Strategy C



(b) Strategy E

Figure 8.7: Geographic distribution of Strategies C and E across languages in the sample. Each subfigure represents the relative frequency of a specific strategy.

8.2 Testing the Synchronic Hypothesis

The Synchronic Hypothesis proposes that inverse-scope constructions are less frequent cross-linguistically than they would be in the absence of a bias for scope transparency. This hypothesis builds on the premise that, if languages lacked such a bias, they would typically use the standard negation construction in 'Not all X are Y' sentences, even when this results in inverse scope. In most languages, standard negation involves a configuration where the subject precedes the negation marker. As a result, when the subject is a universal quantifier (QP), the standard negation construction often yields inverse scope by default.

To avoid inverse scope, languages need to employ alternative strategies, such as using a different negative construction or allowing word orders that explicitly indicate transparent scope relations. This raises the expectation that, if no bias for scope transparency existed, inverse-scope constructions would be as frequent as, or behave similarly to, standard negation constructions for sentences with non-scope-bearing subjects (e.g., definite noun phrases). Testing this hypothesis requires comparing the observed frequency of inverse-scope constructions across languages to this baseline expectation.

This section evaluates the Synchronic Hypothesis by examining the cross-linguistic frequency of inverse-scope constructions and identifying patterns that support or challenge the proposed bias for scope transparency.

To test the hypothesis, we first establish a baseline for each language based on the relative position of a nominal subject and the negative marker in standard negation. This baseline reflects the expected strategy for expressing 'Not all X are Y' propositions under the null hypothesis of no bias for scope transparency.

For example, in English, the negative marker typically follows the subject in standard negation, as in *The kids didn't like the movie*. This means that the expected strategy is A because the subject precedes the negative marker. In contrast, in verb-initial languages, the negative marker often precedes the subject. This leads to an expected strategy of A. In some cases, languages exhibit variability in their pragmatically unmarked word order. For such languages, the expected strategy may be classified as A/C, indicating that either strategy could arise depending on the choice of word order pattern.

For the purpose of statistical analysis, the expected strategies are simplified into A and non-A. Cases classified as A/C are treated as non-A, as they imply that the standard negation construction allows avoiding inverse scope.

This baseline is then compared to the observed data to determine whether Strategy A is attested in the corpus for each language. The comparison treats the presence of Strategy A as a binary property: it is either attested or not. By aligning the expected strategy (based on the baseline) with the observed data, we can evaluate whether the frequency of Strategy A deviates systematically from what would be predicted under the null hypothesis. This binary approach simplifies the analysis while directly addressing the central hypothesis regarding the prevalence of inverse-scope constructions.

To test whether the observed frequency of Strategy A differs significantly from the expected baseline, I use the McNemar test, a statistical method designed to analyze paired nominal data. The test is particularly appropriate here because each language provides paired observations of expected and observed strategies. By focusing on differences between these related conditions, the McNemar test evaluates whether discrepancies—cases where the expected and observed strategies differ—are symmetric or show a directional bias.

In this study, the McNemar test is used to determine whether the observed frequency of Strategy A is significantly lower than the expected frequency under the null hypothesis of no bias for scope transparency.

To perform the test, the data are organized into a 2×2 contingency table, with the rows representing expected strategies (A or non-A) and the columns representing observed strategies (A attested or unattested). The test focuses on the two off-diagonal cells, which capture cases where there is a mismatch between expected and observed strategies:

(i) $n_{A\neg A}$: Languages where the expected strategy is A but Strategy A is unattested.

(ii) $n_{\neg AA}$: Languages where the expected strategy is non-A but Strategy A is attested.

The null hypothesis (H_0) assumes that these mismatches are symmetric ($n_{A\neg A} = n_{\neg AA}$), reflecting no directional bias. The alternative hypothesis (H_1) predicts that more languages exhibit an absence of Strategy A despite its expected occurrence than those where it appears unexpectedly ($n_{A\neg A} > n_{\neg AA}$). This pattern would be consistent with a bias against Strategy A.

To evaluate the Synchronic Hypothesis, the McNemar test was applied separately for each macro-area, following Dryer (1989). This approach allows for a more fine-grained analysis of whether the hypothesized bias against inverse-scope constructions appears consistently across linguistic macro-areas. To control for genealogical bias, one language was randomly selected from each genus, and the test was run over 1,000 iterations. This procedure ensures that the results are not driven by overrepresentation of specific language families within a macro-area.

For each iteration, the McNemar test compared the observed and expected strategies for the sampled languages, focusing on mismatches where languages deviated from the baseline. To adopt a conservative approach, the highest p-value across the 1,000 iterations was recorded for each macro-area. This provides a stringent estimate of whether the observed distribution of Strategy A deviates significantly from the expected baseline, even under the most favorable conditions for the null hypothesis.

Table 8.1 summarizes the mean contingency table values across 1,000 iterations for each macro-area, along with the total number of genera sampled (since each random subsample includes only one language per genus) and the highest observed p-value in each case. The key values for the McNemar test are in the third ($n_{A\neg A}$) and fourth ($n_{\neg AA}$) columns, which represent the off-diagonal cells of the contingency table.

Across all macro-areas, the former values are consistently higher on average ($n_{A\neg A} > n_{\neg AA}$), aligning with the alternative hypothesis. However, the results indicate that the bias against Strategy A is statistically significant only in Eurasia and Africa, where the maximum p-values across iterations remained below the threshold of 0.05 ($p = 0.027$ for Eurasia and

Macro-area	n_{AA}	$n_{A\neg A}$	$n_{\neg AA}$	$n_{\neg A\neg A}$	Total number of genera	Maximum p-value
Africa	14.7	9.3	0	2	26	0.023
Eurasia	16.7	11.3	0.274	0.726	29	0.027
North America	4.14	3.19	1	7.67	16	1
Papunesia	9.4	7.6	0.52	2.48	20	0.13
South America	10.2	4.31	1.49	2	18	1

Table 8.1: Mean contingency table values and maximum p-values across 1,000 iterations for each macro-area

$p = 0.023$ for Africa). These findings support the Synchronic Hypothesis in these two macro-areas, as the observed frequency of Strategy A is significantly lower than expected under the null hypothesis. In contrast, no significant effect was found in the remaining macro-areas.

When considering all languages from each macro-area, including multiple languages from the same genus, significant results were also obtained for Papunesia ($p < 0.001$) and North America ($p = 0.016$), but not South America ($p = 0.182$). However, the potential influence of genealogical bias in such analyses should be acknowledged.

The non-significant results for Papunesia, South America, and North America in the conservative analysis might stem from the limited number of genera sampled in these macro-areas. Papunesia, South America, and North America are represented by 20, 18, and 16 unique genera, respectively, compared to 29 in Eurasia and 26 in Africa. This smaller sample size could reduce statistical power, making it harder to detect significant effects even if the hypothesized bias is present.

Furthermore, Table 8.1 shows that across all macro-areas, languages in which Strategy A is expected are more likely to actually employ this strategy than to lack it ($n_{AA} > n_{A\neg A}$). While these differences are unlikely to be statistically significant, they suggest that the preference for scope transparency does not exert strong pressure against Strategy A, since otherwise we would expect languages that retain Strategy A to be in the minority ($n_{AA} < n_{A\neg A}$). Instead, the findings suggest that Strategy A might be more frequently maintained than abandoned.

Beyond the quantitative results, the contingency tables also help identify unexpected instances of Strategy A ($n_{\neg AA}$)—languages where this strategy is attested even though it does not correspond to the usual word order in standard negation constructions. In the 217-language sample, only five languages exhibit this pattern: Breton (Indo-European; Celtic), Central Aymara (Aymaran), Eastern Bolivian Guaraní (Tupian), Fijian (Austronesian; Oceanic), and Upper Necaxa Totonac (Totonacan).

In Breton, matrix clauses typically follow V2 word order, but in negative clauses, NegVS(O) order is often preferred (Schafer 1995; Kennard 2014), as illustrated in (347).

(347) Breton (Kennard 2014, 7)

a. ne gan ket ar baotr-ed.
NEG sing.3SG NEG DET boy-PL
'The boys do not sing.'

b. ar baotr-ed ne gan-ont ket mat.
DET boy-PL NEG sing-3PL NEG good
'The boys do not sing well.'

Similarly, both word order patterns are available in 'Not all X are Y' sentences, resulting in either Strategy C (348a) or Strategy A (348b). The availability of Strategy A in Breton may be attributed to the influence of French—its main contact language—where Strategy A is dominant.

(348) Breton

a. Aotre am-eus da ober peb tra, med n'-eo ket mad peb tra din.
permission 1SG-be.PRS to do.INF all thing but NEG-be.PRS.3SG NEG good all thing to.1SG
'I have permission to do everything, but not everything is good for me.'
[Kenvreuriez ar brezoneg, 1 Cor. 6:12; 1988]

b. Aotreet eo peb tra, med peb tra n'-eo ket mad evidom.
permit.PTCP be.PRS.3SG all thing but all thing NEG-be.PRS.3SG NEG good for.1PL
'Everything is permitted, but not everything is good for us.' [ibid, 1 Cor. 10:23]

In Fijian, standard negation is formed using a semi-auxiliary verb, *sega*, which negates a complement clause containing the lexical predicate (Dixon 1988, 40). The subject pronoun of the complement clause may optionally raise to the main clause, as illustrated in (349) (Dixon 1988, 280).

(349) Fijian (Dixon 1988, 280)

a. au sega ni la'o.
1SG NEG COMP go
'I'm not going.'

b. e sega ni=u la'o.
3SG NEG COMP=1SG go
'I'm not going.' (lit. 'It is not that I'm going.')

In 'Not all X are Y' sentences, the most frequent strategy is Strategy D, where the quantifier remains in the embedded clause, as demonstrated in (350a). However, there is also a single instance of Strategy A, where the QP raises to the main clause, as shown in (350b).[3]

(350) Fijian

a.

Dou	sa	sega	ni	savasava	kece.
2PL	ASP	NEG	COMP	clean	all

'You are not all clean.' [*Fijian New Translation*, John 13:11; 2010]

b.

ia	na	ka	kece	sa	sega	ni	veivakatatataki	cake.
but	ART	thing	all	ASP	NEG	COMP	build	up

'but not everything builds up.' [ibid, 1 Cor. 10:23]

In Central Aymara (Hardman et al. 1988, 229-302) and Eastern Bolivian Guaraní (González 2005, 250-252), standard negation sometimes features a clause-initial negative marker. This is also the most frequent pattern in 'Not all X are Y' sentences in the corpus, leading to Strategy B. However, in both languages, a single instance of Strategy A is also attested.

Finally, Upper Necaxa Totonac is a verb-initial language with a preference toward VSO (Beck 2004, 92). However, corpus data suggest that universal QPs tend to precede the verb. As a result, while Strategy C aligns with standard negation overall, the mere presence of a subject QP appears to influence word order—independently of scope transparency. Consequently, Strategies B and A are attested in the corpus, whereas Strategy C is absent.

In summary, the results provide partial support for the Synchronic Hypothesis. The significant findings for Eurasia and Africa, even under the conservative analysis of considering the maximal p-value obtained across iterations, suggest that a bias against inverse-scope constructions is operative in these macro-areas. The additional significant results for Papunesia and North America, when considering all available languages, further indicate that the bias may be more widespread, albeit obscured by limited sampling in the conservative analysis.

The following chapter shifts focus from the distribution of strategies to the structural mechanisms that give rise to them, with particular attention to constructions that maintain scope transparency. By analyzing the syntactic means used to implement these strategies, it lays the groundwork for the diachronic analysis in Chapters 10 and 11. While Chapter 7 identified the structures attested cross-linguistically and Chapter 8 examined their geographical distribution and frequency, Chapter 9 investigates the processes through which these structures emerge and the factors driving their development.

[3] This construction closely resembles the Kinyarwanda raising construction in (257), which is discussed in §7.1.

Chapter 9

Pathways to scope transparency

The strategies used to express 'Not all X are Y' propositions across languages typically emerge through one of two pathways: they are either adapted from pre-existing constructions or developed as dedicated forms. This chapter investigates both mechanisms, exploring how languages repurpose existing structures or innovate new ones to ensure scope transparency.

Section 9.1 examines cases where existing constructions—such as word order alternations, expletive inversion, and clefts—are recruited to avoid inverse scope. While these constructions are often associated with information structure or discourse effects, they become conventionalized as grammatical strategies for maintaining scope transparency.

Section 9.2 turns to innovation of dedicated constructions, showing how some languages develop constituent negation constructions whose primary function is to negate subject QPs. These constructions are argued to originate from NEG-stripping structures, which explains the structural properties outlined in §7.2.1.

Lastly, Section 9.3 revisits a longstanding observation in the study of semantic universals: the apparent absence of lexicalized forms meaning 'not all'. While most languages rely on compositional strategies to express this meaning, the findings reveal an extremely rare counterexample, demonstrating that lexicalization of 'not all' is not impossible but subject to strong diachronic constraints.

9.1 Repurposing existing constructions

9.1.1 Word order

In many languages, existing word order patterns are repurposed to convey 'Not all X are Y' propositions in a way that ensures scope transparency. These word orders are often associated with discourse-related factors in other contexts. However, when recruited for the purpose of avoiding inverse scope, they frequently lose their original pragmatic function and become conventionalized as neutral constructions.

In some languages, this reordering of elements is particularly striking, as the recruited structure is otherwise marked or restricted to specific discourse functions. A clear example of this can be seen in Serbian-Croatian-Bosnian, where the most natural way to express a 'Not all X are Y' proposition follows a VS word order, as in (351a). Crucially, this word order is not driven by information structure considerations, as evidenced by the fact that similar sentences without negation or without a universal quantifier are pragmatically marked. For instance, (351b) is marked compared to the default SV alternative (*Svi moji prijatelji puše*), unless the quantifier is in focus, conveying a maximality inference. Likewise, (351c) is also marked relative to the SV alternative (*Moji prijatelji ne puše*), even when answering a question such as "Who doesn't smoke?" (Milica Denić, p.c.).

(351) Serbian-Croatian-Bosnian (Milica Denić, p.c.)

a. Ne puše svi moji prijatelji.
neg smoke-prs.3pl all.nom.pl my.nom.pl.m friend.nom.pl
'Not all my friends smoke.'

b. Puše svi moji prijatelji.
smoke-prs.3pl all.nom.pl my.nom.pl.m friend.nom.pl
'All my friends smoke.'

c. Ne puše moji prijatelji.
neg smoke-prs.3pl my.nom.pl.m friend.nom.pl
'My friends don't smoke.'

With transitive verbs, the neutral word order in 'Not all X are Y' sentences is NegVSO, as demonstrated in (352a). However, outside of this specific context, this word order remains marked. For instance, (352b) can be used in response to a prior statement like "Not all of your friends eat meat, right?", but it is odd out of context (Milica Denić, p.c.).

(352) Serbian-Croatian-Bosnian (Milica Denić, p.c.)

a. Ne jedu svi moji prijatelji meso.
neg eat-prs.3pl all.nom.pl my.nom.pl.m friend.nom.pl meat.acc
'Not all my friends eat meat.'

b. Jedu svi moji prijatelji meso.
eat-prs.3pl all.nom.pl my.nom.pl.m friend.nom.pl meat.acc
'All my friends do eat meat.'

These examples suggest that the VS(O) word order is systematically recruited in this language to ensure scope transparency, rather than being driven by information-structural constraints. This pattern is particularly prevalent in the Balkan linguistic area, where Strategy C serves as the primary strategy in Greek, Serbian-Croatian-Bosnian, and Slovenian—languages that do not make use of Strategy B.

Crucially, the availability of flexible word order alone does not guarantee the naturalness of this strategy. Even in languages with relatively free word order, Strategy C is not automatically viable unless it has been conventionalized for this function. This is evident in Russian, which exhibits a comparable degree of word order flexibility to the aforementioned Balkan languages (Levshina 2020, 65), yet (353a) is markedly odd in Russian. Instead, Russian speakers prefer the negated QP construction (Strategy B), as shown in (353b), highlighting the role of conventionalization in the availability of Strategy C.

(353) Russian (David Erschler, p.c.)

a. ?Ne edjat vse moi druz'ja mjáso.
NEG eat-PRS.3PL all.NOM.PL my.NOM.PL friend.NOM.PL meat.ACC
(Intended: 'Not all of my friends eat meat.')

b. Ne vse moi druz'ja edjat mjáso.
NEG all.NOM.PL my.NOM.PL friend.NOM.PL eat-PRS.3PL meat.ACC
'Not all of my friends eat meat.'

Icelandic provides another example of a construction that is typically associated with discourse-related effects but is repurposed as a neutral strategy for expressing 'Not all X are Y' propositions. Clause-initial negation is relatively common in Icelandic (Callegari and Angantỳsson 2023, 46), but in most contexts, it retains a pragmatic function. This is illustrated in (354), where clause-initial *ekki* 'not' contributes to contrast or emphasis.

(354) Icelandic

a. Ekki kann ég að tala rússnesku.
NEG can I to speak Russian
'I certainly can't speak Russian.' (Holmberg 2000, 450)

b. Ekki hefur Haraldur búið á Akureyri.
NEG has Harold lived in Akureyri
'It doesn't seem that Harold has lived in Akureyri.' or:
'I can't believe that Harold has lived in Akureyri!' (Thrainsson 2007, 343)

In contrast, in 'Not all X are Y' sentences, clause-initial negation in Icelandic functions differently. Unlike other Germanic languages, Icelandic lacks Strategy B (see §6.2.4), making Strategy C the primary means of avoiding inverse scope. Crucially, in this context, clause-initial negation no longer carries its usual pragmatic implications and instead serves as a neutral syntactic pattern. This is demonstrated in (355), where *ekki* appears clause-initially in a dependent clause, a context where negative fronting is otherwise less common (Callegari and Angantỳsson 2023, 46).

(355) Icelandic
Það er fullkomlega eðlilegt að ekki séu allir
it be.PRS.3SG completely natural.NOM.SG.N that NEG be.PRS.SBJV.3PL all.NOM.PL.M
sammála um mál af þessari stærðargráðu
agree.NOM.PL.M about issue.ACC.SG of this.DAT.SG.F magnitude.DAT
'It is completely natural that not everyone agrees on an issue of this magnitude.'
[LCC: News; 2020]

This shift from a pragmatically marked construction to a conventionalized, neutral word order mirrors the pattern observed in the Balkan languages, though it takes a different syntactic form. In Icelandic, negation is fronted not for contrast or emphasis, but simply as a means of structuring the sentence in a way that prevents inverse scope.

Clause-initial negative particles are also found in other Scandinavian languages, as well as in non-Germanic neighboring languages such as Finnish (Lindström 2007). However, outside of Icelandic, this construction typically retains its original pragmatic function in 'Not all X are Y' sentences and does not undergo the same kind of conventionalization into a neutral strategy for ensuring scope transparency.

In these languages, clause-initial negation is often associated with rejecting an explicit or implicit assumption attributed to the interlocutor (Vilkuna 2015, 473), a function that Lindström (2007) terms responsive negation. While we have seen that in Icelandic, this construction loses its pragmatic force in 'Not all X are Y' sentences, this shift does not occur in Finnish. In Finnish, clause-initial negation retains the same discourse function as it does in other contexts, as illustrated in (296). As a result, the neutral way to express 'Not all X are Y' in Finnish is Strategy A, as in (356) (Maria Vilkuna, p.c.).

(356) Finnish (Maria Vilkuna, p.c.)
Kaikki-lla ei ole koira-a.
everyone-ADE NEG.3SG be.CNG dog-PART
'Not everyone has a dog.'

This difference underscores the role of conventionalization in shaping the distribution of Strategy C across languages. While both Icelandic and Finnish exhibit clause-initial negation, only Icelandic has conventionalized it as a neutral means of avoiding inverse scope, whereas in Finnish, it remains pragmatically marked.

In summary, word order patterns that typically serve discourse-related functions can be repurposed as neutral constructions when recruited for expressing 'Not all X are Y' propositions. However, this neutralization is not automatic. Cross-linguistic variation shows that while constituent order flexibility may provide the structural basis for Strategy C, its viability ultimately depends on whether it has been conventionalized for this specific function.

9.1.2 Expletive inversion constructions

Expletive inversion constructions involve an expletive pronoun in preverbal position, followed by an inverted verb-subject order. In English, these constructions commonly occur with unaccusative verbs of existence, appearance, or motion, as illustrated in (357).

(357) There came a sudden stream of cold air through the door, along with two men in blue, from NSTAR. [COCA: Fiction; 2014]

As discussed in §7.5.2, languages like German (358) and Icelandic (359) also employ such constructions for expressing 'Not all X are Y' propositions—classified as a subtype of Strategy E.

(358) German
Es wohnen ja nicht alle Grünenwähler in den reichen Speckgürteln
it live.PRS.3PL indeed NEG all Green.voters in the rich commuter.belts
der Großstädte.
of.the big.cities
'Not all Green party voters live in the wealthy suburbs of the big cities.'
[LCC: News; 2023]

(359) Icelandic
Það komast ekki allir í eitt mót.
it get.PRS.3PL NEG all.NOM.PL.M into one.ACC.SG.N competition.ACC
'Not everyone can get into one competition.' [LCC: News; 2020]

Expletive inversion constructions are typically associated with lack of topicality: they are used in thetic sentences, which do not divide the clause into a topic and a comment (Sasse 1987; Lambrecht and Polinsky 1998; Lambrecht 2000). In this respect, they serve a similar discourse function to subject inversion in languages like Italian, where postverbal subjects often appear in contexts that introduce new referents and other all-new statements. The main difference is that in languages requiring an overt subject, such as English or French, an expletive pronoun (e.g., *there* or *il* 'it') is inserted in preverbal position to satisfy syntactic requirements. In contrast, in languages with null subjects, like Italian, the verb can appear sentence-initially without an overt expletive. Despite these surface differences, both strategies share the function of structuring sentences in a way that backgrounds the subject, aligning with their use in thetic statements.

In some languages, expletive inversion constructions can be used with certain quantifiers, but not with universal ones. For instance, in French, *beaucoup* 'many' can be used in (360a), and this construction can also be negated (360b). In contrast, *tous* 'all' is unacceptable in this construction, whether in affirmative (360c) or negative (360d) contexts.

(360) French (Keny Chatain, p.c.)
a. Il vient beaucoup de linguistes dans ce restaurant ces temps-ci.
it come.PRS.3SG many of linguists in this restaurant these times-here
'Many linguists come to this restaurant these days.'
(modified from example (47a) in Lambrecht 2000, 648)
b. Il ne vient pas beaucoup de linguistes dans ce restaurant ces temps-ci.
it NEG come.PRS.3SG NEG many of linguists in this restaurant these times-here
'Not many linguists come to this restaurant these days.'
c. *Il vient tous les linguistes dans ce restaurant ces temps-ci.
it come.PRS.3SG all the.PL linguists in this restaurant these times-here
(Intended: 'All the linguists come to this restaurant these days.')

d. *Il ne vient pas tous les linguistes dans ce restaurant ces
it NEG come.PRS.3SG NEG all the.PL linguists in this restaurant these
temps-ci.
times-here
(Intended: 'Not all the linguists come to this restaurant these days.')

A similar restriction is found in Swedish, where expletive inversion is possible with *många* 'many' (361a) but not with *alla* 'all' (361b).

(361) Swedish
a. Det kom inte så många av Riskkollegiets medlemmar till detta evenemang.
it come.PST NEG so many of Riskkollegiet.GEN members to this event
'Not so many of Riskkollegiet's members came to this event.' [web example[1]]
b. *Det kom inte alla av Riskkollegiets medlemmar till detta evenemang.
it come.PST NEG all of Riskkollegiet.GEN members to this event
(Intended: 'Not all of Riskkollegiet's members came to this event.')
(Alice Bondarenko, p.c.)

The contrast between 'many' and 'all' appears to reflect the well-established distinction between weak and strong quantifiers (Milsark 1977): weak quantifiers like 'many' can occur in existential constructions (362a), whereas strong quantifiers like 'all' are generally excluded (362b).[2]

(362) a. There are many linguists in this restaurant.
b. *There are all the linguists in this restaurant.

This restriction is consistent with the behavior of expletive inversion constructions in languages like French and Swedish, where only weak quantifiers are permitted. However, it does not fully account for the cross-linguistic variation observed in expletive inversion. Notably, universal quantifiers are acceptable in such constructions in German (358) and Icelandic (359), suggesting that additional syntactic or semantic factors play a role in licensing these structures across languages.

Interestingly, even in German, the acceptability of universal quantifiers in expletive inversion constructions is not uniform. In affirmative contexts, these constructions tend to be marked, at least with animate subjects. While the negative example in (329) is natural and relatively neutral, its affirmative counterpart requires focus on the quantifier. For ex-

[1] `https://www.humiliationstudies.org/documents/evelin/Risknytt_1_2005.pdf`. Accessed on 15 October 2024.

[2] Other types of subject inversion constructions in English are compatible with universal quantifiers, as illustrated in (i).

(i) a. First in the procession came all the men of the village, venerable in character and age.
[COHA: Non-fiction/Academic; 1875]
b. Struggling up the frozen path to the cave came all the animals that God had created.
[COHA: Fiction; 1922]

ample, this construction can be used to reject an implicit assumption implied by a previous statement, as in (363) (Anne Mucha, p.c.).[3]

(363) German (Anne Mucha, p.c.)
A: Nur die Grünenwähler, die in den reichen Speckgürteln der Großstädte leben, haben kein Verständnis für die Probleme der Landbevölkerung.
'Only those Green Party voters who live in the wealthy subsurbs of big cities don't understand the problems faced by the rural population.'
B: Aber es wohnen ja ALLE Grünenwähler in den reichen Speckgürteln
but it live.PRS.3PL indeed all Green.voters in the rich commuter.belts
der Großstädte!
of.the big.cities
'But all Green party voters live in the wealthy suburbs of the big cities!'

This asymmetry in German parallels the distribution of universal quantifiers in clefts: they are more common in negated clefts and usually require contrastive focus in affirmative clefts (see §9.1.3).

In contrast, Icelandic allows expletive inversion with a universal QP even in affirmative contexts, as demonstrated in (364). This difference indicates that the restrictions observed in German do not extend uniformly to all languages that permit expletive inversion.

(364) Icelandic
Mamma, það mættu allir foreldrarnir á fimleikaæfinguna nema
mom it show.up.PST.3PL all parents.DEF to gymnastics.practice.ACC except
þú
you
'Mom, all the parents showed up at the gymnastics practice except you' [web example[4]]

The fact that German and Icelandic permit expletive inversion with a universal QP appears to be linked to a broader structural property of expletive constructions in these languages. It has been argued that these languages employ topic expletives, which are positioned in [Spec, CP], as opposed to subject expletives—found in English and certain other languages—which are located in [Spec, TP] (see Biberauer 2010, and references therein).

[3] Expletive inversion constructions involving a modal auxiliary, such as (i), may have a broader distribution than those with a lexical verb, such as (363).

(i) German
Es müssen alle mitwirken, um in diesen Zeiten der klimatischen Veränderung unsere
it must.3PL all cooperate in.order.to in these times the.GEN climatic.GEN change.GEN our
Umwelt und unser kostbarstes Gut, unser Wasser, zu schützen
environment and our most.precious good our water to protect
'Everyone must cooperate in order to protect our environment and our most precious resource, our water, in these times of climate change' [web example: `https://www.this-magazin.de/artikel/tis_Es_muessen_alle_mitwirken_-3771183.html`. Accessed on 15 October 2024.]

This structural difference accounts for certain syntactic restrictions: while English allows expletives in subject inversion constructions and embedded clauses (365), German does not (366). The crucial distinction lies in the fact that, in English, the expletive is situated in [Spec, TP], meaning that it can be preceded by a verb undergoing V-to-C movement or by a complementizer in C^0. In contrast, the expletive in German resides in [Spec, CP], preventing it from appearing in postverbal position or following a complementizer.

(365) English (von Fintel 1992, 5)
  a. There was a cat in the garden.
  b. Was there a cat in the garden?
  c. John said that there was a cat in the garden.

(366) German (von Fintel 1992, 6)
  a. Es kam ein Jäger in den Wald.
     it came a hunter in the forest
     'There came a hunter into the forest.'
  b. *Kam es ein Jäger in den Wald?
     (Intended: 'Did there come a hunter into the forest?')
  c. *Hans sagte daß es ein Jäger in den Wald kam.
     (Inteded: 'Hans said that there came a hunter into the forest.')

French and Swedish pattern similarly to English in this respect: both allow subject inversion with an expletive pronoun, as demonstrated in (367) and (368), respectively. This suggests that their expletive pronouns occupy [Spec, TP] rather than [Spec, CP].

(367) French
  Vient-il beaucoup de linguistes dans ce restaurant ces temps-ci?
  come.PRS.3SG-it many of linguists in this restaurant these times-here
  'Do many linguists come to this restaurant these days?'

(368) Swedish
  Tyvärr kom det inte många medlemmar
  unfortunately come.PST it NEG many members
  'Unfortunately, not many members came.' [web example[5]]

Taken together, these observations indicate a cross-linguistic correlation: languages with topic expletives tend to permit universal QPs in expletive inversion constructions, whereas languages with subject expletives generally do not. Instead, in the latter languages, expletive inversion is typically restricted to weak quantifiers such as 'many'.

However, a counterexample to this generalization comes from 16th-century Romansh. A translation of the New Testament into the Upper Engadine dialect of Romansh features an

[5] `https://www.facebook.com/story.php/?story_fbid=649315194056648&id=100069345187695&_rdr`. Accessed on 20 October 2024.

expletive inversion construction with a universal QP on three occasions, as exemplified in (369).

(369) Romansh (Upper Engandine)
Mu è nun haun brichia tuots ubedieu alg euangeli.
but EXPL NEG1 have.PRS.3PL NEG2 all.PL.M obey.PTCP to.the.SG.M gospel
'But not everyone obeyed the gospel.' [Bifrun, Rom. 10:16; 1560]

Crucially, Romansh exhibits properties of a subject expletive language according to the criteria discussed above. For instance, in (370), the expletive pronoun appears in a subject inversion construction, similar to English (365), French (367), and Swedish (368). This suggests that the expletive occupies [Spec, TP], aligning it with other subject expletive languages despite its divergence in allowing universal QPs in expletive inversion.

(370) Romansh (Upper Engandine)
Nu sun è forza dudesch huras dalg di?
NEG be.PRS.3PL EXPL surely twelve hours of.the.SG.M day
'Are there not twelve hours of daylight?' [Bifrun, John 11:9; 1560]

This observation suggests that cross-linguistic variation in the availability of universal QPs in expletive inversion may not be directly tied to the distinction between expletive subjects and expletive topics.

Moreover, we have seen that even in German, universal QPs in expletive inversion constructions tend to be marked in affirmative contexts (363). In contrast, weak quantifiers such as 'many' are fully acceptable in this construction, as shown in (371). Thus, while the distinction between 'many' and 'all' in expletive inversion constructions is not as categorical in German (and possibly absent in Icelandic), German still exhibits a similar pattern to French and Swedish, where 'many' is more acceptable in this structure than 'all'.

(371) German
Es wohnen viele ältere Menschen in diesem Bereich.
it live.PRS.3PL many older people in this area
'Many elderly people live in this area.' [web example[6]]

Thus, it appears that German and Icelandic have extended the expletive inversion construction to include universal QPs—primarily in negative contexts in the case of German—much like how negated clefts have been extended to express 'Not all X are Y' propositions in many languages (see §9.1.3). This extension appears to operate independently of the typical discourse function of expletive inversion, which is associated with thetic statements and the backgrounding of the subject.

[6] https://buergerbeteiligung.potsdam.de/content/verbesserung-der-wohnqualitaet-waldstadt-1-erich-wein
Accessed on 20 October 2024.

9.1.3 Clefts and other focus constructions

While the previous subsections examined how certain word order patterns and expletive inversion constructions can be recruited to ensure scope transparency, this subsection turns to another strategy: clefts and related focus constructions. Unlike the previous strategies, which primarily involve reordering elements within a clause, focus constructions repurpose complex syntactic constructions.

Clefts are widely used across languages to mark focus and establish a clear division between given and new information. However, in many languages, negated clefts also serve as a means of expressing 'Not all X are Y' propositions. In some languages, this usage is conventionalized to the point that negated clefts become a dominant strategy for avoiding inverse scope.

Negated clefts with QP are found across many languages, but their frequency and function vary considerably. In present-day English, such constructions are relatively rare and tend to appear in generic statements rather than episodic contexts. Examples (372) and (373) illustrate this pattern, as they convey general claims rather than statements about a particular group in a given situation.

(372) It is not everybody who can live up to this [Phillips, Matt. 19:11; 1958]

(373) Look on the bright side. It's not every student who has access to their teacher 24 hours a day. [COCA: TV; 2001]

By contrast, in Danish, negated clefts are much more frequent and are not restricted to generic statements. They are used productively in episodic contexts, as shown in (374), where the cleft construction expresses a claim about a specific event. Furthermore, negated clefts are about as common as negated QP constructions (Strategy B) in this language (see §6.2.3), suggesting that clefts are not necessarily tied to a specific discourse function in 'Not all X are Y' sentences.

(374) Danish

Der	var	i	alt	80	tilmeldte,	og	faktisk	var	det	ikke	alle,	der
there	be.PST	in	total	80	registered.PL	and	actually	be.PST	it	NEG	all	who

troppede	op.
show.PST	up

'There were a total of 80 registrants, and in fact not everyone turned up.'
[LCC: News; 2023]

Another noteworthy pattern is that universal QPs are generally more acceptable in negated clefts than in affirmative clefts. As shown in (375), universal quantifiers are typically unacceptable as cleft phrases in neutral statements, a pattern noted by Kiss (1998). However, these constructions become significantly more acceptable when used to express a correction or a strong contrast (Brunetti 2004; Dufter 2009), as illustrated in (376).

(375) ?It's every student who has access to their teacher 24 hours a day.

(376) In this case, it is EVERYONE who is being discriminated against. (Dufter 2009, 97)

This asymmetry suggests a broader generalization: quantificational expressions appear to be more acceptable as cleft phrases in downward-entailing environments—that is, in negated clefts or when the quantificational expression itself is negative. For example, *many* is permitted in a negated cleft (377a), but it is markedly odd in an affirmative cleft (377b). A similar pattern emerges with adverbial quantifiers: *often* is far more frequent in negated clefts (378a) than in affirmative clefts (378b), as evidenced by corpus data from COCA, where it appears 271 times in negated clefts but only 4 times in affirmative ones.

(377) a. You know, it's not many girls that would spend the night with a guy and not only not mind him calling another girl the next morning but still be up for round two. [COCA: TV; 2005]

b. ?It's many girls that would spend the night with a guy and not only not mind him calling another girl the next morning but still be up for round two.

(378) a. It's not often that Mark Dantonio's football team looks overmatched. [COCA: Magazine; 2017]

b. It's often that we're seeing ads that they don't know how they possibly got there. [COCA: Spoken; 2018]

Similarly, inherently negative adverbs such as *seldom* and *rarely* are naturally accommodated as cleft phrases, as seen in (379) and (380).

(379) Most sentencing rests in the prosecutor's hands and it is seldom that the defendant wants to take a risk on sentencing [COCA: Academic; 2002]

(380) It is rarely that you would arrive across someone who's not considering the way in which he appears or perhaps the way he appears about the streets, within the office, and in the opposite gatherings that he attends. [COCA: Blog; 2012]

Given that affirmative clauses are generally more frequent in natural discourse, the disproportionate association between negation and quantificational expressions in clefts is unlikely to be accidental.

I propose that this pattern has a functional basis: negated clefts provide a structural means for negation to precede the QP without relying on constituent negation or VS word order. In contrast, when negation is absent, there appears to be little reason to position QPs in cleft constructions, and I am not aware of any language where clefts serve as the primary strategy for expressing affirmative quantificational statements.[7] Crucially, this motivation is independent of the typical focus-marking function of clefts, suggesting that their role in 'Not all X are Y' sentences is primarily syntactic rather than discourse-driven.

[7] QPs qualify as cleft phrases in affirmative clefts in Italian (Brunetti 2004) and Cypriot Greek (Agouraki 2010; Kanikli 2016), but these are marked constructions used in contrastive contexts.

A potential challenge to this account comes from negative adverbs such as *seldom* (379) and *rarely* (380), which can appear as cleft phrases despite the absence of an overt negation marker. I suggest that this pattern may arise through analogy with constructions like *not often* (378a), where negation is explicit.

So far, the discussion has centered on negated clefts and their role in ensuring scope transparency. We now turn to another type of focus construction that serves a similar function but has different properties: wide-scope negation constructions, such as the Mandarin *bu shì* construction and the Japanese *wake de wa nai* construction.

While these constructions share certain properties with clefts, they differ in a crucial way: they are not necessarily restricted to marking narrow focus on a particular constituent. Instead, they can also signal broad focus over the entire clause. This is illustrated in Mandarin, where the copula *shì*—which also functions as a focus marker—can associate either with a subclausal constituent, as in (381a), yielding an interpretation similar to an English *it*-cleft, or with the entire clause, as in (381b), an option unavailable in English *it*-clefts.

(381) Mandarin

a. Bú shì [zhāngsān]$_F$ yào lái.
NEG COP Zhangsan will come
'It's not Zhangsan that will come.' (Jin and Chen 2022, 319)

b. Bu shi [yifu tai shou]$_F$, shi [ni tai pang le]$_F$.
NEG COP clothing too slim COP 2SG too fat PART
'It's not that the clothing is too small, it's just that you're too big.'
(Paul and Whitman 2008, 424)

The key question, then, is how *shì* interacts with focus in 'Not all X are Y' sentences such as (382). Specifically, does it associate with the QP (narrow focus), as in (381a), or with the entire clause (broad focus), as in (381b)?

(382) Mandarin (Fan 2017, 46)
Bu-shi mei-pi ma dou tiaoguo-le liba.
NEG-COP every-CLF horse all jump-over-ASP fence
'Not every horse jumped over the fence.'

I argue that in certain cases, *shì* can associate with the entire clause rather than exclusively with the QP. This is evident in (383), which closely parallels (381b). Further support for this analysis comes from the absence of a scalar inference in (383): if the QP were focused, it would necessarily trigger scalar alternatives, resulting in a scalar inference (see Amiraz 2022).

(383) Mandarin (Danfeng Wu, p.c.)
Bú shì měigè rén dōu chídào le, shì nǐ dào de tài zǎo le. Qítā
NEG COP every person all be.late PRT COP you arrive DE too early PRT other

rén dōu shì zhǔnshí dào de.
people all COP on.time arrive DE
'It's not that everyone was late, it's that you arrived too early. Everyone else arrived on time.'

However, it is important to note that while (383) is pragmatically marked, (382) is a neutral statement. This contrast suggests that (382) more plausibly involves narrow focus on the QP, whereas (383) exhibits broad focus on the entire clause.

A similar pattern is observed in Japanese. According to Yoshimura (2013), the primary function of *wake de wa nai* is to object to an inference made by the interlocutor, as exemplified in (384), or to reject an inference that is implicitly suggested by prior discourse.

(384) Japanese (Yoshimura 2013, 49)
Context: Taro was assigned to an overseas branch of his company and asked his girlfriend, Hanako, to come with him, but she refused. Taro asked Hanako why she wants to stay in Tokyo, and she replies:
Tokyo ni nokori-tai wake-de-wa na-i no. Issho ni ike na-i
Tokyo in stay-want conclusion-COP-CNTR NEG-PRS PRT together go.HYP NEG-PRS
dake na no.
only COP PRT
'It is not that I want to stay in Tokyo. I just can't go with you.'

However, when used as Strategy D (385), this construction is pragmatically neutral and does not carry its usual discourse function (Akitaka Yamada, p.c.). In other words, a construction that typically serves to challenge an inference is instead repurposed to ensure scope transparency.

(385) Japanese (Akitaka Yamada, p.c.)
Seito-tati-no zenin-ga siken-ni uka-tta wake-de-wa na-i.
student-ASS-GEN all-NOM exam-DAT pass-PST conclusion-COP-CNTR NEG-PRS
'Not all of the students passed to exam.'
(lit. 'It is not the conclusion that all of the students passed to exam.')

At the same time, *wake de wa nai* can also be used in 'Not all X are Y' sentences in its original pragmatic function. Similar to the Mandarin *shì* construction in (383), *wake de wa nai* can associate with broad focus, allowing the scalar inference to be canceled. This is illustrated in (386).

(386) Japanese (Yusuke Yagi, p.c.)
Tatoe anata ga 30-pun mat-tei-ta to shi-temo, zen'in ga okure-ta
even.if you NOM 30-minute wait-ASP-PST QUOT do-even all NOM be.late-PST

wake-de-wa na-i.
conclusion-COP-CNTR NEG-PRS
'Even if you waited for half an hour, that doesn't mean everyone was late. [You just got here too early—everyone else arrived on time.]'

Overall, the data show that focus constructions can be repurposed as strategies for ensuring scope transparency in 'Not all X are Y' sentences. While clefts and wide-scope negation constructions differ in their syntactic properties, both allow negation to precede (or, at least, be higher than) the QP without requiring constituent negation or VS word order. The extent to which these constructions become conventionalized, however, varies across languages. In some cases, they remain pragmatically marked, while in others, they develop into neutral strategies for expressing 'Not all X are Y' propositions.

9.2 Innovating dedicated constructions

The previous section demonstrated how existing constructions—ranging from word order alternations to focus-marking structures—are repurposed to maintain scope transparency in 'Not all X are Y' sentences. In many cases, these constructions originate in discourse-related functions but become conventionalized as neutral grammatical strategies for avoiding inverse scope.

However, not all languages rely on the repurposing of pre-existing structures. Instead, some develop dedicated constructions specifically for expressing 'Not all X are Y' propositions. This section explores this process of innovation, focusing on constituent negation constructions that directly target subject QPs.

Although negated QP constructions are usually new grammatical developments, they do not emerge in isolation. Instead, I argue that they typically develop through the reanalysis of an existing pattern of constituent negation, tracing their origins to NEG-stripping structures.

Stripping, also known as bare argument ellipsis, is a coordinate structure in which only one constituent of the elided clause remains overt (Konietzko 2016). In NEG-stripping, this remnant is negated, as illustrated in (387a). A related construction, known as negative contrast ellipsis (Kolokonte 2008), lacks a coordinating conjunction, as shown in (387b). Both of these constructions share similarities with contrastive negation structures, such as *not X but Y* (McCawley 1991; Toosarvandani 2013; Silvennoinen 2017; Wu 2021), where the negated constituent appears first, as in (387c).

(387) a. Some of the students passed, but not all of them.
b. Some of the students passed—not all of them.
c. But everyone sort of expected that there would be another large program waiting in the wings, the large Constellation Program that would keep **not all of the workers but a large piece of the workers** from the shuttle program. [COCA: Spoken; 2012].

To the best of my knowledge, it remains an open question whether languages distinguish between these three types of constructions—specifically, whether some languages permit only a subset of them and whether the same negative marker is consistently used across all cases. Consequently, while the following discussion centers on NEG-stripping, these related constructions may be equally relevant.

NEG-stripping constructions tend to allow for a variety of remnant types, as illustrated in (388). In contrast, constituent negation in free-standing clauses is far more restricted, as demonstrated in (389). Thus, in English, while NEG-stripping permits negation of a proper noun or an adverb, among the constituent types examined here, only QPs remain fully acceptable in free-standing constituent negation constructions.

(388) a. Most of them are behind bars or dead, but not all of them. [COCA: TV; 2012]
b. Some patients hide out while waiting for their hair to grow, but not John. [COCA: Spoken; 1993]
c. She was always the first to greet me, but not today. [COCA: Fiction; 2018]

(389) a. Not all of them are behind bars.
b. *Not John hides out while waiting for his hair to grow.
c. *Not today was she the first to greet me.

This contrast between NEG-stripping and constituent negation in free-standing clauses in English is mirrored in Modern Greek. Just as English allows a variety of remnants in NEG-stripping but restricts constituent negation in independent clauses to QPs, Greek permits QPs to appear as remnants in NEG-stripping, as in (390a), but does not allow constituent negation of subject QPs in free-standing clauses, as shown in (390b).

(390) Modern Greek
a. Esís íste katharí, allá óxi óli.
you are clean but NEG all
'You are clean, though not all of you.' [Filos, John 13:10; 1994]
b. *Óxi óli íste katharí.
NEG all are clean
(Intended: 'Not all of you are clean.')

This pattern suggests a strong cross-linguistic connection between NEG-stripping and negated QP constructions. While NEG-stripping appears to be widely available, the emergence of a dedicated negated QP construction is much more restricted. Based on the findings of this study, I propose the following universal:

(391) *Universal 1*
If a language has a negated QP construction, it also allows NEG-stripping with a remaining QP, but not necessarily vice versa.

This generalization is motivated by the observation that, across languages, negated QP constructions are never attested in the absence of NEG-stripping. However, the reverse does not hold: there are languages, such as Modern Greek, that allow NEG-stripping with a negated QP but lack a dedicated negated QP construction in free-standing clauses.

Another key observation, previously discussed in §7.2.1, concerns the type of negative markers used in these constructions:

(392) *Universal 2*
If a language has a negated QP construction, the negative marker that is used in this construction is almost always the same marker that is used in NEG-stripping.

Taken together, these generalizations suggest that negated QP constructions do not emerge independently but rather develop from a pre-existing ellipsis construction. As a result, they inherit certain properties—most notably, the type of negative marker—from the source construction.

As noted in §7.2.1, two potential counterexamples to the universal in (392) are found in the sample: Portuguese and Standard Malay.

In present-day Portuguese, the negated QP construction employs the negative marker 'not even, nor', as shown in (393b), whereas NEG-stripping features *não*, as in (393a), which is also the default marker for standard negation. This apparent divergence raises questions about the historical development of these constructions.

(393) Portuguese

a. Ora vós estais limpos, mas não todos.
now you be.PRS.2PL clean but NEG all
'And you are clean, though not all of you.'
[*Almeida Revista e Corrigida*, John 13:10; 1968]

b. Nem todos estais limpos.
NEG all be.PRS.2PL clean
'Not all of you are clean.' [ibid, John 13:11]

However, evidence from earlier Portuguese suggests that *não* and *nem* were once interchangeable in negated QP constructions, as seen in the original Almeida translation from 1681 (394). This indicates that the current distribution of *nem* may be a relatively recent innovation.

(394) Portuguese

a. Naõ todos saõ capazes desta palavra.
NEG all are capable of.this word
'Not everyone can accept this word' [Almeida, Matt. 19:11; 1681]

b. Todas as cousas me sam licitas, mas nem todas as cousas convem;
all the things to.me are lawful but NEG all the things appropriate
'All things are lawful for me, but not all things are beneficial.'
[ibid, 1 Cor. 6:12]

Both variants are also attested in 16th-century texts, as demonstrated in (395).

(395) Portuguese

a. Nem todos os que tẽ viola sam tangedores.
NEG all those who have guitar are players
'Not all those who have guitars are players.' [Corpus do Português; 1569-70]

b. e não todos os que sinificão qualidade acabão em mente
and NEG all.PL those which mean.PRS.3PL quality finish.PRS.3PL in *mente*
porque ja agora não diremos prestesmente.
because now anymore NEG say.FUT.1PL quickly.
'[And we can notice that adverbs ending in *-mente* indicate quality.] And not all those that indicate quality end in *-mente* because now we no longer say *prestesmente*.' [Corpus do Português; 1536]

The earliest attestation I found in the corpus, from the 15th century, employs *nom*, a historical variant of *não* (396).

(396) Portuguese

Desto se segue que nom todo bem fazer que o seruo faz
from.this REFL follow.PRS.3SG that not every good do.INF that the servant do.PRS.3SG
he seruiço.
be.PRS.3SG service
'From this it follows that not every good deed that the servant does is service.'
[Corpus do Português; 1430-1443]

These examples support the hypothesis that *não* was the original marker in negated QP constructions, while the emergence and eventual predominance of *nem* represents a later development. This scenario aligns with the generalization in (392), as the same negative marker was initially used in both negated QP constructions and NEG-stripping. However, further research is needed to clarify the diachronic trajectory of this shift from *não* to *nem* in negated QP constructions.

A similar issue arises in Malay. Standard negation is expressed by the pre-verbal negative particle *tidak*, as shown in (397a), while nominal predicates are negated with *bukan*, as in (397b) (Kroeger 2014, 137-138).

(397) Standard Malay (Kroeger 2014, 137-138)

a. Mereka tidak menolong kami.
3PL NEG help 1PL.EXCL
'They didn't help us.'

b. Dia bukan guru.
3SG NEG teacher
'She is not a teacher.'

In NEG-stripping, *bukan* is consistently used, as illustrated in (398).

(398) Malay
Kamu memang bersih tetapi bukan semua kamu.
you indeed clean but NEG all you
'And you are clean, though not all of you.' [*Alkitab Versi Borneo*, John 13:10; 2016]

However, in negated QP constructions, both *tidak* and *bukan* are attested, as shown in (399).

(399) Malay
a. Tidak semua orang dapat menerima ajaran ini.
NEG all person can receive teaching this
'Not all people can accept this teaching.' [*Alkitab Versi Borneo*, Matt. 19:11; 2016]
b. kerana bukan semua orang mempunyai iman.
because NEG all person have faith
'because not all people have faith.' [ibid, 2 Thess. 3:2]

Further research is needed to determine whether this variation reflects a later stage in the diachronic development of negated QP constructions, akin to the shift observed in Portuguese.

So far, the discussion has centered on the origins of negated QP constructions in NEG-stripping. Another crucial aspect of their grammatical behavior concerns their interaction with negative concord.

Negative concord occurs when two elements that are capable of independently marking negation combine in a sentence without yielding double negation. Instead, they contribute to a single semantic negation. This phenomenon is most commonly associated with negative indefinites, such as Italian *nessuno* 'nobody', which must co-occur with clausal negation when appearing postverbally, as shown in (400).

(400) Italian (Zeijlstra 2022, 29)
Gianni non ha telefonato a nessuno.
Gianni NEG has called to nobody
'Gianni didn't call anybody.'

Interestingly, I observe that there is no systematic correlation between negative concord with negative indefinites and negative concord with negated universal QPs. Even languages that exhibit strict (obligatory) negative concord with negative indefinites—such as many Slavic languages—typically do not extend this pattern to negated QPs. For instance, in Russian, negated QPs occur without clausal negation, regardless of their position relative to the verb, as shown in (401).

(401) Russian
a. Ne vse mal'čiki videli Mašu.
NEG all boys saw Masha-ACC
'Not all the boys saw Masha.' (Borschev et al. 2006)

b. Odnako v eto poverili ne vse.
however in this believe.PST.PFV.PL NEG everyone
'However, not everyone believed this.' [LCC: News; 2022]

Negative concord with negative indefinites is widespread cross-linguistically, occurring in at least 170 languages out of a 206-language sample in WALS (Haspelmath 2013). Haspelmath (1997) provides a diachronic explanation for this phenomenon, arguing that negative indefinites often arise through grammaticalization from non-negative indefinites, which initially require clausal negation to render a clause negative. For example, in English, *anybody* must co-occur with clausal negation in a sentence like *John didn't call anybody* because it is an NPI and cannot independently express negation. Over time, NPIs like *anybody* may become restricted to negative contexts and reanalyzed as inherently negative, leading to their use without clausal negation in negative replies. However, even at this stage of grammaticalization, these elements continue to co-occur with clausal negation due to the persistence of negative concord as a grammaticalized pattern.

In contrast, I argue that negative concord with negated universal QPs originates from a different diachronic pathway, specifically in languages where standard negation involves a bipartite negative marker. In such languages, negated QP constructions mirror the structure of standard negation by employing both components of the bipartite marker.

For instance, in Afrikaans, standard negation is expressed using the negative particle *nie*, which follows the finite verb or auxiliary, while a second instance of *nie* appears clause-finally (Donaldson 1993, 402).[8] This is exemplified in (402a). In negated QP constructions, the first *nie* directly modifies the QP, while the second *nie* retains its clause-final position, as illustrated in (402b). Importantly, this pattern mirrors the structure of NEG-stripping (402c), reinforcing the generalization in (392) that negated QP constructions tend to retain the negative marker used in ellipsis.

(402) Afrikaans

a. Ek ken nie daardie man nie.
I know NEG that man NEG
'I don't know that man.' (Donaldson 1993, 402)

b. Maar nie almal sal van HAIRSPRAY hou nie.
but NEG everyone will of Hairspray like NEG
'But not everyone will like Hairspray.' [LCC: News; 2016]

c. Julle wat hier by My is, is skoon, maar nie almal nie.
you.PL who here by me be.PRS be.PRS clean but NEG all NEG
'You who are here with me are clean, but not all (of you).'
[*Bybel vir Almal*, John 13:10; 2008]

A similar pattern is observed in other languages that feature bipartite negation. Early Modern French, for example, exhibits negative concord in negated QP constructions, where

[8] In certain contexts, only one instance of *nie* is required (Donaldson 1993, 401-402).

both negative particles are stacked in the pre-QP position, as shown in (403a). This same ordering is also found in NEG-stripping (403b).

(403) Early Modern French

a. | Non | point | ung | chascũ | qui | me | dit | Sire | Sire | entrera |
---|---|---|---|---|---|---|---|---|---|
| NEG1 | NEG2 | one | everyone | who | to.me | say.PRS.3SG | Lord | Lord | enter.FUT.3SG |

au	royaulme	des	cieulx
to.the.SG.M	kingdom	of.the.PL	heaven

'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Lefèvre, Matt. 7:21; 1523 (1525)]

b. | Et | vous | estes | netz: | main | non | point | tous. |
---|---|---|---|---|---|---|---|
| and | you.PL | be.PRS.2PL | clean.PL.M | but | NEG1 | NEG2 | all.PL.M |

'And you are clean, though not all (of you).' [ibid, John 13:10]

Early Modern Dutch likewise exhibits negative concord in negated QP constructions, with the preverbal negative clitic *en* co-occurring with the negative particle *niet*, as seen in (404a) (see Zeijlstra (2004) for an analysis of Jespersen's Cycle in Dutch). However, unlike Afrikaans and French, NEG-stripping in Dutch does not preserve both negative elements. Instead, the negative clitic, which lacks a suitable host in the reduced structure, is elided along with the rest of the clause, leaving only *niet*, as demonstrated in (404b).

(404) Early Modern Dutch

a. | Niet | een yegelijck, | die | tot | my | segt, | Heere, | Heere, | en | sal |
---|---|---|---|---|---|---|---|---|---|
| NEG | everyone | who | to | me | say.PRS.3SG | Lord | Lord | NEG | will.PRS.SG |

ingaen	in='t	Coninckrijcke	der	hemelen
enter	in=the.SG.N	kingdom	of	heaven

'Not everyone who says to me 'Lord, Lord' will enter into the kingdom of heaven.'
[*Statenvertaling*, Matt, 7:21, 1637]

b. | Ende | ghy lieden | zijt | reyn, | doch | niet | alle. |
---|---|---|---|---|---|---|
| and | you people | be.PRS.2PL | clean | but | NEG | all |

'And you are clean, but not all (of you).' [ibid, John 13:10]

Crucially, this suggests that negative concord in negated QP constructions in Early Modern Dutch is not inherited from NEG-stripping but rather extended from standard negation. In contrast, 16th-century French maintains a direct structural parallel between negated QP constructions and NEG-stripping. Notably, the stacking of *non point* in French cannot be explained as an extension from standard negation, since the latter involves the preverbal clitic *ne*. Meanwhile, in Afrikaans, the relationship between standard negation, NEG-stripping, and negated QP constructions remains less clear, as both inheritance from ellipsis and extension from standard negation are plausible diachronic scenarios.

A key question in languages with bipartite negation is which of the two negative markers directly modifies the QP in negated QP constructions. Cross-linguistically, it is typically—but not always—the newer negative marker that attaches to the QP, while the older negative marker retains its usual position in the clause. This pattern is exemplified by Early Modern Dutch, where the novel negative element is the one that appears with the QP.

Multiple negative markers typically emerge through Jespersen's Cycle. According to Breitbarth et al. (2020, 77–92), reinforcing elements tend to first develop into adverbs marking narrow-focus negation and constituent negation in coordinate structures before eventually being reanalyzed as part of the neutral standard negation construction. This diachronic trajectory helps explain why these novel negative markers tend to be the ones modifying the QP.

Estonian presents an interesting case in that it exhibits negative concord in negated QP constructions (405a) even though the negative particle *mitte* is no longer used in standard negation (Tamm 2015), as illustrated in (405b). Similarly, *mitte* remains in use as an optional element in the formation of negative indefinites, where it co-occurs with the negative auxiliary *ei*, as shown in (405c).

(405) Estonian

a. Siiski mitte kõik ei ole saanud kuulekaks evangeeliumile.
however NEG all.NOM.PL NEG be.PRS.CNG become.PTCP.PERF obedient.TRANSL.SG gospel.ALL.SG
'However, not all have become obedient to the gospel.' [Bible Society, Rom. 10:16; 1997]

b. Me ei karda raamatu-id
we.NOM NEG fear.CNG book-PART.PL
'We are not afraid of books' (Tamm 2015)

c. (Mitte) keegi ei tea sellest.
NEG someone NEG know.CNG it.ELA
'No one knows about this.' (Pook and Lindström 2024, 193)

Notably, in line with the generalization in (392), *mitte* also appears in NEG-stripping constructions, where it does not co-occur with *ei*, as demonstrated in (406).

(406) Estonian
Ka teie olete puhtad, kuid mitte kõik.
also you are clean but NEG all
'And you are clean, though not all of you.' [Bible Society, John 13:10; 1997]

Thus, as with negative concord involving negative indefinites, the distribution of negative concord with negated QPs is best understood through a diachronic lens. However, the crucial difference lies in the source construction: while negative concord with negative indefinites typically arises from the grammaticalization of NPIs that originally required clausal negation, negative concord with negated QPs appears to develop in languages where standard negation involves a bipartite negative marker. This historical trajectory accounts for the lack of correlation between the two types of negative concord and explains why negative concord with negated QPs is far less widespread cross-linguistically.

9.3 The non-lexicalization of 'not all': A near-absolute universal

The phenomenon of the non-lexicalization of 'not all' was first systematically identified by Horn (1972, 1989), who observed that while many languages lexicalize the meanings corresponding to the A-, E-, and I-corners of the Aristotelian square of opposition (see Figure 9.1)—namely, 'all,' 'none', and 'some'—they consistently lack a lexical item for the O-corner, 'not all'. For example, English expresses this meaning using compositional forms like *not all* or *not every* rather than a single lexical item like **nall*.

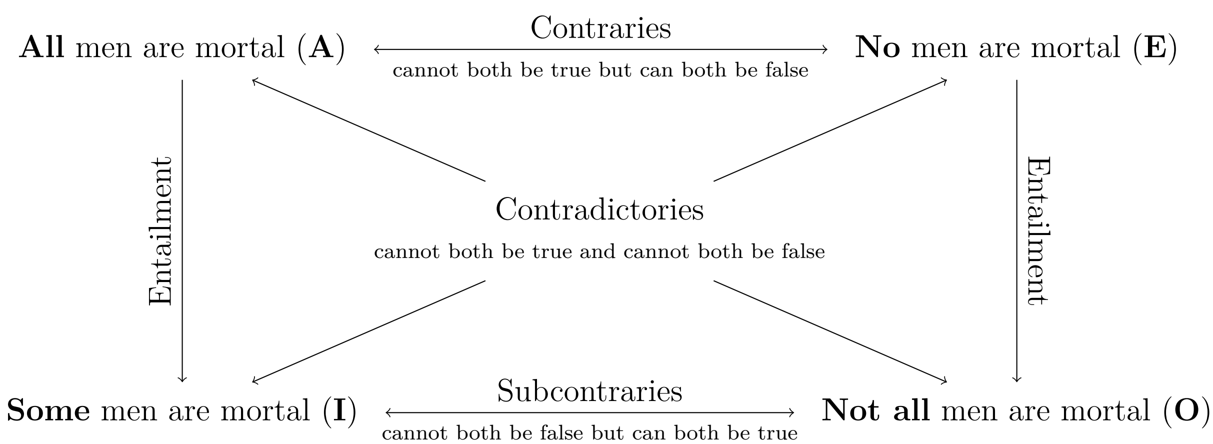


Figure 9.1: The Aristotelian square of opposition

This paradigmatic gap is not confined to quantifiers over individuals but extends to other logical operators, including temporal quantifiers (**nalways* 'not always') and connectives (**nand* 'not and'), making it a robust cross-linguistic phenomenon.

The systematic absence of O-lexicalization, often referred to as the "*nall* problem" or the "missing O-corner", has long been regarded as a universal feature of natural languages. However, for Horn's generalization to make testable predictions, it is essential to define what it means for an expression to be considered lexicalized. This is often implicitly assumed in the literature to refer to a single word. Yet, the notion of a "word" lacks a universally applicable definition across languages and is often based on orthographic conventions rather than intrinsic linguistic properties (see Haspelmath 2023, and references therein).

Indeed, my dataset includes several languages with orthographic words that mean 'not all'. For example, in Central Guerrero Nahuatl (Uto-Aztecan), the word *xnochimej* translates to 'not all' but is composed of the negative clitic *x=*, the quantifier *nochi* 'all', and the plural marker *-mej*. Despite being orthographically represented as a single word, this form remains semantically transparent.

An alternative approach to defining lexicalization is to require that forms be monomorphemic. However, this criterion is likely too restrictive, as it would exclude forms like English *someone* or French *quelqu'un* 'someone', which are morphologically complex but generally considered lexicalized.

Following Jackendoff (2013) and Haspelmath (2024), I define a lexical item as a form or construction that must be stored in the mental lexicon because its meaning is not predictable from the meanings of its components, if those components exist. This definition encompasses idiomatic expressions, such as *kick the bucket*, whose meanings cannot be inferred from their individual parts, as well as semantically idiosyncratic forms like *someone*, which is not compositionally derived in the same way as the phrase *some person*.

In addition to fully lexicalized forms, one can consider partial lexicalizations, where the meaning is predictable from the meanings of the components, but the specific association is idiosyncratic. For example, the meaning of *inaccurate* can be derived from the negative prefix *in-* and the adjective *accurate*, but speakers must know that the correct form is *inaccurate* rather than *unaccurate*. This idiosyncrasy is not governed by a general morpho-phonological rule but instead reflects the etymological origins of these forms: both *in-* and *accurate* are derived from Latin, while *un-* originates in Old English.

With these criteria in mind, Horn's universal can be understood as the observation that languages systematically lack lexical items meaning 'not all', whose meanings are not compositionally derived from their parts. Under this definition, *xnochimej* in Central Guerrero Nahuatl is not considered a counterexample. This raises the question of why languages avoid lexicalizing the O-corner ('not all') while often lexicalizing the other three corners of the square of opposition. Scholars have proposed various explanations for this gap, broadly falling into two categories: communicative and diachronic approaches.

The communicative approach posits that the I-corner ('some') blocks the need for the lexicalization of the O-corner because it typically conveys the same meaning once scalar implicatures are calculated. Under this view, only three out of the four corners of the square of opposition need to be lexicalized to differentiate between three mutually exclusive and collectively exhaustive situations: 'All X are Y', 'No X are Y', and 'Some, but not all, X are Y'. Languages consistently lexicalize the I-corner rather than the O-corner due to the inherent complexity of negative compared to positive expressions (Horn 1972, 1989; Katzir and Singh 2013; Uegaki 2023; Züfle and Katzir 2022), the typically greater informativity of unstrengthened I-statements (Enguehard and Spector 2021), or a principle of communicative stability (Bar-Lev and Katzir 2023).

However, as Hoeksema (1999) points out, if the lexicalization of I-forms renders O-forms redundant, why are negated QP constructions such as *not every* or *not all* still permissible and pragmatically useful? Moreover, some languages have multiple lexical items corresponding to one of the other corners of the square of opposition, raising the question of why languages exhibit such parsimony specifically when it comes to the O-corner.

To address this, Hoeksema (1999) proposes a diachronic explanation for the non-lexicalization of 'not all'. He argues that 'Not all X are Y' propositions are not sufficiently frequent in discourse to trigger the univerbation of negated QP constructions like *not all* into a single lexical item such as *nall*. By contrast, Zeijlstra (2022, 166–170) suggests that the issue is not the low frequency of these propositions but rather the fact that the quantifier in negated QP constructions is typically focused, which precludes morpho-phonological univerbation.

While the theoretical explanations discussed above—both communicative and diachronic—offer frameworks for understanding the systematic absence of lexicalized O-corner ('not all') forms, they rest on generalizations that, to date, have not been subjected to rigorous empirical testing. Horn's universal, despite its prominence in the literature, has largely remained an untested hypothesis, and its validity across a wide range of languages has been assumed rather than verified.

This study represents the first large-scale empirical investigation of Horn's universal, utilizing a diverse language sample of 217 languages to systematically evaluate whether 'not all' is indeed absent as a lexicalized form across languages. While the primary focus of this study is the phenomenon of inverse scope and the strategies languages employ to express 'Not all X are Y' propositions, it also indirectly tests whether languages have lexicalized forms for 'not all', even though this is not its central aim.

Remarkably, the results reveal a single counterexample, suggesting that the non-lexicalization of 'not all' is not an absolute universal but rather a near-absolute one. This finding directly bears on the theoretical accounts discussed earlier: while they explain the rarity of O-lexicalizations, the existence of even one counterexample indicates that these explanations must allow for specific pathways that enable exceptions under certain conditions.

As discussed in Section 7.7.1, Garifuna (Arawakan) presents a counterexample with a fixed form meaning 'not all', which appears to have originated from a negative verbal quantifier. This is illustrated in (342), repeated below as (407).

(407) Garifuna
Má-su-tiña gayára ja-bágari-dun má-mari-ga
NEG-be.all-T3PL be.able VBLZ-life-VBLZ NEG-spouse-GA
'Not everyone can remain single.' [Wycliffe Bible Translators, Matt. 19:11; 2012]

The negative form *másutiña* (NEG-be.all-T3PL) is unique in the language sample in that it appears to lack a corresponding affirmative form. The expected affirmative counterpart *gásutiña* (AFF-be.all-T3PL) does not occur in the New Testament translation or, as far as I can determine, anywhere else on the internet. Given the significantly higher frequency of affirmative 'All X are Y' sentences compared to negative 'Not all X are Y' sentences, this absence is unlikely to be accidental. Instead, I suggest that *másutiña* is likely historically derived from a stative verb meaning 'be all', which ceased to be used in its affirmative form but was preserved as a fixed negative form.

Crucially, the form *másutiña* is not transparently composed of a negation marker and a universal quantifier, even though this appears to have been its historical lexical source. Instead, it resembles negative quantifiers that were historically derived from the combination of a negation marker and a quantificational element but are no longer analyzed as complex forms. For instance, English *none* originates from Proto-Germanic **nainaz*, which itself derives from **ne* 'not' + **ainaz* 'one, some'. Although the original negative marker is still reflected in the form as *n*, *none* is no longer compositionally analyzed as a combination of negation and an existential quantifier.

Thus, since the meaning of *másutiña* cannot be compositionally derived from independently meaningful parts, it qualifies as a lexical item, which must be stored in the mental lexicon.

Another form that can be argued to be partially lexicalized is Portuguese *nem todos* 'not all'. As noted in §9.2, the negated QP construction employs the negative marker *nem* 'not even, nor', which does not function as a general marker of negation in other contexts. Instead, Portuguese typically uses *não* for clausal negation, NEG-stripping, and coordination constructions. This suggests that the use of *nem* in negated QP constructions represents a distinct function. One could imagine a scenario where *nem* ceases to be used as a focus particle or conjunction due to the grammaticalization or borrowing of a new form, yet is retained exclusively in negated QP constructions. In such a hypothetical situation, *nem todos* would no longer be semantically transparent and could instead be reanalyzed as a fixed form.

Together, these examples illustrate that while the lexicalization of 'not all' is exceedingly rare, it is not universally ruled out. These findings align with the diachronic perspective, which posits that such lexicalization is not impossible but highly unlikely due to the processes that typically govern lexicalization patterns. However, they also reveal important differences between the pathways available for the lexicalization of 'not all' and those more commonly attested for other negative quantifiers.

Unlike 'not all', negative quantifiers and negative indefinites, which lexicalize the E-corner of the square of opposition ('none'), frequently emerge through well-documented diachronic pathways. For example, negative indefinites often arise from a process in which a generic noun or minimizer initially functions as an NPI and is later reanalyzed as inherently negative. A classic example of this is French *personne*, which originally meant 'person' but was reanalyzed as 'nobody'. By contrast, as noted by Hoeksema (1999) and Zeijlstra (2022), this pathway is unavailable for 'not all' because universal quantifiers do not undergo a stage as NPIs. Without this intermediate step, the reanalysis necessary for grammaticalization is effectively precluded. Additionally, as discussed earlier, the univerbation of a negated QP into a single lexical form, such as *nall*, is also improbable due to morpho-phonological constraints that limit such processes.

As a result, while univerbation and the reanalysis of NPIs are commonly attested cross-linguistically for the E-corner, these pathways are largely absent for the O-corner. Instead, a more plausible pathway for the lexicalization of 'not all' involves the loss of either the negative marker or the universal quantifier in other contexts, leaving the construction meaning 'not all' as its sole remaining function. This shift enables a previously complex form to be reanalyzed as a fixed expression, as one of its components no longer operates independently in the language. However, this type of change remains exceptionally rare cross-linguistically, highlighting the structural and diachronic barriers to lexicalizing the O-corner.

So far, the discussion has centered on pathways that begin with a form transparently meaning 'not all' and potentially evolve into a lexical item like *nall*. However, this raises a broader question: could there be other lexical sources for 'not all' that bypass these constraints?

To illustrate this point, consider constructions like *Not only X but (also) Y*, as in (408a). In English, these constructions are superficially similar to negated QP constructions because the negative marker precedes the scope-bearing operator rather than occurring in its typical position in standard negation. Traditional grammar categorizes *Not only X but (also) Y* as a correlative conjunction, akin to *either…or* and *both…and*. In some languages, *not only* can also occur without the second conjunct, as shown in (408b). However, this usage is likely significantly less frequent cross-linguistically than the correlative conjunction, much like negated QP constructions are less common than constituent negation in NEG-stripping and contrastive coordination constructions (see §9.2).

(408) a. Not only Perez but other Panamanians have suffered from gross administrative neglect on the part of their own authorities. [COCA: Magazine; 1999]

b. Not only Phelps is getting rich. [COCA: Newspaper; 2008]

Notably, *Not only X but (also) Y* constructions can arise from a variety of lexical sources. In some languages, these constructions involve neither a negative marker nor a focus particle corresponding to 'only' in affirmative contexts.

For instance, in Japanese, this meaning is expressed through constructions such as *bakari/dake de naku* or *bakari/dake ka*, where *dake* and *bakari* are focus particles meaning 'just' or 'only'. While the former construction contains a negative marker (*naku*), the latter does not—it is derived from a rhetorical question, 'Is it just?' (Martin 2004, 367). Similarly, in Bengali, this meaning is conveyed using *æke to…tar upôr*, which literally translates as 'One [thing is that…], on top of that, …' (Thompson 2012, 235).[9]

This variation in lexical sources for *Not only X but (also) Y* constructions stands in sharp contrast to the structure of complex 'not all' constructions. According to the findings of this study, 'not all' constructions are consistently semantically transparent, invariably involving both a negative marker and a universal quantifier. This raises the question of why languages appear less innovative in lexicalizing 'not all' constructions compared to others, such as 'not only'. For example, one might expect languages to develop 'not all' forms from emphatic expressions like *far from everyone*, as in (409), since these expressions entail 'Not all X are Y'. Yet, such grammaticalization pathways do not seem to occur across languages.

(409) a. But **far from everyone** is interested in alternative opinions. [NOW: US; 2022]

[9] It remains an open question whether such non-transparent constructions are exclusively restricted to correlative conjunctions like (408a) or whether they can also be applied to cases like (408b). If the former is true, the non-transparency of these constructions is not directly comparable to the semantic transparency of 'not all' constructions, as discussed in the next paragraph.

b. And **far from everyone** injured actually sues. [NOW: US; 2022]

c. To its backers, this offers an opportunity to communicate Biggie's essence in a way even the most raw bootleg can't – though as with so many things web3, **far from everyone** will see the upside. [NOW: US; 2022]

In summary, the diachronic approach provides a compelling explanation for why certain grammaticalization pathways that are available for the E-corner ('none') are unavailable for the O-corner ('not all'). For instance, universal quantifiers do not pass through an intermediate stage as negative polarity items, precluding the reanalysis that often leads to the emergence of negative indefinites. Nevertheless, this study identifies alternative pathways for the lexicalization of 'not all', such as the loss of one of its components (the negative marker or the universal quantifier) in other contexts, leaving the construction meaning 'not all' as its sole function. While this process can lead to reanalysis as a fixed expression, it appears to occur far less frequently across languages. Additionally, there may be other historical accidents or language-specific developments that give rise to lexicalized or partially lexicalized forms of 'not all', but the results of this study suggest that such scenarios are exceedingly rare.

Another intriguing question for future research is why the lexical sources of complex 'not all' constructions are so restricted. Languages seem to exhibit a notable lack of creativity in this domain. While various expressions imply or even entail the truth of a 'Not all X are Y' proposition, these expressions do not appear to grammaticalize into 'not all' constructions—whether complex or lexicalized. Understanding this limitation may shed light on the principles that govern the lexicalization of logical expressions.

Part IV

The rise and fall of inverse-scope constructions

Chapter 10

Diachronic results

This chapter explores the diachronic evolution of strategies for expressing propositions of the form 'Not all X are Y'. While previous chapters explored the synchronic tendency toward scope transparency, the diachronic analysis focuses on the historical processes and mechanisms that drive these shifts over time. By examining how specific strategies emerge, decline, or persist across centuries, this chapter offers a deeper understanding of the forces shaping the distribution of inverse-scope and transparent-scope constructions—including the bias for scope transparency.

The aim of this chapter is to test the Diachronic Hypothesis, stated again here for clarity in (410). This hypothesis builds on the idea that languages are shaped by a gradual pressure toward expressing scope relations transparently, but it also acknowledges that inverse-scope constructions can arise as a byproduct of unrelated grammatical changes.

(410) *Diachronic Hypothesis*

(i) In a given language, the relative frequency of transparent-scope constructions tends to increase over time.

(ii) Conversely, inverse-scope constructions become more frequent only when the dominant transparent-scope alternative is lost due to broader grammatical changes, prompting the language to rely on the standard negation construction.

To investigate this hypothesis, the chapter examines two contrasting diachronic pathways. The first pathway, presented in (411), examines how inverse-scope constructions are replaced by transparent-scope strategies over time to give rise to the observed patterns discussed and analyzed in Chapters 7 and 9.

(411) *Pathway of decline of inverse-scope constructions*

Stage 1 Inverse-scope constructions are frequently employed, possibly even the dominant strategy for expressing 'Not all X are Y' propositions.

Stage 2 The language innovates a transparent-scope strategy. Over time, the frequency of the novel construction increases, while the frequency of the older inverse-scope construction decreases.

Stage 3 The inverse-scope construction either falls out of use or becomes rare and mostly restricted to formulaic expressions.

The second pathway, presented in (412), focuses on how inverse-scope constructions emerge when transparent-scope alternatives are lost due to broader grammatical changes.

(412) *Pathway of emergence of inverse-scope constructions*

Stage 1 Inverse-scope constructions are rare or not in use.

Stage 2 The dominant transparent-scope construction is lost due to broader grammatical changes.

Stage 3 The language resorts to using the standard negation construction, often resulting in inverse scope.

This chapter examines how these pathways unfold in the histories of Germanic and Romance languages, two subfamilies that are particularly well-represented in the diachronic data available through the NT corpus. This dataset not only allows us to trace shifts in the relative frequencies of inverse-scope and transparent-scope constructions over time but also to investigate the mechanisms driving these changes, including the emergence of new transparent-scope strategies and the influence of broader syntactic developments. By comparing the trajectories of Germanic and Romance languages, this chapter seeks to uncover patterns that either support or challenge the Diachronic Hypothesis.

One key observation is that transparent-scope strategies often displace inverse-scope constructions, but the transition is rarely abrupt. Instead, these shifts typically occur in stages, beginning with the innovation of a transparent-scope strategy and followed by a gradual decline in the use of the older inverse-scope construction. However, the rate of this process varies significantly across languages. Furthermore, the findings reveal that inverse-scope constructions can re-emerge due to broader grammatical changes, not necessarily as a last resort but rather in variation with a transparent-scope strategy.

Broadly speaking, Germanic and Romance languages have followed similar pathways—largely aligning with those predicted in (411) and (412). However, some notable differences set the two families apart. Most Romance languages, though not all, adopted Strategy B (negated QP constructions) relatively quickly, largely abandoning Strategy A. In contrast, many Germanic languages retained Strategy A as a secondary or even primary strategy well after Strategy B had emerged. One possible explanation is that Strategy A was more deeply entrenched in many Germanic languages, having reached a higher frequency before the rise of Strategy B. This may have contributed to its greater diachronic stability. However, the available quantitative data are insufficient to confirm this hypothesis, leaving the question open for future research.

The structure of this chapter is as follows: the first section examines the evolution of strategies in Germanic languages, tracing their historical development from Old Germanic to the present day (§10.1). The second section focuses on Romance languages, with particular attention to the transition from Latin to Medieval and Modern Romance varieties (§10.2). Each section begins with a diachronic grammatical overview of the relevant language family, highlighting key changes in word order, negation, and universal quantifiers. These grammatical developments provide essential context for understanding the transitions between different strategies for expressing 'Not all X are Y' propositions. Each section concludes with a summary and a discussion of the overarching trends observed in the respective language family.

The next chapter synthesizes the findings of this chapter and explores their implications for the Scope Transparency Hypothesis. While this analysis confirms the existence of the diachronic patterns predicted by the hypothesis, it also underscores the variability and complexity of these changes across languages.

10.1 Germanic languages

10.1.1 Overview

Diachronic data for the Germanic languages date back to the 9th–10th centuries for West Germanic and the 13th–14th centuries for North Germanic. Most of the data are drawn from NT translations, sometimes supplemented by examples from other contemporary sources to provide a more comprehensive empirical picture. While the focus is on languages with historical data, some languages with only synchronic data are included because they help illuminate broader patterns within Germanic.

The Germanic language sample includes both official languages, such as English, German, Dutch, Swedish, and Danish, as well as minority varieties, including Scots, Pennsylvania German, Hutterite German, Plautdietsch, Eastern Low German, and West Low German, as shown in Table 10.1. Scots, closely related to English, is primarily spoken in the Scottish Lowlands and northern Ulster in Ireland. Pennsylvania German, Hutterite German, and Plautdietsch are minority varieties maintained by Anabaptist communities in North America, originating from dialects of Rhenish Franconian, Bavarian, and Eastern Low German, respectively. Finally, Eastern and West Low German represent regional dialects spoken in parts of northern Germany and neighboring countries.

The diachronic trajectories reveal significant shifts in the distribution of strategies over time. In the earliest attested stages, Strategies C and A alternated, likely reflecting word order changes associated with the gradual adoption of V2 or SVO patterns. Strategy C, primarily tied to NEG-V1 constructions, declined as this word order pattern became obsolete, paving the way for Strategy A to emerge as the dominant strategy in many languages.

In the late medieval period, some Germanic languages developed Strategy B constructions. However, these were not fully conventionalized and appeared only sporadically for several centuries. By the 19th and 20th centuries, Strategy B had become dominant in

Language group	Language	Years
Anglic	English	990–2011
	Scots	1904–2016
High German	German	9th century–2015
	Hutterite German	2016
	Luxemburgish	2017
	Pennsylvania German	2013
Low German	Old Saxon and Middle Low German	9th–15th centuries
	Eastern Low German	1929
	Plautdietsch	2003
	West Low German	2019
Macro-Dutch	Dutch	13th century–2013
	Afrikaans	1953–2000
North Germanic	Swedish	1526–2015
	Danish	1550–1997
	Icelandic	1584–2007
	Faroese	1823–1937
	Norwegian	1921–2018

Table 10.1: Germanic language sample

most Germanic languages, with notable exceptions: Icelandic, where Strategy C persists to be dominant, and Swedish and Faroese, which reverted to Strategy A as their dominant strategy.

Today, Strategy A is rare and primarily found in formulaic expressions in English, German, Dutch, Danish, and Icelandic. It is moderately frequent in Norwegian and Afrikaans and remains highly frequent in Swedish and Faroese.

10.1.2 Key grammatical properties

Word order The dominant word order pattern in early Germanic languages, including Old English, Old High German, and Old Norse, is Verb-Second (V2) in main clauses. Subordinate clauses, on the other hand, often follow a verb-final pattern (Cichosz et al. 2016, 52-56).

The V2 pattern positions the finite verb in the second slot of the clause, regardless of the element occupying the first position. This allows for SVO order when the subject appears clause-initially, but also subject-verb inversions, such as OVS or Adv-V-S, when other constituents are fronted. However, pronominal subjects often remain in preverbal position even when another constituent appears clause-initially, diverging from the stricter V2 configurations seen with nominal subjects (Cichosz et al. 2016, 52-56, 114-115).

The Verb-First (V1) pattern is less frequent than other patterns, although it is relatively common in poetry. In prose, it often marks transitions or emphasizes new information. The

V1 pattern is also associated with various factors, such as negation, existential constructions, motion verbs, and verbs of saying (Cichosz et al. 2016, 121-125).

Modern English stands out among the Germanic languages as the only one to have completely lost the V2 pattern. Other Germanic languages, including German, Dutch, and the Scandinavian languages, have retained V2 in main clauses, though with some variation in subordinate clause word order.

Negation In Old Germanic languages, negation was primarily expressed by a pre-verbal negative clitic inherited from Proto-Germanic **ni*. Word order patterns influenced the position of this clitic, with NEG-V1 constructions being prevalent in earlier stages but gradually giving way to new configurations due to changes in syntax, such as the rise of V2 and SVO orders.

In Old English, the most frequent word order pattern in negative clauses was V1 word order, with the negative clitic *ne* immediately preceding the finite verb, as demonstrated in (413).[1]

(413) Old English (van Kemenade 1997, 96)

Ne	mihte	se	deað	him	genealæcan
NEG	can.PST.3SG	the	death	3SG.DAT	approach.PTCP.PST

'Death could not have approached him'

As in Old English, V1 word order was also the most frequent pattern in main negative clauses in Old Saxon (Breitbarth and Jäger 2018, 148). However, in Old High German, negative clauses were only weakly associated with V1 word order (Cichosz et al. 2016, 146). The distribution of word order patterns in Old High German varied significantly by text. For instance, Otfried's *Evangelienbuch* contains about twice as many instances of V1 as V2 in negated clauses, while other texts, such as Isidor (about 800), show significantly more V2 clauses, with approximately three to four times as many V2 as V1 instances in negative declarative clauses (Breitbarth and Jäger 2018, 145). This preference for V2 is even stronger in later texts, such as Notker's *Psalter* (early 11th century), where V2 occurs roughly 20 times more frequently than V1 in negative clauses.

These patterns highlight the gradual decline of NEG-V1 constructions and the increasing dominance of V2 and SV word orders as Old Germanic languages transitioned into their later stages.

Another significant change in the expression of negation that emerged during the late Old Germanic and early Middle Germanic periods was the innovation of post-verbal negative markers through Jespersen's Cycle. Jespersen's Cycle outlines a process of grammatical renewal in the expression of standard negation. In Stage 1, negation is expressed by a single negative marker. In some contexts, this marker is reinforced by an additional element. This

[1] Cichosz (2020) found that out of 3,793 negative main clauses in Old English, 46.2% exhibited V1 word order. This was followed by 24.7% with XVS word order, where X is a clause-initial constituent other than the subject. Overall, 70.9% of negative main clauses had VS word order, compared to only 21.8% with SV word order, while the remaining instances either lacked a subject or could not be classified.

new element may become grammaticalized as a fully-fledged negative marker, resulting in bipartite negation in Stage 2. Gradually, the new marker takes over as the primary expression of negation, while the original marker is weakened and eventually lost. Stage 3 marks the complete disappearance of the older negative marker, leaving the newer element as the sole negator. This process reflects a broader pattern of language change, whereby older forms are eroded and replaced by innovative elements (Willis et al. 2013).

Jespersen's Cycle is particularly well-documented in the West Germanic languages. In Stage 1, negation was marked by the pre-verbal clitic *ne*/*ni*. During Stage 2, additional negative adverbs were introduced, leading to the development of bipartite negation. The following examples from Middle English, Middle High German, Middle Low German, and Middle Dutch illustrate this transitional stage:

(414) Middle English (Ingham 2013, 133)
Þu [...] Keiser, nauest nawt þis strif rihtwisliche idealet
you emperor NEG.have NEG this contest justly assigned
'You [...] Emperor, have not fairly apportioned this contest'

(415) Middle High German (Breitbarth and Jäger 2018, 152)
"Ich en-wil es niht erwinden", sprach aber der chune man.
I NEG-want it NEG omit said but the brave man
'I do not want to omit it, said the brave man.'

(416) Middle Low German (Breitbarth and Jäger 2018, 156)
dar en sculle wii se nicht ane hinderen
there NEG shall we them NEG from bar
'we shall not bar them from it'

(417) Middle Dutch (Breitbarth 2013, 193)
Want ic ne wille niet, broeder, dat ghi onwetende sijt
because I NEG want NEG brother that you unknowing be
'Because I do not want you to be unknowing, brother'

By Stage 3, the pre-verbal clitic was lost, and the newer negative particle became the dominant marker of negation. This is reflected in the modern forms, such as *not* in English, *nicht* in German, and *niet* in Dutch.[2] For formal analyses of Jespersen's Cycle in West

[2]Afrikaans, however, exhibits a distinct development of bipartite negation, where the negative marker *nie* is repeated at the end of the clause, as shown in (i).

(i) Afrikaans (Biberauer 2012, 6)
Ek is nie ryk nie.
I be.PRS NEG rich NEG
'I'm not rich.'

This system differs from the typical progression of Jespersen's Cycle and is thought to originate from the reanalysis of a clause-final anaphoric negator (Biberauer 2009, 2012).

Germanic languages, see Zeijlstra (2004); Wallage (2008); van Gelderen (2011); Breitbarth et al. (2020); Zeijlstra (2022), among others.

The North Germanic languages exhibit a somewhat distinct trajectory, undergoing two cycles of negation renewal. In early Old Norse, the pre-verbal clitic *ne* was sometimes reinforced by the post-verbal clitic *-at*/*-a*, as illustrated in (418a). By the 9th century, the pre-verbal *ne* disappeared, leaving the post-verbal clitic as the primary marker of negation, as shown in (418b).

(418) Old Norse (Willis et al. 2013, 10)

a. ef Gunnarr ne kømr=að
   if Gunnarr NEG come.PRS.3SG=NEG
   'if Gunnarr does not come'

b. gaft=at=tu ástgiafar
   give.PST=NEG=2SG love.presents
   'You did not give love presents.'

This post-verbal clitic was eventually replaced by the negative particle *eigi*, exemplified in (419), which evolved into modern negators like *inte* (Swedish), *ikke* (Danish and Norwegian), *ekki* (Icelandic), and *ikki* (Faroese).

(419) Old Norse (Willis et al. 2013, 11)
   Eigi má ek þat vita.
   NEG can.PRS.1SG I that know.INF
   'I cannot know that.'

Universal quantifiers All Germanic languages have retained Proto-Germanic **allaz* 'all', which continues to serve as a general universal quantifier. However, innovations of distributive universal quantifiers, such as 'every', also emerged across the Germanic languages.

Haspelmath (1995, 370-371) illustrates this development through English *each* and *every*, German *jeder* 'every', and Dutch *elk* 'every'. These forms lack synchronic internal structure but can be traced back to combinations involving Proto-Germanic wh-determiners and particles such as **ajw-* ('ever') and **ga-* ('together'). For example:

- Proto-Germanic **ajw-hwalik* ('ever-which') developed into Modern English *each* and Dutch *elk* 'every'. Additionally, Modern English *every* emerged from the strengthening of *each* by the addition of *ever*, resulting in a compounded form (*ever* + *each* > *every*).

- Proto-Germanic **ajw-ga-hwalik* ('ever-together-which') evolved into Modern German *jeglich* 'any, every'.

- Proto-Germanic **ajw-hweþar* ('ever-which of the two') gave rise to Modern German *jeder* 'every'.

In the North Germanic languages, distributive universal quantifiers are typically derived from Proto-Germanic **hwarjaz* 'which, what [of many]' and **hwarjazuh* 'each'. These forms evolved into Old Norse *hverr* ‘each, every’ (also ‘who, which’), which served as a basis for similar forms in its descendants, such as Modern Icelandic *hver* and Modern Swedish *varje*.

10.1.3 Old Germanic

The earliest translation of the New Testament into a Germanic language is the Gothic Bible (Wulfila Bible), completed in the 4th century. This translation, rendered into Gothic (East Germanic), closely adheres to the Greek source text in relevant verses. For example, in (420), the negative marker *ni* precedes the universal quantifier in a B construction. However, this may simply reflect a calque from the Greek source, making it uncertain whether this construction actually existed in this language.

(420) Gothic
akei ni allai ufhausidedun aiwaggeljon
but NEG all.PL obey.PST.3PL gospel.DAT
‘But not all obeyed the gospel’ [*Gothic Bible*, Rom. 10:16; 4th century]

Later Germanic translations, dating from the 9th to 10th centuries, are available in Old High German, Old English, and Old Saxon (Old Low German) (Gow 2012, 199-206). In contrast, no extant translations of the New Testament or gospels into Scandinavian languages exist before the Reformation era in the 16th century (Ejrnæs 2012, 240). Consequently, this subsection will focus on the West Germanic languages.

In translations of the NT into Old Germanic languages, Old English exclusively features Strategy C (3 instances), and Old Saxon also employs only Strategy C (1 instance). In contrast, Old High German exhibits greater variability, with examples of Strategy A (2 instances), Strategy B (2 instances), and a single case of a C/E construction. However, the occurrence of Strategy B in Old High German is limited to a specific translation and may reflect a calque from the Latin source rather than an established linguistic pattern. By contrast, Strategy A is attested in several other Old High German texts. Overall, Strategies C and A dominate this period, consistent with the word order variability in negative clauses discussed in the previous subsection.

Strategy A Strategy A is attested in two relevant verses from Otfrid of Weissenburg’s *Evangelienbuch*, as shown in (421).[3]

(421) Old High German
a. In hímil al ni géngit (joh iz gót ni hengit,
in heaven.ACC all.NOM.SG.N NEG go.PRS.3SG and it.ACC god.NOM NEG allow.PRS.3SG
iz wirdit noh giwéizit!) thaz mih drühtin
it.NOM become.PRS.3SG still show.PTCP.PST that.NOM.SG.N me.ACC Lord

[3] Old High German and Old Saxon data were extracted from the Referenzkorpus Altdeutsch (REA), version 1.2 (Zeige et al. 2022), of the Deutsch Diachron Digital (DDD) project, through the platform ANNIS (Krause and Zeldes 2016).

heizit
call.PRS.3SG
'Not everyone goes to heaven (nor does God allow it, it will yet be shown!) who call me Lord.' [*Evangelienbuch*, Ev. 2, chap. 23, lines 19-20]

b. álle man ni-ntnéinent thaz thínu
all.NOM.PL.M man.NOM.PL NEG-deny.PRS.3PL that.ACC.SG.N your.NOM.PL.N
wort giméinent.
word.NOM.PL mean.PRS.3PL
'Not everyone refuses your teaching.'
[*Evangelienbuch*, Ev.3, chap. 10, line 36]

The *Evangelienbuch* is a retelling of the gospels, composed between 863 and 871 in the Rhenish Franconian dialect (Gow 2012, 203). Written in rhyming couplets, it deviates significantly from its Latin source. Note that the choice of Strategy A in these verses may have been influenced by the constraints of rhyme rather than purely linguistic considerations. Additionally, the second example (421b) appears to be an original addition by Otfrid and does not correspond to the Latin source.

Another Late Old High German text that employs Strategy A is Notker's translation of *De Interpretatione*, based on Boethius's Latin rendering of Aristotle's work. In this translation, two relevant examples utilize Strategy A, while the Latin source text employs Strategy B.[4]

(422) a. Latin
quoniam non omnis potestas oppositorum est
since NEG every power contrary.GEN.PL be.PRS.3SG
'because not every power is susceptible of contraries' [Boethius, *De interpretatione*]

[4]The broader context of example (422) is provided below to clarify its meaning:

(i) Nevertheless it is evident that not every thing which can "be," and can "walk," is capable also of the opposites, for in some cases this is not true. In the first place, in those things which are potent irrationally, as fire is calorific, and has irrational power; rational powers then are those of many things, and of the contraries; but not all irrational powers, for, as we have said, fire cannot heat, and not heat, nor such other things as always energize. Yet even some irrational powers can at the same time receive opposites; but this has been stated by us, **because not every power is susceptible of contraries**, not even such as are predicated, according to the same species. [*De interpretatione*]

Similarly, the context of example (423) is included for clarity. Note that the term *contradiction* in this instance refers to contradictory negation rather than a contradictory proposition:

(ii) That there is then one affirmation contradictorily opposed to one negation, and what these are, has been shown, also that there are other contraries, and what they are, **and that not every contradiction is true or false**, and why and when it is true or false. [*De interpretatione*]

b. Old High German
uuánda álle máhte oppositorum nîeht máhtîge ne-sint
because all powers contrary.GEN.PL NEG capable NEG-are
'because not every power is susceptible of contraries' [Notker, *De interpretatione*]

(423) a. Latin
et quoniam non omnis uera uel falsa contradictio
and since NEG every true or false contradiction
'and that not every contradiction is true or false' [Boethius, *De interpretatione*]

b. Old High German
Únde dáz álle uuíderchétâ nîeht ne-skéident uuâr únde lúgi.
and that all contradictions NEG NEG-true are or false
'and that not every contradiction is true or false' [Notker, *De interpretatione*]

However, Strategy A is not exclusive to Old High German. Although Strategy C appears to have been the dominant strategy in Old English, Strategy A was also in use during Late Old English. The examples in (424) are found in the writings of Ælfric, composed only a decade after the Wessex Gospels and likewise written in the West Saxon dialect.

(424) Old English

a. forþon þe ge ealle ne cunnon þæt l understandan
because you all NEG can the Latin understand
'[It is fitting for us, bishops, that we open to you, priests, in the English language the book-lore that our canon and also the Christ's book prescribe to us;] as not all of you can understand Latin.'
[Ælfric, *First Old English letter for Wulfstan*; early 11th century]

b. forðan þe hi ealle ne beoð of gode
because they all NEG are of god
'[Now is it to be known, that we should not at all trust too much to dreams,] because they are not all from God; [some dreams are in truth from God, even as we read in books, and some are from the devil for some deceit, [seeking] how he may pervert the soul]'
[Ælfric, *Lives of Saints*, 1:466, line 404; 996-997]

The use of Strategy A in (424) raises the question of why NEG-V1 word order (Strategy C) was not employed, especially since it could have easily avoided inverse scope. One possibility is that NEG-V1 word order was significantly less frequent in certain genres. For instance, Cichosz (2020) reports that 49.2% of negative main clauses in the Wessex Gospels feature NEG-V1 word order. By contrast, I tentatively observe that Ælfric's letters, where example (424a) appears, seem to show a tendency toward SV word order in negative main clauses. A detailed quantitative analysis of these texts is left for future research.

This apparent difference in word order may reflect a genre-specific pattern rather than a broader linguistic shift. Supporting this, other works by Ælfric display different distributions; for example, in his *Supplemental Homilies*, NEG-V1 accounts for 45.3% of negative main clauses (Cichosz 2020). Thus, the apparently lower frequency of NEG-V1 in Ælfric's letters may be influenced by the genre of the text rather than signaling a general trend or authorial preference.

Ælfric's letters, in this respect, bear some resemblance to Middle English texts, where the NEG-V1 pattern gradually declined (Wallage 2012). However, it is not necessarily the case that Ælfric's letters are more telling of the state of the spoken language in this period. In fact, Cichosz (2020, 371–372) found that NEG-V1 was more common in quoted speech than in non-speech text. Therefore, further research is needed to fully understand the role of genre in shaping the distribution of NEG-V1 in Late Old English.

Strategy B Strategy B is attested only in a prose translation of Tatian's *Diatessaron* (an early gospel harmony), which was based on the Latin version and produced around 830 at Fulda in the East Franconian dialect (Gow 2012, 202). This translation is generally regarded as being heavily dependent on the Latin word order (see Cichosz et al. 2016, 26 and references therein). Indeed, it employs the same strategies as Latin in the relevant verses, as demonstrated in (425). Notice, however, that this example instantiates negative concord, which is absent in the Latin source. While this slight divergence suggests some adaptation in the translation process, it remains uncertain whether Strategy B was genuinely in use in Old High German or if it simply reflects the influence of the Latin source.

(425) Old High German
nalles alle ni gifahant thiz uuort.
NEG all NEG seize this word
'Not everyone grasp this word' [*Tatian*, Matt. 19:11; around 830]

Jäger (2008, 92–103) analyzes *nalles* as a negative focus particle, primarily used in NEG-stripping, as in (426), and in contrastive coordination.

(426) Old High German (Jäger 2008, 101)
huuanda ir uns uuard chiboran, nalles imu selbemu
because he us was born NEG him self
'because he was born for us, not for himself'

It is plausible that this negative particle could also occur in NEG-stripping constructions like (427), as languages often allow greater flexibility for constituent negation in ellipsis contexts (see §9.2). However, this does not imply that *nalles* could appear in an independent clause as part of a B construction. Without further examples similar to (425) from other sources, it remains uncertain whether Old High German genuinely employed a B construction.

(427) Old High German
Inti ir birut subre, nalles alle.
and you are clean not all
'And you are clean, though not all of you' [*Tatian*, 155.21-22, around 830]

Strategy C Strategy C is attested in several texts. In Old English, the *Wessex Gospels*, composed around 990 in the West Saxon dialect, include three examples of ‘Not all X are Y’ sentences, all of which employ Strategy C, as shown in (428). In these instances, the negative clitic *ne* appears clause-initially, directly preceding the finite verb. This word order aligns with the most frequent pattern observed in main negative clauses in Old English (see §10.1.2).

(428) Old English

a. Ne gæð ælch þæra on heofene riche þe cwyþ
NEG go.PRS.3SG every those.GEN into heaven.GEN.SG kingdom that say.PRS.3SG
to me drihten drihten
to 1SG.DAT Lord Lord
‘Not everyone who says to me, “Lord, Lord,” will enter the kingdom of heaven’
[*Wessex Gospels*, Matt. 7:21; about 990]

b. Ne under-foð ealle men þis word
NEG receive.PRS.3PL all men this word
‘Not everyone receives this word’ [ibid, Matt. 19:11]

c. ne sende ge ealle clæne.
NEG be.PRS.PL you all clean
‘You are not all clean.’ [ibid, John 13:11]

Strategy C is also attested in Old Saxon in the *Heliand*, a gospel harmony composed between 830–840 in alliterative verse. This text contains only one relevant instance of a ‘Not all X are Y’ sentence, which employs Strategy C. Similar to Old English, V1 word order is the most frequent pattern in main negative clauses in the *Heliand* (Breitbarth and Jäger 2018, 148). Consequently, the use of Strategy C in this context aligns with the prevailing syntactic tendencies of Old Saxon.

(429) Old Saxon

Ne cumat thea alle te himile, thea
NEG come.PRS.3PL those.NOM.PL.M all.NOM.PL.M to heaven.DAT those.NOM.PL.M
[the] hîr hrôpat te mi [manno] te mundburd.
who here call.PRS.3PL to me.ACC men.GEN.PL to protection.ACC
‘Not all will reach Heaven who here hail me Protector, their Master and Lord.’
[*Heliand*, chap. 23, lines 1915-1916; around 830-840]

In contrast, no instances of Strategy C are found in Old High German translations of the New Testament from this period. However, an example of Strategy C does appear in Notker’s translation of *Aristotelis Categoriae* (430b), which is based on Boethius’s Latin version (430a). Notably, Notker diverges from the Latin source, which employs Strategy B, by using Strategy C instead.

(430) a. Latin

Non	ergo	omnia	qualia	recipiunt	magis	et	minus.
NEG	therefore	all	qualities	receive	more	and	less

‘Therefore, not all qualities receive more and less.’

[Boethius, *Aristotelis Categoriae*]

b. Old High German

Fóne	díu	ne-sínt	nîeht	álliu	qualia	mêr	únde	mín.
from	this	NEG-are	NEG	all	qualities	more	and	less

‘Therefore, not all qualities receive more and less.’

[Notker, *Aristotelis Categoriae*]

Summary Old English, Old Saxon, and Old High German appear to have lacked Strategy B, apart from the potentially anomalous case of the Tatian translation into Old High German. Consequently, the choice of negation strategies in these languages was between Strategy A (SV word order) and Strategy C (VS word order).

The choice between Strategies C and A appears to be influenced by general word order patterns in negative clauses. The relatively lower frequency of Strategy C in Old High German compared to Old English and Old Saxon may reflect broader differences in word order. As discussed in §10.1.2, negative clauses in Old High German are only weakly associated with V1 word order. Consequently, Strategy A seems to have been more prevalent than Strategy C in Late Old High German.

The alternation between Strategies A and C seems to reflect broader constraints on word order rather than a distinct syntactic treatment of ‘Not all X are Y’ sentences. There is no evidence to suggest that these sentences diverged in word order from other negative clauses where the subject is not a scope-bearing element.

The available evidence suggests that Strategy C might have been the older pattern, while Strategy A rose in prominence as word order in main negative clauses began to shift. As the NEG-V1 pattern declined, Strategy A became increasingly common, aligning more closely with the emerging preference for V2 or SVO word order. This diachronic trajectory appears to be supported by variation across the West Germanic languages: in Old High German, where NEG-V1 word order was already relatively rare, Strategy A seems to have been more prevalent than in Old English or Old Saxon, although the evidence remains limited.

By the Middle Germanic period, V1 clauses had become increasingly rare, eventually all but disappearing. This transition and its implications will be discussed further in the next subsection.

10.1.4 Middle Germanic

Corpus examples from this period include New Testament translations into Middle English, Middle High German, Middle Low German, and Middle Dutch, as well as examples from other literary sources. Additionally, a few relevant examples are found in Icelandic and Old Norse sagas.

Figure 10.1 shows the distribution of strategies in Middle Germanic translations of the New Testament, defined here as covering the years 1000–1500 for comparability, even though this does not align precisely with the conventional periodization of each language.[5]

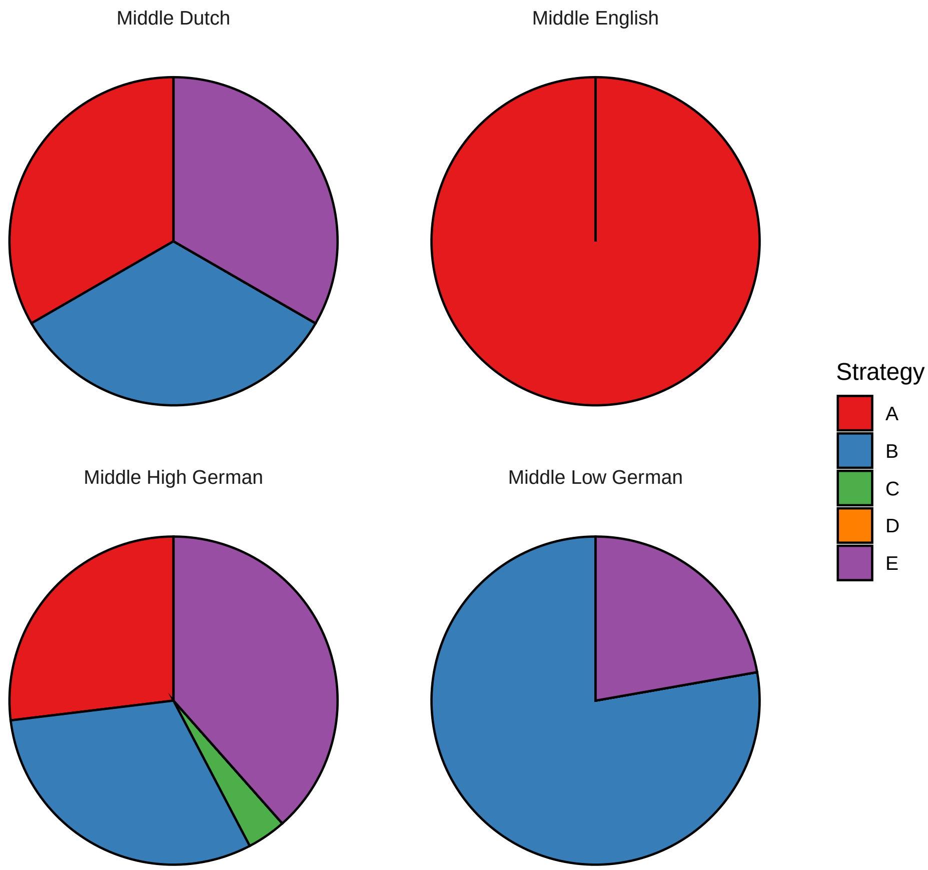


Figure 10.1: The distribution of strategies in translations of the NT into Middle Germanic languages (11th-15th centuries)

During this period, Strategy C nearly disappears, while Strategy A becomes more prominent. However, Strategy A is no longer attested in Middle High German and Middle Low German translations from the latter half of the 15th century.

An important innovation of this period is the emergence of Strategy B, first appearing around the 13th–14th centuries. This strategy involves negative particles developed through Jespersen's Cycle (see §10.1.2) and is also attested in Middle English in other sources.

Another major difference observed between Old and Middle Germanic languages is the gradual disappearance of Strategy C, driven by the decline of the NEG-V1 pattern.

[5]The quantitative data for English exclude John Wycliffe's 14th-century translation. As discussed in §5.2, this translation differs significantly from other contemporary translations in the Western tradition due to its strong dependence on Latin word order, which compromises its representativeness.

Strategy A With the decline of Strategy C, Strategy A emerges as the dominant strategy in Middle English.[6] The earliest examples of Strategy A in Middle English that I have identified date to the 13th century. One such example, (431a), appears in *Ancrene Wisse*, a guide for anchoresses (a form of Christian monasticism) composed in the early 13th century.[7] Additional instances can be found in the *South English Legendary* (431b)-(431d), a collection of saints' lives likely written in the latter half of the 13th century.

(431) Middle English

a. Ah alle ne mahe nawt halden a riwle
but all NEG can.PRS.PL NEG obey.PTCP.PST one rule
'But not everyone can observe a single rule, [and they need not and should not observe the outer rule in the same way.]' [*Ancrene Wisse*, early 13th century]

b. Al ore couent n-is nouȝt here
all our covenant NEG-is NEG here
'Not all our covenant is here'
[*South English Legendary: Life of St. Brendan*, late 13th century]

c. þat euerech ne mai nouȝt at is feste : ane day habbe, i-wis.
that each NEG may NEG at his fest one day have indeed
'[One reason for celebrating All Saints' Day is] that not every saint can have their own feast day, indeed.'
[*South English Legendary: All Saints' Day*, late 13th century]

d. ase ech ne may nouȝth allone þe [*al.* ane] feste habbe in þe ȝere
as each NEG may NEG allone the a feast have in the year
'as not every one may have their own feast day in the year.'
[ibid]

By the 14th century, if not earlier, Strategy A appears to have become the dominant strategy in Middle English. This is illustrated by two representative examples: one from Robert Manning's *The Story of England* (432a), and another from Geoffrey Chaucer's *The Canterbury Tales* (432b).[8]

(432) Middle English

a. **Al ys nought soþ**, ne nought al lye,
Ne al wysdam, ne al folye
'[Many wonders are told of King Arthur…] Not all is true, nor all lie, nor all

[6]Middle English data were extracted from the Corpus of Middle English Prose and Verse (McSparran et al. 2006).

[7]This example features a *same*-type predicate (see §2.3), which limits its usefulness as standalone evidence for the use of Strategy A.

[8]Note that when *ne* appears before the QP, it functions as a negative conjunction ('nor') rather than as a marker of constituent negation. For example, in (432b), the first occurrence of *ne* in *Ne every appul that is fair at eye ne is nat good* serves as a conjunction, as reflected in the Modern English translation.

wisdom, nor all folly'

[Manning, *De Story of Englande*, lines 10,587-10,588; 1338]

b. **But al thyng which that shineth as the gold**
Nis nat gold, as that I have herd it told;
Ne every appul that is fair at eye
Ne is nat good, what so men clappe or crye.
'But everything that glitters is not gold, as often I have heard the saying told; not every apple pleasing to the eye is good, though men may praise it to the sky.'

[Chaucer, *The Canterbury Tales*, "The Canon's Yeoman's Tale," lines 962-965; 1387–1400]

Additional examples of Strategy A are found in New Testament translations from this period, including two verses cited in *Life of Soul* (433a)-(433b) and an anonymous 14th-century translation (433c), only part of which survives.

(433) Middle English

a. al þilke þat seyȝen to me lord lord ne schul not entren in to þe
all those that say.PRS.PL to me Lord Lord NEG shall NEG enter.INF in to the
kyngdome of heuene
kingdom of heaven
'Not everyone who says to me, "Lord, Lord," will enter into the kingdom of heaven' [*Life of Soul*; about 1400]

b. alle men ne take not þis word
'Not everyone can accept this teaching' [ibid]

c. Alle þinges beþ lefful to me, bote **alle þinges ne beþ noȝt spedful to me**.
'All things are lawful for me, but not all things are beneficial.'
[*A Fourteenth Century English Biblical Version*, 1 Cor. 6:12]

In addition to Middle English, Strategy A is also attested in the Middle Dutch *Liège Harmony* (434), a gospel harmony composed around 1250 (Gow 2012, 207).

(434) Middle Dutch
Alle de liede en weten den sin van desen warden nit
all the people NEG know the meaning of these word NEG
'Not everyone can accept this teaching' [*Liège Harmony*, around 1250]

Strategy A is also employed several times in the *Augsburg Bible* (1350), a Middle High German translation, as exemplified in (435a). The *Mentelin Bible*, the first printed Bible in German, also uses Strategy A (435b). Although published in 1466, the *Mentelin Bible* is based on a 14th-century translation, likely originating from Nuremberg. While this translation is considered poor in style and accuracy (Gow 2012, 210), it is not dependent on the Latin word order in the relevant verses.

(435) Middle High German

a. Alle ding mir zimlich sint, aber alle ding nicht fugent.
all things to.me befitting are but all things NEG are.suitable
'All things are lawful for me, but not all things are beneficial.'
[*Augsburg Bible*, 1 Cor. 6:12, 1350]

b. All fachent sy nit das wort
all grasp they NEG this word
'Not everyone grasps this word' [*Mentelin Bible*, Matt. 19:11; 1466]

By contrast, 15th-century translations into High German (*Ottheinrich Bible*, 1430) and Low German (*Lübeck Bible*, 1494) do not employ Strategy A, except in the excluded verses (see §5.5). While this does not necessarily indicate that Strategy A had completely disappeared from these languages by this period, it suggests that its decline occurred considerably earlier than in English and Dutch.

Strategy B The earliest instance of Strategy B in Germanic languages that I have identified appears in the Middle Dutch *Liège Harmony* (436). Notably, this example involves negative concord, in contrast to the Middle English examples discussed below.

(436) Middle Dutch
Nit alle die mi heeten Here, Here, en selen comen te hemelrike
NEG all who me call Lord Lord NEG will come to kingdom.of.heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Liège Harmony*, around 1250]

In Middle English, instances of Strategy B can be traced back to at least the second half of the 14th century. Interestingly, the majority of examples I identified appear in the writings of John Wycliffe (437) or a treatise attributed to him (438), although some examples are found in other works as well, such as *The Testament of Love* (439). This concentration of examples in a limited set of texts might indicate that Strategy B was not as widely conventionalized as Strategy A, which is employed by a broader range of authors.

(437) Middle English

a. For ellis þe Fadir hadde þis Goost, and Crist hadde not þis same Goost, and so not al þat þe Fadir haþ had Crist as verre God.
'For otherwise the Father would have this Spirit, and Christ would not have this same Spirit, and so **not all that the Father has would Christ have as true God**.' [*Select English Works of John Wyclif*, late 14th century]

b. And so not al þat ben fed þus shal come to þe blisse of hevene.
'And thus, not all who are fed in this way shall come to the bliss of heaven.' [ibid]

(438) Middle English

a. Not alle prestis ar had for prelats, for þe name makiþ not þe bischop, but þe lif.
'Not all priests are considered prelates, for the title does not make the bishop, but the way of life.' [*An apology for Lollard Doctrines*, late 14th century]

b. Noght alle bischoppis in name ar bischoppis in dede
'Not all who are bishops by title are truly bishops' [ibid]

(439) Middle English

Hereby may (quod she) lightly ben knowe that **not al thinges to be, is of necessitè**, though god have hem in his prescience. For som thinges to be, is of libertè of wil.
'By this (said she), it can be easily known that not all things are necessarily destined to happen, even though God has foreknowledge of them, for some things occur through the freedom of will.' [*The Testament of Love*, mid to late 14 century]

In Middle High German, the *Augsburg Bible* from 1350 uses Strategy B only once in (440a). However, this strategy becomes more frequent in 15th-century translations. For example, the *Ottheinrich Bible* from 1430, which is largely based on the *Augsburg Bible*, adopts Strategy B in (440b), replacing the use of Strategy A in the corresponding verse of the *Augsburg Bible* (435a). Notably, from the 15th century onward, clear instances of Strategy A are absent in High German translations, marking a significant shift toward the use of Strategy B.

(440) Middle High German

a. wann nicht alle die uz israhel daz die sint israhelite
for NEG all those from Israel that those are Israelites
'For not all Israelites truly belong to Israel.' [*Augsburg Bible*, Rom. 9:6; 1350]

b. Alle ding mir zimlich sind: aber nicht alle ding fürgent
all things to.me befitting are but NEG all things are.suitable
'All things are lawful for me, but not all things are beneficial.'
[*Ottheinrich Bible*, 1 Cor. 6:12, 1430]

Strategy C Despite the decline of the NEG-V1 pattern, instances of Strategy C persist in Early Middle English poetry, as shown in (441).

(441) Middle English

a. Neren hii alle noht cnihtes
NEG.were they all NEG knights
'They were not all knights' [*Layamon's Brut*, around 1190-1215]

b. Nis noht al soþ ne al les; þat many men seggeþ
NEG.is NEG all sooth nor all lies that many men say
'[It is] not all sooth nor all falsehood that minstrels sing (lit. 'many men say')]'
[ibid]

c. Ne beoþ nowt ones alle sunne
NEG are NEG of.one.kind all sins
'Not all sins are equal, [as they are of two types: one arises from the desire of the flesh, the other from the disposition of the spirit.]'
[*The Owl and the Nightingale*, 12-13th century]

An isolated instance of Strategy C also appears in Middle High German (442). Notice, however, that this example involves V1 word order without the pre-verbal negative clitic, unlike the Middle English examples.

(442) Middle High German
bevahent niht alle ditz wort
grasp NEG all this word
'Not everyone grasp this word' [*Augsburg Bible*, Matt. 19:11; 1350]

While the West Germanic languages were losing Strategy C during this period, Icelandic innovated a distinct Strategy C construction. In example (443), from the *Bandamanna saga* (13th–14th century), the negative particle *eigi* is fronted to clause-initial position and followed by the verb. Unlike the Strategy C constructions found in the Old West Germanic languages, where the negative marker was the pre-verbal clitic inherited from Proto-Germanic, Icelandic employs the new negative particle that emerged through Jespersen's Cycle. As a result, Strategy C in Icelandic follows V2 word order, whereas in the Old West Germanic languages, Strategy C typically relied on V1 word order. This type of construction later appears in other Scandinavian languages and even in some West Germanic languages. However, among the languages examined in this study, it remains most common in Icelandic.

(443) Icelandic
Eigi er enn öllu skemmt frændi.
NEG is still all ruined kinsman
'All is not lost yet, kinsman' [*Bandamanna saga*, chap. 11; 13th–14th century]

Another type of Strategy C construction found in Icelandic features XVS word order, where X is a clause-initial constituent other than the subject or the negative particle. For example, in (444), the adverb *þá* 'then' appears in the initial position, followed by the verb in the second position, adhering to the V2 word order pattern. Unlike in (443), the negative particle remains in its usual post-verbal position in (444).

(444) Icelandic
Þá voru enn eigi komnir allir þeir sem henni höfðu liði heitið.
then were still NEG come all those who her had help promised
'but not all those had come yet who had promised aid to Asdis'
[*Grettis saga*, chap. 83; 14th century]

Strategy E Strategy E is relatively frequent in this period. It is often employed when the subject is a personal pronoun, as exemplified in (445), or a definite noun phrase. Notably, these contexts remain common for Strategy E in modern Germanic languages.

(445) Middle Dutch
Ghi-ne sijt nit suuer alle.
you-NEG are NEG clean all
'Not all of you are clean.' [*Liège Harmony*, around 1250]

However, Strategy E is also attested with expletive subject pronouns, as demonstrated by the examples in (446) from Middle English. These instances involve an extraposed relative clause, where the subject pronoun can be interpreted as either expletive or cataphoric. It is important to note that these are not *it*-clefts, as the lexical predicate—semantically the scope of the universal quantifier—appears in the main clause rather than in the dependent clause.

(446) Middle English
a. Hit is not al gold, that glareth.
'Not all that glitters is gold.' [*The House of Fame*, line 272; 1374-1385]
b. Hit is not al golde that Shynyth as golde
'Not all that shines like gold is gold'
[*Three prose versions of the Secreta Secretorum*; 1338]
c. It is nat al sothe thynge that semeth
'Not all that seems [true] is true'
[*The Romaunt of the Rose*, line 7544; mid or late 14th century]

Strategy E is also attested in North Germanic languages. For instance, (447) provides an example from an Old Norse saga dating to the late 13th century.

(447) Old Norse
at Völsungar væri eigi allir dauðir.
that Völsungs were NEG all dead
'[but now will I away from the land and go meet the sons of Hunding, and do them to wit] that the Volsungs are not all dead.'
[*Völsunga saga*, chap. 17; late 13th century]

Summary Strategy C largely fell out of use during this period in the West Germanic languages, with Strategies A and E emerging as the dominant strategies. However, Icelandic stands out for innovating a distinct Strategy C construction. It remains unclear whether similar developments occurred in other Scandinavian languages due to a lack of sufficient data.

Meanwhile, Strategy B began to appear across the Germanic languages during this period but did not yet become fully conventionalized. This is evidenced by its uneven distribution across texts and its complete absence from some translations. These patterns suggest that while Strategy B was emerging as a viable option, it had not yet achieved widespread or systematic use.

10.1.5 Early Modern Germanic

Figure 10.2 illustrates the distribution of strategies used in translations of the New Testament into Early Modern Germanic languages, defined here as spanning the years 1500–1700 for the sake of comparability.

Strategy A continues to play a significant role during this period, with the notable exceptions of German and Icelandic. In contrast, Strategy B, despite its introduction several centuries earlier, has not achieved widespread adoption and is either absent or only sporadically attested in certain languages.

Strategy E appears frequently in translations from this era, which may be partially attributed to the influence of Martin Luther's seminal translation into Early New High German, where this strategy is particularly prominent. A similar influence is evident in the use of Strategy C, as will be discussed shortly.

Strategy A Strategy A remains the dominant strategy in English. The first complete translation of the New Testament into Early Modern English, produced by William Tyndale in 1526, was highly influential on subsequent translations, including the King James Version from 1611. Tyndale's translation demonstrates a clear preference for Strategy A (448a), a preference that is also evident in the King James Version (448b).

(448) Early Modern English

a. He sayde unto them: all men cannot awaye with that saynge: but they to whom it is geven. [Tyndale, Matt. 19:11; 1526]

b. But hee said vnto them, All men cannot receiue this saying, saue they to whom it is giuen. [KJV, Matt. 19:11; 1611]

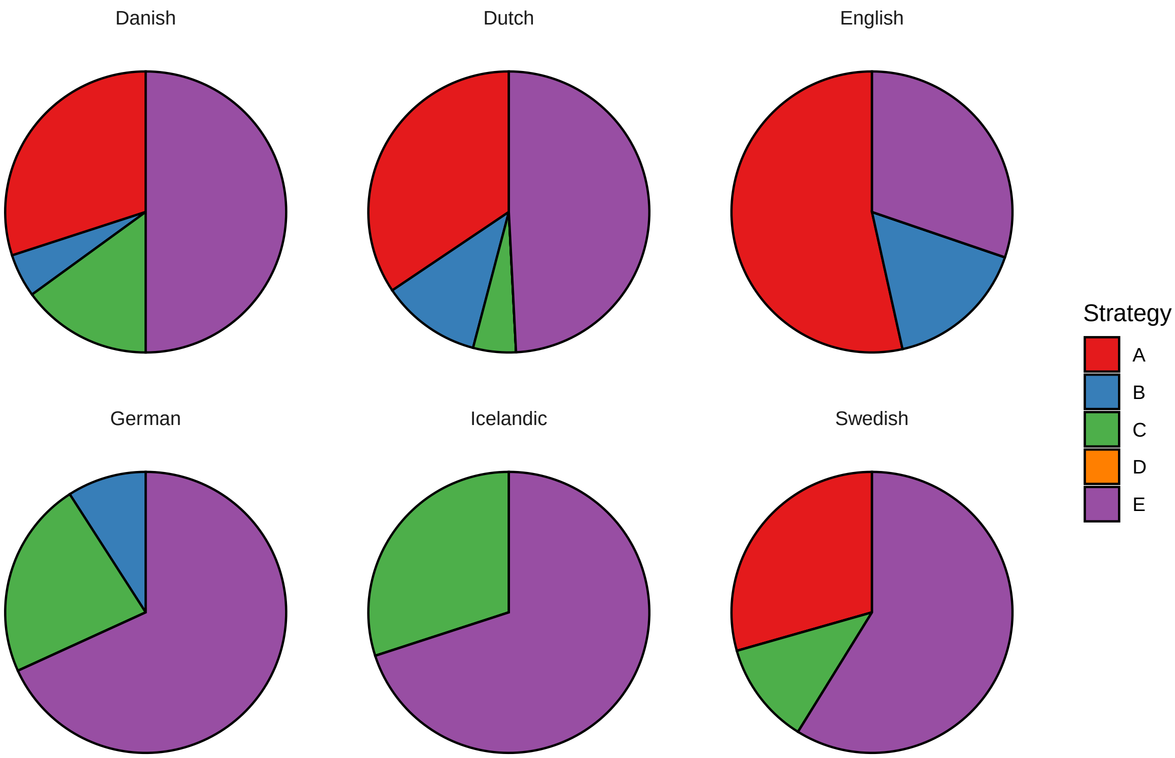


Figure 10.2: The distribution of strategies in translations of the NT into Early Modern Germanic languages (16th-17th centuries)

Strategy A is also prevalent in Dutch, as illustrated by representative examples from the *Biestkens Bible* from 1560 and the official *Statenvertaling* from 1637 in (449a) and (449b), respectively.

(449) Early Modern Dutch

a. Maer alle man en heeft de wetenheyt niet.
but all man NEG has the knowledge NEG
'But not everyone possesses this knowledge.' [*Biestkens Bible*, 1 Cor. 8:7; 1560]

b. Alle en vatten dit woort niet
all NEG understand this word NEG
'Not everyone understands this word' [*Statenvertaling*, Matt. 19:11; 1637]

Danish and Swedish also make frequent use of Strategy A, as demonstrated in (450) and (451), respectively. The Danish *Bible of Christian III* (1550), while heavily influenced by

Luther's translation, diverges from it by employing Strategy A, suggesting that this strategy was established in Danish.

(450) Danish
Men huer Mand haffuer icke denne Visdom
but every man has NEG this wisdom
'But not everyone possesses this knowledge.' [*Bible of Christian III*, 1 Cor. 8:7; 1550]

(451) Swedish
Hwar man taghr̃ icke thetta oordhit
every man takes NEG this word
'Not everyone takes this word' [*Gustav Vasa Bible*, Matt. 19:11; 1526]

In contrast, Strategy A is completely absent in Martin Luther's 1522 translation into Early New High German and in the Icelandic *Guðbrandsbiblía* from 1584. While earlier German translations attest to the use of this strategy, the limited data available for Icelandic makes it unclear whether the strategy had been lost in Icelandic or whether it was never widely used in the language to begin with.

Strategy B Strategy B tends to function as a secondary strategy during this period. In English, this strategy appears only once in Tyndale's translation, specifically in the rendering of Matthew 7:21 (452a). This verse is frequently associated with Strategy B cross-linguistically, possibly due to its heavy subject (see §5.5). Later translations, such as the *Geneva Bible* from 1560 and the *King James Version* from 1611, also employ Strategy B solely in this verse (452b), following Tyndale's precedent.

In contrast, the *Douay–Rheims Bible*, a Catholic translation from 1582, uses Strategy B more frequently, almost as often as Strategy A, as demonstrated in (452c). This stands in contrast to Tyndale's and the KJV's treatments of the same verse (448a)-(448b).

(452) Early Modern English
a. Not all they thatt say unto me, Master, Master, shall enter in to the kyngdome off heven [Tyndale, Matt. 7:21; 1526]
b. Not euery one that saith vnto me, Lord, Lord, shall enter into the kingdome of heauen: but he that doth the will of my father which is in heauen. [KJV, Matt. 7:21; 1611]
c. Who said to them: Not al take this word, but they to whom it is giuen. [*Douay–Rheims Bible*, Matt. 19:11; 1582]

The *Douay–Rheims Bible*, unlike Protestant translations based on the Greek text, derives from the Latin *Vulgate*. However, the *Vulgate* closely follows Greek word order in the relevant verses (see §5.5), so the differences in strategy choices cannot be attributed to its influence. It is possible that Strategy B had not yet become fully conventionalized, leading to a less consistent application across writers during this period.

Similarly to English, Strategy B is only sporadically attested in Dutch. Some translations avoid it entirely, such as the *Leuvense Bijbel* from 1548 and the *Biestkens Bijbel* from 1560. Others use it exclusively in Matthew 7:21, including the *Liesvelt Bijbel* from 1526 [1542], the *Deux-Aes Bijbel* from 1562, and the *Statenvertaling* from 1637. The only exception is the *De Vorsterman Bijbel* from 1528 [1531], which employs Strategy B multiple times. This strategy typically involved negative concord, as in (453a), but sometimes only the negative particle *niet* appears, as in (453b).

(453) Early Modern Dutch

a. Niet alle dye tot mi segghen, HERE HERE, en sullen int rijck der hemelen comen
NEG all those to me say Lord Lord NEG will in.the kingdom of.the heaven come
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Liesvelt Bijbel*, Matt. 7:21; 1526 [1542]]

b. Niet een yeghelick die tot my seyt: Heere, Heere, sal int Koninckrijcke der Hemelen komen
NEG one every those to me say Lord Lord will in.the kingdom of.the heaven come
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Deux-Aes Bijbel*, Matt. 7:21; 1562]

Strategy B is notably absent in Martin Luther's 1522 translation into Early New High German, despite its attestation in earlier High German translations. This raises the question of whether its omission reflects dialectal variation or a deliberate stylistic choice by Luther.

One possibility is that Luther's dialect did not naturally permit the use of Strategy B. Support for this interpretation may come from a comparison with Johann Eck's 1537 translation, a Catholic response to Luther's work. While Eck's translation generally follows Luther in the relevant verses, it includes two instances of Strategy B, one of which is illustrated in (454). Eck's translation reflects linguistic features characteristic of the Bavarian dialect, whereas Luther's translation was primarily influenced by the Upper Saxon dialect and the Saxon Chancery language (Leppin 2024). This contrast suggests that the absence of Strategy B in Luther's translation may be tied to regional variation within Early New High German.

(454) Early New High German
Nit ain jeglicher der zu mir sagt: Herr Herr würdt eingan in das himelreich
NEG one any who to me say Lord Lord will enter in the kingdom.of.heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Eck, Matt. 7:21; 1537]

Alternatively, Strategy B may have been available in Luther's dialect, but he chose not to use it for reasons that remain unclear. It is possible that he considered it unsuitable or inconsistent with his stylistic preferences, though the rationale for this choice cannot be determined. Regardless of the explanation, the absence of Strategy B in Luther's translation highlights that the strategy was not yet fully conventionalized at the time.

Translations into Scandinavian languages also largely avoid Strategy B, with only a single instance attested in one of the Danish translations (455).

(455) Danish
Thi icke alle som ere aff Israel disse ere Israel
because NEG all who be.PRS of Israel these be.PRS Israel
'For not all who are from Israel, these are Israel.' [Resens, Rom: 9:6; 1607]

Strategy C Strategy C is attested in all languages examined except English, which had lost V2 word order. In some instances, the negative particle is fronted to clause-initial position, enabling VS word order. This construction had already been attested in Icelandic a few centuries earlier (443) and is documented during this period in Icelandic (456), Swedish (457), and even German (458). Some contemporary translations into Dutch and Danish follow Luther in this specific verse, but this construction does not appear elsewhere in these languages, in contrast to its wider use in Icelandic and Swedish.

(456) Icelandic
Allt leyfist mér, en eigi batar það mig allt.
everything is.permitted to.me but NEG benefits it me everything
'All things are lawful for me, but not all things are beneficial.'
[*Guðbrandsbiblía*, 1 Cor. 6:12; 1584]

(457) Swedish
Icke wardher hwar och en kommande in j himmelrike som sägher till mich/ Herre, Herre/
NEG becomes everyone coming into kingdom.of.heaven who says to me Lord Lord
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Gustav Vasa Bible*, Matt. 7:21; 1526]

(458) Early New High German
Nicht ist alles fleysch eynerley fleysch
NEG is all flesh same flesh
'Not all flesh is alike' [Luther, 1 Cor. 15:39; 1522]

A more common variation of Strategy C involves XVS word order, where a constituent other than the negative particle is fronted to the clause-initial position. This is exemplified in German (459).

(459) Early New High German
das wortt fasset nit yderman
the.ACC.SG.N word.ACC.SG grasp.PRS.3SG NEG everyone
'Not everyone grasps this word' [Luther, Matt. 19:11; 1522]

Strategy E Strategy E is particularly frequent during this period. In English, it often appears in a floating quantifier construction, as illustrated in (460a), and occasionally involves an extraposed relative clause, as shown in (460b).

(460) Early Modern English

a. But they have nott all obeyed to the gospell. [Tyndale, Rom. 10:16; 1526]

b. For they are not all israhelites which cam off Israel [ibid, Rom. 9:6]

At least in German, the prevalence of Strategy E is not solely driven by an effort to avoid inverse scope; it is also influenced by the frequent use of floating quantifiers in affirmative sentences. In a corpus of Middle High German, Fuß and Hinterhölzl (2023) identified 38 instances of a nominative *alles* 'all, everything' co-occurring with the expletive *es*, as illustrated in (461), compared to only two occurrences of *alles* in clause-initial subject position. This widespread use of floating quantifiers likely reduced the need for Strategies A and B, both of which rely on pronominal quantifiers in subject position.

(461) Middle High German (Fuß and Hinterhölzl 2023)
Iz was allez uerloren
it was all lost
'All was lost'

Luther's translation also features expletive inversion constructions, a distinct type of Strategy E (see §7.5.2), as illustrated in (462). In this construction, the verb agrees with the low subject rather than the expletive pronoun.

(462) Early New High German

a. Es werden nicht alle/ die zu mir sagen/ Herr herr/ yñ das hymel reych komenn
it will.3PL NEG all who to me say.3PL Lord Lord into the heaven kingdom come.INF
'Not everyone who says to me, "Lord, Lord," will enter into the kingdom of heaven' [Luther, Matt. 7:21; 1522]

b. Es hat aber nicht yderman das wissen
it have.3SG but NEG everyone the.ACC.SG.N knowledge
'But not everyone has the knowledge.' [ibid, 1 Cor. 8:7]

In Dutch, a superficially similar construction occasionally occurs, involving a non-expletive pronoun that can be analyzed as cataphoric in examples like (463a). Additionally, there is an instance that could be interpreted as using a generic pronoun (463b), but this construction does not appear to have become conventionalized.

(463) Early Modern Dutch

a. Sij en sullen alle int rijc der hemelen niet comen die my segghen Heere Heere
they NEG will.3PL all into kingdom of.the heaven NEG come.INF who to.me say.3PL Lord Lord
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven' [*Leuvense Bijbel*, Matt. 7:21; 1548]

b. Sij en vaten alle dit woert niet
they NEG understand.3PL all this word NEG
'Not everyone understands this word' [ibid, Matt. 19:11]

In Icelandic, both an expletive inversion construction (464a) and a construction with a non-expletive pronoun, similar to those found in Dutch (464b), are attested.

(464) Icelandic

a. Því það eru eigi allt Íraelsmenn sem af Írael eru
because it be.PRS.3PL NEG all.NOM.SG.N Israelites who of Israel be.PRS.3PL
'Because not all Israelites are of Israel' [*Guðbrandsbiblía*, Rom. 9:6; 1584]

b. Þeir munu eigi allir, sem til mín segja: Herra, herra, - innganga í Guðs ríki
they will.3PL NEG all.NOM.PL.M who to me say.PRS.3PL Lord Lord enter.INF into God's kingdom
'Not everyone who says to me, "Lord, Lord," will enter God's kingdom' [ibid, Matt. 7:21]

Summary Strategy A remains a major strategy during this period, with the notable exceptions of German and Icelandic, where it is unattested. Strategy E is even more frequently attested and appears to have gained significant prominence, likely influenced in part by the significant impact of Luther's translation, which favored this strategy.

Strategies B and C are also attested but function primarily as secondary strategies. Strategy B, though present in certain translations, tends to appear sporadically in specific verses. Strategy C, on the other hand, is associated with specific constructions, such as XVS or negative-fronting structures, and its use varies across languages.

Overall, this period reflects a continuation of trends observed in Middle Germanic languages. The apparent greater variability may be attributed more to the size and diversity of the corpus than to any substantive linguistic change.

10.1.6 Late Modern Germanic

Overview Figures 10.3 and 10.4 illustrate the distribution of strategies in New Testament translations into Late Modern Germanic varieties, divided into two periods. It is important to note that some languages are represented only in the earlier period, which should be considered when interpreting trends across these time frames.

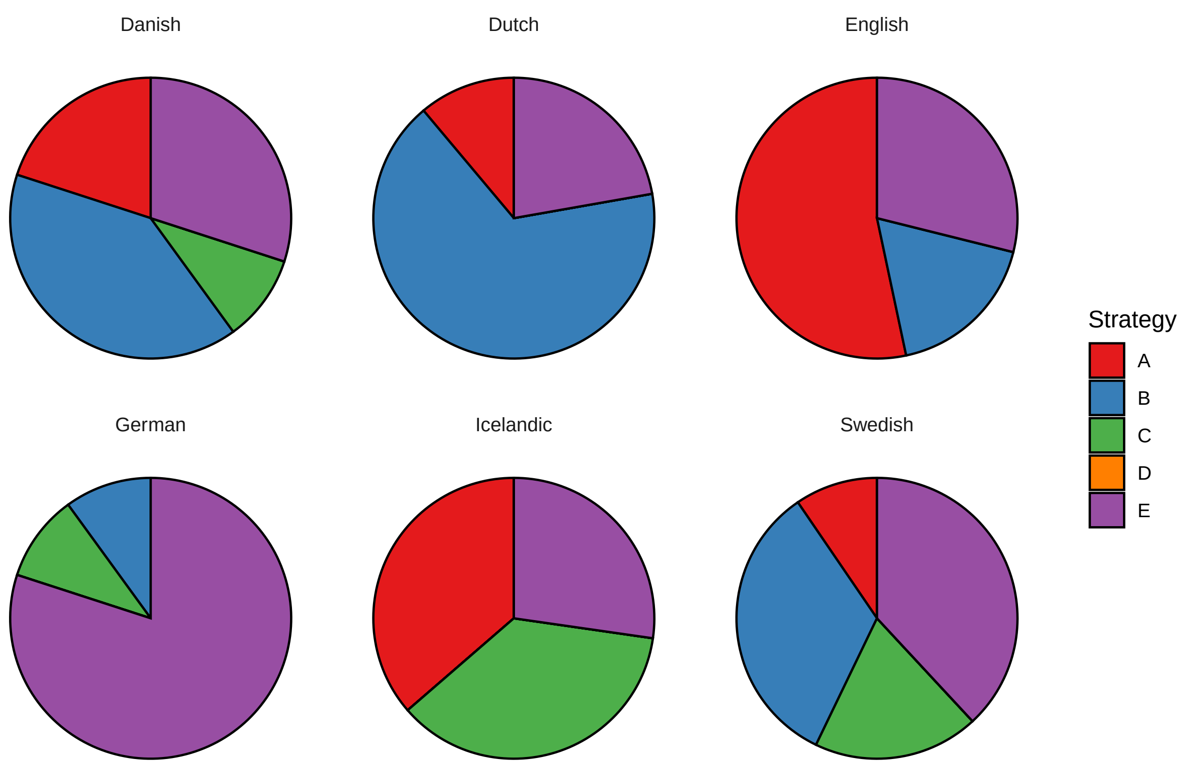


Figure 10.3: The distribution of strategies in translations of the NT into Late Modern Germanic languages (18th-19th centuries)

A more comprehensive view of the current state of these languages is provided in Figure 10.5, which displays the distribution of strategies in corpora of online news from the past decade. For most languages, a corpus of one million sentences from the online news section of the Leipzig Corpora Collection was analyzed, yielding approximately 500–1,000 instances of 'Not all X are Y' sentences per language. Further details about the corpus and the methodology can be found in Chapter 6.

The 18th and 19th centuries saw an increase in the frequency of Strategy B and a decline of Strategy A in Danish, Dutch, Swedish, and likely German. In contrast, Strategy A

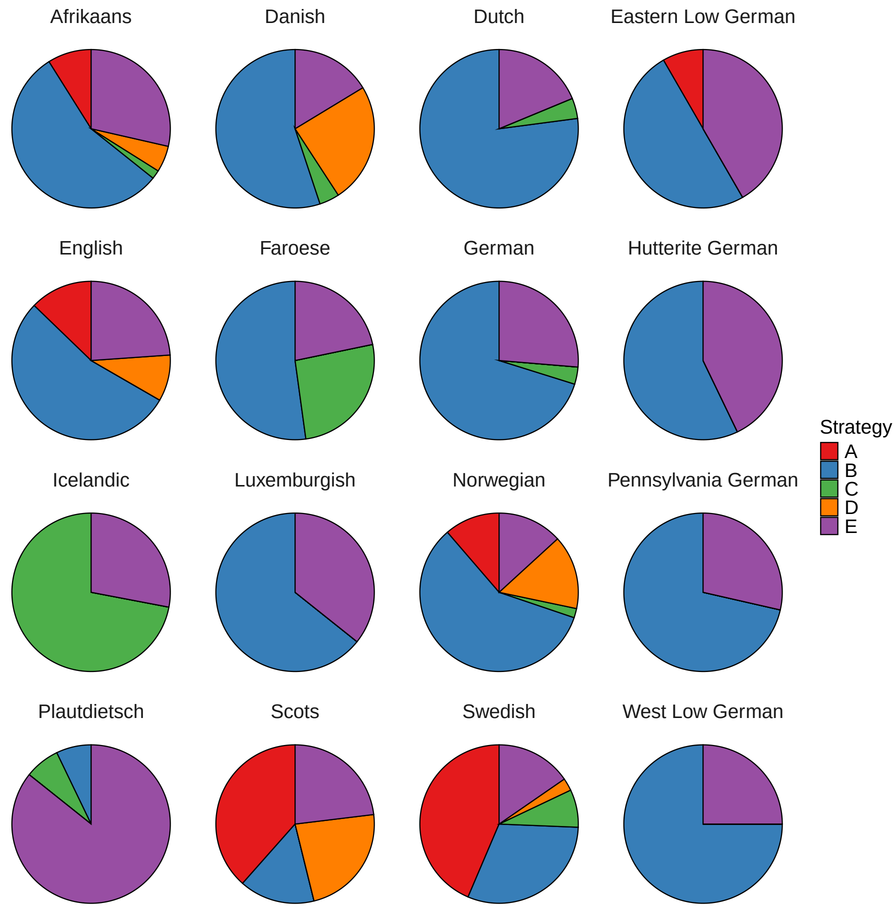


Figure 10.4: The distribution of strategies in translations of the NT into Late Modern Germanic languages (20th-21st centuries)

remained dominant in English, with Strategy B continuing as a secondary strategy. Icelandic exhibited a surprising rise in the frequency of Strategy A, which was not attested in an earlier translation. This shift may be linked to the diachronic instability of Strategy C constructions involving negative fronting, which is also observed later in Faroese.

During the 20th and 21st centuries, Strategy B gradually became the dominant strategy in most, though not all, Germanic languages. Meanwhile, Strategy A declined significantly and is no longer attested in several varieties. Additionally, Strategy D emerged as an important strategy, particularly in Scandinavian languages, as well as in English and Scots.

The main exceptions to these trends are Plautdietsch, Swedish, Faroese, and Icelandic. Scots also appears as an exception in Figure 10.4, but this is misleading because most data points come from a 1904 translation. At that time, Strategy A was still very frequent in English as well.

In Plautdietsch, Strategy E is the dominant strategy, reflecting a distribution similar to that observed in certain earlier varieties of German.

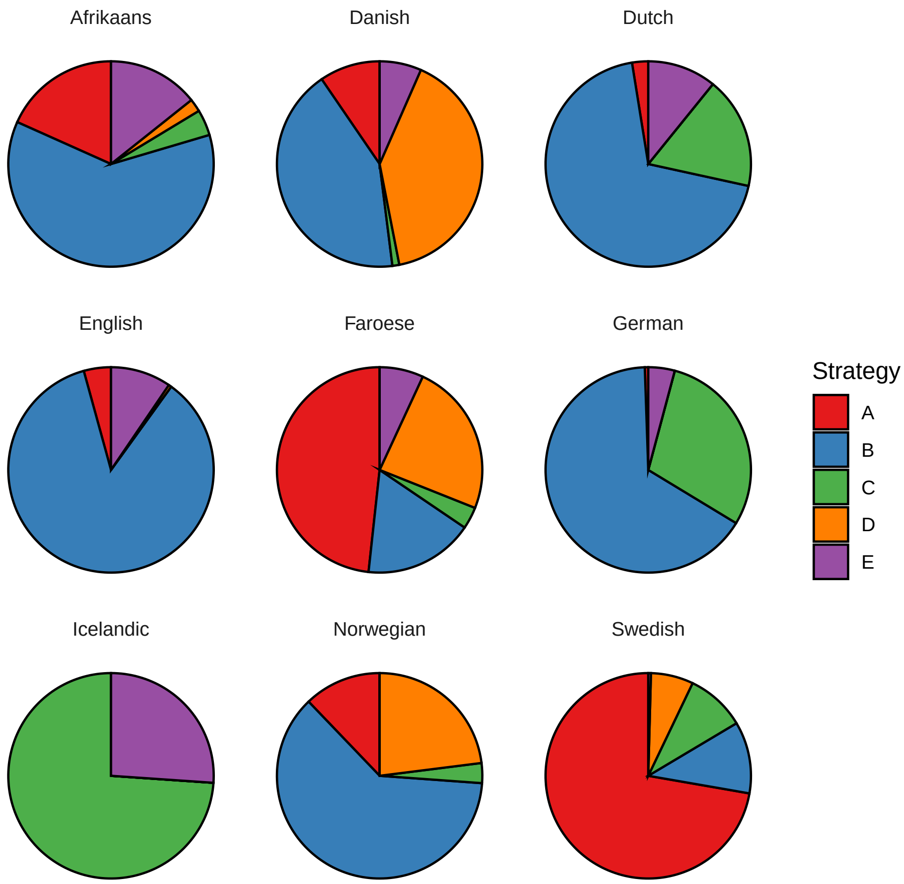


Figure 10.5: The distribution of strategies in the online news section of the Leipzig Corpora Collection for present-day Germanic languages

In Swedish, translations from the late 19th and early 20th centuries predominantly employed Strategy B, aligning with other Mainland Scandinavian languages. However, in contemporary Swedish, Strategy A has become the dominant strategy, while Strategy B has taken a secondary role. This transition may be related to the replacement of the negative particle *icke* with *inte*, which coincided with a significant decline in the frequency of Strategy B.

In Faroese, translations from the first half of the 20th century primarily employed Strategy B and Strategy C, the latter involving the fronting of the negative particle, similar to Icelandic. However, this type of Strategy C became obsolete in later periods, likely contributing to the rise of Strategy A in Faroese.

Finally, in Icelandic, Strategy C constructions with negative fronting were already well-established by the early 20th century and remain the dominant strategy today. Icelandic is unique among the Germanic varieties surveyed here in that it still lacks Strategy B.

Strategy A: Decline During the 18th and 19th centuries, Strategy A was rarely used in Dutch (465) and Danish (466), and it does not appear at all in translations into German or Swedish. In East Low German, Strategy A is attested only once, in an early 20th-century translation (467).

(465) Dutch
Alle dingen zijn mij geoorloofd, maar alle dingen zijn niet nuttig
all things are to.me allowed but all things are NEG beneficial
'All things are lawful for me, but not all things are beneficial.'
[*Voorhoeve New Testament*, 1 Cor. 6:12; 1877]

(466) Danish
Men Alle have ikke den Kundskab.
but all have NEG the knowledge
'But not everyone possesses this knowledge.'
[*Frederik VI's New Testament*, 1 Cor. 8:7; 1819]

(467) East Low German
Oewerst alltauhop hebben s' des Insichten nich.
but all have they the insights NEG
'But not everyone possesses this knowledge.' [Voß, 1 Cor. 8:7; 1929]

In contrast, Strategy A remained the dominant strategy in English throughout the 18th and 19th centuries. For instance, the *Quaker Bible* of 1764, single-handedly produced by the self-taught translator Anthony Purver, predominantly employs Strategy A (468). Similarly, an 1808 translation by Charles Thomson, one of the Founding Fathers of the United States, and the *Living Oracles*, both published in the United States, employ Strategy B only once (both in Matthew 7:21, similar to Tyndale's version (452a) and the KJV (452b)). In other cases, these translations predominantly use Strategy A, as demonstrated in (469) and (470). Notably, the KJV employs Strategy E in Romans 10:16, suggesting that these later translations were not heavily reliant on earlier versions.

(468) But all have not obeyed the Gospel [Purver, Romans 10:16; 1764]

(469) But all have not duly hearkened to the good news [Thomson, Rom. 10:16; 1808]

(470) Nevertheless, all have not obeyed the good tidings
[*Living Oracles*, Rom. 10:16; 1826]

Instances of Strategy A can also be found in other literary sources from this period, as illustrated in (471) and (472).

(471) "Hunters!" repeated Edward — "But why must you have hunters? **Every body does not hunt**." Marianne coloured as she replied, "But most people do."'
[Austen, *Sense and Sensibility* chap. 17, 1811]

(472) a. "All men must die," said a voice quite close at hand; "**but all are not condemned to meet a lingering and premature doom**, such as yours would be if you perished here of want." [Brontë, *Jane Eyre*, chap. 28; 1847]

b. "**All have not your powers**, and it would be folly for the feeble to wish to march with the strong." [ibid, chap. 34]

Around 1900, we begin to see several translations in which Strategy A is used only sporadically or not at all. For example, Strategy A is entirely absent in a 1900 translation of the Pauline Epistles by William Gunion Rutherford. However, other translations from this period continue to use Strategy A relatively frequently, such as the *The Modern Speech New Testament* by Richard Francis Weymouth (473), the *Twentieth Century New Testament* (474) and J. B. Phillips's *New Testament in Modern English* (1958) (475).

(473) But all believers do not recognize these facts. [Weymouth, 1 Cor. 8:7; 1903]

(474) And yet every one did not give heed to the Good News [TCNT, Rom. 10:16; 1904]

(475) Yet all who have heard have not responded to the Gospel. [Phillips, Rom. 10:16; 1958]

Notably, Strategy A is not restricted to British English during this period; it is also attested in American translations, such as (476).

(476) and that I may be delivered from unreasonable and wicked men; for all do not hold the faith. [Montgomery, 2 Thess. 3:2; 1924]

By the late 20th century, Strategy B had clearly become the dominant strategy in English. Most translations from this period onwards avoid Strategy A entirely.

Interestingly, however, Strategy A appears to have been retained in some dialects of English. This is evident in the *Common English Bible* from 2011, which contains several instances of Strategy A (477). This translation, produced by a team of 120 translators primarily from the United States, does not provide enough information to identify the specific dialect represented by these examples. Importantly, however, speakers of Mainstream American English generally find these examples unnatural.

(477) a. But everyone hasn't obeyed the good news. [CEB, Rom. 10:16; 2011]

b. Everything is permitted, but everything isn't beneficial. Everything is permitted, but everything doesn't build others up. [ibid, 1 Cor. 10:23]

c. Pray too that we will be rescued from inappropriate and evil people since everyone that we meet won't respond with faith. [ibid, 2 Thess. 3:2]

Thus, the 20th century appears to have been a transitional period for English, during which Strategy A was largely lost in most dialects but retained in a few.

Similarly, Strategy A experienced a significant decline or even disappeared in other languages, such as Dutch and several varieties of High and Low German. By contrast, Strategy

A remains stable as a secondary strategy in Afrikaans and Norwegian, and to a lesser extent in Danish. This pattern is reminiscent of the use of Strategy A in English during the first half of the 20th century. Thus, the rate of decline of Strategy A varies significantly across these languages, yet the overall trends are broadly similar.

Stratey A: Re-emergence The main exceptions to the decline of Strategy A are Swedish, Faroese, and 19th-century Icelandic.

In Swedish, Strategy B was the dominant strategy in an 1894 translation by Paul Petter Waldenström and the official 1917 Church Bible, where Strategy A is entirely unattested.

During this period, the negative particle *icke* remained the preferred form in formal registers. Over the course of the 20th century, however, *icke*—cognate with Danish and Norwegian *ikke* and Icelandic *ekki*—was replaced by *inte*, derived from the negative indefinite *enkti* 'nothing, no' (Hellquist 1922, 275; cited in Breitbarth et al. 2020).

Following this shift, the frequency of Strategy B declined, and Strategy A re-emerged as the dominant strategy. This shift is evident in modern New Testament translations, such as the *Bible 2000* (478a) and the *Swedish Folk Bible* (478b), both of which make minimal use of Strategy B and predominantly employ Strategy A. The same preference is reflected in contemporary online news data.

(478) Swedish

a. Alla kan inte tillägna sig detta
all can NEG acquire REFL this
'Not everyone can accept this' [*Bibel 2000*, Matt. 19:11; 1999]

b. Men alla ville inte lyda evangeliet.
but all wanted NEG obey the.gospel
'But not everyone wanted to obey the gospel'
[*Svenska Folkbibeln*, Rom. 10:16; 1998]

This suggests that *inte* did not inherit the full syntactic distribution of *icke*, rendering its use in Strategy B constructions somewhat pragmatically marked. Consequently, Strategy A re-emerged as the dominant strategy, since it is based on standard negation.

In Icelandic, Strategy A does not appear in translations from either the 16th century or the 20th–21st centuries. However, it is attested four times in a 19th-century translation, as shown in (479).

(479) Icelandic
En allir hafa ekki þessa þekkingu
but all have NEG this knowledge
'But not everyone possesses this knowledge.' [*Viðeyjarbiblía*, 1 Cor. 8:7; 1841]

The repeated use of Strategy A in the *Viðeyjarbiblía* is surprising, given its absence in earlier translations. Notably, another text from this period also employs Strategy A twice in the same paragraph (480a)-(480b), before switching to Strategy C in a paraphrase of

Matthew 7:21 (480c). This suggests that the use of Strategy A in the *Viðeyjarbiblía* (479) is representative of its time.

(480) Icelandic

a.

En	með	því	allir	menn	vilja	ekki	þýðast	sannleikann
but	with	that	all	men	want	NEG	to.be.interpreted	the.truth

'But because not everyone wants to understand the truth'
[*Hugleiðingar um höfuðatriði kristinnar trúar*, chap. 46, p. 396; 1839]

b.

hún	er	söfnuður,	sem	allir	eru	ekki	í
she	is	congregation	that	all	are	NEG	in

'it [the Christian Church] is a congregation which not everyone is in' [ibid]

c.

Og	með	því	að	ekki	komast	allir	í	himnaríki,	sem	segja:	Herra! Herra!
and	with	that	that	NEG	get	all	in	kingdom.of.heaven	who	say	Lord Lord

'And because not everyone who says "Lord, Lord," will enter the kingdom of heaven' [ibid]

One possible explanation for the use of Strategy A in 19th-century Icelandic is that Strategy C had not yet become as conventionalized as it is today. Present-day Icelandic is unusual among Scandinavian languages in its frequent use of Strategy C with negative fronting. In earlier Icelandic translations, however, this construction was far less prevalent. For instance, the *Guðbrandsbiblía* (1584) contains only one instance of Strategy C with negative fronting, while the *Viðeyjarbiblía* (1841) includes three such instances. Other occurrences of Strategy C in these texts involve XVS word order without negative fronting. By comparison, the 1908 translation published by the Icelandic Bible Society features eight instances of Strategy C with negative fronting, and the 2007 Bible Society translation contains ten. These trends suggest that Strategy C became more conventionalized in Icelandic relatively recently, with its earlier distribution being more limited, similar to its historical use in Swedish.

However, this does not account for the difference between the *Guðbrandsbiblía* (1584), which does not employ Strategy A, and the *Viðeyjarbiblía* (1841), where it appears multiple times. One possibility is that the absence of Strategy A in the *Guðbrandsbiblía* is coincidental, and this strategy was in fact present in Icelandic during that period, as in Swedish and Danish. Another possibility is that an actual linguistic change occurred between these periods, marked by a growing tendency to use pronominal quantifiers in subject position instead of floating quantifiers. If such a shift took place, it would have favored the adoption of Strategy A over Strategy E. Testing this hypothesis, however, lies beyond the scope of the present study.

The emergence of Strategy A in 19th-century Icelandic may parallel its development in present-day Faroese. New Testament translations from the first half of the 20th century into

Faroese predominantly employed Strategies B and C, as illustrated in (481).[9] This suggests that Faroese historically shared common features with both Icelandic and Danish, reflecting its geographical position between these two speech communities.

(481) Faroese

Alt	er	loyvt,	men	ikki	er	alt	gagnligt.	Alt	er	loyvt,	men
everything	is	allowed	but	NEG	is	everything	useful	everything	is	allowed	but

ikki	alt	byggir	upp.
NEG	everything	builds	up

'All things are lawful, but not all things are beneficial. All things are lawful, but not all things build up.' [Dahl, 1 Cor. 10:23; 1937 (1961)]

However, in present-day Faroese, Icelandic-like negative fronting has apparently become pragmatically restricted, much like in Swedish (see §9.1.1). As a result, Strategy A appears to have replaced Strategy C, becoming the dominant strategy in Faroese, as demonstrated in (482). Strategy B, by contrast, remains somewhat restricted, similarly to its status in Swedish.

(482) Faroese

Men	øll	fingu	ikki	pengar.
but	everyone	get.PST.PL	NEG	money.ACC

'But not everyone received money.'

In summary, the re-emergence of Strategy A in Swedish, 19th-century Icelandic, and present-day Faroese appears to have been driven by distinct factors in each context. In Swedish, the replacement of the negative marker *icke* with *inte* during the 20th century apparently made constituent negation (Strategy B) less natural or pragmatically marked, paving the way for Strategy A to become the dominant strategy. By contrast, in 19th-century Icelandic and present-day Faroese, the instability of the negative fronting construction, on which Strategy C relied, appears to have contributed to the rise of Strategy A. In Icelandic, Strategy C only became fully conventionalized in the 20th century, while in Faroese, Icelandic-like negative fronting has largely disappeared in contemporary usage, allowing Strategy A to take its place as the dominant strategy.

Strategy B Strategy B is not particularly prevalent in the two available translations into English and German from the 18th century. This period saw a decline in the production of new translations compared to the Reformation era, which had been characterized by the publication of seminal Protestant and Catholic translations reflecting the theological and cultural shifts of the time.

In English, Purver's translation from 1764 contains three instances of Strategy B, though two involve ellipsis, as exemplified by the first sentence in (483). These are not considered

[9] For a detailed analysis of the differences between the two existing NT translations into Faroese, see §6.2.5.

true instances of Strategy B, as ellipsis constructions tend to be more permissive in allowing constituent negation (see §9.2).

(483) All Things are lawful for me, but not all Things profitable: all Things are lawful for me, but all Things do not edify. [Purver, 1 Cor. 10:23; 1764]

In German, a 1763 revision of the Catholic *Dietenberger Bible*, originally published in 1534, demonstrates a clear preference for Strategy E, much like Luther's 1522 translation. This preference may reflect a broader tendency toward the use of floating quantifiers rather than pronominal ones, as illustrated in (484), where the floating quantifiers are employed in both affirmative and negative contexts. Notably, this example is not derived from either Luther's or the original *Dietenberger Bible*.

(484) German

Es	ist	mir	wohl	alles	erlaubt,	aber	es	nutzt	nicht	alles.
it	be.PRS.3SG	1SG.DAT	indeed	all.N	allow.PTCP	but	it	use.PRS.3SG	NEG	all.N

Es	ist	mir	wohl	alles	erlaubt;	aber	es	ist	nicht	alles	auferbaulich.
it	be.PRS.3SG	1SG.DAT	indeed	all.N	allow.PTCP	but	it	be.PRS.3SG	NEG	all.N	edifying

'Everything is lawful for me, but not everything is beneficial. Everything is lawful for me, but not everything is edifying.' [*Dietenberger Bible*, 1 Cor. 10:23; 1763 edition]

It appears that Strategy B rose to prominence around the late 19th and early 20th centuries.

In English, Strategy B remained relatively restricted throughout the 19th century. Even self-proclaimed literal translations, such as *Young's Literal Translation* (1862), predominantly employed Strategies A and E. Strategy B was only occasionally used, as illustrated in (485). Thus, the relative frequency of Strategy B in English during this period remained comparable to its usage in the 16th and 17th centuries.

(485) And it is not possible that the word of God hath failed; for not all who are of Israel are these Israel; [Young, Rom 9:6; 1862]

In the first half of the 20th century, translations into English show considerable variation in their use of Strategy B, suggesting a transitional period. Some translations, such as William Gunion Rutherford's 1900 version and the *American Standard Version* (1901), employ Strategy B frequently. In contrast, it remains rare in other translations, such as the *Twentieth Century New Testament* (1904), where it is used only once. It is only in the second half of the 20th century that Strategy B becomes a frequent feature across all English translations.

(486) Nevertheless, to this gospel not all have hearkened. [Rutherford, Rom. 10:16; 1900]

(487) All things are lawful; but not all things are expedient. All things are lawful; but not all things edify. [ASV, 1 Cor. 10:23; 1901]

(488) I am far from suggesting that God's words have not been fulfilled. Not all who are descended from Israel are true Israelites. [TCNT, Rom. 9:6; 1904]

In other languages, Strategy B seems to have become dominant earlier than it did in English. In Danish, for example, Strategy B is attested four times in *Frederick VI's New Testament* from 1819, as in (489), where earlier translations employed Strategy A. By the time of the official 1907 translation, Strategy B had already emerged as the dominant strategy.

(489) Danish
Alt er mig tilladt, men ikke Alt er nyttigt; Alt er mig
all.N be.PRS 1SG.DAT allow.PTCP but NEG all.N be.PRS useful all.N be.PRS 1SG.DAT
tilladt, men ikke Alt opbygger.
allow.PTCP but NEG all.N build.up.PRS
'Everything is lawful for me, but not everything is beneficial. Everything is lawful for me, but not everything builds up.'
[*Frederick VI's New Testament*, 1 Cor. 10:23; 1819]

In German, Strategy B emerges as the dominant strategy in the *Elberfeld Bible*, as shown in (490), first published in 1855 and accessed here through a 1905 edition.

(490) German
Aber nicht alle haben dem Evangelium gehorcht.
but NEG all.NOM.PL have.PRS.3PL the.DAT.SG.N gospel.DAT.SG obey.PTCP
'But not all have obeyed the gospel.' [*Elberfeld Bible*, Rom. 10:16; 1855 (1905)]

In Dutch, the *Voorhoeve New Testament* (1877) demonstrates a preference for Strategy B, as illustrated in (491).

(491) Dutch
Alles is geoorloofd, maar niet alles is nuttig; alles is
all.N be.PRS.3SG allow.PTCP but NEG all.N be.PRS.3SG useful all.N be.PRS.3SG
geoorloofd, maar niet alles sticht.
allow.PTCP but NEG all.N build.up.PRS.3SG
'Everything is lawful, but not everything is beneficial. Everything is lawful, but not everything builds up.' [*Voorhoeve New Testament*, 1 Cor. 10:23; 1877]

In Swedish, Strategy B is also the dominant strategy, as seen in an 1894 translation by Paul Petter Waldenström, exemplified in (492), and the official 1917 Church Bible.

(492) Swedish
Allt är tillåtet, men icke allt är nyttigt; allt är tillåtet,
all.N be.PRS allow.PTCP but NEG all.N be.PRS useful all.N be.PRS allow.PTCP

men icke allt uppbygger.
but NEG all.N build.upPRS
'Everything is lawful, but not everything is beneficial. Everything is lawful, but not everything builds up.' [Waldenström, 1 Cor. 10:23; 1894]

In sum, the dominance of Strategy B in most Germanic languages is a relatively recent development, emerging over the past 100–200 years, with the timeline varying by language. Icelandic is unusual among the Germanic languages in that it entirely lacks Strategy B.

Strategy C In Icelandic, Strategy C has been the dominant strategy for over a century, as evidenced by its prevalence in the 1908 translation by the Icelandic Bible Society, as in (493a). It remains the dominant strategy in the 2007 Bible Society translation, as shown in (493b), and in contemporary online news corpora. Typically, the negative particle *ekki* is fronted to clause-initial position.

(493) Icelandic

a. Alt er leyfilegt, en ekki er alt
all.NOM.SG.N be.PRS.3SG allow.PTCP.NOM.SG.N but NEG be.PRS.3SG all.NOM.SG.N
gagnlegt. Alt er leyfilegt, en ekki
beneficial.NOM.SG.N all.NOM.SG.N be.PRS.3SG allow.PTCP.NOM.SG.N but NEG
uppbyggir alt.
build.up.PRS.3SG all.NOM.SG.N
'Everything is lawful, but not everything is beneficial. Everything is lawful, but not everything builds up.' [Icelandic Bible Society, 1 Cor. 10:23; 1908]

b. Ekki hafa allir þessa þekkingu.
NEG have.PRS.3PL all.NOM.PL.M this.ACC.SG.F knowledge.ACC
'But not everyone possesses this knowledge.'
[Icelandic Bible Society, 1 Cor. 8:7; 2007]

Strategy C with negative fronting is also attested in Swedish, as shown in (494), though it is considered archaic (Alice Bondarenko, p.c.).

(494) Swedish
Inte skall var och en som säger 'Herre, Herre' till mig komma in i himmelriket
NEG will everyone who say Lord Lord to me come into the.kingdom.of.heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Svenska Folkbibeln*, Matt. 7:21; 1996]

Strategy D Strategy D becomes an important strategy in many Germanic languages during the 19th and 20th centuries.

In English, Strategy D is attested at least as early as the 19th century, as shown in (495).

(495) "Ah!–she understands what she has to do,–nobody better," rejoined Leah significantly; "**and it is not every one could fill her shoes**—not for all the money she gets." [Brontë, *Jane Eyre*, chap. 17; 1847]

By the first half of the 20th century, Strategy D appears in several New Testament translations published in Britain and the United States, as illustrated in (496)-(499). It appears to be in free variation with Strategy B.

(496) It is not all who are sprung from Israel that are Israel [Rutherford, Rom. 9:6; 1900]

(497) "It is not every man," He replied, "who can receive this teaching, but only those on whom the grace has been bestowed. [Weymouth, Matt. 19:11; 1903]

(498) He said to them, "It is not everyone who can accept that, but only those who have a special gift. [Goodspeed, Matt. 19:11; 1923]

(499) "It is not everybody who can live up to this," replied Jesus, "–only those who have a special gift. [Phillips, Matt. 19:11; 1958]

However, Strategy B had become clearly dominant in English, and Strategy D was used much less frequently. Nevertheless, isolated instances of Strategy D appear in translations such as the *New English Bible* (500) and the *New Revised Standard Version* (501).

(500) and that we may be rescued from wrong-headed and wicked men; for it is not all who have faith. [NEB, 2 Thess. 3:2; 1970]

(501) It is not everyone, however, who has this knowledge. [NRSV, 1 Cor. 8:7; 1989]

In Scandinavian languages, Strategy D is more prominent today, particularly in Danish (502), Norwegian (503), Faroese (504), and Swedish (505), with decreasing relative frequency.

(502) Danish

Det	er	ikke	enhver,	som	kan	forstå	og	tage	til sig,	hvad	jeg	siger
it	is	NEG	everyone	who	can	understand	and	take	to oneself,	what	I	say

'Not everyone can understand and accept what I say'
[*Bible in Everyday Danish*, Matt. 19:11; 1992]

(503) Norwegian

Men	det	er	ikke	alle	som	har	denne	kunnskapen.
but	it	is	NEG	all	who	have	this	knowledge

'But not everyone possesses this knowledge.' [*Bokmål 1978/1985*, 1 Cor. 8:7; 1978/1985]

(504) Faroese

Men	tað	eru	ikki	allar	bygdirnar	sum	hava	sæð
but	it	be.PRS.PL	not	all.NOM.PL.F	villages.NOM.PL.DEF	which	have.PRS.PL	see.SUP

hendan vøksturin.
this.ACC.SG.M growth.ACC
'But not all the villages have seen this growth.' [LCC, News; 2020]

(505) Swedish
Det är inte alla som är så bekanta med den svenska politiken.
it be.PRS NEG everyone who be.PRS so familiar with the Swedish politics
'It is not everyone who is that familiar with Swedish politics.' [LCC, News; 2022]

Additionally, Strategy D serves as a secondary strategy in Afrikaans (506).

(506) Afrikaans
Dit is nie al die mense wat kan verstaan wat Ek gesê het nie
it is NEG all the people who can understand what I said have NEG
'Not everyone can understand what I said' [*Bybel vir Almal*, Mt 19:11; 2008]

Strategy D also appears to be frequent in Scots, though the available data is limited. A 2016 translation of the canonical gospels into Ulster Scots employs Strategy D twice (507a)-(507b) and Strategy B only once (507c).

(507) a. It's **no iverie** yin that says tae me, 'Loard, Loard,' that'll git ïntae tha Kïngdom o Heiven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven' [Ulster-Scots Language Society, Matt. 7:21; 2016]
b. Jesus saed, "It's **no iveriebodie** can hannle that, jist thaim that's halpt bi God.
'Jesus said, "Not everybody can handle that, but only those who are helped by god.' [ibid, Matt. 19:11]
c. Fer he knowed wha wud bethray hïm, an that wus fer why he saed that **no iverie** last yin o thaim wus clain.
'For he knew who would betray him, and that was why he said that not every single one of them was clean.' [ibid, John 13:11]

Strategy E Strategy E is the dominant strategy only in Plautdietsch. A 2003 translation demonstrates a strong preference for Strategy E (508a)-(508b), with Strategy B appearing only once, in the second sentence of (508c). In this respect, Plautdietsch resembles earlier German varieties before Strategy B became dominant.

(508) Plautdietsch
a. Daut sent nich aules Israeliten, dee von Israel häastaumen
it are NEG all Israelites who of Israel descended
'For not all Israelites truly belong to Israel.'
[*De Plautdietsche Bibel*, Rom. 9:6; 2003]

b. Oba de Menschen weeten daut nich aula.
but the people know this NEG all
'But not everyone possesses this knowledge.' [ibid, 1 Cor. 8:7]

c. Wie haben to aules Frieheit, oba daut deent nich aules toom
we have to everything permission but it serves NEG everything to.the
gooden. Wie haben to aules Frieheit, oba nich aules but opp.
good we have to everything permission but NEG everything builds up
'Everything is lawful for us, but not everything is beneficial. Everything is lawful for us, but not everything builds up.' [ibid, 1 Cor. 10:23]

Summary Strategy A experienced a significant decline in most Germanic languages during the 19th and 20th centuries, albeit at varying rates. However, it re-emerged in Swedish, Faroese, and 19th-century Icelandic (though it later declined in the latter) due to broader grammatical changes.

Meanwhile, Strategy B rose to prominence across many Germanic languages during the same period, particularly in Danish, Dutch, German, and Swedish, eventually becoming the dominant strategy in English as well. In Swedish, however, its prevalence diminished over the course of the 20th century.

Strategy C remains dominant in Icelandic, which is unique among the Germanic languages in lacking Strategy B. Although this strategy was also attested in Swedish and Faroese, it eventually became obsolete in both.

Strategy D gained prominence as a secondary strategy in several languages, particularly in the Scandinavian languages, English, and Scots. In English, but not in the other languages, this strategy has largely been supplanted by the dominant Strategy B.

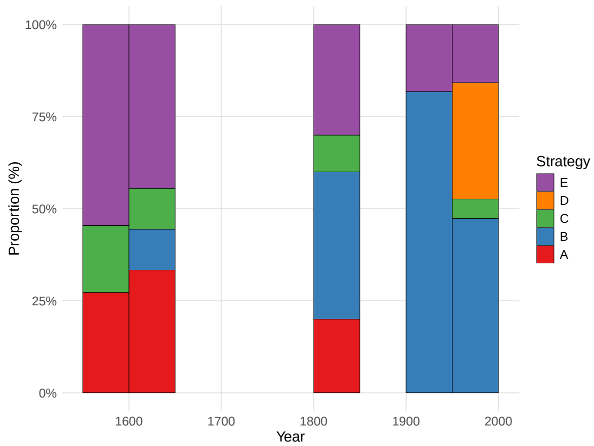
Finally, Strategy E emerged as the dominant strategy exclusively in Plautdietsch, reflecting patterns observed in earlier German varieties.

10.1.7 Summary and discussion

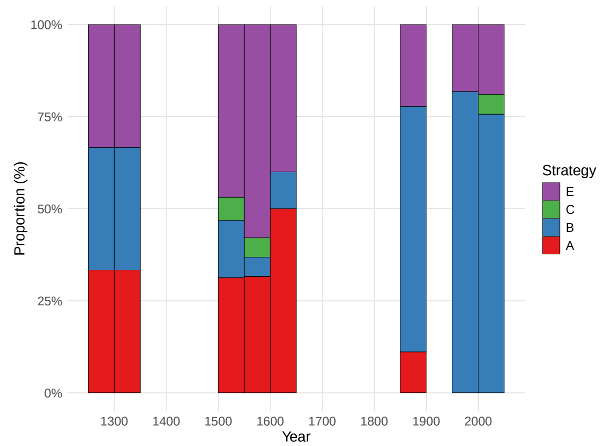
Figure 10.6 illustrates diachronic changes in the distribution of strategies across five Germanic languages with a long and relatively continuous recorded history in the corpus. Missing bars indicate periods for which no corpus data is available. Additionally, translations from the 13th and 14th centuries into Dutch and English provide only a limited amount of data, increasing uncertainty regarding the relative frequencies of strategies.

As the figure shows, there is a considerable time lag between the initial introduction of Strategy B and the point at which it becomes the dominant strategy. During this transitional period, Strategy A typically remains an important strategy, and even the dominant one. Furthermore, the decline of Strategy A is generally monotonic. However, Swedish—along with Faroese and 19th-century Icelandic, as discussed below—presents a notable exception, as Strategy A experiences a resurgence due to broader grammatical changes.

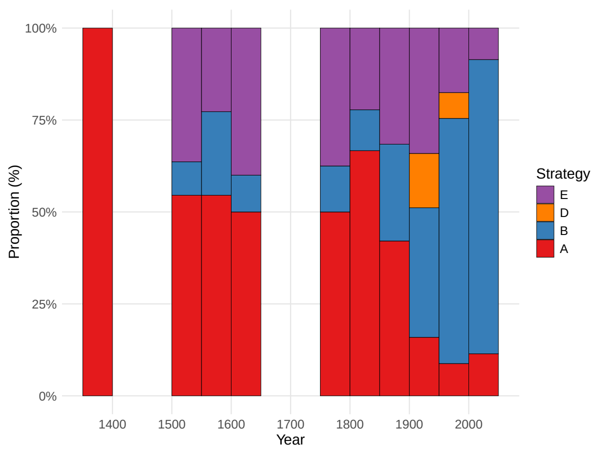
Another notable pattern is the prevalence of Strategy B and the absence of Strategy A in High German during the first half of the 15th century, compared to their distribution



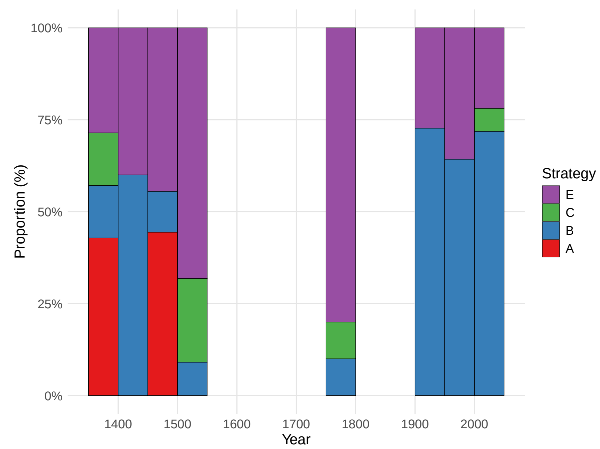
Danish



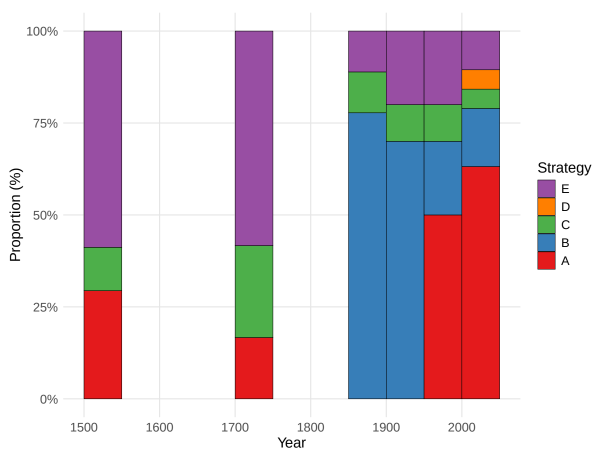
Dutch



English



German



Swedish

Figure 10.6: Distribution of strategies over time across representative Germanic varieties

in the second half of the century. As discussed in §10.1.4, the *Mentelin Bible*, printed in 1466, was likely based on an earlier manuscript from the 14th century, which explains its closer resemblance to the data from that period. However, it remains an open question why Strategy B is frequently attested in 15th-century texts but later recedes. This may reflect instability and variation during the early stages of the grammaticalization of Strategy B.

Strategy A emerged in Germanic languages at two distinct points in time. The first, documented in West Germanic and possibly extending to North Germanic, occurred during the shift away from the NEG-V1 word order pattern, which was characteristic of Old West Germanic languages. As the use of NEG-V1 declined, the dominant Strategy C (characterized by VS word order) became less available. Without an established Strategy B construction at this time, Strategy A, which follows the most frequent SV word order in main negative clauses, became the dominant strategy.

The second emergence of Strategy A occurred in Swedish and Faroese during the 20th century and in Icelandic during the 19th century, although in Icelandic, this was only a transitional phase.

In Swedish, the re-emergence of Strategy A coincided with the replacement of the negative particle *icke* by *inte*. While it remains unclear why this shift led to a significant decline in the use of Strategy B, it appears that Strategy A—which is based on clausal negation—regained prominence as a result of this process and became the dominant strategy in Swedish.

In Faroese and 19th-century Icelandic, the reappearance of Strategy A is tied to the diachronic instability of the NEG-V1 pattern, which forms the basis of Strategy C. Notably, this construction did not originate from the NEG-V1 pattern found in Old Germanic but developed independently centuries later, primarily in Scandinavian languages. As seen earlier in Old West Germanic, this shift encouraged the adoption of Strategy A, which is rooted in SV word order. However, the outcomes in these languages diverged: in Icelandic, the NEG-V1 pattern underlying Strategy C eventually stabilized, solidifying Strategy C as the dominant strategy. In contrast, Faroese abandoned the NEG-V1 construction, allowing Strategy A to become the dominant strategy in contemporary usage.

In most Germanic languages, Strategy A remained stable following its initial emergence. In English, for instance, Strategy A likely developed during Late Old English, around the 10th century, and persisted as the dominant strategy until the late 19th century. In contrast, High German appears to have been the first Germanic language to lose Strategy A, with its decline beginning as early as the 15th–16th centuries. This shift in German was influenced not only by a preference for scope transparency but also by an increasing tendency to favor floating quantifiers over pronominal quantifiers in subject position.

The earliest instances of Strategy B can be traced back to the 13th–14th centuries, when some Germanic languages began innovating negated QP constructions that incorporated the newly developed negative particles. However, these constructions were sporadic and did not become fully conventionalized until much later. During the 16th–18th centuries, Strategy B was relatively infrequent, with Strategies A and E dominating in most translations.

It was not until the 19th century that Strategy B began to gain prominence in several Germanic languages, including German, Dutch, Swedish, and Danish. In English, by contrast, Strategy B did not become the dominant strategy until the 20th century, overtaking Strategies A and D during this period.

The remaining strategies play a less prominent role as dominant strategies. Icelandic stands out among the Germanic languages for its lack of Strategy B and its continued reliance on Strategy C, which remains the dominant strategy to this day.

Strategy D gained prominence as an important strategy during the 19th and 20th centuries in several Germanic languages, particularly in Scandinavian varieties, English, and Scots. While Strategy D has largely been supplanted by Strategy B in English, it remains relatively common in Danish, Norwegian, Faroese, Swedish, and Scots today.

Finally, Strategy E is employed today as the dominant strategy only in Plautdietsch. This preference mirrors patterns observed in earlier German varieties before Strategy B became widespread.

Overall, the diachronic trajectories observed across Germanic languages lend credence to the Diachronic Hypothesis outlined in (410). In accordance with the pathway in (412), Strategy A emerged as a result of broader grammatical changes, often tied to shifts in word order or, in the case of Swedish, the replacement of the negative marker. Despite mirroring the standard negation constructions in these languages, Strategy A generally declined over time, gradually giving way to transparent-scope alternatives, as predicted by the pathway in (583).

This decline unfolded at varying rates across different languages but followed a broadly similar direction, except in cases where grammatical changes disrupted this trajectory, leading to distinct outcomes. In some languages, Strategy A has nearly disappeared or has become confined to formulaic expressions. In others, the decline has been slower, and Strategy A continues to function as a secondary strategy. This diachronic trend suggests that, while languages may not always exhibit a synchronic preference for transparent-scope strategies, bias toward scope transparency has influenced their evolution in the long term.

10.2 Latin and Romance languages

10.2.1 Overview

The Romance language sample includes both official languages, such as French, Spanish, and Italian, as well as minority varieties, including Romansh, Friulian, Ladin, Galician, Piemontese, and Occitan, as shown in Table 10.2. Romansh, Friulian, and Ladin are spoken in the Alpine regions of Switzerland and Italy. Galician is a language spoken in northwestern Spain, which is closely related to Portuguese. Piemontese and Occitan are regional languages primarily spoken in northwestern Italy and southern France, respectively.

As with the Germanic language sample (see §10.1), the data are primarily drawn from NT translations, while other contemporary sources are occasionally referenced.

The diachronic trajectories of Latin and Romance languages reflect a cyclic change in the distribution of strategies. Strategy B was the dominant strategy in Latin, but it was later nearly or entirely lost due to the grammaticalization of new universal quantifiers from

Language group	Language	Years
Gallo-Rhaetian	French	13th century–1987
	Romansh	16th–19th centuries
	Friulian	1860–2000
	Ladin	present (elicitation)
West Ibero-Romance	Spanish	13th century–1973
	Portuguese	1681–2000
	Galician	1861–1989
Italian Romance	Italian	14th century–2008
Gallo-Italian	Venetian	13th–19th centuries
	Piemontese	1834
Occitanic	Occitan	13th century–2016
Catalan	Catalan	1832–2000
Southern Romance	Corsican	1861–2005
Eastern Romance	Romanian	1668–1998

Table 10.2: Romance language sample

the Latin adjective *totus* 'whole', which was initially incompatible with constituent negation (Strategy B). As a result of this process, Strategies A and C took on the role of fallback strategies, as they involve clausal negation—aligning with the pathway in (584). With the re-emergence of Strategy B in most varieties, Strategies A and C were largely sidelined, establishing Strategy B as the dominant strategy, consistent with the pathway in (583). Today, Strategies A and C remain prominent primarily in languages where Strategy B is absent.

10.2.2 Key grammatical properties

Word order Classical Latin is characterized by a highly flexible word order, enabled by its rich case marking and verbal inflection. Despite this flexibility, the most frequent word order is SOV. All other word order permutations are equally possible, although they are often associated with information structure considerations. However, some scholars have argued that SVO was also frequent, particularly in spoken registers, whereas SOV was the conservative pattern found in formal prose (see Ledgeway 2012, 59-62, and references therein).

In the Late Latin period, the word order continued to be relatively flexible, although the proportion of verb-final clauses diminished compared to the Classical Latin period. Often, the verb appears in clause-medial position, resulting in either SVO or OVS order (Herman 2000, 86).

The verb-medial order of Late Latin evolved into a Verb-Second (V2) pattern in Early Romance, where the verb raises to C^0 and is preceded by an element fronted to a high

	Matrix Clauses			Complement Clauses		
	V1	V2	V3+	V1	V2	V3+
Old French	0%	75%	25%	0%	100%	0%
Old Occitan	8%	53%	49%	24%	64%	12%
Old Spanish	1%	92%	7%	50%	40%	10%
Old Venetian	24%	59%	17%	8%	84%	8%

Table 10.3: Comparison of verb position in matrix clauses and complement clauses across Medieval Romance varieties, based on Wolfe (2018)

position in the left periphery (Ledgeway 2012, 65). This development is illustrated by the example in (509) from Old French.[10]

(509) Old French (Ledgeway 2012, 66)
$[_{CP}$ Sa venoison$_2$ fist$_1$ $[_{IP}$ t_1 a l' ostel porter $t_2]]$
his goods make.PST.3SG to the hostel carry.INF
'he had his goods carried to the hostel'

Table 10.3, based on Wolfe (2018), compares the frequency of verb positions in several Medieval Romance varieties in selected texts.

Due to the high frequency of subject fronting to the clause-initial topic position, pre-verbal subjects were later reanalyzed as occupying a dedicated subject position in [Spec, IP]. Thus, V2 word order constitutes a transitional phase between the original Latin SOV basic word order and the SVO word order of Modern Romance (Ledgeway 2012, 65-69).

Negation In Latin, standard negation is expressed by the pre-verbal negative particle *non*, as demonstrated in (510).

(510) Latin (Larrivée 2017, 452)
Non loquor
NEG1 speak.PRS.1SG
'I am not speaking'

Many Romance languages retained the pre-verbal negative particle, as demonstrated by Spanish (511).

(511) Spanish (Larrivée 2017, 452)
(Yo) no hablo
I NEG1 speak.PRS.1SG
'I am not speaking'

[10]The presence of a V2 stage in the history of Romance languages is a matter of debate. For further discussion, see Benincà (2006), Wolfe (2018), and Poletto (2014), among others.

In some varieties, the negative particle sometimes fuses with a clitic pronoun, exemplified in Venetian in (512).

(512) Venetian (Larrivée 2017, 453)
No-l dorme gnente.
NEG1-3SG.NOM sleep.PRS.3SG nothing
'He doesn't sleep at all.'

Other Romance languages innovated negative markers through a Jespersen's Cycle. For example, Old French developed post-verbal negative particles derived from minimizers, such as *pas* 'step', *point* 'point', and *mie* 'crumb' (Hansen 2013, 56). In contemporary Standard French, the post-verbal negative particle *pas* still co-occurs with the pre-verbal clitic *ne*, as shown in (513).

(513) Standard French (Larrivée 2017, 453)
Je ne parle pas.
I NEG1 speak.PRS.1SG NEG2
'I'm not speaking.'

In spoken registers, only the post-verbal particle *pas* is employed, as shown in (514).

(514) Colloquial French (Larrivée 2017, 453)
Je parle pas.
I speak.PRS.1SG NEG
'I'm not speaking.'

In Catalan, the post-verbal negative particle is optional, as shown in (515).

(515) Catalan (Espinal 1991, 34)
La Maria no va (pas) aprovar.
the Maria NEG1 PST NEG2 pass
'Maria did not pass.'

Some Romance varieties also employ a clause-final negative marker, as demonstrated in Venetian in (516).[11]

(516) Venetian (Larrivée 2017, 454)
No la go miga magnada NO!
NEG1 3SG.F.ACC have.PRS.1SG NEG2 eat.PTCP NEG3
'I really did not eat it!'

According to Zanuttini (1997) and Zeijlstra (2004, 2022), pre-verbal negative markers in Romance (NEG1) are Neg^0 heads, and they project a NegP. On the other hand, post-verbal

[11] For analyses of the discourse functions associated with NEG2 and NEG3 particles in Romance languages, see, among others, Espinal (1993); Schwenter (2005); Hansen and Visconti (2009); Squartini (2017); for syntactic accounts, see Zanuttini (1997); Poletto et al. (2008); Ledgeway and Schifano (2023).

negative particles (NEG2 and NEG3) are usually phrasal, and they do not occupy the head position of a NegP. Negative markers that are syntactic heads are always adjacent to the verb, as they take the *v*P as their complement, whereas phrasal negative markers may appear further away from the verb. For an analysis of the status of the negative particle in Classical and Late Latin, see Gianollo (2016).[12]

A common diagnostic for the type of negative marker is the *why not* test: only phrasal negative markers can appear in this construction, since *why* is a phrasal adverb, and heads cannot adjoin to XPs (Merchant 2006). For instance, in Italian, the clausal negation marker *non* cannot be used in this context because it is a head. Instead, the language employs the negative response particle *no*, as shown in (517). In contrast, French *pas* is phrasal and hence can appear in the *why not* construction, as demonstrated in (518).

(517) Italian (Merchant 2006)
  a. *Perche non?
  b. Perche no?

(518) French (Merchant 2006)
  Pourquoi pas?

Universal quantifiers The primary universal quantifier in Latin is *omnis.* This item has been retained by Italian and other Romance languages of Italy, but it ceased to be used in most other Romance languages. Even in Italian, *ogni* is restricted to distributive predication and can only combine with singular count nouns.

In all Romance languages, the most frequent universal quantifier is *tous* (French) / *tutti* (Italian) / *todos* (Spanish) (Vincent 2017, 748). These items are derived from the Latin adjective *totus* 'whole'. In Latin, *totus* occurs mostly in the singular. Even in the plural, it does not express distributive universal quantification but rather quantification over parts of an entity, as demonstrated in (519), similarly to the English adjective *whole* (Haspelmath 1995, 364-365). However, the development of *totus* into a proper universal quantifier has already began in Late Latin (Martzloff 2014; Pinkster 2015), as shown in (520).

(519) Classical Latin

Pervigilat	noct-es	tota-s.
remain.awake	night-ACC.PL	whole-ACC.PL

'S/he remains awake during entire nights.'
(Plaut. *Aul.* 1.1.33, cited in Haspelmath 1995, 365)

(520) Late Latin

Nam	totos	istos	hostes	tuos	statim
for	whole.ACC.M.PL	those.ACC.M.PL	enemy.ACC.PL	your.ACC.M.PL	immediately

[12]See Gianollo (2020) For a broader perspective on the syntactic status of negative markers across languages.

captivos habebis.
captive.ACC.M.PL have.FUT.2SG
'For you will soon have all those enemies of yours in bonds.'
(Apul. *Met.*. 7.12.2, cited in Pinkster 2015, §11.38)

Other universal quantifiers are derived from free-choice determiners, such as Latin *quisque* 'every' (< *quis* 'who, which' + *que* 'and, also') and Rumanian *fiecare* 'every(one)' (< *care* 'who, which' + *fie* be.SBJV.3SG') (Haspelmath 1995, 370). There are also universal quantifiers like Italian *ciascuno* 'every(one)', which are derived from the distributive preposition *katá*, borrowed from Greek into Late Latin (Haspelmath 1995, 377).

10.2.3 Latin

Strategy B appears to have been the dominant strategy in Latin throughout its recorded history, as evidenced by representative examples from Old Latin (521), Classical Latin (522), and Late Latin (522).[13]

(521) Old Latin
Non omnis aetas, Lyde, ludo convenit.
NEG every age Lydus.VOC school.DAT be.fit.PRS.3SG
'Not every age, Lydus, is suited for school.'
[Plautus, *Bacchides*, Act I, Scene 2; about 200 BCE]

(522) Classical Latin
Sed non omnia, quaecumque loquimur, mihi videntur ad artem
but NEG everything whatever say.PRS.1PL 1.SG.DAT seem.PRS.PASS.3PL to art.ACC
et ad praecepta esse revocanda.
and to lesson.PL.ACC be.PRS.INF recall.PTCP
'But to my mind not everything that we say need be reduced to theory and rule.'
[Cicero, *De Oratore*, bk. II; 55 BCE]

(523) Late Latin
non enim omnes in priore uita geometrae fuerunt
NEG because all.PL.NOM in prior.SG.ABL life geometrician.PL.NOM be.PERF.3PL
'For not all have been geometricians in their previous life'
[Augustine, *De Trinitate*, bk. 12, chap. 15; about 400-428 CE]

The Vulgate predominantly employs Strategy B as well, as shown in (524). However, it is important to note that the Vulgate closely mirrors the Greek source text in the relevant verses.

(524) Sed non omnes obediunt Evangelio.
but NEG all.PL.NOM obey.PRS.3PL gospel.DAT
'But not all obey the gospel' [*Vulgate*, Rom. 10:16; late 4th century]

[13] For a detailed analysis of negated QP constructions in Latin, see Tierney (2018).

10.2.4 Medieval Romance

Table 10.4 presents the distribution of strategies in translations of the NT into Medieval Romance varieties. The numbers are represented as counts rather than relative frequencies due to the small amount of data. The dominant strategies are typically A (French and Occitan) or C (Spanish and Tuscan), whereas Strategy B—the dominant strategy in Latin—is only sporadically attested.

	A	B	C	E
French	3	0	0	0
Occitan	6	1	1	1
Spanish	1	0	3	1
Tuscan	0	0	2	1
Venetian	1	1	0	0

Table 10.4: The distribution of strategies in translations of the NT into Medieval Romance varieties (13th-14th centuries)

Among the translations accessible to me, Strategy B appears only once in Old Venetian (525) and Old Occitan (526). In both cases, this strategy is employed in Matthew 7:21, which is the verse most strongly associated with this strategy, possibly due to the length of the subject (see §5.5).

(525) Old Venetian
Non ogno homo che dice: – Misser, misser – entrarà el lo regno
NEG every man that say.PRS.3SG Lord Lord enter.FUT.3SG to the kingdom
de Dio
of heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Venetian Harmony*, Matt. 7:21; 13th century]

(526) Old Occitan
No totz hom que ditz a mi, Senher, Senher, no intrara el
NEG all person that say.PRS.3SG to me Lord Lord NEG enter.FUT.3SG in.the
regne dels cels
kingdom of.the heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Nouveau Testament de Lyon*, Matt. 7:21; 13th century]

In (526), note that the negated QP co-occurs with clausal negation. This pattern is typically characteristic of languages with a bipartite negative marker (see §7.2.2), which is not the case for Old Occitan. I have no clear explanation for the use of this construction in this verse; it may even be the result of a scribal error. Regardless, the fact that this construction is used only once in the *Nouveau Testament de Lyon*, a complete translation of the New Testament, suggests that it was probably not fully established as a conventional pattern.

The rarity of Strategy B in Medieval Romance suggests that the Latin Strategy B construction was not directly extended from *omnis* to universal quantifiers derived from *totus*. In Latin, *totus* functioned as an adjective, modifying nouns without altering the syntactic category of the phrase—it remained a determiner phrase (DP), not a quantifier phrase (QP) (see §10.2.2). Medieval Romance languages, like their modern descendants, likely did not allow constituent negation of regular DPs.[14] Consequently, *totus*-modified phrases, which retained the properties of a DP during the early stages of their grammaticalization, were also incompatible with constituent negation.

This incompatibility meant that Strategy B, which worked with *omnis*, could not be directly inherited for *totus*-quantifiers in Medieval Romance. Instead, Romance languages had to innovate a new version of Strategy B for these quantifiers, as their grammaticalization advanced and *totus* began to function as a true quantifier.

Unlike quantifiers derived from *totus*, those descended from *omnis* apparently did not need to reinvent Strategy B. However, it remains an open question whether there was continuity in the use of Strategy B between Latin and those Medieval Romance languages that retained *omnis*, such as Old Venetian (525).

Since Strategy B was not yet fully conventionalized with quantifiers derived from *totus*, the dominant strategies in Medieval Romance were A and C, which involve clausal negation. The choice between these strategies seems to depend, at least in part, on whether a language allowed verb-initial (V1) clauses. To illustrate this, we begin with Old French, which provides a clear example of how word order constraints influenced the availability of Strategy A.

As Table 10.3 shows, Old French did not permit V1 clauses, though it was not strictly SVO. For instance, in (527), the subject follows the verb in a typical V2 clause.

(527) Old French

Lors	monta	Jhesus	en	la	montaigne
then	ascend.PST.3SG	Jesus	in	the.SG.F	mountain

‘Then Jesus went up on the mountain’ [*Old French Bible*, Matt. 5:1, 1235-1260]

For Strategy C to be available in Old French, an element needed to be fronted to clause-initial position, thereby enabling V2. In the absence of such an element, Strategy C was not viable, and the language instead resorted to Strategy A. Evidence for this preference comes from a 13th-century translation of the gospels, which employs Strategy A in all three instances, as illustrated in (528).

[14] Modern Romance languages generally prohibit constituent negation of focused DPs, as seen in examples like the ungrammatical Spanish sentence **No Juan comió una manzana* (Intended: ‘It wasn’t Juan who ate an apple’). By extrapolation, this restriction can be interpreted as a continuation of patterns from earlier stages of these languages.

(528) Old French

a. Tuit cil qui me dient: 'Sire, sire,' n'-enterront mie el
all those who to.me say.PRS.3PL Lord Lord NEG1-enter.FUT.3PL NEG2 in.the.SG.M
regne des ciels
kingdom of.the.PL heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Old French Bible*, Matt. 7:21; 1235-1260]

b. Tuit vos n'-estes mie net.
all you NEG1-be.PRS.2PL NEG2 clean
'Not all of you are clean.' [ibid, John 13:11]

Other Romance languages allowed V1 clauses, making Strategy C a viable alternative to Strategy A, as demonstrated in Old Spanish (529), Old Occitan (530), and Old Tuscan (531). These constructions are reminiscent of the use of Strategy C in certain present-day Balkan languages, such as Modern Greek, Serbian-Croatian-Bosnian, and Slovenian (see §7.3.1). Similar to those Balkan languages, the NegVS(O) word order is significantly less common in clauses where the subject is a regular DP but occurs frequently in 'Not all X are Y' sentences.

(529) Old Spanish

a. No prenden todos esta palaura
NEG take.PRS.3PL all this.SG.F word
'Not everyone takes this word'
[*Biblia prealfonsí*, Matt. 19:11; mid-13th century]

b. Mas no obedecen todos a-la preygacion.
but NEG obey.PRS.3PL all to-the.SG.F preaching
'But not all obey the preaching.' [ibid, Rom. 10:16]

(530) Old Occitan
No prendo tuit aquesta paraula
NEG take.PRS.3PL all this.SG.F word
'Not everyone takes this word'
[*Nouveau Testament de Lyon*, Matt. 19:11; 13th century]

(531) Old Tuscan

a. Non enterrà nel regno de' cieli ogni persona che mi
NEG enter.FUT.3SG to.the.SG.M kingdom of heaven every person that to.me
dice: messere, messere
say.PRS.3SG Lord Lord
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Tuscan Harmony*, Matt. 7:21; 14th century]

b. Non possono tutti adempiere questa parola
NEG can.PRS.3PL all carry.out.INF this.SG.F word
'Not everyone can carry out this word' [ibid, Matt. 19:11]

Alongside Strategy C, these languages also employ Strategy A. In Old Occitan, Strategy A is even the dominant strategy, similarly to Old French, as in (532).

(532) Old Occitan
Mas tuit no obezisso a l'-avangeli.
but all NEG obey.PRS.3PL to the-gospel
'But not all obey the gospel'
[*Nouveau Testament de Lyon*, Rom. 10:16; 13th century]

In Old Spanish, on the other hand, Strategy C is the dominant strategy, but Strategy A is also occasionally employed, as shown in (533).[15]

(533) Old Spanish
Ca tod aquel que dira a mi, Sennor, Sennor, no entrara
because every that that say.FUT.3SG to me Lord Lord NEG enter.FUT.3SG
en el regno de los cielos;
to the kingdom of the heavens
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Biblia prealfonsí*, Matt. 7:21; mid-13th century]

Strategy A is also attested in another source from this period (534).

(534) Old Spanish
ca todas las mançanas non son dulces
because all.PL.F the.PL.F apples NEG be.PRS.3PL sweet.PL.F
'["Everything in its own season," said the knave,] "for all apples are not sweet; [therefore it is suitable that we prepare ourselves for whatever happens."]'
[*Livro del cavallero Cifar*, about 1300]

In Old Venetian, Strategy A is attested once (535), alongside an instance of Strategy B (525).[16]

(535) Old Venetian
Ogne homo no entende questa parola
every man NEG understand.PRS.3SG this.SG.F word
'Not everyone understands this word'
[*Venetian Harmony*, Matt. 19:11; 13th century]

Finally, Strategy E is exemplified in (536).

[15] In addition to the three instances of Strategy C, two instances involve V1 word order but are classified as C/E (see §7.6.2.

[16] The *Tuscan Harmony* (14th century) contains only two relevant verses, both of which employ Strategy C (531). However, later translations of the New Testament into Italian sometimes use Strategy A (see §10.2.5), suggesting that Strategy A may have also been available in Old Tuscan.

(536) Old Tuscan
voi non siete mondi tutti.
you NEG be.PRS.2PL clean.PL.M all.PL.M
'You are not all clean.' [*Tuscan Harmony*, John 13:11; 14th century]

In sum, Strategy B seems to have been either rare or unavailable in several Medieval Romance languages during the 13th-14th centuries. Consequently, the dominant strategies typically involved clausal negation, resulting in Strategy A or C, depending on the word order. Old French opted for Strategy A because it did not allow V1 clauses. Despite the tendency toward V2 word order, other languages also allowed V1 clauses, thus enabling the use of Strategy C. However, Strategy C was typically in alternation with Strategy A. This might indicate that Strategy C, which involved a divergence from the typical word order, was not fully conventionalized. Indeed, it was soon abandoned in most varieties in favor of Strategy B, as outlined in the next subsection.

10.2.5 Early Modern Romance

Figure 10.7 presents the distribution of strategies in translations of the NT into Early Modern Romance varieties. Strategy B becomes the dominant strategy in most languages, with the notable exception of French, where Strategy A remains the most frequent strategy. Strategy C loses ground over this period, whereas Strategy E becomes relatively frequent in some languages. Notably, Strategy A is still occasionally employed in this period by languages that would later lose it altogether.

The emergence and spread of Strategy B acted as a catalyst for change in the use of Strategies A and C, which were the dominant strategies in Medieval Romance varieties. Other changes, such as the loss of V2 (except in Romansh) and the consolidation of an SVO order, do not seem to play a major role, since the crucial factor is the choice between clausal negation (Strategies A and C) and constituent negation (Strategy B).

Strategy A Despite the increased use of Strategy B, Strategy A remains attested in nearly all Romance varieties during this period. In French, Strategy A continues to be the dominant strategy, as demonstrated in (537).

(537) French

a. Tous ne sont point capables de ce mot
all.PL.M NEG1 be.PRS.3PL NEG2 capable.PL of this.SG.M word
'Not everyone is capable of this word' [Lefèvre, Matt. 19:11; 1523 (1525)]

b. Tous ceux qui me disent: Seigneur, Seigneur, n'-entreront
all.PL.M those.PL.M who to.me say.PRS.3PL Lord Lord NEG1-enter.FUT.3PL
pas *pour cela* dans le royaume du ciel
NEG2 for that in the.SG.M kingdom of.the.SG.M heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven

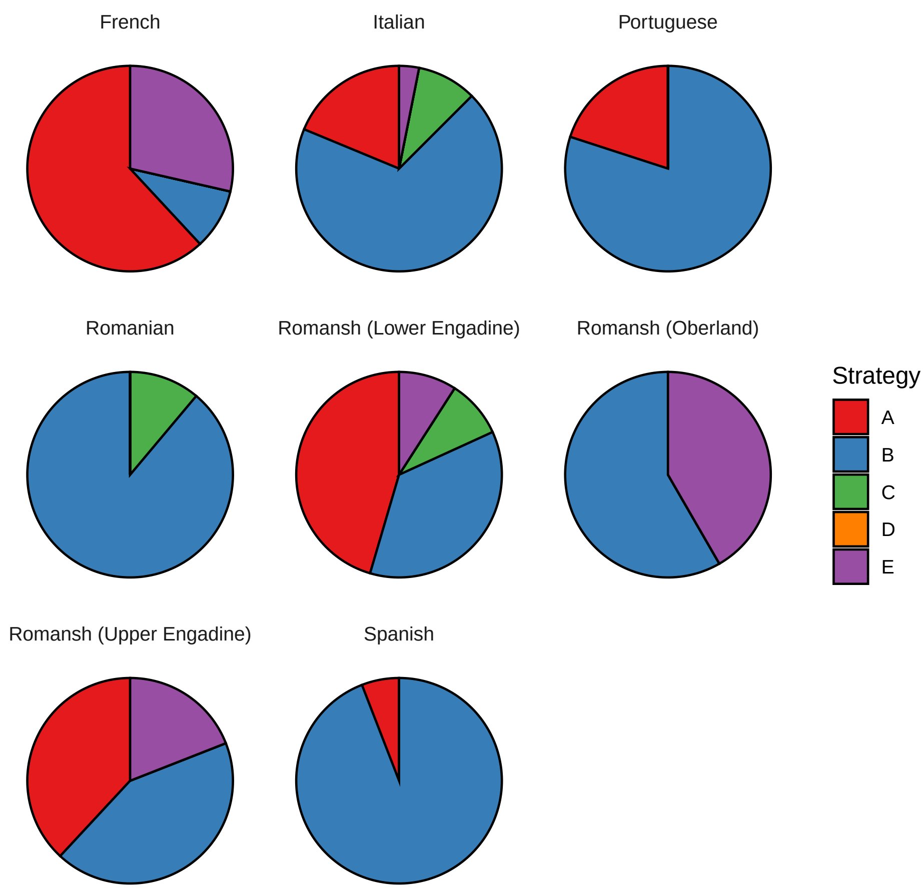


Figure 10.7: The distribution of strategies in translations of the NT into Early Modern Romance varieties (15th-17th centuries)

on that account'
[*Nouveau Testament de Mons*, Matt. 7:21; 1667, italics in the original]

Strategy A is also frequently employed in the Upper Engandine and Lower Engandine dialects of Romansh (538)-(539).

(538) Romansh (Upper Engandine)

a. Tuottas chioses sun à mi licitas, mu tuottas chioses nun sũ nüzzaifles.
all.PL.F things be.PRS.3PL to me permitted.PL.F but all.PL.F things NEG1 be.PRS.3PL beneficial.PL
'All things are lawful, but not all things are beneficial.'
[Bifrun, 1 Cor. 10:23; 1560]

b. Ma tuots nun haun obedieu à l' Evangeli.
but all.PL.M NEG1 have.PRS.3PL obey.PTCP to the gospel
'But not all obeyed the gospel.' [Gritti, Rom. 10:16; 1640]

(539) Romansh (Lower Engandine)
Tuots nun sun capabels da quaista chiausa chia vus dichais
all.PL.M NEG1 be.3PL capable.PL of this.SG.F thing which you say.PRS.2PL
'Not everyone is capable of what you are saying' [Dorta, Matt. 19:11; 1679]

In Italian, Strategy A is relatively frequent in the *Malermi Bible* from 1471 (540a), the first printed bible in this language. The *Malermi Bible* is partly based on pre-existing vernacular translations, though it is primarily a new translation by Nicolò Malermi (Leonardi et al. 2012, 284-285). The prevalence of strategy A in this text may reflect the influence of earlier versions. In later translations, however, Strategy A becomes less common. A shift away from Strategy A can already be seen in Antonio Brucioli's 1532 translation, which strongly favors Strategy B and does not employ Strategy A at all. Interestingly, however, a 1558 revision of Brucioli's translation by an anonymous editor from Geneva replaces one instance of Strategy B in the original version (544b) with Strategy A (540b). Similarly, a 1607 translation by Giovanni Diodati employs Strategy A twice, as demonstrated in (540c), though it clearly favors Strategy B.[17]

(540) Italian

a. tutti nõ pigliano questa parola
all.PL.M NEG take.PRS.3PL this.SG.F word
'Not everyone takes this word' [Malermi, Matt. 19:11; 1471 (1490)]

b. tutti nõ son capaci di questa parola
all.PL.M NEG be.PRS.3PL capable.PL of this.SG.F word
'Not everyone is capable of this word'
[Anonymous revision of Brucioli, Matt. 19:11; 1558]

c. Ma tutti non hanno ubbidito all'-Evangelio
but all.PL.M NEG have.PRS.3PL obey.PTCP to.the.SG.M-gospel
'But not all have obeyed the good news' [Diodati, Rom. 10:16; 1607]

As in later Italian translations, Strategy A is relatively rare in Spanish and Portuguese. In Spanish, it is attested only once in an anonymous translation from the 15th century (541a).

[17] It is worth considering whether these differences reflect dialectal variation, as Brucioli was a Florentine, Malermi was from Venice, and Diodati was from Geneva. This may suggest that Strategy A was more common in certain northern varieties than in the Tuscan dialect, which forms the basis of Standard Italian, though this remains speculative.

In the *Biblia del Oso* from 1569, Strategy A is unattested, except in a construction with a *same*-type predicate (541b).

(541) Spanish

a.

mas	todos	non	obedeçen	al	euangelio
but	all.PL.M	NEG	obey.PRS.3PL	to.the.SG.M	gospel

‘But not all obey the gospel’ [Anonymous, Rom. 10:16; 15th century]

b.

Toda	carne	no	es	la	misma	carne
all.SG.F	flesh	NEG	be.PRS.3SG	the.SG.F	same.SG.F	flesh

‘Not all flesh is the same flesh.’ [*Biblia del oso*, 1 Cor. 15:39, 1569]

Strategy A is also employed in Portuguese in (542).

(542) Portuguese

Todas	as	cousas	me	sam	licitas,	mas	todas	as	cousas
all.PL.F	the.PL.F	things	to.me	be.PRS.3PL	lawful.PL.F	but	all.PL.F	the.PL.F	things

naõ	sam	convenientes:	Todas	as	cousas	me	sam	licitas,
NEG	be.PRS.3PL	beneficial.PL	all.PL.F	the.PL.F	things	to.me	be.PRS.3PL	lawful.PL.F

mas	todas	as	cousas	naõ	edificam.
but	all.PL.F	the.PL.F	things	NEG	build.up.PRS.3PL

‘All things are lawful for me, but not all things are beneficial. All things are lawful for me, but not all things build up.’ [Almeida, 1 Cor. 10:23; 1681]

In contrast, Strategy A is entirely unattested in translations into Romanian and the Oberland dialect of Romansh. While this absence may be accidental, it is also possible that Strategy A had already fallen out of use by the time these translations were produced in the mid-17th century. Alternatively, Strategy A may never have been used in these varieties; however, this seems less likely given the prevalence of Strategy A across Romance varieties.

Strategy B By this period, Strategy B is already the dominant or at least a widely used strategy in all the surveyed varieties, except French. In Spanish, while it is unattested in a 13th-century translation (see §10.2.4), it becomes the most frequent strategy in a collection of 15th-century translations from miscellaneous sources, as exemplified in (543a). Strategy B is also by far the dominant strategy in the *Biblia del oso* from 1569 (543b).

(543) Spanish

a.

non	todos	toma(n)	esta	palabra
NEG	all.PL.M	take.PRS.3PL	this.SG.F	word

‘Not everyone takes this word’ [Anonymous, Matt. 19:11; 15th century]

b.

No	todos	son	capaces	deste	negocio
NEG	all.PL.M	be.PRS.3PL	capable.PL	of.this.SG.M	matter

‘Not everyone is capable of this matter.’ [*Biblia del oso*, Matt. 19:11; 1569]

In Italian, the *Malermi Bible* from 1471 employs Strategy B only twice, as exemplified in (544a), while favoring Strategies A and C. In contrast, Antonio Brucioli's 1532 translation strongly favors Strategy B, as shown in (544b), much like contemporary Spanish translations. A similar pattern is found in Giovanni Diodati's 1607 translation, which also exhibits a clear preference for Strategy B (544c).

(544) Italian

a. Ma non tutti obediscono al euangelio
but NEG all.PL.M obey.PRS.3PL to.the.SG.M gospel
'But not all obey the gospel' [Malermi, Rom. 10:16; 1471 (1490)]

b. Non tutti sono capaci di questo detto.
NEG all.PL.M be.PRS.3PL capable.PL of this.SG.M saying
'Not everyone is capable of this saying.' [Brucioli, Matt. 19:11; 1532 (1544)]

c. Non tutti son capaci di questa cosa che voi dite
NEG all.PL.M be.PRS.3PL capable.PL of this.SG.F thing that you say.PRS.2PL
'Not everyone is capable of what you say' [Diodati, Matt. 19:11; 1607]

Strategy B is also the dominant strategy in 17th-century translations into Portuguese (545) and Romanian (546).[18]

(545) Portuguese

a. naõ todos saõ capazes desta palavra
NEG all.PL.M be.PRS.3PL capable.PL of.this.SG.F word
'Not everyone is capable of this word' [Almeida, Matt. 19:11; 1681]

b. Todas as cousas me sam licitas, mas nem todas as cousas convem
all.PL.F the.PL.F things to.me be.PRS.3PL lawful.PL.F but NEG all.PL.F the.PL.F things be.useful.PRS.3PL
'All things are lawful for me, but not all things are beneficial.' [ibid, 1 Cor. 6:12]

(546) Romanian

Nu toți curați sînteț.
NEG all.PL.M clean.PL.M be.PRS.2PL
'Not all of you are clean.' [*Bucharest Bible*, John 13:11; 1668 (1688)]

Notably, Strategy B is also attested in a 16th-century translation into French, even though this strategy is unavailable in present-day Standard French (see §10.2.6). A 1523 translation by Jacques Lefèvre d'Étaples employs Strategy B twice (547). Notice that both negative markers precede the QP in this construction.[19]

[18] In Portuguese during this period, there is free variation between the negative markers *não* and *nem* in negated QP constructions. See §9.2 for further discussion.

[19] These Strategy B constructions also appear in the 1535 *Bible d'Olivétan*, a revision of Lefèvre's version by Pierre Robert Olivétan.

(547) French

a. Non point ung chascũ qui me dit Sire Sire entrera
NEG1 NEG2 one everyone who to.me say.PRS.3SG Lord Lord enter.FUT.3SG
au royaulme des cieulx
to.the.SG.M kingdom of.the.PL heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Lefèvre, Matt. 7:21; 1523 (1525)]

b. Car non point tous ceulx qui sont de Israel: iceulx
because NEG1 NEG2 all.PL.M those.PL.M who be.PRS.3PL of Israel those.PL.M
sont Israelites.
be.PRS.3PL Israelites
'For not all those who are from Israel are Israelites.' [ibid, Rom. 9:6]

Crucially, this construction is not limited to NT translations. It is also attested in two other contemporary French texts, both translated from Latin (548)-(549). While one might suspect that these instances are mere calques from Latin, this explanation is unlikely in at least one case: in (549), the Latin source text does not employ Strategy B in the corresponding passage (*sed **non** continuo, **quicumque** principes vel duces sunt, etiam reges dici possunt*). This suggests that Strategy B was not merely a direct borrowing from Latin but was instead a viable, if relatively uncommon, option in 16th-century French.[20]

(548) French

Et de faict, que veut Dieu autre chose par ce propos: D'Isaac prendra son nom ta semence? sinon declarer ouuertement que **non pas tous ceux qui sont nays d'Abraham selon la chair, sont quant & quant enfans de Dieu** & par consequent heritiers des promesses, mais que ceux qui ont la foy par laquelle Abraham obtint la promesse de Dieu, ceux là, di-ie, & non autres sont la semence d'Abraham.

'What, therefore, does he intend when he says: 'Through Isaac shall your seed be named,' except to explain clearly that **not all those who have been born from Abraham according to the flesh are automatically sons of God** and consequently inheritors of the promises, but those belong to the seed of Abraham who are people of faith, inasmuch as it was through faith that Abraham deserved the promise of God.' [Anonymous translation of *Paraphrases of Erasmus*; 1563][21]

(549) French

Combien que celuy qui est Roy puisse aussi estre appellé Prince, a cause de la principauté qu'il a de commander: & conducteur, pource qu'il conduit l'armee: **Mais non pas tous ceux qui sont Princes ou conducteurs peuuent aussi incontinent estre appellez Roys**, ce que fut cest Aristobule.

[20]In other places, the text cited in (548) employs Strategy A, as in *Car tous ceux qui sont descendus d'Israel, ne sont pas pourtant vrays Israelites* 'For not all who are descended from Israel are true Israelites.'

[21]English translation by John B. Payne, Albert Rabil Jr, and Warren S. Smith Jr.

'Although a king himself may be called a prince, from his principality in governing, and a leader, because he leads the army, **but it does not follow that all who are princes and leaders may also be called kings**, as that Aristobulus was.' [Anonymous translation of Augustine's *City of God*, bk. 18, chap. 45; 1570][22]

However, Strategy B did not become conventionalized in French, and it does not appear in later NT translations, such as a 1667 translation by Antoine Le Maistre and Louis-Isaac Lemaistre de Sacy. Thus, the occurrence of Strategy B in 16th-century French texts instantiates a failed (abortive) innovation—a phenomenon discussed by (Kroch 1989; Postma 2010; Brook 2024), among others.

In Romansh, Strategy B typically involves the novel negative marker (NEG2). In the Upper Engandine and Lower Engandine dialects, these constructions contain the negative marker *brichia*, derived from a minimizer meaning 'crumb' (Willis et al. 2013, 13), as demonstrated in (550)-(551). In this period, clausal negation is expressed by the pre-verbal negative marker *nun* (NEG1), whereas *brichia* (NEG2) only optionally co-occurs with it in these texts, similarly to the optionality of *pas* in present-day Catalan (see §10.2.2). The fact that *brichia* does not co-occur with *nun* in Strategy B is surprising, since *brichia* typically cannot negate a clause by itself. Many languages exhibit negative concord in situations where the negative marker modifying the QP normally does not occur on its own (see §7.2.2), but this is not attested in Romansh.

(550) Romansh (Upper Engandine)

a. Tuottas chioses sun à mi licitas, mu brichia tuottas chiosas sun nüzzaiules
all.PL.F things be.PRS.3PL to me permitted.PL.F but NEG2 all.PL.F things be.PRS.3PL beneficial.PL
'All things are lawful for me, but not all things are beneficial.'
[Bifrun, 1 Cor. 6:12; 1560]

b. Brichia tuotta chiarn eis la medesima chiarn
NEG2 all.SG.F flesh be.PRS.3SG the.SG.F same.SG.F flesh
'Not all flesh is the same flesh.' [Gritti, 1 Cor. 15:39; 1640]

(551) Romansh (Lower Engandine)
Brichia tuots eschet nets.
NEG all.PL.M be.PRS.2PL clean.PL.M
'Not all of you are clean.' [Dorta, John 13:11; 1679]

In the Oberland dialect of Romansh, the negative marker *bucca* (NEG2), possibly derived from Latin *bucca* 'mouth(ful)' (Willis et al. 2013, 13), is employed in both Strategy B (552) and clausal negation. The older negative marker *nun* has a very restricted distribution in this translation, appearing mostly in the fixed form *scha nun* 'if not'.

[22] English translation by Marcus Dods.

(552) Romansh (Oberland)
Bucca tuts pon prender ent quei plaid
NEG all.PL.M can.PRS.3PL take.INF in this.SG.M word
'Not everyone can accept this word' [Gabriel, Matt. 19:11; 1648]

Strategy C Unlike Strategy A, which remained a frequent strategy in several varieties, Strategy C ceased to be a dominant strategy. In Italian, Strategy C is employed three times in Malermi's version from 1471, exemplified in (553), but not in Brucioli's or Diodati's versions, published in 1532 and 1607, respectively.

(553) Italian
Non entrara nel regno di cieli ogni homo che a me dice
NEG enter.FUT.3SG in.the.SG.M kingdom of heaven every man that to me say.PRS.3SG
signor signor
lord lord
'Not every man who says to me 'Lord, Lord' will enter the kingdom of heaven.'
[Malermi, Matt. 7:21; 1471 (1490)]

Strategy C is also attested once in Romanian (554).[23]

(554) Romanian
Nu-l cuprind toț cuvîntul acesta
NEG-3SG.N.ACC understand.PRS.3PL all.PL.M word.ACC.SG.DE this.SG.N
'Not everyone understands this word.' [*Bucharest Bible*, Matt. 19:11; 1668 (1688)]

It appears, then, that Strategy C was largely abandoned in favor of Strategy B, which does not require V1 word order.

Strategy E Strategy E becomes an important strategy in some languages. In French, it involves a floating quantifier construction, as exemplified in (555). Other languages usually omit the subject in this construction, as demonstrated in Spanish (556). Such instances are classified as C/E (see §7.6.2), and are not included in the count.

[23]There is also an instance of Strategy C in the Lower Engadine dialect of Romansh (i), but it involves V2 word order, triggered by the clause-initial adverb *eir* 'also'.

(i) Romansh (Lower Engandine)
Et eir, per quai chi sun semm d'-Abraham, *nun* sun els *però* tuots
and also because that be.PRS.3PL seed of-Abraham NEG be.PRS.3PL they however all.PL.M
iffaunts
children
'Neither because they are the seed of Abraham are they all [his] children.' [Dorta, Rom. 9:7, 1679, italics in the original]

(555) French

a. Mais ilz n-obeissent point tous a l-euangile.
but they NEG1-obey.PRS.3PL NEG2 all.PL.M to the-gospel
'But they do not all obey the gospel.' [Lefèvre, Rom. 10:16; 1523 (1525)]

b. Vous n'-estes pas tous purs.
you NEG1-be.PRS.2PL NEG2 all.PL.M pure.PL.M
'You are not all pure.' [*Nouveau Testament de Mons*, John 13:11; 1667]

(556) Spanish
No soys limpios todos.
NEG be.PRS.2PL clean.PL.M all.PL.M
'You are not all clean.' [*Biblia del oso*, John 13:11; 1569]

One of the translations into the Upper Engandine dialect of Romansh also employs an expletive inversion construction, another type of Strategy E construction (see §7.5.2). This is exemplified in (557).

(557) Romansh (Upper Engandine)
Mu è nun haun brichia tuots ubedieu alg euangeli.
but EXPL NEG1 have.PRS.3PL NEG2 all.PL.M obey.PTCP to.the.SG.M gospel
'But not everyone obeyed the gospel.' [Bifrun, Rom. 10:16; 1560]

Summary This period is characterized by a significant increase in the frequency of Strategy B in most varieties. This is accompanied by a gradual decline in the frequency of Strategy A and a drop in the use of Strategy C. Strategy E becomes an important strategy in some languages, but it never becomes the dominant one.

10.2.6 Late Modern Romance

Figures 10.8 and 10.9 present the distribution of strategies in translations of the NT into Late Modern Romance varieties, divided into periods. Notice that certain languages have translations only from one of these periods, so the overall trends are not directly comparable between the two time frames. Additionally, these figures exclude languages for which only partial translations are available, but these are occasionally mentioned in the text.

In most languages, Strategy B is by far the dominant strategy. However, notable exceptions include French, Corsican, and Occitan, where Strategy B is absent (though see qualifications below), as well as Piemontese, where it is less prevalent (although the available data are limited to the 19th century). Meanwhile, Strategy D (negated clefts) begins to emerge in some languages. In contrast, Strategy A declines and even disappears in most varieties.

Strategy A In the 19th century, Strategy A is attested in translations into several Romance varieties. In French, it remains the dominant strategy, as exemplified in (558).

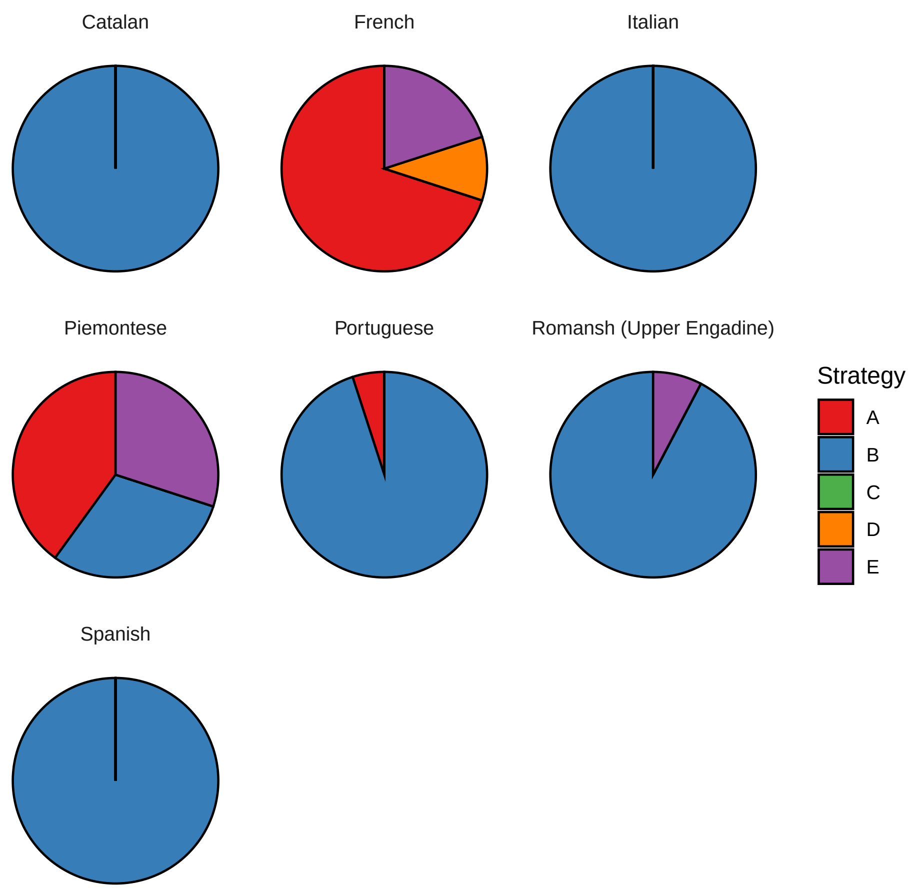


Figure 10.8: The distribution of strategies in translations of the NT into Late Modern Romance varieties (18th-19th centuries)

(558) French

Tous	ne	comprennent	pas	cette	parole
all.PL.M	NEG1	understand.PRS.3PL	NEG2	this.SG.F	saying

'Not everyone understands this saying.' [Segond, Matt. 19:11; 1880]

Piemontese is another language where Strategy A is frequently employed during this period (559). However, its prevalence in Piemontese is puzzling, given that Strategy B is also available.

(559) Piemontese

Ma	tutti	a	l'han	nen	ubbid-i	a-l	evangeli
but	all	SBJ.3	have.PRS.3PL	NEG	obey-PTCP	to-the	gospel

'But not all have obeyed the gospel' [Bert and Geymet, Rom. 10:16; 1834]

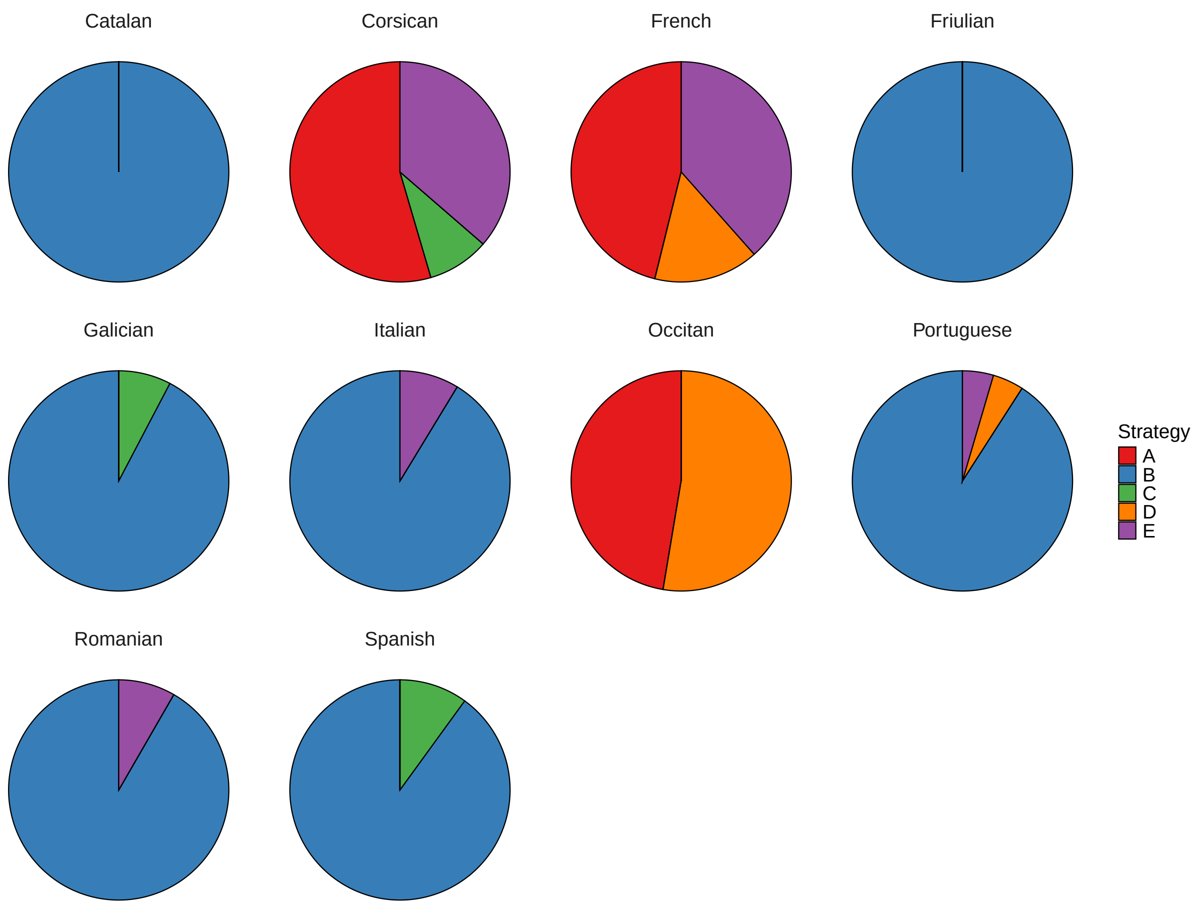


Figure 10.9: The distribution of strategies in translations of the NT into Late Modern Romance varieties (20th-21st centuries)

In Portuguese, a 1848 revision of Almeida's translation replaces instances of Strategy A (542) with Strategy B (560a). Curiously, however, another verse that had featured Strategy B in the original translation was altered into Strategy A in the revised edition (560b). On the other hand, a 1771 translation by António Pereira de Figueiredo does not contain instances of Strategy A.

(560) Portuguese

a. Todas as cousas me são licitas, mas nem todas as
all.PL.F the.PL.F things to.me be.PRS.3PL lawful.PL.F but NEG all.PL.F the.PL.F
cousas convém: todas as cousas me são licitas,
things be.PRS.3PL beneficial.PL all.PL.F the.PL.F things to.me be.PRS.3PL
mas nem todas as cousas edificão.
lawful.PL.F but NEG all.PL.F the.PL.F things build.up.PRS.3PL
'All things are lawful for me, but not all things are beneficial. All things are

lawful for me, but not all things build up.'
[Revision of Almeida, 1 Cor. 10:23; 1848]

b. todos não estais limpos.
all.pl.m neg be.prs.2pl clean.pl.m
'Not all of you are clean.' [ibid, John 13:11]

Strategy A is also attested in translations of Matthew from the 19th century into Corsican (561), Friulian (562), and Venetian (563).[24]

(561) Corsican
Tutti quelli, che mi dicunu, Signore, Signore, nun intrarannu in
all.pl.m those.pl.m the to.me say.prs.3pl Lord Lord neg enter.fut.3pl in
lu regnu di li celi
the.sg.m kingdom of the.pl heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Matthew in Corsican*, Matt. 7:21, 1861]

(562) Friulian
Duçh no capìssin chèste peràule
all.pl.m neg understand.prs.3pl this.sg.f word
'Not everyone understands this word' [Pietro dal Pozzo, Matt. 19:11; 1860]

(563) Venetian
Tuti no i capisse sta parola
all.pl.m neg sbj.3pl understand.prs.3 this.sg.f word
'Not everyone understands this word' [Fontana, Matt. 19:11; 1859]

[24]These translations are part of a series of translations of Matthew into Romance varieties of Italy. They are apparently based on Martini (1771), which employs Strategy B:

(i) Italian

a. Non tutti quelli, che a me dicono, Signore, Signore, entreranno nel
neg all.pl.m those.pl.m thay to me say.prs.3pl Lord Lord enter.fut.3pl in.the.sg.m
regno de'-cieli
kingdom of-heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Martini, Matt. 7:21; 1771 (1817/1832)]

b. Non tutti capiscono questa parola
neg all.pl.m understand.prs.3pl this.sg.f word
'Not everyone understands this word' [ibid, Matt. 19:11]

There are also contemporary translations of Matthew into Romance varieties of French (Frainc-Comtou, Guernésiais, and Picard), based on the French version by Louis-Isaac Lemaistre de Sacy. These translations follow the French version in employing Strategies E and A, hence they do not provide independent evidence for the use of Strategy A, as their choice of strategy might be influenced by their source text.

In translations from the 20th and 21st centuries, Strategy A is attested in French, Occitan (565), and Corsican (566).[25]

(564) French
Tous les hommes ne sont pas capables d'-accepter cet
all.PL.M the.PL.M men NEG1 be.PRS.3PL NEG2 capable.PL of-accept.INF this.SG.M
enseignement
teaching
'Not all men are capable of accepting this teaching.'
[*Bible en français courant*, Matt. 19:11; 1987]

(565) Occitan
a. Totes, praquò, an pas obeit à la bona novèla.
all.PL.M however have.PRS.3PL NEG obey.PTCP to the.SG.F good.SG.F news
'However, not everyone has obeyed the good news.'
[Cubaynes, Rom. 10:16; 1956]
b. Ça que la, aquela coneissença, totes l'-an pas.
however that.SG.F knowledge all.PL.M 3SG.ACC- have.PRS.3PL NEG
'However, not everyone possesses this knowledge.' [Larzac, 1 Cor. 8:7; 2016]

(566) Corsican
Ma tutti ùn hanu ubbiditu à a bona nutìzia.
but all.PL.M NEG have.PRS.3PL obey.PTCP to the.SG.F good.SG.F news
'But not everyone has obeyed the good news.' [Dubois, Rom. 10:16; 2005]

However, evidence suggests that Strategy A is also declining in French. While this strategy remains common and acceptable in Standard French, it appears to be becoming less natural in Colloquial French. For instance, in (567), the absence of the pre-verbal negative marker *ne* (NEG1)—a hallmark of informal speech—makes this construction slightly unnatural or stylistically marked, suggesting a clash between a construction associated with formal language (Strategy A) and informal grammatical features, specifically the absence of *ne* (Philippe Schlenker, p.c.). This indicates that Strategy A may be increasingly associated with the standard register and gradually disappearing from spoken French.

[25] The prevalence of Strategy A in translations into Occitan and Corsican may be partly influenced by French, the prestige variety in the regions where these languages are spoken. In Occitan, a 1956 translation by Canonge Juli Cubaynes favors Strategy A, whereas a 2016 translation by Joan Roqueta-Larzac predominantly employs Strategy D, though both strategies appear in each text. It remains an open question whether this difference reflects an actual change over time or simply a greater degree of independence from the French versions.

In Corsican, a 2005 translation by Christian Dubois was published in a bilingual edition alongside the French text of *La Nouvelle Bible Segond*. The Corsican translation closely follows the French version in the relevant verses, which may explain the high frequency of Strategy A.

(567) Colloquial French (Philippe Schlenker, p.c.)
Tout le monde va pas venir.
everyone FUT.3SG NEG2 come.INF
'Not everyone will come.'

The decline of Strategy A is even more pronounced in Quebec French, where speakers report that they do not use this construction and instead associate it with Metropolitan French (Justin Royer and Vincent Rouillard, p.c.).

Strategy B While most Romance languages share a preference for Strategy B, they sometimes differ in how these constructions are formed. For instance, earlier translations into the Upper Engandine and Lower Engandine dialects of Romansh employed the novel negative marker *brichia* (NEG2) in Strategy B, rather than the older negative marker *nun* (NEG1)—as shown in (550)-(551)—even though *brichia* is only optional in clausal negation. By contrast, Catalan employs the older negative marker *no* (NEG1) in Strategy B, rather than the optional novel negative marker *pas* (NEG2). This is demonstrated in (568).

(568) Catalan

a. No tots son capassos d'-aixó
NEG all.PL.M be.PRS.3PL capable.PL of-this
'Not everyone is capable of this' [Prat, Matt. 19:11; 1832]

b. No tothom és capaç d'-això que dieu
NEG everyone be.PRS.3SG capable of-this that say.PRS.2PL
'Not everyone is capable of what you say'
[*Bíblia Evangèlica Catalana*, Matt. 19:11; 2000]

19th-century Venetian stacks the negative markers *no* (NEG1) and *minga* (NEG2), as shown in (569). This is similar to the form of Strategy B in 16th-century French (547). It is possible that this construction was not fully conventionalized in Venetian, given that *minga* is used only 15 times within this text, meaning that *no* almost always appears by itself, as in (563).

(569) Venetian
No minga tuti colori, che me dise: Paron, Paron, i
NEG1 NEG2 all.PL.M those.PL.M that to.me say.PRS.3PL Lord Lord SBJ.3PL
vegnirà drento del regno in cielo
come.FUT.3 into the.SG.M kingdom of heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Fontana, Matt. 7:21; 1859]

In the Upper Engandine dialect of Romansh, *brichia* was largely replaced by *na* in Strategy B (570). The primary function of *na* is as a negative response particle particle, whereas standard negation is expressed by the pre-verbal negative particle *non* (NEG1). Thus, both *bricha* (the form spelled as *brichia* in earlier texts) and *na* are employed in constituent negation constructions, including Strategy B, but *na* is considerably more frequent than *bricha* in this text.

(570) Romansh (Upper Engandine)
Na tuots comprendan quaist pled
NEG all.PL.M understand.PRS.3PL this.SG.M word
'Not everyone understands this word' [Menni, Matt. 19:11; 1861]

As previously mentioned, Strategy B is not available in Standard French. However, some speakers do accept instances of this strategy. For example, Baunaz (2011) provides several examples of Strategy B, including (571).

(571) French (Baunaz 2011, 113)
Pas tous les garçons n'-ont bu de coca.
NEG all the boys NEG-have drank of coke
'Not all the boys drank coke.'

Speaker judgments on this construction vary. Several speakers I consulted found it unacceptable, while others accepted it, though some rejected the use of *ne* (which is required in Standard French), considering the construction strictly colloquial.

This construction appears to be more acceptable when the quantifier functions as a pronoun rather than as a determiner. For example, some speakers find (572a) natural but consider (572b) somewhat odd. Others, however, accept both constructions, much like (571).

(572) Colloquial French (Philippe Schlenker, p.c.)

a. Pas tout le monde va venir.
NEG2 everyone FUT.3SG come.INF
'Not everyone will come.'

b. %Pas tous les étudiants vont venir.
NEG2 all.PL.M the.PL.M students FUT.3PL come.INF
'Not all of the students will come.'

It remains an open question why the function of the quantifier affects acceptability. One possibility is that grammaticalization plays a role: negated QP constructions may be more advanced with pronouns due to a frequency effect. Since *tout le monde* is a pronoun that functions as a single unit, it is likely to occur more frequently in discourse compared to a determiner combined with a common noun (e.g., *tous les étudiants*). As a result, the collocation *pas tout le monde* may be more familiar or feel more natural to speakers than *pas tous les NP*, which has an open slot that can accommodate various nouns, making it feel less idiomatic.

Alternatively, the difference may be due to the availability of an alternative construction. The determiner-based structure in (572b) has a well-formed alternative in Strategy E—*Les étudiants vont pas tous venir* ('The students aren't all going to come'). By contrast, no equivalent alternative exists for (572a), which may explain why speakers find it more acceptable.

Speakers of Quebec French, on the other hand, have reported that Strategy B is commonly used in spoken language (Justin Royer and Vincent Rouillard, p.c.). Further research is needed to determine the sociolinguistic factors that govern this variation across the Francophone world.

Strategy C Strategy C is sporadically attested in some languages, such as Spanish (573) and Corsican (574).

(573) Spanish
y no desempeñan todos los miembros la misma función
and NEG perform.PRS.3PL all.PL.M the.PL.M members the.SG.F same.SG.F function
'And not all members perform the same function.'
[*Biblia de Jerusalén*, Rom. 12:4; 1973]

(574) Corsican
a. Nun capiscunu tutti sta parola
NEG understand.PRS.3PL all.PL.M this.SG.F word
'Not everyone understands this word' [*Matthew in Corsican*, Matt. 7:21; 1861]
b. Ùn lu capiscenu micca tutti st'-insignamentu
NEG1 it understand.PRS.3PL NEG2 all.PL.M this-teaching
'Not everyone understands this teaching'
[*Les Évangiles en Corse*, Matt. 19:11; 1994]

Among the varieties surveyed in this study, the only one where Strategy C appears to be the dominant strategy is the Nones dialect of Ladin. This variety lacks Strategy B and instead predominantly employs Strategy C (Enrico Flor, p.c.), as exemplified in (575).

(575) Ladin (Nones dialect, Enrico Flor, p.c.)
a. No a magnà zena tuti i popi (no).
NEG have eaten dinner all the boys NEG
'Not all the boys have eaten dinner.'
b. No g'è ci tuti (no)
NEG there.is/are here all NEG
'Not everyone is here.'

It is possible that Ladin shares this areal feature with certain Balkan languages, such as Slovenian (see §7.3.1), due to language contact. However, it could also be a coincidence. Friulian, another Romance variety spoken in a region between Ladin and Slovenian, currently favors Strategy B as its dominant strategy—at least in the dialect represented by my data. It remains an open question whether other Friulian dialects predominantly use Strategy C, which would support the hypothesis that Ladin's preference for Strategy C is part of a broader areal pattern.

Strategy D Some languages adopted Strategy D in this period. In Metropolitan French, Strategy D is sporadically attested. A 1880 translation by Louis Segond shows a preference for Strategy A, but it also contains an instance of Strategy D (576a).[26] In a 1910 revision, this construction is replaced by Strategy E (576b).

(576) French

a. Ce n'-est pas quiconque me dit: Seigneur, Seigneur!
this NEG1-be.PRS.3SG NEG2 anyone to.me say.PRS.3SG Lord Lord
qui entrera dans le royaume des cieux
who enter.FUT.3SG in the.SG.M kingdom of.the.PL heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Segond, Matt. 7:21; 1880]

b. Ceux qui me disent: Seigneur, Seigneur! n'-entreront
those.PL.M who to.me say.PRS.3PL Lord Lord NEG1-enter.FUT.3PL
pas tous dans le royaume des cieux
NEG2 all.PL.M in the.SG.M kingdom of.the.PL heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[Segond, Matt. 7:21; 1910]

Strategy D is more frequently employed in the 20th-century *Bible en français courant* (577), though Strategies A and E remain more common.

(577) French

a. Ce ne sont pas tous ceux qui me disent: "Seigneur,
this NEG1 be.PRS.3PL NEG2 all.PL.M those.PL.M who to.me say.PRS.3PL Lord
Seigneur", qui entreront dans le Royaume des cieux
Lord who enter.PRS.3PL in the.SG.M kingdom of.the.PL heaven
'Not everyone who says to me, "Lord, Lord," will enter the kingdom of heaven'
[*Bible en français courant*, Matt. 7:21; 1982 (1997)]

[26] Strategy D is also attested in other contemporary sources, such as in (i).

(i) French
Ce ne sont pas tous les dommages éprouvés dans ces
this NEG1 be.PRS.3PL NEG2 all.PL.M the.PL damages experience.PTCP.PL.M in these.PL
pays qui seront réparés; ce serait impossible, et ce ne
countries which be.FUT.3PL redress.PTCP.PL.M this be.COND.3SG impossible and this NEG1
serait pas juste.
be.COND.3SG NEG2 just.SG.F
'Not all the damage suffered in these countries will be redressed; it would be impossible, and it would not be fair.'
[Larnaude, *La guerre et la réparation des dommages*; 1916]

b. Car ce n'-est pas tout le monde qui accepte
because this NEG1-be.PRS.3SG NEG2 all.SG.M the.SG.M world who agree.PRS.3SG
de croire.
of believe.INF
'for everyone agrees to believe.' [ibid, 2 Thess. 3:2]

Two speakers of Metropolitan French I consulted (Angèle Bernard and Keny Chatain, p.c.) did not find these examples entirely natural. In contrast, Strategy D is considerably more frequent in Quebec French, as illustrated by (578).[27]

(578) Quebec French
D'ailleurs, c'est souvent les "mansplanners" qui nous font cadeau de la douce expression «Ce **ne** sont **pas tous** les hommes qui…». Ce **ne** sont **pas tous** les Allemands qui sont des nazis. Ce **ne** sont **pas tous** les chemins qui mènent à Rome, et ce **ne** sont **pas tous** les épisodes de 30 vies qui ont été écrits par Fabienne.
'Moreover, it is often the "mansplainers" who give us the gift of the sweet expression "Not all men…". Not all Germans are Nazis. Not all the roads lead to Rome, and not all the episodes of *30 vies* were written by Fabienne.' [web example[28]]

Even though Strategy D is restricted in Metropolitan French, it is not entirely absent. It is not easy to pinpoint its typical function, but it seems to be more compatible with generic statements about abilities and propensities than with episodic statements, reminiscent of the use of this strategy in present-day English (see §10.1.6). This is demonstrated in (579).

(579) Metropolitan French (Benjamin Spector, p.c.)
Ce ne sont pas tous les enfants qui en sont
this NEG1 be.PRS.3PL NEG2 all.PL.M the.PL.M children who PART be.PRS.3PL
capables.
capable.PL
'It's not every child who is capable of that.'

Finally, Strategy D is also attested once in Brazilian Portuguese (580).

(580) Brazilian Portuguese
Não é toda pessoa que me chama de "Senhor, Senhor" que
NEG be.PRS.3SG all.SG.F person that to.me call.PRS.3SG as Lord Lord that

[27]The *Bible en français courant* was produced through international collaboration, raising the possibility that the verses featuring Strategy D (577) were translated by speakers of Quebec French or another variety where Strategy D is frequently employed.

[28]Bonenfant, Rosalie. *Le Journal de Québec.* 22 September 2017, `https://www.journaldequebec.com/2017/09/22/pourquoi-je-ne-suis-pas-feminisme`. Accessed on 10 November 2024.

entrará	no	Reino	do	Céu
enter.FUT.3SG	in.the.SG.M	kingdom	of	heaven

'Not everyone who calls me, "Lord, Lord," will enter the kingdom of heaven'
[*Nova Tradução na Linguagem de Hoje*, Matt. 7:21; 2000]

Summary Over this period, Romance languages that had previously adopted Strategy B generally lost Strategy A. However, Strategy A remains frequently used in Standard French, Corsican, and Occitan, which lack Strategy B. Strategy C is only sporadically attested in most varieties but remains the dominant strategy in the Nones dialect of Ladin.

10.2.7 Summary and discussion

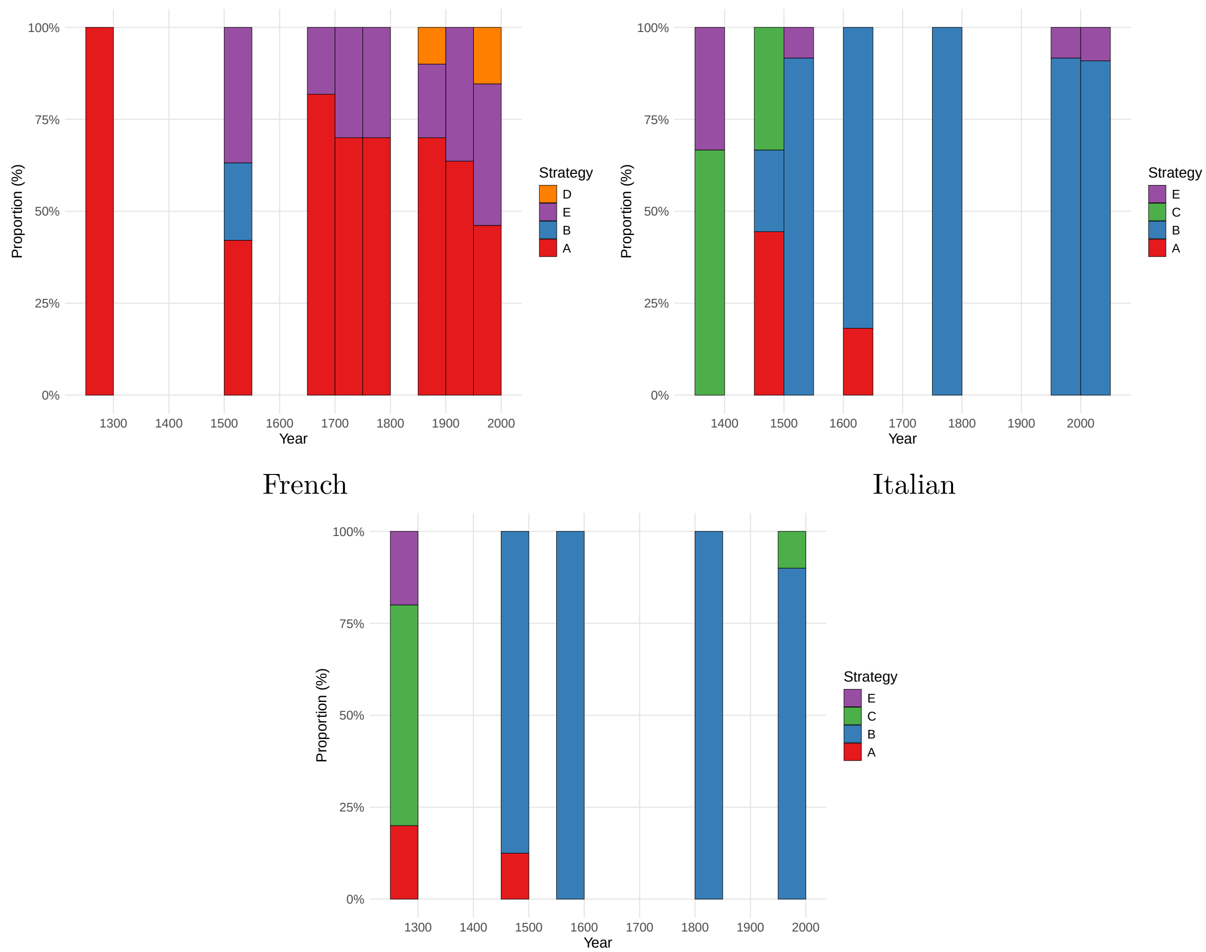
The evolution of strategies throughout the history of Latin and Romance languages strongly reflects the impact of competition between strategies. As predicted by the pathway in (412), Strategy A emerged in Medieval Romance as a result of a broader grammatical change: the grammaticalization of new universal quantifiers derived from *totus*. Conversely, the decline and eventual loss of Strategy A in most varieties was typically driven by the adoption of Strategy B, rather than by broader structural changes, aligning with the pathway in (411).

Figure 10.10 illustrates changes over time in the distribution of strategies across three Romance varieties with a long and relatively continuous recorded history in the corpus. Missing bars indicate periods for which no corpus data is available. Additionally, translations from the 13th and 14th centuries provide only a limited amount of data, increasing uncertainty regarding the relative frequencies of strategies. However, these periods are included here as they represent a stage prior to the adoption—or at least the conventionalization—of Strategy B.

As the figure shows, Strategy A was never as frequent in Italian and Spanish as in French, and it gradually disappeared in the former languages following the introduction of Strategy B.

Notably, there is no clear correlation between the adoption of Strategy B and whether a language underwent Jespersen's Cycle. One might expect that languages with newly grammaticalized negative markers (NEG2) would take longer to develop Strategy B, as a novel negator might initially have a more restricted syntactic distribution. However, some languages that developed new negators do not appear to have been late adopters of Strategy B. For instance, Strategy B is attested in the Upper Engadine dialect of Romansh as early as the 16th century, where it involves the novel negative marker *brichia* (550a), which was still optional in clausal negation at the time. Similarly, 19th-century translations into Piemontese and Lombard show Strategy B constructions featuring the NEG2 markers *nen* < 'nothing' and *mi(n)ga* < 'crumb' (Parry 2013, 78), respectively.

Conversely, some languages that retained the original Latin NEG1 marker, such as Corsican and the Nones dialect of Ladin, do not employ Strategy B. This suggests that the absence of Strategy B is not strongly correlated with whether a language underwent Jespersen's Cycle.



French

Italian

Spanish

Figure 10.10: Distribution of strategies over time across representative Romance varieties

But should this be surprising? According to Breitbarth et al. (2020), emphatic reinforcers undergo a transitional stage on their path to becoming clausal negation markers, during which they are reanalyzed as negative polarity adverbs (NPAs)—a type of NPI that appears in non-argument positions and reinforces negation. For example, in English, *one bit* functions as an NPA because it can occur outside argument positions, as in *I don't like it one bit.* In contrast, *one red cent* remains a regular NPI since it can only be used as an argument, as in *I wouldn't pay one red cent for this.*

Breitbarth et al. (2020) argue that NPAs are frequently used to express constituent negation, as in *Not only X but (also) Y* constructions, as illustrated in (581).

(581) Old Saxon (Breitbarth et al. 2020, 48)

thuo	niet	ekir	iro	selon	neuen	ok	[...]
then	NEG	only	3PL.GEN	soul.PL	but	also	

'then not only their soul, but also...'

From this perspective, newly grammaticalized negative markers are not necessarily limited to clausal negation and may also appear in negated QP constructions, as they develop from NPAs, which are often used to express constituent negation.

The findings from this study suggest that the specialization of a negative marker for constituent negation (specifically in negated QP constructions) is at least partially independent from its grammaticalization as a marker of clausal negation. In some varieties of Romansh, for example, the novel negative marker was already employed in negated QP constructions before it had fully grammaticalized as a clausal negation marker. This suggests that clausal negation and constituent negation should be seen as distinct functions of negative markers, even though they are expressed by the same elements in many languages.[29]

[29] This issue relates to the broader debate on the nature of semantic negation: the monoguist approach posits a single type of negation corresponding to the logical connective, while the ambiguist approach allows negation to semantically compose with non-propositional constituents. For further discussion, see Horn (1985, 1989); Collins and Postal (2014); Bar-Asher Siegal (2025), among others.

Chapter 11

General discussion

The previous chapter tested the Diachronic Hypothesis, which is repeated here for convenience as (582).

(582) *Diachronic Hypothesis*

- **(i)** In a given language, the relative frequency of transparent-scope constructions tends to increase over time.
- **(ii)** Conversely, inverse-scope constructions become more frequent only when the dominant transparent-scope alternative is lost due to broader grammatical changes, prompting the language to rely on the standard negation construction.

In Chapter 1, the interaction between these two diachronic trajectories was metaphorically compared to a game of snakes and ladders, illustrating the inherent asymmetry in how inverse-scope and transparent-scope constructions are adopted within a language over time. Each square in the game represents a possible balance between these two types of constructions. The bottom square corresponds to a language that relies exclusively on inverse-scope constructions, while the top square represents a language that relies solely on transparent-scope constructions. Notably, this "game" has no definitive endpoint, emphasizing the dynamic nature of these diachronic processes.

A bias toward scope transparency gradually nudges the language closer to the top square. This progression can sometimes be slow and incremental, but at other times it occurs more rapidly, akin to a swift "climb" up a ladder. However, at any point, the language may lose its dominant transparent-scope construction due to various grammatical changes, causing it to "slide down a snake" and increase the relative frequency of inverse-scope constructions. Importantly, there is no inherent force pulling languages toward the bottom square. Such a shift can still occur, but only as an indirect consequence of other, independently motivated grammatical changes.

The asymmetry between the decline and emergence of inverse-scope constructions is tied to their underlying causal factors, rather than the relative probability of each type of change. These probabilities are influenced by various factors, such as the development of a novel universal quantifier that might be incompatible with constituent negation, or the obsolescence of a particular word order pattern in negative clauses. In other words, there are numerous “snakes” that could lead to the loss of a transparent-scope strategy. However, the likelihood of such events occurring is not the central issue when evaluating evidence for a bias toward scope transparency. What matters most is understanding the causal mechanisms driving these changes.

Evaluating whether the evidence supports this hypothesis is a nuanced task, as determining whether a change stems from broader grammatical shifts often involves interpretation. Moreover, there is a significant risk of selectively rationalizing exceptions to fit the hypothesis. Therefore, the aim of this chapter is to present the evidence both supporting and challenging the hypothesis, ultimately providing a verdict—one that, by its nature, remains subjective.

The primary argument in favor of the hypothesis is that most Germanic and Romance languages exhibit a similar pattern: Strategy A, which reached its peak during the High and Late Middle Ages (approximately 1000 to 1500), gradually declined as transparent-scope strategies became more prevalent. This decline led to different outcomes across languages: in some, Strategy A disappeared entirely; in others, it survived primarily in formulaic expressions such as *All is not lost*; and in yet others, it remained relatively productive as a secondary strategy.

The decline and eventual loss of inverse-scope constructions typically follows the innovation of a transparent-scope strategy, as outlined in (583), rather than occurring as a direct consequence of broader grammatical changes.

(583) *Pathway of decline of inverse-scope constructions*

Stage 1 Inverse-scope constructions are frequently employed, possibly even the dominant strategy for expressing ‘Not all X are Y’ propositions.

Stage 2 The language innovates a transparent-scope strategy. Over time, the frequency of the novel construction increases, while the frequency of the older inverse-scope construction decreases.

Stage 3 The inverse-scope construction either falls out of use or becomes rare and mostly restricted to formulaic expressions.

The cross-linguistic variation within Romance languages aligns with the pathway of decline outlined in (583). Broadly speaking, Romance languages that developed Strategy B gradually abandoned Strategy A, often within a relatively short time frame. In contrast, languages that did not adopt Strategy B—such as Metropolitan French and Corsican—continue to employ Strategy A frequently. Additionally, there are Romance languages where neither Strategy A nor B is dominant. For example, Ladin (Nones dialect) primarily uses Strategy C, while Occitan favors Strategy D.

In Germanic languages, the situation is notably more complex. As in Romance languages, Strategy B became the dominant strategy in most Germanic languages, while Strategies C and D rarely emerged as dominant alternatives. However, unlike in Romance, the rise of Strategy B in Germanic languages resulted in a much slower decline in the use of Strategy A.

In fact, Strategy A continues to persist in many Germanic languages, often as a secondary strategy. Remarkably, there are even cases where Strategy A remains the dominant strategy today, despite the presence of Strategy B, as seen in languages such as Swedish and Faroese.

The reasons why Romance languages were generally more eager adopters of Strategy B than Germanic languages remain unclear. One possibility is that Strategy A reached a higher frequency in many Germanic languages, which may have contributed to its greater diachronic stability. However, the available quantitative data are insufficient to substantiate this hypothesis, leaving the question open to further investigation.

Crucially, in all of these languages, using the standard negation construction results in Strategy A. This indicates that the shift away from Strategy A was not driven by broader grammatical changes but rather by an increasing preference for transparent-scope strategies. To the best of my knowledge, no independent grammatical changes during this period rendered Strategy A less preferred. Instead, the decline of Strategy A follows the development of alternative, transparent-scope strategies.

Some languages took a long time to phase out Strategy A. In English, for instance, Strategy A is attested as early as the 10th century and remained the dominant strategy well into the 19th century. It was only in the second half of the 20th century that Strategy A experienced a significant decline in most English dialects. Today, many speakers find it odd outside of echo denial contexts and formulaic expressions.

Similarly, transparent-scope strategies often took significant time to gain prominence. In English, Strategy B emerged around the 14th century but remained relatively infrequent until the late 19th century. However, during the 20th century, the frequency of Strategy B rose sharply, becoming the dominant strategy within just a few decades. This trajectory gives the relative frequency of Strategy B in English the shape of a long-tailed S-curve, roughly mirroring the decline of Strategy A. In other languages, similar transitions occurred much more quickly, while in some, the shift toward transparent-scope strategies remains incomplete.

The transitions in English are visualized in Figure 11.1. Missing bars indicate periods for which no corpus data is available. As the figure illustrates, Strategy A began its decline during the late 19th century or early 20th century, coinciding with a gradual rise in Strategy B and the brief prominence of Strategy D, which later became highly restricted.

The prolonged survival of Strategy A, even in the presence of transparent-scope strategies, can be attributed to the effect of diachronic persistence. As long as Strategy A maintains a relatively high usage frequency, children continue to acquire and use it. Only when its frequency falls below a certain threshold does generational transmission become disrupted, leading to the eventual decline of Strategy A (see Sankoff 2019, for a discussion of the role of transmission in variation and language change).

More broadly, this pattern aligns with the phenomenon of linguistic obsolescence, where declining forms continue to persist at low frequencies for an extended period, producing a characteristic long-tailed S-curve in their frequency distribution (Sankoff and Wagner 2020; Brook 2024). Obsolescent forms often retain footholds through stylistic or pragmatic specialization, slowing their eventual disappearance. Within the context of this study, Strategy

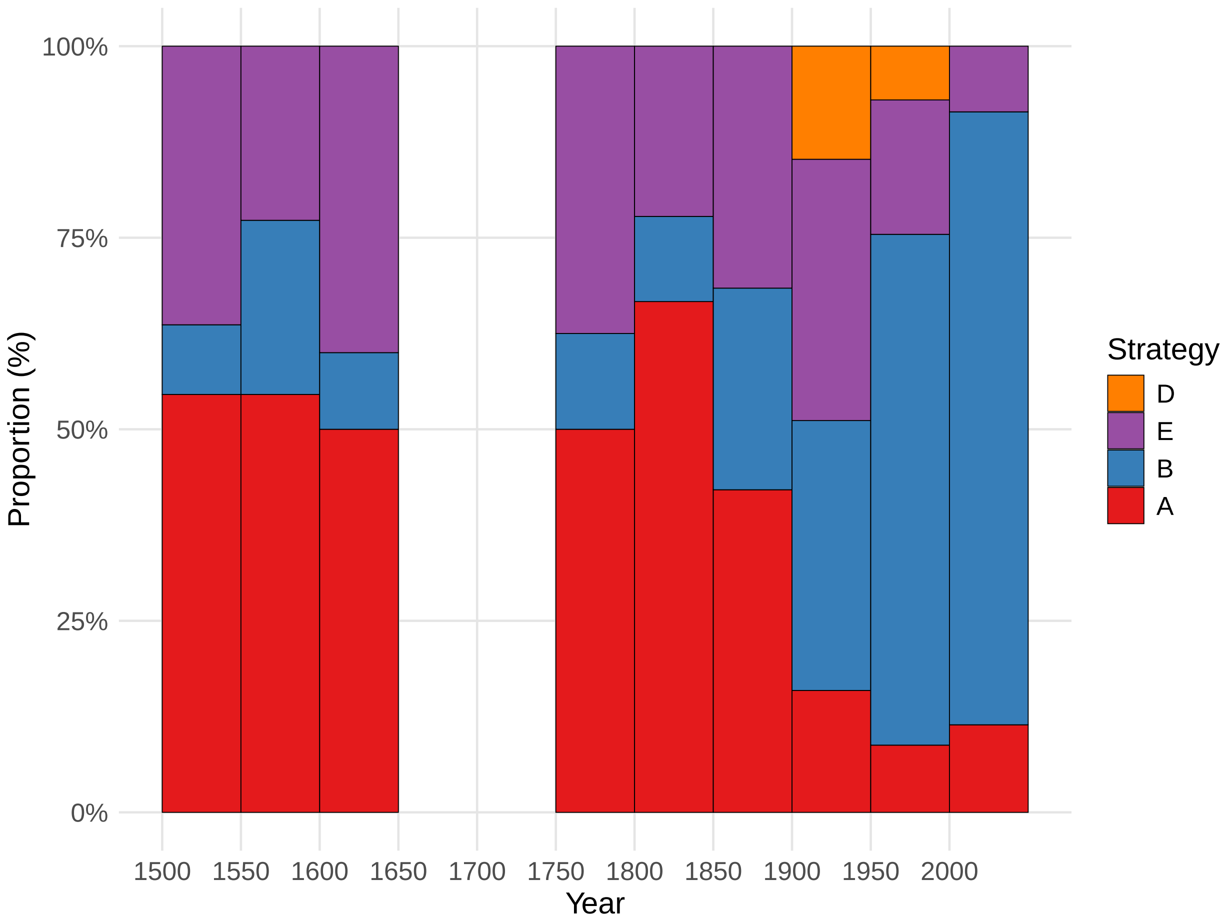


Figure 11.1: The distribution of strategies in translations of the NT into English since the 16th century

A exemplifies this process, persisting in specific languages through its association with formulaic expressions, such as *All is not lost*, or its use in contrastive contexts (see Chapter 6).

Conversely, a newly emerging transparent-scope strategy may face syntactic or pragmatic restrictions that limit its frequency. For example, in colloquial French, some speakers find Strategy B acceptable when the quantifier functions as a pronoun but not when it serves as a determiner (572). Similarly, clefts or specific word order patterns may retain the pragmatic markedness often associated with these constructions. Over time, these restrictions can be relaxed, allowing the transparent-scope strategy to become more frequent, but this process can take considerable time.

Thus, the pathway of decline of Strategy A largely aligns with the diachronic pathway outlined in (583). The (re-)emergence of Strategy A was predicted to follow the pathway below:

(584) *Pathway of emergence of inverse-scope constructions*

Stage 1 Inverse-scope constructions are rare or not in use.

Stage 2 The dominant transparent-scope construction is lost due to broader grammatical changes.

Stage 3 The language resorts to using the standard negation construction, often resulting in inverse scope.

The observed processes of the emergence of Strategy A are largely compatible with the pathway outlined in (584). However, the emergence of Strategy A is not necessarily a last resort. Rather than being necessitated by the loss of other options, Strategy A often arises in variation with a transparent-scope alternative.

For example, in Old Germanic languages, Strategy A developed as an alternative to Strategy C. While Strategy C became restricted following the decline of the NEG-V1 pattern, it did not disappear entirely. Since this pattern remains available, one might expect speakers to favor Strategy C—similar to present-day Icelandic, where Strategy C is the dominant strategy even though this word order is not the usual word order in other types of negative clauses. In other words, the emergence of Strategy A in Old Germanic seems to align with broader changes in word order but is not strictly mandated by them. Instead, the limited data seems to suggest that the choice between Strategies A and C reflects the broader distribution of NEG-V1 and V2 patterns in negative main clauses.

Similarly, the emergence of Strategy A in Faroese was not an inevitable consequence of the decline of Strategy C. Unlike Icelandic, Faroese also had Strategy B at its disposal. However, Strategy B appears to have been restricted in some way—similar to its use in present-day Swedish—which prevented it from filling the gap left by the decline of Strategy C.

In Swedish, the proposed explanation for the re-emergence of Strategy A is that Strategy B became restricted following the shift from the negative particle *icke* to *inte*. The exact reason for this restriction remains unclear. Both *icke* and *inte* share a similar lexical source as neuter forms of a negative indefinite and are fully grammaticalized. Furthermore, given that Germanic languages typically do not permit constituent negation of most constituents—except in coordination and ellipsis constructions—there is little independent evidence to suggest that *inte* has a more restricted syntactic distribution than *icke* (aside from compounds such as *icke-lingvister* 'non-linguists').

A comparable issue arises in 19th-century Icelandic, where Strategy A first appears in the available data. I have argued that the NEG-V1 pattern was not yet fully conventionalized during this period and was apparently restricted in a manner similar to its use in Swedish at the time. This restriction, combined with a broader shift across Germanic languages toward favoring pronominal quantifiers over adverbial (floating) ones, likely contributed to the rise of Strategy A. However, it remains unclear whether Icelandic underwent a similar shift in its use of pronominal quantifiers. The available data, which focuses on 'Not all X are Y' sentences, is insufficient to determine whether this type of broader change took place.

Another potential caveat concerns the emergence of Strategy A in Medieval Romance. According to the proposed scenario, newly grammaticalized universal quantifiers derived from Latin *totus* 'whole' were initially incompatible with constituent negation. However, Strategy A is also observed in constructions where the universal quantifier is derived from Latin *omnis* 'all'. In such cases, we would expect a continuity of its use in negated QP constructions from Latin into those Medieval Romance varieties that retained reflexes of *omnis*. Why, then, do inverse-scope constructions appear with descendants of *omnis*? In Section 10.2, I proposed that these constructions may have emerged by analogy with those involving quantifiers derived from *totus*. Nevertheless, further research is necessary to validate this hypothesis.

Language contact represents another explanatory factor that must be considered. Within Europe, the distribution of Strategy A in present-day languages exhibits notable areal clustering in two distinct regions. The first encompasses the area of France, where Strategy A is currently attested not only in French but also in Corsican, Occitan, and Breton. The second region broadly corresponds to the Nordic countries—including Finnish and Estonian but excluding Icelandic. Strategy A is especially frequent in the eastern part of this region—in particular Finnish, Estonian, and Swedish—but considerably less frequent in Norwegian and Danish, which prefer Strategies B and D. Faroese constitutes an interesting exception: despite historically extensive contact with Danish, it predominantly employs Strategy A. This suggests that the re-emergence of Strategy A in Faroese occurred independently of language contact.

It is challenging, however, to disentangle the influence of internal grammatical factors from external linguistic contact. The above-mentioned languages spoken in France all share two key characteristics: they lack Strategy B and exhibit a stronger tendency toward SVO word order compared to related languages outside this region. These features alone create favorable conditions for the emergence of Strategy A, though other transparent-scope strategies, such as Strategy D in Occitan, remain available. Consequently, it is unclear whether these languages adopted Strategy A directly through contact with French or whether its emergence was indirectly facilitated by their shared grammatical properties—particularly the absence of Strategy B. Most likely, both factors play a role.

The situation in the Nordic region is more complex. Until the early 20th century, Strategy A was in decline across Mainland Scandinavian languages (Swedish, Norwegian, Danish), paralleling developments observed in other Germanic languages such as English and Dutch (in German, this decline occurred much earlier). The unexpected re-emergence of Strategy A as the dominant strategy in contemporary Swedish thus raises the question of language contact. While the revival of Strategy A in Faroese clearly cannot be attributed to influence from Danish (which prefers Strategies B and D), a contact-based explanation is harder to exclude for Swedish, given that Strategy A is also dominant in Finnish, a neighboring language. However, although contact with Finnish may have contributed to the renewed dominance of Strategy A in Swedish, it seems unlikely to have been the primary cause. Majority languages typically exert greater influence on minority languages than the reverse (Thomason 2001), and Swedish, being the majority language in Sweden, would not typically be expected to adopt grammatical features from Finnish, a minority language. Consequently, internal grammatical developments in Swedish are more likely candidates for the re-emergence and current dominance of Strategy A.

More broadly, we may ask whether language contact between Germanic and Romance languages influenced the distribution of strategies. Within Europe, Strategy B is generally the dominant transparent-scope strategy, whereas Strategies C and D show more limited regional clustering. Notably, Strategy B emerged in several Germanic and Romance languages simultaneously during the late Medieval period, suggesting that areal diffusion may have significantly contributed to its adoption and spread. Crucially, this emergence is unlikely to result from broader grammatical changes such as Jespersen's Cycle, given that Strategy B often involves negative particles directly inherited from Latin in Romance languages, rather than newly grammaticalized forms.

Moreover, Strategy A persisted for an extended period in French and, to a lesser degree, Romansh, both languages that historically experienced substantial contact with Germanic languages. For Romansh, any influence from German—the primary Germanic language with which it was in contact—could only have been indirect, since German lost Strategy A relatively early compared to other Germanic languages. The relatively low frequency of Strategy B in Early New High German might have contributed to Romansh's delayed adoption of this strategy, but this remains highly speculative, as Strategy B is already attested in the earliest NT translation available to me, dating back to the 16th century. Similarly, the prolonged absence of Strategy B in French could theoretically be linked to the slower diffusion of Strategy B across northwestern Europe; however, French appears to have been leading rather than following this trend.

The absence of Strategy B in French also raises a fundamental question: why do some expected linguistic changes fail to materialize? Rosemeyer and Grossman (2021) argue that historical linguistics often focuses on explaining why a change occurred, but it is equally important to ask why certain developments did not take place—known as the actuation problem (Weinreich et al. 1968). They propose that systematic blocking effects—whether due to language-specific pragmatic constraints, discourse patterns, or broader structural factors—can prevent even well-attested grammaticalization pathways from taking hold. Their study of the FINISH > ANTERIOR grammaticalization pathway demonstrates that, even when a construction aligns with a widespread cross-linguistic pattern, it may fail to conventionalize if the necessary pragmatic inferences do not become entrenched in speakers' linguistic behavior.

However, in the case of Strategy B in French and several neighboring Romance varieties, no such blocking effects can be readily identified. Moreover, French and its neighboring varieties are situated in a linguistic area where Strategy B is common. Yet, despite prolonged contact with languages that have adopted this strategy, Strategy B did not spread to certain languages. The reason for this divergence remains unclear, highlighting an important gap in our understanding of why certain changes do not occur.

Overall, the predictions of the Diachronic Hypothesis are supported by the results. However, the findings suggest that the bias for scope transparency exerts a relatively weak effect. First, inverse-scope constructions can persist for an extended period even when transparent-scope alternatives are available. Second, the adoption of Strategy A does not necessarily require a language to face insurmountable constraints. When a dominant transparent-scope strategy becomes restricted due to broader grammatical changes, a language may begin using Strategy A even if the pre-existing transparent-scope strategy is not entirely lost. In other cases, the language has the option of shifting to another available transparent-scope strategy but instead chooses to adopt Strategy A. Thus, the use of Strategy A is neither a last resort synchronically nor a last resort when it first emerges in a language.

Part V

Conclusion

Chapter 12

Conclusion

As the results of this study have already been summarized in Chapter 1, this chapter focuses on discussing the broader implications of these findings, highlighting the methodological contributions, and outlining prospects for future research.

Theoretical implications This study has demonstrated that scope transparency functions as a weak universal force, favoring the innovation and the increased reliance on transparent-scope strategies. However, this bias is not strong enough to override all other pressures at the synchronic level—in particular, persistence effects, where a pre-existing inverse-scope construction remains in use even after a transparent-scope alternative has emerged.

A key finding of this research is that the cross-linguistic variation in the availability of inverse scope is not entirely predictable based on a language's grammatical properties or its inventory of transparent-scope strategies. While certain syntactic features—such as word order or the type of negative marker—can influence which strategies a language is more likely to have at its disposal, they do not fully determine whether a language favors transparent or inverse-scope constructions.

Instead, each language exists in a synchronic state reflecting its unique diachronic trajectory. While the bias for scope transparency gradually encourages a shift toward greater reliance on transparent-scope strategies, this process is neither unidirectional nor deterministic. Broader grammatical changes can disrupt or even reverse this progression, leading to the re-emergence of inverse-scope constructions. Crucially, the timing of these grammatical shifts influences a language's synchronic state through persistence effects, meaning that the distribution of strategies in a given language is not solely determined by its current grammatical properties.

This has important implications for understanding how soft syntactic constraints shape languages (see §3.1.1). In the specific case of scope transparency, its effects can sometimes be directly observed in synchronic data (Bobaljik and Wurmbrand 2012). However, as this study demonstrates, persistence effects can obscure these influences, making them undetectable in many languages. Consequently, the impact of scope transparency must be inferred from cross-linguistic distributional asymmetries and recurrent diachronic trajectories rather than from language-specific patterns alone. Thus, this study underscores the necessity of integrating cross-linguistic and diachronic perspectives when investigating universal biases, as doing so reveals how these biases shape long-term language evolution without necessarily producing uniform or immediately observable outcomes.

More broadly, this study contributes to our understanding of competition in the syntax-semantics interface. Most constraint-based models of synchronic competition assume that candidate sets are restricted to PFs derived from the same LF (Müller and Sternefeld 2001), which means that inverse-scope constructions and negated QP constructions would not directly compete since the latter involve constituent rather than clausal negation (see §3.1.1). However, this study demonstrates that diachronic competition does not operate in the same way. Instead, the evidence suggests that constructions expressing the same meaning can block each other over time, even when they differ substantially in their syntactic structure.

In particular, this study argues that for two constructions to compete diachronically, it is sufficient that they encode the same truth-conditional meaning and involve the same scope-bearing elements (a universal quantifier and negation). By contrast, no evidence was found for a blocking effect between logically equivalent constructions that differ in their quantificational expressions—such as those where an existential quantifier takes scope over negation, yielding a 'Some X are not Y' statement, which is truth-conditionally equivalent to 'Not all X are Y' (see §2.1).

A related issue concerns how universal biases are reflected in typological distributions. This study finds that inverse-scope constructions (Strategy A) are more common cross-linguistically than any individual transparent-scope alternative. Moreover, when considering which strategy serves as the dominant one in a given language, Strategy A consistently ranks either first or second across all macro-areas (see §8.1). This raises an important question: how can the universality of the bias toward scope transparency be reconciled with the widespread presence of inverse-scope constructions?

It has long been recognized that typological distributions do not always reflect underlying linguistic preferences. For example, despite arguments that languages exhibit a universal bias toward SVO word order, SOV languages remain equally common cross-linguistically (Dryer 2013a). Such discrepancies are typically attributed to inheritance from ancestral languages (Gell-Mann and Ruhlen 2011; Maurits and Griffiths 2014; Hammarström 2015).

Suppose a theory predicts a preference for Type A over Type B, yet cross-linguistically, languages of Type B are just as common. Maslova (2000) identifies two key factors that can skew current distributions through inheritance: (i) proto-languages of Type B may have spread and diversified more extensively than those of Type A, with daughter languages tending to preserve their ancestral type; or (ii) there may have been more proto-languages of Type B to begin with, and this initial asymmetry continues to shape the present distribution.

This insight has led researchers to adopt a dynamic approach to typology, which focuses on transition rates between Type A and Type B rather than static distributions. The key idea is that even if Type B languages are prevalent today due to inheritance, analyzing transition probabilities can reveal whether shifts from Type B to Type A are more frequent than the reverse. If so, this provides evidence for an underlying linguistic bias toward Type

A, effectively neutralizing the confounding effects of historical inheritance (Maslova 2000; Dunn et al. 2011; Jäger and Wahle 2021).

The findings of this study point to a different competing pressure. Rather than being a product of inheritance from proto-languages, the prevalence of inverse-scope constructions arises from cyclic disruptive changes. While a universal bias may gradually favor a shift toward transparent-scope strategies, Strategy A benefits from being the fallback strategy when broader grammatical changes interfere.

What makes the phenomenon examined here special is that it involves constructions that are deeply intertwined with more fundamental linguistic features, such as word order, negation strategies, and the inventory of universal quantifiers. These constructions do not exist in isolation but as part of a larger grammatical system with multiple interdependent components. Consequently, changes in one area of the system can reshape the distribution of strategies, sometimes overriding the bias for scope transparency and influencing whether a language ultimately favors transparent-scope or inverse-scope strategies.

The recurrent rise and fall of inverse-scope constructions can be understood in relation to attractor states—structural patterns that languages tend to converge on over time and rarely abandon unless an unlikely combination of influencing factors comes into play (Haig 2018; Schmidtke-Bode 2019; Seržant and Moroz 2022). While languages generally evolve toward scope transparency, accidental changes that push them in the opposite direction occur frequently enough to prevent the emergence of an attractor state. Instead, languages remain in a state of fluctuation, where the distribution of strategies tends to shift over time rather than stabilizing into a preferred long-term pattern.

This highlights the importance of understanding the causal factors driving diachronic change. While the dynamic approach provides estimates of transition rates, it does not explain what triggers these shifts. As a result, it cannot distinguish between functionally motivated changes and those that arise as incidental byproducts of developments in other areas of the linguistic system. However, a key limitation of the diachronic analysis in this study is that it relies on the subjective interpretation of causal mechanisms underlying these changes.

Another approach employed in this study is to analyze synchronic distributions, but rather than simply comparing the relative prevalence of different types—for instance, whether or not inverse scope is available—it evaluates the observed frequency of a given type against an independently established baseline. This method allows for objective statistical analysis but comes with its own limitations: it presupposes a specific causal model for determining the expected baseline, which may not always provide a fully accurate or unbiased point of comparison.

Finally, the disruptive effect of cyclic changes in the broader linguistic system further demonstrate that language change can sometimes lead to structurally "worse" outcomes in specific grammatical domains. An important question raised by Enke et al. (2024) is whether such changes can always be explained as unwelcome byproducts of local improvements in other areas of the linguistic system. The findings of this study support the view that an increased reliance on constructions violating scope transparency is typically triggered by shifts occurring in other grammatical domains. Crucially, there are no clear instances of a language adopting Strategy A for no apparent reason, nor of a language abandoning a transparent-scope strategy without broader grammatical restructuring.

Methodological contributions This dissertation expands the scope of semantic typology beyond the lexical domain to investigate cross-linguistic variation in the expression of propositional meaning. Traditionally, semantic typology has focused on lexical semantics, including kinship systems, color terms, and body part nomenclature (see Evans 2010, for a review), as well as inventories of logical expressions like indefinite pronouns (Haspelmath 1997) and quantifiers (Gil 1992; Bach et al. 1995; Keenan and Paperno 2012; Paperno and Keenan 2017). By contrast, this study examines how languages encode a specific type of proposition—namely, 'Not all X are Y'—on a scale that has not been previously achieved.

Furthermore, it introduces an innovative use of a parallel corpus to examine a phenomenon that has traditionally been studied through grammaticality judgments and elicitation. By adopting this corpus-based approach, the study significantly expands the cross-linguistic coverage, not only increasing the number of languages studied but also enhancing the typological diversity of the sample. Thus, it demonstrates that a phenomenon previously described primarily in a small set of well-documented languages can be systematically investigated on a much broader scale. Moreover, the use of corpus data allowed for the collection of diachronic evidence, shedding light on the dynamics of change in this semantic domain—an aspect that would have remained inaccessible through grammaticality judgments alone.

Additionally, this study contributes to corpus-based typology by validating the use of NT translations for investigating semantic phenomena (see Chapter 6). While parallel corpora—including NT translations—have been previously used in typological research (see §5.1), their reliability for studying semantic properties remains an open question. By comparing the distribution of strategies in NT translations with that found in online news corpora for nine present-day Germanic languages, the analysis demonstrates that, despite its relatively small size and restricted genre, the NT corpus provides a reliable source of data for analyzing subtle semantic phenomena. These findings highlights the potential of parallel corpora as valuable tools for cross-linguistic and diachronic research from a meaning-based perspective, where the objective is to examine how different languages encode a specific type of meaning.

Parallel corpora also hold significant potential for the study of semantic universals. While the primary focus was on cross-linguistic variation in the availability of inverse scope, it also indirectly tested the semantic universal proposed by Horn (1972) regarding the non-lexicalization of 'not all'. Notably, it uncovered a single counterexample to this generalization, which had long been considered an absolute universal (see §9.3). This finding underscores the value of parallel corpora in investigating semantic universals, particularly in cases where identifying potential counterexamples is akin to looking for a needle in a haystack (assuming the needle is even there to be found).

Distinguishing between a lexicalization that is truly absent across languages and one that is merely rare requires an exceptionally large and diverse language sample. Grammars and dictionaries typically do not provide negative evidence regarding what meanings are not encoded in a given language, unless their absence is typologically unusual. In contrast, parallel corpora offer a systematic and scalable method for probing such questions, making them an especially promising tool for identifying cross-linguistic patterns and testing claims of absolute universality.

Future directions This study raises several compelling questions for future research.

One immediate question concerns how a strategy evolves when it either increases in frequency or declines over time. A natural expectation is that such changes involve an expansion or contraction of the range of contexts in which the construction is used, much like the process of grammaticalization (Hopper and Traugott 1993, among others). But what determines which contexts form the core usage of a strategy and which are more peripheral?

For example, both Strategy A and Strategy D declined in English over the 20th century. While both are pragmatically marked relative to more frequent strategies, such as Strategies B and E, their distributions are not identical. Beyond their presence in specific formulaic expressions—such as *All is not lost* (Strategy A) and *It's not every day that…* (Strategy D)—each strategy serves distinct discourse functions. Strategy A is strongly associated with echo denial and, outside of this context, appears to be more compatible with generic statements. Strategy D also tends to occur in generic contexts but in a subtly different way.

When not used for echo denial, Strategy A often carries a rhetorical effect, reinforcing the idea that the rejected statement is patently false. This is exemplified in the naturally occurring example in (585), previously discussed in §5.4. In this example, the assertion is uninformative because no one is expected to seriously believe that every email the speaker writes is a great work of art. The use of Strategy A seems to highlight its status as given information.

(585) I do think one of the messages of A.I. is that most people make middling work, and middling work is easy to replace. **Every email I write is not a great work of art**, you know.
[*The Ezra Klein Show*, "Will A.I. Break the Internet? Or Save It?", 5 April 2024]

By contrast, while Strategy D is also associated with generic statements in present-day English, it does not appear to carry the same rhetorical effect as Strategy A. Although it can be broadly characterized as more emphatic than Strategy B—the neutral strategy in English—it more often introduces new information, highlighting exceptions to a previously implied or assumed universal generalization. This is illustrated in (586).

(586) Boehner: The big bad House Republican budge would discard everything under the sun according to my friends across the aisle, would still require a $5 trillion increase in the debt ceiling over the next 10 years. Why? Because of the great big demographic bubble baby boomers like me that are going to retire and continue retiring for the next 20, 25 years. It's a big challenge.
Burnett: **So it's not every increase in the debt ceiling that you're going to say has to be matched one for one**. It's the point you're making with this one

to try to force overall reform, willing to borrow for your plan?

[COCA: Spoken; 2012]

Why did Strategies B and D gravitate toward their respective discourse functions as their distribution became more restricted? More broadly, is the division of labor between different strategies only shaped by language-specific developments, or do certain inherent properties of these strategies guide the relationship between form and discourse function? For instance, is the discourse function of Strategy D connected to the broader function of negated clefts across languages?

Answering these questions requires a more detailed examination of corpus data, including historical corpora that can shed light on how these strategies evolved over time. While this study has focused on tracking changes in the relative frequencies of different strategies, future research should aim to uncover the factors influencing their selection in specific contexts and how these preferences shift over time. A key limitation of using a parallel corpus is that it constrains the range of contexts in which this type of meaning appears. As a result, it cannot reveal the factors that determine the choice between different strategies, such as syntactic, semantic, pragmatic, or processing-related constraints that may guide speakers' and writers' preferences. To fully address these issues, future studies should incorporate corpora containing original texts and spoken data, allowing for a more nuanced investigation of how these various factors interact in shaping the distribution of these strategies.

This need for a broader empirical foundation aligns with the concerns raised by Miestamo et al. (2022), who highlight the gap between proposed usage-based explanations for cross-linguistic variation and actual empirical studies of language use. While typologists often that discourse preferences shape grammatical patterns, these claims frequently remain untested. Investigating the choice between different strategies for expressing 'Not all X are Y' in corpora of original texts and spoken data would provide precisely the kind of empirical discourse-based evidence Miestamo et al. (2022) advocate, allowing for a clearer understanding of how syntactic, semantic, and pragmatic factors interact in real language use.

Another promising avenue for future research is to further strengthen the empirical foundation of the Scope Transparency Hypothesis from a psycholinguistic perspective. While experimental studies have investigated inverse scope in other types of constructions, *all…not* constructions have received comparatively little attention (see §3.2.2). To establish that inverse scope in *all…not* constructions is intrinsically associated with processing difficulty, it would be crucial to demonstrate this effect in languages where inverse-scope constructions are frequently used. The reasoning behind this approach is that evidence for a processing bias must come from a language where the bias runs counter to usage frequency; otherwise, processing difficulty could simply reflect lack of exposure rather than an inherent cognitive cost (Bickel et al. 2015; Sauppe et al. 2023).

An experimental study could also help address the question raised in Section 2.4: does inverse scope arise from a mismatch between scope relations and word order, or between scope relations and hierarchical structure? When these two factors conflict, which one takes precedence? This question ultimately boils down to whether *all…not* constructions incur processing difficulty in languages where the canonical subject position is argued to be structurally lower than negation.

Furthermore, if inverse scope is primarily determined by the relationship between scope relations and word order—as argued in this study—why do some verb-final languages favor wide-scope negation constructions (Strategy D) over *all…not* constructions (Strategy A), even though this does not alter the relative order of the QP and the negative marker? This pattern is particularly evident in Japanese (316) and Korean (318), both of which overwhelmingly prefer Strategy D over Strategy A, the latter being highly restricted.

In this study, I did not analyze these wide-scope negation constructions as involving inverse scope because they contain overt cues indicating that negation is in a structurally high position. This contrasts with *all…not* constructions, where the hierarchical relations must be inferred either from word order or based on additional diagnostics (see §2.4). The key question, then, is whether these overt cues reduce or even eliminate the processing difficulty associated with a mismatch between scope relations and word order. If they do not, what motivates the use of Strategy D over the structurally simpler Strategy A? Addressing these questions could provide deeper insights into how processing constraints shape syntactic structures across languages.

Appendix: New Testament translations cited in this work

Table A.1 lists the New Testament translations cited throughout this study. The titles given here are the commonly used names of the translations rather than their original full titles, which are often lengthy or not distinctive enough to identify the translation. For example, the first edition of the *King James Version* (1611) bore the title "THE HOLY BIBLE, Conteyning the Old Testament, AND THE NEW: Newly Translated out of the Originall tongues: & with former Translations diligently compared and reuised, by his Maiesties Speciall Comandement".

For each translation, the table provides both the year of the original publication and the edition from which the corpus data were drawn. Since the text itself may differ across editions, an effort was made to use the earliest available edition for each translation. When two editions share the same name but differ in content, the edition year is included parenthetically, for example, *Louis Segond (1880)* versus *Louis Segond (1910)*.

Language	Title	Original publication	Edition
Afrikaans	South African Bible Society (1953)	1933	1953
	South African Bible Society (1983)	1983	1983
	Nuwe Lewende Vertaling	2006	2006
	Bybel vir Almal	2008	2008
	Die Bybel 2020 Edition	2020	2020
Basque	Elizen arteko Biblia	1983	1983
Belarusian	Vasil Syargeevich Syomukha	2003	2003
Breton	Guillaume Le Coat	1897	2011
	Alfred Llewellyn Jenkins Revision	1897	1897
	Kenvreuriez ar brezoneg	1988	1988
Bulgarian	Tsarigrad	1871	1871
	Easy-to-Read Version	2000	2000
	Contemporary Bulgarian Translation	2013	2013
Catalan	Josep Melcior Prat i Colom	1832	1832
	Bíblia Evangèlica Catalana	2000	2000
Corsican	Il Vangelo di S. Matteo	1861	1861
	4 Evangiles en Corse	1994	2005
	Christian Dubois	2005	2005
Czech	Bible 21	2009	2009
Danish	Christian 2.s danske Bibel	1524	1524
	Christian 3.s danske Bibel	1550	1550

Language	Title	Original publication	Edition
	Hans Poulsen Resens	1607	1607
	Resen–Svane	1647	1838
	Frederik VI's Nye Testamente	1819	1819
	Den autoriserede oversættelse	1907	1907
	Bibelen på hverdagsdansk	1992	1992
Dutch	Liège Harmony	13th century	13th century
	Stuttgart Harmony	1332	1332
	Liesvelt Bible	1526	1542
	Vorsterman Bible	1528	1531
	Leuven Bible	1548	1548
	Biestkens Bible	1560	1560
	Deux-Aes Bible	1562	1562
	Statenvertaling	1637	1637
	Gerbrand Vissering	1854	1854
	Voorhoeve New Testament	1877	1877
	Nederlands Bijbelgenootschap	1951	1951
	Nieuwe Bijbelvertaling	2004	2004
English	Wessex Gospels	990	990
	14th century biblical version	14th century	14th century
	Wycliffe Bible	1382	1395
	Tyndale Bible	1526	1526
	Geneva Bible	1557	1560
	Douay–Rheims Bible	1582	1582
	King James version	1611	1611
	Quaker Bible	1764	1764
	Thomson's Translation	1808	1808
	Living Oracles	1826	1826
	Young's Literal Translation	1862	1862
	Darby Bible	1890	1890
	William Gunion Rutherford	1900	1900
	American Standard Version	1901	1901
	Weymouth New Testament	1903	1913
	Ferrar Fenton	1903	1905
	Twentieth Century New Testament	1904	1904
	Moffatt, New Translation	1922	1922
	An American Translation	1923	1923
	Helen Barrett Montgomery	1924	1924
	Bible in Basic English	1941	1965
	Phillips New Testament in Modern English	1958	1961
	New English Bible	1970	1970
	The Living Bible	1971	1971
	New Living Translation	1996	1996
	The Passion Translation	2011	2011
	Common English Bible	2011	2011
Estonian	Bible Society	1997	1997
Faroese	Evangelium Sankta Matthæus aa Førisk	1823	1823
	Jákup Dahl	1923	1961
	Victor Danielsen	1937	1937
Finnish	Pyhä Raamattu	1938	1938
	Uusi kirkkoraamattu	1992	1992

Language	Title	Original publication	Edition
	Raamattu Kansalle	2012	2012
French	Old French Bible	1235-1260	1235-1260
	Jacques Lefèvre d'Étaples	1523	1523
	Bible d'Olivétan	1535	1535
	Nouveau Testament de Mons	1667	1667
	Martin	1696	1744
	Ostervald	1724	1779
	Louis Segond (1880)	1880	1880
	Louis Segond (1910)	1880	1910
Friulian	Lu Vanzèli seònd S. Matìe	1860	2021
	Bibie in Linie	1997	2011
Galician	El Evangelio segun San Mateo	1861	1861
	A Biblia Galega	1989	2001
German	Tatian, Gospel Harmony	9th century	9th century
	Otfrid von Weißenburg	10th century	10th century
	Augsburg Bible	1350	1350
	Ottheinrich Bible	1430	1430
	Mentelin Bible	1466	1466
	Lübeck Bible	1494	1494
	Luther Bible	1522	1522
	Eck Bible	1534	1534
	Dietenberger Bible	1534	1763
	Elberfelder	1855	1905
	Schlachter Bible	1951	1951
	BasisBibel	2010	2010
	Neue Genfer Übersetzung	2011	2011
	Hoffnung für alle	2015	2015
Hungarian	Szent István Társulati Biblia	1973	1973
	Magyar Bibliatársulat	1975	2014
Hutterite German	Hutterisch New Testament	2016	2016
Icelandic	Gudbrand Bible	1584	1584
	Viðeyjarbiblía	1841	1841
	Icelandic Bible Society (1908)	1908	1908
	Icelandic Bible Society (2007)	2007	2007
Irish	An Bíobla Naomhtha	1602	1817
Italian	Malermi	1490	1490
	Antonio Brucioli	1532	1544
	Anonymous revision of Brucioli	1558	1558
	Diodati	1607	1607
	Martini	1771	1771
	Bibbia CEI (1974)	1974	1974
	Bibbia CEI (2008)	1974	2008
Koine Greek	Novum Testamentum Graece (Nestle-Aland 28)	1st-2nd centuries	2012
Latin	Biblia Sacra Vulgata	4th century	2007
Latvian	Latviešu Valoda	1965	1997
Lithuanian	Bible Society of Lithuania	1999	1999
Low German	Heliand	10th century	10th century
	Kölner Bibel	1478	1478
	Ernst Voss	1929	1960
	De Bibel JHF	1987	1987

Language	Title	Original publication	Edition
Luxemburgish	Luxemburgish Bible	2017	2017
Macedonian	The Dynamic New Testament	1999	1999
	Macedonian Standard Bible	2006	2006
Mirandese	Evangelhos em Mirandês	2014	2014
Modern Greek	Today's Greek Version	1985	1985
	Filos Pergamos	1994	1994
Northern Tosk Albanian	ABS Version	1994	1994
	Interconfessional Version	2007	2007
Norwegian	Fyrebilsbibelen	1921	1921
	Det Norsk Bibelselskap	1930	1930
	Norwegian Bible Society (1985)	1978	1985
	En Levende Bok	2005	2005
	Guds Ord	2017	2017
	Guds Ord Hverdagsbibelen	2018	2018
Occitan	Nouveau Testament de Lyon	13th century	13th century
	Lis Evangèli	1903	1903
	Cubaynes	1956	1956
	Larzac	2016	2016
Pennsylvania German	Di Heilich Shrift	2013	2013
Piedmontese	'L Testament Neuv Dë Nossegnour Gesu Crist	1834	1834
Plautdietsch	De Plautdietsche Bibel	2003	2003
Polish	Millennium Bible	1965	1965
Portuguese	Almeida	1681	1681
	Periera	1790	1821
	Ameida Revista	1848	1848
	Almeida Revista e Corrigida	1968	1968
	Nova Tradução na Linguagem de Hoje	2000	2000
Romanian	Cornilescu	1924	1924
	Biblia Sinodală	1998	1998
Romansh (Lower Engadine)	Jacob Dorta à Vulpera	1679	1679
Romansh (Oberland)	Lucius Gabriel	1648	1648
Romansh (Upper Engadine)	Jachiam Tütschett Bifruns	1560	1560
	Johan L. Gritti	1640	1640
	J. Menni	1861	1861
Russian	Synodal Bible	1876	1876
	Bible Translation Institute	2015	2015
Scots	Smith's New Testament	1904	1904
	Tha Fower Gospels ïn Ulstèr-Scots	2016	2020
Scottish Gaelic	Ewen MacEachen	1875	1875
	Eadar-theangachadh Ùr	2009	2009
Serbian-Croatian-Bosnian	Karadžić	1847	1847
	Sveto pismo Staroga i Novoga uvita	1861	1861
	Šarić	1942	1942
	Biblija: suvremeni hrvatski prijevod	2002	2002
Slovak	The Roháček version	1936	1936
	Ecumenical	2008	2008
	Evanjelický preklad	2015	2015
Slovenian	István Küzmics (1848)	1771	1848
	István Küzmics (1926)	1771	1926
	Jubilejni prevod Nove zaveze	1985	1985

Language	Title	Original publication	Edition
	Slovenski standardni prevod	1996	1996
Spanish	Biblia prealfonsí	13th century	13th century
	BNM	15th century	15th century
	Biblia del Oso	1569	1569
	Félix Torres Amat	1824	1824
	Biblia de Jerusalen	1973	1973
Swedish	Gustav Vasa (1526)	1526	1526
	Gustav Vasa (1541)	1541	1541
	Karl XII:s bibel	1703	1703
	Waldenström (1894)	1894	1894
	Waldenström (1921)	1894	1921
	1917 års bibelöversättning	1917	1917
	Svenska Folkbibeln	1998	1998
	Bibel 2000	2000	2000
	nuBibeln	2015	2015
Ukrainian	Kulish	1906	1906
	Ohienko	1962	1962
Venetian	Gospel Harmony	13th century	13th century
	Il Vangelo di S. Matteo	1859	1859
Welsh	Y Beibl Cymraeg Newydd	1988	1988

Table A.1: New Testament translations cited in this work

Bibliography

Agouraki, Y. (2010). *It*-clefts and stressed operators in the preverbal field of Cypriot Greek. *Lingua 120*(3), 527–554.

Amiraz, O. (2019). Semi positive polarity items. In M. Baird and J. Pesetsky (Eds.), *Proceedings of North East Linguistic Society (NELS) 49*, Volume 1, pp. 49–62. Amherst, MA: GLSA.

Amiraz, O. (2020). Non-maximality and homogeneity: Parallels between collective predicates and absolute adjectives. In M. Franke, N. Kompa, M. Liu, J. L. Mueller, and J. Schwab (Eds.), *Proceedings of Sinn und Bedeutung 24*, Volume 1, pp. 66–83. Osnabrück/Berlin: Osnabrück University and Humboldt University of Berlin.

Amiraz, O. (2022). Not all scalar inferences are alike: The effect of existential presuppositions. In M. Degano, T. Roberts, G. Sbardolini, and M. Schouwstra (Eds.), *Proceedings of Amsterdam Colloquium 23*, pp. 8–14. Amsterdam: University of Amsterdam.

Anderson, C. (2004). *The structure and real-time comprehension of quantifier scope ambiguity.* Ph. D. thesis, Northwestern University, Evanston, IL.

Ariel, M. (2002a). The demise of a unique concept of literal meaning. *Journal of Pragmatics 34*(4), 361–402.

Ariel, M. (2002b). Privileged interactional interpretations. *Journal of Pragmatics 34*(8), 1003–1044.

Ariel, M. (2008). *Pragmatics and grammar.* Cambridge: Cambridge University Press.

Bach, E., E. Jelinek, A. Kratzer, and B. H. Partee (Eds.) (1995). *Quantification in natural languages.* Dordrecht: Kluwer Academic Publishers.

Baker, C. L. (1970). Double negatives. *Linguistic Inquiry 1*(2), 169–186.

Bar-Asher Siegal, E. A. (2021). The formation and the cognitive knowledge of literary languages: The case of Hebrew and Aramaic in the Middle Ages. *Hebrew Studies 62*, 343–363.

Bar-Asher Siegal, E. A. (2024). A formal approach to reanalysis and the early semantic stability hypothesis: Exploring the test case of the negative counterfactual marker *ʾilmale* in Hebrew and Aramaic. *Linguistics*.

Bar-Asher Siegal, E. A. (2025). Two types of negation: Semantics, distribution and history. In E. A. Bar-Asher Siegal, N. Boneh, E. Grossman, and A. Rubinstein (Eds.), *Language Change: Theoretical and Empirical Perspectives*. Springer Nature.

Bar-Asher Siegal, E. A. and K. De Clercq (2019). From negative cleft to external negator. In M. Bouzouita, A. Breitbarth, L. Danckaert, and E. Witzenhausen (Eds.), *Cycles in language change*, pp. 228–248. Oxford University Press.

Bar-Lev, M. E. (2021). An implicature account of homogeneity and non-maximality. *Linguistics and Philosophy 44*, 1045–1097.

Bar-Lev, M. E. and R. Katzir (2023). Communicative stability and the typology of logical operators. *Linguistic Inquiry*, 1–23.

Barbiers, S., J. Van der Auwera, H. Bennis, E. Boef, G. De Vogelaer, and M. Van der Ham (2008). *Syntactic atlas of the Dutch dialects*, Volume 2. Amsterdam: Amsterdam University Press.

Barchas-Lichtenstein, J. (2012). Garifuna quantification. In E. Keenan and D. Paperno (Eds.), *Handbook of quantifiers in natural language*, Volume 1, pp. 165–225. Springer.

Baunaz, L. (2011). *The grammar of French quantification*. Springer.

Beck, D. (2004). *A grammatical sketch of Upper Necaxa Totonac*. Munich: Lincom Europa.

Beck, S. (1996). Quantified structures as barriers for LF movement. *Natural Language Semantics 4*(1), 1–56.

Beghelli, F. and T. Stowell (1997). Distributivity and negation: The syntax of each and every. In A. Szabolcsi (Ed.), *Ways of Scope Taking*, pp. 71–107. Dordrecht: Springer.

Benincà, P. (2006). A detailed map of the left periphery of medieval Romance. In R. Zanuttini, H. Campos, E. Herburger, and P. H. Portner (Eds.), *Crosslinguistic research in syntax and semantics: Negation, tense and clausal architecture*, pp. 53–86. Washington, D.C.: Georgetown University Press.

Bentz, C. and R. Ferrer-i-Cancho (2016). Zipf's law of abbreviation as a language universal. In I. Y. Christian Bentz, Gerhard Jäger (Ed.), *Proceedings of the Leiden Workshop on Capturing Phylogenetic Algorithms for Linguists*, pp. 1–4. Tübingen: Universität Tübingen.

Biberauer, T. (2009). Jespersen off course? the case of contemporary Afrikaans negation. In E. van Gelderen (Ed.), *Cyclical change*, pp. 91–132. John Benjamins Publishing Company.

Biberauer, T. (2010). Semi null-subject languages, expletives and expletive *pro* reconsidered. In T. Biberauer, A. Holmberg, I. Roberts, and M. Sheehan (Eds.), *Parametric variation: Null subjects in minimalist theory*, pp. 153–199. Cambridge: Cambridge University Press.

Biberauer, T. (2012). Competing reinforcements: When languages opt out of Jespersen's cycle. In A. M. van Kemenade and N. de Haas (Eds.), *Proceedings of Historical Linguistics 19*, pp. 1–30. John Benjamins Publishing Company.

Bickel, B., J. Nichols, T. Zakharko, A. Witzlack-Makarevich, K. Hildebrandt, M. Rießler, L. Bierkandt, F. Zúñiga, and J. B. Lowe (2023). The AUTOTYP database (v1.1.1).

Bickel, B., A. Witzlack-Makarevich, K. K. Choudhary, M. Schlesewsky, and I. Bornkessel-Schlesewsky (2015). The neurophysiology of language processing shapes the evolution of grammar: Evidence from case marking. *PLoS One 10*(8), e0132819.

Bobaljik, J. D. (1995). *Morphosyntax: The syntax of verbal inflection.* Ph. D. thesis, Massachusetts Institute of Technology, Cambridge, MA.

Bobaljik, J. D. and S. Wurmbrand (2012). Word order and scope: Transparent interfaces and the 3/4 signature. *Linguistic Inquiry 43*(3), 371–421.

Boeckx, C. (2001). Scope reconstruction and A-movement. *Natural Language and Linguistic Theory 19*(3), 503–548.

Borschev, V., E. V. Paducheva, B. H. Partee, Y. G. Testelets, and I. Yanovich (2006). Sentential and constituent negation in Russian BE-sentences revisited. In *Proeecings of Formal Approaches to Slavic Linguistics (FASL) 14*, pp. 50–65. Ann Arbor, Mich: Michigan Slavic Publications.

Bott, O. and F. Schlotterbeck (2015). The processing domain of scope interaction. *Journal of Semantics 32*(1), 39–92.

Brasoveanu, A. and J. Dotlačil (2015). Sentence-internal *same* and its quantificational licensors: A new window into the processing of inverse scope. *Semantics and Pragmatics 8*, 1–52.

Breitbarth, A. (2013). Negation in the history of Low German and Dutch. In D. Willis, C. Lucas, and A. Breitbarth (Eds.), *The history of negation in the languages of Europe and the Mediterranean, Volume I: Case studies*, pp. 190–238. Oxford University Press.

Breitbarth, A. and A. Jäger (2018). History of negation in High and Low German. In A. Jäger, G. Ferraresi, and H. Weiss (Eds.), *Clause structure and word order in the history of German*, pp. 181–219. Oxford University Press.

Breitbarth, A., C. Lucas, and D. Willis (2020). *The history of negation in the languages of Europe and the Mediterranean, Volume II: Patterns and processes.* Oxford University Press.

Brook, M. (2024). Obsolescence and abortive innovations in variationist approaches to language change. *Language and Linguistics Compass 18*(4), e12516.

Brunetti, L. (2004). *A unification of focus*. Padua: Unipress.

Buell, L. (2008). VP-internal DPs and right-dislocation in Zulu. *Linguistics in the Netherlands 25*(1), 37–49.

Büring, D. (1997). The great scope inversion conspiracy. *Linguistics and Philosophy 20*(2), 175–194.

Büring, D. (2003). On D-trees, beans, and B-accents. *Linguistics and Philosophy 26*, 511–545.

Büring, D. (2016). *Intonation and meaning*. Oxford: Oxford University Press.

Callegari, E. and Á. Angantỳsson (2023). Non-subject initial clauses and the left periphery in Icelandic: A distributional approach. *Working Papers in Scandinavian Syntax 109*, 41–62.

Carden, G. (1970). A note on conflicting idiolects. *Linguistic Inquiry 1*(3), 281–290.

Carnie, A. and E. Guilfoyle (2000). *The syntax of verb initial languages*. New York, NY: Oxford University Press.

Carstens, V. and M. Diercks (2009). Parameterizing case and activity: Hyper-raising in bantu. In S. Kan, C. Moore-Cantwell, and R. Staubs (Eds.), *Proceedings of North East Linguistic Society (NELS) 40*, pp. 99–118. Amherst, MA: GLSA.

Chen, S. Y. and B. van Tiel (2021). Every ambiguity isn't syntactic in nature: Testing the Rational Speech Act model of scope ambiguity. In A. Ettinger, E. Pavlick, and B. Prickett (Eds.), *Proceedings of the Society for Computation in Linguistics (SCiL) 2021*. Association for Computational Linguistics.

Cheng, L. L.-S. (1995). On *dou*-quantification. *Journal of East Asian Linguistics 4*(3), 197–234.

Cheng, L. L.-S. (2008). Deconstructing the *shi…de* construction. *The Linguistic Review 25*(3–4), 235–266.

Chierchia, G. (2006). Broaden your views: Implicatures of domain widening and the "logicality" of language. *Linguistic Inquiry 37*(4), 535–590.

Chierchia, G. (2013). *Logic in grammar: Polarity, free choice, and intervention*. Oxford University Press.

Cichosz, A. (2020). Negation and verb-initial order in Old English main clauses. *Journal of English Linguistics 48*(4), 355–381.

Cichosz, A., J. Gaszewski, and P. Pęzik (2016). *Element order in Old English and Old High German translations*. John Benjamins Publishing Company.

Cirillo, R. (2009). *The syntax of floating quantifiers: Stranding revisited*. Netherlands Graduate School of Linguistics.

Collins, C. (2020). Outer negation of universal quantifier phrases. *Linguistics and Philosophy 43*(3), 233–246.

Collins, C. and P. M. Postal (2014). *Classical NEG raising: An essay on the syntax of negation*, Volume 67. Cambridge, MA: MIT press.

Conroy, A., S. Fults, J. Musolino, and J. Lidz (2008). Surface scope as a default: The effect of time in resolving quantifier scope ambiguity. Poster presented at the 21st CUNY Conference on Sentence Processing.

Constant, N. (2012). English rise-fall-rise: A study in the semantics and pragmatics of intonation. *Linguistics and Philosophy 35*, 407–442.

Crnič, L. (2014). Non-monotonicity in NPI licensing. *Natural Language Semantics 22*, 169–217.

Crnič, L. (2019). *Any*: Logic, likelihood, and context (pt. 2). *Language and Linguistics Compass 13*(11), e12353.

Davies, M. (2008). The Corpus of Contemporary American English (COCA). Available online at `https://www.english-corpora.org/coca/`.

Davies, M. (2010). The Corpus of Historical American English (COHA). Available online at `https://www.english-corpora.org/coha/`.

Davies, M. (2016). Corpus of News on the Web (NOW). Available online at `https://www.english-corpora.org/now/`.

Davies, M. (2018). The iWeb Corpus. Available online at `https://www.english-corpora.org/iWeb/`.

Decker, K. (2005). *The song of Kriol: A grammar of the Kriol language of Belize*. SIL International.

Diesing, M. (1997). Yiddish VP order and the typology of object movement in Germanic. *Natural Language and Linguistic Theory 15*(2), 369–427.

Dixon, R. M. W. (1988). *A grammar of Boumaa Fijian*. University of Chicago Press.

Donaldson, B. C. (1993). *A grammar of Afrikaans*. De Gruyter Mouton.

Dryer, M. S. (1989). Large linguistic areas and language sampling. *Studies in Language 13*(2), 257–292.

Dryer, M. S. (2013a). Order of subject, object and verb (v2020.4). In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Zenodo.

Dryer, M. S. (2013b). Position of negative morpheme with respect to subject, object, and verb (v2020.4). In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Zenodo.

Dryer, M. S. and M. Haspelmath (Eds.) (2013). *WALS Online (v2020.3)*. Zenodo.

Dufter, A. (2009). Clefting and discourse organization: Comparing Germanic and Romance. In A. Dufter and D. Jacob (Eds.), *Focus and background in Romance languages*, pp. 83–122. John Benjamins Publishing Company.

Dunn, M., S. J. Greenhill, S. C. Levinson, and R. D. Gray (2011). Evolved structure of language shows lineage-specific trends in word-order universals. *Nature 473*(7345), 79–82.

Durrleman, S. (2008). *The syntax of Jamaican Creole: A cartographic perspective*. John Benjamins Publishing Company.

Ejrnæs, B. (2012). The Bible in the languages of Scandinavia. In R. Marsden and E. A. Matter (Eds.), *The New Cambridge History of the Bible*, New Cambridge History of the Bible, pp. 239–250. Cambridge University Press.

Enguehard, É. and B. Spector (2021). Explaining gaps in the logical lexicon of natural languages: A decision-theoretic perspective on the square of Aristotle. *Semantics and Pragmatics 14*(5), 1–31.

Enke, D. W., L. M. Hyman, J. Nichols, G. Seiler, T. Weber, and A. Hölzl (Eds.) (2024). *Language change for the worse*. Berlin: Language Science Press.

Espinal, M. T. (1991). Negation in Catalan: Some remarks with regard to *no pas*. *Catalan Working Papers in Linguistics 1*, 33–63.

Espinal, M. T. (1993). The interpretation of *no pas* in catalan. *Journal of Pragmatics 19*(4), 353–369.

Etxepare, R. and M. Uribe-Etxebarria (Forthcoming). Crosslinguistic variation in constituent negation. In J.-W. Zwart, H. Demirdache, and M. Arche (Eds.), *Footprints of phrase structure*. Amsterdam: John Benjamins Publishing Company.

Evans, N. (2010). Semantic typology. In J. J. Song (Ed.), *The Oxford Handbook of Linguistic Typology*. Oxford University Press.

Fan, L. (2017). No “chinese-speaking phase” in chinese children’s early grammar–a study of the scope between negation and universal quantification in mandarin chinese. *Lingua 185*, 42–66.

Fodor, J. D. (1982). The mental representation of quantifiers. In S. Peters and E. Saarinen (Eds.), *Processes, beliefs, and questions*, pp. 129–164. Dordrecht: D. Reidel Publishing Company.

Fuß, E. and R. Hinterhölzl (2023). On the historical development of pronouns referring to situations: The rise of pre-finite ‘expletives’ in German. *Journal of Historical Syntax 7*(2), 1–54.

Futrell, R., K. Mahowald, and E. Gibson (2015). Quantifying word order freedom in dependency corpora. In J. Nivre and E. Hajičová (Eds.), *Proceedings of the third international conference on dependency linguistics (Depling 2015)*, pp. 91–100. Uppsala: Uppsala University.

Gell-Mann, M. and M. Ruhlen (2011). The origin and evolution of word order. *Proceedings of the National Academy of Sciences 108*(42), 17290–17295.

Giannakidou, A. (1998). *Polarity sensitivity as (non)veridical dependency*. John Benjamins Publishing Company.

Gianollo, C. (2011). Native syntax and translation effects: Adnominal arguments in the Greek and Latin New Testament. *Oslo Studies in Language 3*(3), 75–101.

Gianollo, C. (2016). The Latin system of negation at the syntax-semantics interface. *Rivista di Grammatica Generativa 38*, 115–135.

Gianollo, C. (2020). The morpho-syntactic nature of the negative marker. In V. Déprez, V. M. Déprez, M. T. Espinal, and M. T. E. i Farré (Eds.), *The Oxford handbook of negation*, pp. 117–134. Oxford University Press.

Gianollo, C. (2024). Negative concord and word order in the Greek Bible and New Testament. In G. Di Bartolo and D. Kölligan (Eds.), *Postclassical Greek: Problems and Perspectives*, pp. 187–224. De Gruyter Mouton.

Gil, D. (1992). Scopal quantifiers: Some universals of lexical effability. In M. Kefer and J. van der Auwera (Eds.), *Meaning and grammar: Cross-linguistic perspectives*. Berlin: De Gruyter Mouton.

Givón, T. (1978). Negation in language: Pragmatics, function, ontology. In P. Cole (Ed.), *Pragmatics*, pp. 69–112. New York: Academic Press.

Goldhahn, D., T. Eckart, and U. Quasthoff (2012). Building large monolingual dictionaries at the Leipzig Corpora Collection: From 100 to 200 languages. In *Proceedings of the 8th International Language Resources and Evaluation (LREC'12)*, pp. 759–765. Istanbul: European Language Resources Association (ELRA).

González, H. A. (2005). *A grammar of Tapiete (Tupi-guarani).* Ph. D. thesis, University of Pittsburgh, Pittsburgh, PA.

Gow, A. C. (2012). The Bible in Germanic. In R. Marsden and E. A. Matter (Eds.), *The New Cambridge History of the Bible*, pp. 198–216. Cambridge University Press.

Grygar-Rechziegel, A. (1980). On the function of quantifiers in czech negative sentences. *Studies in Slavic and General Linguistics 1*, 103–123.

Gualmini, A. (2004). Some knowledge children don't lack. *Linguistics 42*(5), 957–982.

Gualmini, A. (2008). The rise and fall of Isomorphism. *Lingua 118*(8), 1158–1176.

Guzmán Naranjo, M. and L. Becker (2022). Statistical bias control in typology. *Linguistic Typology 26*(3), 605–670.

Haig, G. (2018). The grammaticalization of object pronouns: Why differential object indexing is an attractor state. *Linguistics 56*(4), 781–818.

Halpert, C. (2016). Raising parameters. In K.-m. Kim, P. Umbal, T. Block, Q. Chan, T. Cheng, K. Finney, M. Katz, S. Nickel-Thompson, and L. Shorten (Eds.), *Proceedings of the West Coast Conference on Formal Linguistics (WCCFL) 33*, pp. 186–195. Somerville, MA: Cascadilla Press.

Hammarström, H. (2015). The basic word order typology: An exhaustive study. Paper presented at the conference *Diversity Linguistics: Retrospect and Prospect*, Leipzig, 1-3 May 2015. Available at: `https://www.eva.mpg.de/fileadmin/content_files/linguistics/conferences/2015-diversity-linguistics/Hammarstroem_slides.pdf`.

Hanitramalala, R. and I. Paul (2012). Malagasy quantifiers. In E. Keenan and D. Paperno (Eds.), *Handbook of quantifiers in natural language*, Volume 1, pp. 613–645. Springer.

Hansen, M.-B. M. (2013). Negation in the history of French. In D. Willis, C. Lucas, and A. Breitbarth (Eds.), *The history of negation in the languages of Europe and the Mediterranean, Volume I: Case studies*, pp. 51–76. Oxford: Oxford University Press.

Hansen, M.-B. M. and J. Visconti (2009). On the diachrony of "reinforced" negation in French and Italian. In C. Rossari, C. Ricci, and A. Spiridon (Eds.), *Grammaticalization and pragmatics: Facts, approaches, theoretical issues*, pp. 137–171. Brill.

Hardman, M. J., J. Vásquez, and J. e. a. de Dios Yapita (1988). *Aymara: Compendio de estructura fonológica y gramatical.* La Paz: ILCA.

Harris, A. C. and L. Campbell (1995). *Historical syntax in cross-linguistic perspective.* Cambridge: Cambridge University Press.

Hartmann, K. and T. Veenstra (Eds.) (2013). *Cleft structures.* John Benjamins Publishing Company.

Haspelmath, M. (1995). Diachronic sources of 'all' and 'every'. In E. Bach, E. Jelinek, A. Kratzer, and B. H. Partee (Eds.), *Quantification in natural languages*, pp. 363–382. Springer.

Haspelmath, M. (1997). *Indefinite pronouns.* Oxford: Oxford University Press.

Haspelmath, M. (2013). Negative indefinite pronouns and predicate negation (v2020.3). In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online.* Zenodo.

Haspelmath, M. (2021). Explaining grammatical coding asymmetries: Form–frequency correspondences and predictability. *Journal of Linguistics 57*(3), 605–633.

Haspelmath, M. (2023). Defining the word. *Word 69*(3), 283–297.

Haspelmath, M. (2024). Four kinds of lexical items: Words, lexemes, inventorial items, and mental items. *Lexique 34*.

Haurholm-Larsen, S. (2016). *A grammar of Garifuna.* Ph. D. thesis, University of Bern, Bern.

Heine, B. and M. Reh (1984). *Grammaticalization and reanalysis in African languages.* Hamburg: Buske.

Hellquist, E. (1922). *Svensk etymologisk ordbok.* Gleerup.

Hemforth, B. and L. Konieczny (2019). When all linguists did not go to the workshop, none of the Germans but some of the French did: The role of alternative constructions for quantifier scope. In K. Carlson, C. J. Clifton, and J. D. Fodor (Eds.), *Grammatical approaches to language processing*, pp. 167–185. Springer.

Herman, J. (2000). *Vulgar Latin.* University Park, Pa: Penn State Press.

Hoeksema, J. (1987). The logic of exception. In *Proceedings of Eastern States Conference on Linguistics (ESCOL) 4*, pp. 100–113. Columbus, OH: Ohio State University.

Hoeksema, J. (1999). Blocking effects in the expression of negation. *GAGL: Groninger Arbeiten zur germanistischen Linguistik 43*, 43–59.

Holmberg, A. (2000). Scandinavian stylistic fronting: How any category can become an expletive. *Linguistic Inquiry 31*(3), 445–483.

Homer, V. and R. Bhatt (2019). Licensing of PPI indefinites: Movement or pseudoscope? *Natural Language Semantics 27*(4), 279–321.

Homer, V. and R. Bhatt (2020). Restructuring and the scope of negation in Hindi-Urdu. *Glossa: a journal of general linguistics 5*(1), 12.

Hopper, P. J. and E. C. Traugott (1993). *Grammaticalization.* Cambridge: Cambridge University Press.

Horn, L. R. (1972). *On the semantic properties of logical operators in English.* Ph. D. thesis, University of California, Los Angeles.

Horn, L. R. (1985). Metalinguistic negation and pragmatic ambiguity. *Language 61*(1), 121–174.

Horn, L. R. (1989). *A natural history of negation.* Chicago: University of Chicago Press.

Ingham, R. (2013). Negation in the history of English. In D. Willis, C. Lucas, and A. Breitbarth (Eds.), *The history of negation in the languages of Europe and the Mediterranean, Volume I: Case studies*, pp. 119–150. Oxford University Press.

Jackendoff, R. (2013). Constructions in the parallel architecture. In T. Hoffmann and G. Trousdale (Eds.), *The Oxford Handbook of Construction Grammar*, pp. 70–92. Oxford University Press.

Jackendoff, R. S. (1969). An interpretive theory of negation. *Foundations of Language 5*, 218–241.

Jackendoff, R. S. (1972). *Semantic interpretation in generative grammar.* Cambridge, MA: MIT Press.

Jacobs, J. (1984). Funktionale Satzperspecktive und Illokutionssemantik Perspective fonctionnelle de la phrase et sémantique illocutoire. *Linguistische Berichte 91*, 25–58.

Jäger, A. (2008). *History of German negation.* John Benjamins Publishing Company.

Jäger, G. and J. Wahle (2021). Phylogenetic typology. *Frontiers in Psychology 12*, 682132.

Jespersen, O. (1917). *Negation in English and other languages.* Copenhagen: A.F. Høst.

Jin, D. and J. Chen (2022). Towards the copular status of clefts: Evidence from diachrony. *Acta Linguistica Academica 69*(3), 301–333.

Kamali, B. (2011). *Topics at the PF interface of Turkish.* Ph. D. thesis, Harvard University, Cambridge, MA.

Kamali, B. and H. Zeijlstra (2024). Negative dependencies in Turkish. *Languages 9*(11), 342.

Kanikli, A. (2016). The distribution of quantifiers in clefts. *Lingua 171*, 24–36.

Katzir, R. and R. Singh (2013). Constraints on the lexicalization of logical operators. *Linguistics and Philosophy 36*(1), 1–29.

Kayne, R. (1975). *French syntax: The transformational cycle.* Cambridge, MA: MIT Press.

Kayne, R. (1994). *The antisymmetry of syntax.* Cambridge, MA: MIT Press.

Keenan, E. and D. Paperno (Eds.) (2012). *Handbook of quantifiers in natural language*, Volume 1. Springer.

Kelepir, M. (2001). *Topics in Turkish syntax: Clausal structure and scope.* Ph. D. thesis, Massachusetts Institute of Technology, Cambridge, MA.

Kennard, H. J. (2014). The persistence of verb second in negative utterances in Breton. *Journal of Historical Linguistics 4*(1), 1–39.

Key, M. R. and B. Comrie (Eds.) (2023). *The Intercontinental Dictionary Series.* Leipzig: Max Planck Institute for Evolutionary Anthropology.

Kiss, K. É. (1998). Identificational focus versus information focus. *Language 74*(2), 245–273.

Kolokonte, M. (2008). *Bare argument ellipsis and information structure.* Ph. D. thesis, Newcastle University, Newcastle.

Konietzko, A. (2016). *Bare argument ellipsis and focus.* John Benjamins Publishing Company.

Krause, T. and A. Zeldes (2016). Annis3: A new architecture for generic corpus query and visualization. *Digital Scholarship in the Humanities 31*(1), 118–139.

Krifka, M. (1998). Scope inversion under the rise-fall contour in German. *Linguistic Inquiry 29*(1), 75–112.

Križ, M. (2016). Homogeneity, non-maximality, and *all.* *Journal of Semantics 33*(3), 493–539.

Križ, M. (2019). Homogeneity effects in natural language semantics. *Language and Linguistics Compass 13*(11), e12350.

Kroch, A. S. (1979). *The semantics of scope in English.* New York: Garland Pub.

Kroch, A. S. (1989). Reflexes of grammar in patterns of language change. *Language Variation and Change 1*(3), 199–244.

Kroeger, P. (2014). External negation in Malay/Indonesian. *Language 90*(1), 137–184.

Kumar, R. (2006). *The syntax of negation and the licensing of negative polarity items in Hindi*. London: Routledge.

Labov, W. (1975). *What is a linguistic fact?* Lisse: Peter de Ridder Press.

Ladusaw, W. A. (1980). *Polarity sensitivity as inherent scope relations*. New York: Garland Pub.

Lambrecht, K. (2000). When subjects behave like objects: An analysis of the merging of S and O in sentence-focus constructions across languages. *Studies in Language 24*(3), 611–682.

Lambrecht, K. and M. Polinsky (1998). Typological variation in sentence-focus constructions. In *Proceedings of Chicago Linguistic Society (CLS) 33*, pp. 189–206. Chicago, IL: University of Chicago.

Larrivée, P. (2017). Negation and polarity. In A. Dufter and E. Stark (Eds.), *Manual of Romance morphosyntax and syntax*, pp. 449–471. Berlin: De Gruyter Mouton.

Lasnik, H. (1972). *Analyses of negation in English*. Ph. D. thesis, Massachusetts Institute of Technology, Cambridge, MA.

Ledgeway, A. (2012). *From Latin to Romance: Morphosyntactic typology and change*. Oxford: Oxford University Press.

Ledgeway, A. and N. Schifano (2023). Negation and verb-movement in Romance: New perspectives on jespersen's cycle. *Probus 35*(1), 151–211.

Lee, S. (2009). *Interpreting scope ambiguity in first and second language processing: Universal quantifiers and negation*. Ph. D. thesis, University of Hawai'i at Mānoa, Honolulu, HI.

Leonardi, L. et al. (2012). The Bible in Italian. In R. Marsden and E. A. Matter (Eds.), *The New Cambridge History of the Bible*, Volume 2, pp. 268–287. Cambridge University Press.

Leppin, V. (2024). The German Bible from Martin Luther to Johann Eck. In J. P. McNutt and H. J. Selderhuis (Eds.), *The Oxford Handbook of the Bible and the Reformation*, pp. 124–136. Oxford University Press.

Lesage, J. (2020). *A grammar and lexicon of Kam, a Niger-Congo language of central eastern Nigeria*. Ph. D. thesis, INALCO, Paris.

Lesage, J. (2025). Negation in Kam. In M. Miestamo and L. Veselinova (Eds.), *Negation in the world's languages I: Africa*. Berlin: Language Science.

Levshina, N. (2019). Token-based typology and word order entropy: A study based on universal dependencies. *Linguistic Typology 23*(3), 533–572.

Levshina, N. (2020). Efficient trade-offs as explanations in functional linguistics: Some problems and an alternative proposal. *Revista da ABRALIN 19*(3), 50–78.

Levshina, N. (2022). Corpus-based typology: Applications, challenges and some solutions. *Linguistic Typology 26*(1), 129–160.

Lin, J.-W. (1998). On existential polarity *wh*-phrases in Chinese. *Journal of East Asian Linguistics 7*(3), 219–255.

Lindström, J. (2007). Initial clausal negation – a nordic areal feature. In A. Ammann (Ed.), *Linguistics festival*, pp. 31–58. Bochum: Brockmeyer.

Liu, Y., H. Ye, L. Weissweiler, P. Wicke, R. Pei, R. Zangenfeind, and H. Schütze (2023). A crosslingual investigation of conceptualization in 1335 languages. In A. Rogers, J. Boyd-Graber, and N. Okazaki (Eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 12969–13000. Toronto: Association for Computational Linguistics.

Mahajan, A. (2017). Quantification in Hindi. In D. Paperno and E. L. Keenan (Eds.), *Handbook of quantifiers in natural language*, Volume 2, pp. 383–430. Springer.

Martin, S. E. (2004). *A reference grammar of Japanese*. Honolulu, HI: University of Hawai'i Press.

Martzloff, V. (2014). Some remarks on the prehistory of *omnis* and other Latin pronouns and adjectives meaning 'all' or 'whole'. *Journal of Latin Linguistics 13*(2), 211–241.

Maslova, E. (2000). A dynamic approach to the verification of distributional universals. *Linguistic Typology 4*, 307–333.

Maurits, L. and T. L. Griffiths (2014). Tracing the roots of syntax with Bayesian phylogenetics. *Proceedings of the National Academy of Sciences 111*(37), 13576–13581.

May, R. (1985). *Logical form: Its structure and derivation.* Cambridge, MA: MIT Press.

Mayer, T. and M. Cysouw (2014). Creating a massively parallel Bible corpus. In *Proceedings of the international conference on language resources and evaluation (LREC)*, pp. 3158—3163. Reykjavik: European Language Resources Association (ELRA).

Mayr, C. and B. Spector (2010). Not too strong! Generalizing the scope economy condition. In M. Prinzhorn, V. Schmitt, and S. Zobel (Eds.), *Proceedings of Sinn und Bedeutung 14*, pp. 305–321. Vienna: University of Vienna.

McCawley, J. D. (1991). Contrastive negation and metalinguistic negation. In *Proceedings of Chicago Linguistic Society (CLS) 27*, Volume 2, pp. 189–206. Chicago, IL: University of Chicago.

McSparran, F., P. Schaffner, J. Latta, A. Pagliere, C. Powell, and M. Stoeffler (2006). Corpus of Middle English Prose and Verse. Ann Arbor, MI: University of Michigan. Available online at `https://quod.lib.umich.edu/c/cme/`.

Meeuwis, M. (2020). *A grammatical overview of Lingála: Revised and extended edition.* München: Lincom.

Merchant, J. (2006). Why no(t)? *Style 40*(1-2), 20–23.

Miestamo, M. (2005). *Standard negation: The negation of declarative verbal main clauses in a typological perspective*. Berlin: De Gruyter Mouton.

Miestamo, M., D. Bakker, and A. Arppe (2016). Sampling for variety. *Linguistic Typology 20*(2), 233–296.

Miestamo, M., K. Shagal, and O. O. Silvennoinen (2022). Typology and usage: The case of negation. *Linguistic Typology at the Crossroads 2*(1), 121–154.

Milsark, G. (1977). Toward an explanation of certain peculiarities of the existential construction in English. *Linguistic Analysis 3*(1), 1–29.

Müller, G. and W. Sternefeld (2001). The rise of competition in syntax: A synopsis. In G. Müller and W. Sternefeld (Eds.), *Competition in syntax*, pp. 1–68. De Gruyter Mouton.

Munro, P. (2017). Chickasaw quantifiers. In D. Paperno and E. L. Keenan (Eds.), *Handbook of quantifiers in natural language*, Volume 2, pp. 113–201. Springer.

Munro, P. and C. E. Gallagher (2014). Garifuna negatives. In L. Michael and T. Granadillo (Eds.), *Negation in Arawak languages*, pp. 13–53. Brill.

Musolino, J. (1998). *Universal grammar and the acquisition of semantic knowledge: An experimental investigation into the acquisition of quantifier negation interaction in English.* Ph. D. thesis, University of Maryland, College Park, MD.

Musolino, J. (2006). Structure and meaning in the acquisition of scope. In V. van Geenhoven (Ed.), *Semantics in acquisition*, pp. 141–166. Springer.

Musolino, J. and J. Lidz (2006). Why children aren't universally successful with quantification. *Linguistics 44*(4), 817–852.

Naranjo, M. G. and L. Becker (2018). Quantitative word order typology with UD. In *Proceedings of International Workshop on Treebanks and Linguistic Theories (TLT) 17*, pp. 91–104. Linköping: Linköping University Electronic Press.

Neukom-Hermann, A. (2016). *Negation, quantification and scope: A Corpus Study of English and German all...not constructions*. Ph. D. thesis, University of Zurich, Zurich.

Nida, E. A. (1964). *Toward a science of translating: With special reference to principles and procedures involved in Bible translating*. Leiden: Brill.

Nissenbaum, J. W. (2000). *Investigations of covert phrase movement*. Ph. D. thesis, Massachusetts Institute of Technology, Cambridge, MA.

Oh, S.-R. (2006). *Plurality markers across languages*. Ph. D. thesis, University of Connecticut, Storrs, CT.

Oikonomou, D., F. Golcher, and A. Alexiadou (2020). Quantifier scope and information structure in Greek. *Glossa: a journal of general linguistics 5*(1), 81.

Östling, R. (2015). Word order typology through multilingual word alignment. In C. Zong and M. Strube (Eds.), *Proceedings of Association for Computational Linguistics (ACL) 53*, Volume 2, pp. 205–211. Beijing: Association for Computational Linguistics.

Östling, R. (2016). Studying colexification through massively parallel corpora. In P. Juvonen and M. Koptjevskaja-Tamm (Eds.), *The lexical typology of semantic shifts*, pp. 157–176. Berlin: De Gruyter Mouton.

Öztürk, B. (2005). *Case, referentiality, and phrase structure*. Amsterdam: John Benjamins Publishing Company.

Özyıldız, D. (2017). Quantifiers in Turkish. In D. Paperno and E. L. Keenan (Eds.), *Handbook of quantifiers in natural language*, Volume 2, pp. 857–937. Springer.

Paperno, D. and E. L. Keenan (Eds.) (2017). *Handbook of quantifiers in natural language*, Volume 2. Springer.

Park, K. and S. Dubinsky (2020). The effects of focus on scope relations between quantifiers and negation in Korean. In P. Farrell (Ed.), *Proceedings of the Linguistic Society of America (LSA) 2020*, pp. 100–106. New Orleans: Linguistic Society of America.

Parry, M. (2013). Negation in the history of Italo-Romance. In D. Willis, C. Lucas, and A. Breitbarth (Eds.), *The history of negation in the languages of Europe and the Mediterranean, Volume I: Case studies*, pp. 77–118. Oxford: Oxford University Press Oxford.

Pat-El, N. (2006). Syntactical aspects of negation in Syriac. *Journal of Semitic Studies 51*(2), 329–348.

Paul, I. M. (2000). *Malagasy clause structure*. Ph. D. thesis, McGill University Montreal, Canada.

Paul, W. and J. Whitman (2008). *Shi…de* focus clefts in Mandarin Chinese. *Linguistic Review 25*(3-4), 413–451.

Petersen, H. P. (2022). Evidence for the modification of dialect classification of modern spoken Faroese. *European Journal of Scandinavian Studies 52*(1), 43–58.

Pinkster, H. (2015). *Oxford Latin syntax, Volume 1: The simple clause*. Oxford: Oxford University Press.

Poletto, C. (2014). *Word order in old Italian*, Volume 7. Oxford: Oxford University Press.

Poletto, C. et al. (2008). On negative doubling. *Quaderni di lavoro ASIt 8*, 57–84.

Pook, H. and L. Lindström (2024). The use of the indefinite pronoun *keegi* ‘someone’ in Estonian dialects. *Nordic Journal of Linguistics 47*(2), 192–223.

Postma, G. (2010). The impact of failed changes. In A. Breitbarth, C. Lucas, S. Watts, and D. Willis (Eds.), *Continuity and change in grammar*, pp. 269–302. John Benjamins Publishing Company.

Redden, J. E. (1963). *Lingala: Basic course*. Department of State.

Reinhart, T. (2005). *Interface strategies*. Cambridge, MA: MIT Press.

Rognan, H. S. and B. Beekhuizen (2025). A discovery procedure for synlexification patterns in the world’s languages. In M. Hahn, P. Rani, R. Kumar, A. Shcherbakov, A. Sorokin, O. Serikov, R. Cotterell, and E. Vylomova (Eds.), *Proceedings of the 7th Workshop on Research in Computational Linguistic Typology and Multilingual NLP*, pp. 93–113. Vienna: Association for Computational Linguistics.

Rooth, M. (1992). A theory of focus interpretation. *Natural Language Semantics 1*(1), 75–116.

Rooth, M. E. (1985). *Association with focus*. Ph. D. thesis, University of Massachusetts Amherst, Amherst, MA.

Rosemeyer, M. (2012). How to measure replacement: Auxiliary selection in Old Spanish bibles. *Folia Linguistica Historica 33*, 135–173.

Rosemeyer, M. and A. Enrique-Arias (2016). A match made in heaven: Using parallel corpora and multinomial logistic regression to analyze the expression of possession in Old Spanish. *Language Variation and Change 28*(3), 307–334.

Rosemeyer, M. and E. Grossman (2021). Why don’t grammaticalization pathways always recur? *Corpus Linguistics and Linguistic Theory 17*(3), 653–681.

Saeed, J. I. (1982). The syntactic status of quantifiers in Somali. *Bulletin of SOAS 45*(3), 525–545.

Sankoff, G. (2019). Transmission revisited. In D. W. Lightfoot and J. Havenhill (Eds.), *Variable properties in language: Their nature and acquisition*, pp. 155–166. Washington, DC: Georgetown University Press.

Sankoff, G. and S. E. Wagner (2020). The long tail of language change: A trend and panel study of Québécois French futures. *Canadian Journal of Linguistics/Revue canadienne de linguistique 65*(2), 246–275.

Sasse, H.-J. (1987). The thetic/categorical distinction revisited. *Linguistics 25*(3), 511–580.

Sauerland, U. (2005). Don't interpret focus! Why a presuppositional account of focus fails and how a presuppositional account of givenness works. In E. Maier, C. Bary, and J. Huitink (Eds.), *Proceedings of Sinn und Bedeutung 9*, pp. 370–384. Nijmegen: Nijmegen Centre for Semantics.

Sauppe, S., Å. Næss, G. Roversi, M. Meyer, I. Bornkessel-Schlesewsky, and B. Bickel (2023). An agent-first preference in a patient-first language during sentence comprehension. *Cognitive Science 47*(9), e13340.

Schafer, R. (1995). Negation and verb second in Breton. *Natural Language and Linguistic Theory 13*(1), 135–172.

Schmidtke-Bode, K. (2019). Attractor states and diachronic change in Hawkins's "Processing Typology". In K. Schmidtke-Bode, N. Levshina, S. M. Michaelis, and I. Seržant (Eds.), *Explanation in typology: Diachronic sources, functional motivations and the nature of the evidence*, pp. 123–148. Berlin: Language Science Press.

Schnell, S. and N. N. Schiborr (2022). Crosslinguistic corpus studies in linguistic typology. *Annual Review of Linguistics 8*(1), 171–191.

Schwenter, S. A. (2005). The pragmatics of negation in Brazilian Portuguese. *Lingua 115*(10), 1427–1456.

Scontras, G. and L. S. Pearl (2021). When pragmatics matters more for truth-value judgments: An investigation of quantifier scope ambiguity. *Glossa: a journal of general linguistics 6*(1), 110.

Sells, P. and S. Kim (2006). Korean NPIs scope over negation. *Language Research 42*(2), 275–297.

Seržant, I. A. (2019). Weak universal forces: The discriminatory function of case in differential object marking systems. In K. Schmidtke-Bode, N. Levshina, S. M. Michaelis, and I. Seržant (Eds.), *Explanation in typology: Diachronic sources, functional motivations and the nature of the evidence*, pp. 149–178. Berlin: Language Science Press.

Seržant, I. A. and G. Moroz (2022). Universal attractors in language evolution provide evidence for the kinds of efficiency pressures involved. *Humanities and Social Sciences Communications 9*(1), 1–9.

Silvennoinen, O. O. (2017). Not only apples but also oranges: Contrastive negation and register. *Studies in Variation, Contacts and Change in English 19*.

Sjöberg, A. (2023). *Knowledge predication: A semantic typology.* Ph. D. thesis, Stockholm University, Stockholm.

Skirgård, H., H. J. Haynie, H. Hammarström, D. E. Blasi, J. Collins, J. Latarche, J. Lesage, T. Weber, A. Witzlack-Makarevich, M. Dunn, G. Reesink, R. Singer, C. Bowern, P. Epps, J. Hill, O. Vesakoski, N. K. Abbas, S. Ananth, D. Auer, N. A. Bakker, G. Barbos, A. Bolls, R. D. Borges, M. Browen, L. Chevallier, S. Danielsen, S. Dohlen, L. Dorenbusch, E. Dorn, M. Duhamel, F. E. H. Ali, J. Elliott, G. Falcone, A.-M. Fehn, J. Fischer, Y. G. Ate, H. Gibson, H.-P. Göbel, J. A. Goodall, V. Gruner, A. Harvey, R. Hayes, L. Heer, R. E. H. Miranda, N. Hübler, B. H. Huntington-Rainey, G. Inglese, J. K. Ivani, M. Johns, E. Just, I. Kapitonov, E. Kashima, C. Kipf, J. V. Klingenberg, N. König, A. Koti, R. G. A. Kowalik, O. Krasnoukhova, K. L. Lindsey, N. L. M. Lindvall, M. Lorenzen, H. Lutzenberger, A. Marley, T. R. A. Martins, C. M. German, S. van der Meer, J. Montoya, M. Müller, S. Muradoğlu, HunterGatherer, D. Nash, K. Neely, J. Nickel, M. Norvik, B. Olsson, C. A. Oluoch, D. Osgarby, J. Peacock, I. O. Pearey, N. Peck, J. Peter, S. Petit, S. Pieper, M. Poblete, D. Prestipino, L. Raabe, A. Raja, J. Reimringer, S. C. Rey, J. Rizaew, E. Ruppert, K. K. Salmon, J. Sammet, R. Schembri, L. Schlabbach, F. W. P. Schmidt, D. Schokkin, J. Siegel, A. Skilton, H. de Sousa, K. Sverredal, D. Valle, J. Vera, J. Voß, D. W. Smith, T. Witte, H. Wu, S. Yam, J. Y. , M. Yong, T. Yuditha, R. Zariquiey, R. Forkel, N. Evans, S. C. Levinson, M. Haspelmath, S. J. Greenhill, Q. D. Atkinson, and R. D. Gray (2023, March). Grambank v1.0. Dataset.

Song, Y., A. H. Jimenez, and G. Scontras (2021). Cross-linguistic scope ambiguity: An investigation of English, Spanish, and Mandarin. In P. Farrell (Ed.), *Proceedings of the Linguistic Society of America (LSA) 2021*, Volume 6, pp. 572–586. Linguistic Society of America.

Sportiche, D. (1988). A theory of floating quantifiers and its corollaries for constituent structure. *Linguistic Inquiry 19*(3), 425–449.

Squartini, M. (2017). Italian non-canonical negations as modal particles: Information state, polarity and mirativity. In C. Fedriani and A. Sansò (Eds.), *Pragmatic markers, discourse markers and modal particles*, pp. 203–228. John Benjamins Publishing Company.

Stave, M., L. Paschen, F. Pellegrino, and F. Seifart (2021). Optimization of morpheme length: A cross-linguistic assessment of Zipf's and Menzerath's laws. *Linguistics Vanguard 7*(s3), 20190076.

Stenson, N. (2019). *Modern Irish: A comprehensive grammar.* Oxford: Routledge.

Straka, M. and J. Straková (2017). Tokenizing, POS tagging, lemmatizing and parsing UD 2.0 with UDPipe. In *Proceedings of the CoNLL 2017 shared task: Multilingual parsing from raw text to universal dependencies*, pp. 88–99. Vancouver: Association for Computational Linguistics.

Syrett, K., G. Simon, and K. Nisula (2014). Prosodic disambiguation of scopally ambiguous quantificational sentences in a discourse context. *Journal of Linguistics 50*(2), 453–493.

Szabolcsi, A. (1997). Strategies for scope taking. In A. Szabolcsi (Ed.), *Ways of scope taking*, pp. 109–154. Springer.

Szabolcsi, A. (2004). Positive polarity–negative polarity. *Natural Language and Linguistic Theory 22*(2), 409–452.

Taglicht, J. (1985). Some uses of *all* or: The king's horses and other curiosities or: All is not well in our grammar. Unpublished manuscript, The Hebrew University of Jerusalem, Jerusalem.

Tamm, A. (2015). Negation in Estonian. In M. Miestamo, A. Tamm, and B. Wagner-Nagy (Eds.), *Negation in Uralic languages*, pp. 399–432. Amsterdam: John Benjamins Publishing Company.

Thayer, J. H. (1889). *A Greek-English lexicon of the New Testament.* New York: American Book Company.

Thomason, S. G. (2001). *Language contact.* Edinburgh: Edinburgh University Press.

Thompson, H.-R. (2012). *Bengali.* Amsterdam: John Benjamins Publishing Company.

Thrainsson, H. (2007). *The syntax of Icelandic.* Cambridge: Cambridge University Press.

Tierney, M. C. (2018). A syntactic analysis of negation in Latin: Negated quantifier phrases. Senior honors thesis, New York University, New York, NY.

Tilleson, P. (2019). *On bipartite negation.* Ph. D. thesis, University of Minnesota, Minneapolis, Minnesota.

Tobler, A. (1902). *Vermischte Beiträge zur Französischen Grammatik*, Chapter Tout ce qui reluit n'est pas or, pp. 190–196. Leipzig: S. Hirzel.

Tonhauser, J. (2006). *The temporal semantics of noun phrases: Evidence from Guaraní.* Ph. D. thesis, Stanford University, Stanford, CA.

Toosarvandani, M. (2013). Corrective *but* coordinates clauses not always but sometimes. *Natural Language and Linguistic Theory 31*, 827–863.

Tottie, G. and A. Neukom-Hermann (2010). Quantifier-negation interaction in English: A corpus linguistic study of *all... not* constructions. In L. R. Horn (Ed.), *The expression of negation*, pp. 149–185. Berlin: De Gruyter Mouton.

Tunstall, S. L. (1998). *The interpretation of quantifiers: Semantics and processing.* Ph. D. thesis, University of Massachusetts Amherst, Amherst, MA.

Turano, G. (2000). On clitics and negation in Albanian. *Rivista di grammatica generativa 25*, 81–117.

Uegaki, W. (2023). The informativeness/complexity trade-off in the domain of boolean connectives. *Linguistic Inquiry 55*(1), 174–196.

Ura, H. (1994). Varieties of raising and the feature-based theory of movement. *MIT Occasional Papers in Linguistics 7.*

van der Wal, J. and J. Maniacky (2015). How 'person' got into focus: Grammaticalization of clefts in Lingala and Kikongo areas. *Linguistics 53*(1), 1–52.

van Gelderen, E. (2011). *The linguistic cycle: Language change and the language faculty.* Oxford University Press.

van Kampen, J. (2023). Everything is not equal in adult and child Dutch: The scope of universal quantifiers with negation. *Linguistics in the Netherlands 40*(1), 120–136.

van Kemenade, A. (1997). Negative-initial sentences in Old and Middle English. *Studia Anglica Posnaniensia 31*, 91–104.

Viau, J., J. Lidz, and J. Musolino (2010). Priming of abstract logical representations in 4-year-olds. *Language Acquisition 17*(1-2), 26–50.

Vilkuna, M. (2015). Negation in Finnish. In M. Miestamo, A. Tamm, and B. Wagner-Nagy (Eds.), *Negation in Uralic languages*, pp. 457–485. Amsterdam: John Benjamins Publishing Company.

Vincent, N. (2017). Determination and quantification. In A. Dufter and E. Stark (Eds.), *Manual of Romance Morphosyntax and Syntax*, pp. 727–770. De Gruyter Mouton.

von Fintel, K. (1992). The modal-existential construction. Unpublished manuscript, University of Massachusetts Amherst, Amherst, MA.

von Fintel, K. (1999). NPI licensing, Strawson entailment, and context dependency. *Journal of Semantics 16*(2), 97–148.

Wagner, M. (2008). A compositional analysis of contrastive topics. In A. Schardl, M. Walkow, and M. Abdurrahman (Eds.), *Proceedings of North East Linguistic Society (NELS) 38*, pp. 415–428. Amherst, MA: GLSA.

Wälchli, B. (2009). Data reduction typology and the bimodal distribution bias. *Linguistic Typology 13*(1), 77–94.

Wälchli, B. (2014). Algorithmic typology and going from known to similar unknown categories within and across languages. In B. Szmrecsanyi and B. Wälchli (Eds.), *Aggregating dialectology, typology, and register analysis: Linguistic variation in text and speech*, pp. 355–393. De Gruyter Moutonr.

Wälchli, B. and M. Cysouw (2012). Lexical typology through similarity semantics: Toward a semantic map of motion verbs. *Linguistics 50*(3), 671–710.

Wälchli, B. and A. Sjöberg (2024). A law of meaning. *Linguistic Typology at the Crossroads 4*(2), 1–71.

Wallage, P. (2008). Jespersen's cycle in Middle English: Parametric variation and grammatical competition. *Lingua 118*(5), 643–674.

Wallage, P. (2012). Negative inversion, negative concord and sentential negation in the history of english. *English Language and Linguistics 16*(1), 3–33.

Wallis, S. (2020). *Statistics in corpus linguistics research: A new approach.* Oxford: Routledge.

Weinreich, U., W. Labov, and M. Herzog (1968). Empirical foundations for a theory of language change. In W. Lehamn and Y. Malkiel (Eds.), *Directions for historical linguistics*, pp. 95–195. Austin, TX: University of Texas Press.

Wiedenhof, J. (2015). *A grammar of Mandarin.* Amsterdam: John Benjamins Publishing Company.

Wijffels, J., M. Straka, and J. Straková (2023). *UDPipe: Tokenization, parts of speech tagging, lemmatization and dependency parsing with the UDPipe NLP toolkit.* Prague: Charles University. R package version 0.8.11.

Willis, D., C. Lucas, and A. Breitbarth (2013). Comparing diachronies of negation. In *The history of negation in the languages of Europe and the Mediterranean, Volume I: Case studies*, Volume 5, pp. 1–50. Oxford: Oxford University Press.

Wolfe, S. (2018). *Verb second in medieval Romance.* Oxford: Oxford University Press.

Wu, D. (2021). Syntax of negation in corrective *but* sentences. In Ö. Bakay, B. Pratley, E. Neu, and P. Deal (Eds.), *Proceedings of North East Linguistic Society (NELS) 52.* Amherst, MA: GLSA.

Wurmbrand, S. (2008). Word order and scope in German. *GAGL: Groninger Arbeiten zur germanistischen Linguistik 46*, 89–110.

Wurmbrand, S. (2018). The cost of raising quantifiers. *Glossa: a journal of general linguistics 3*(1), 19.

Yatsushiro, K., A. Sugawara, and U. Sauerland (2019). Quantifier scope and intonation in German. In M. M. Brown and B. Dailey (Eds.), *Proceedings of Annual Boston University Conference on Language Development (BUCLD) 43*, pp. 730–743. Cascadilla Press.

Yoshimura, A. (2013). Descriptive/metalinguistic dichotomy?: Toward a new taxonomy of negation. *Journal of Pragmatics 57*, 39–56.

Zanuttini, R. (1997). *Negation and clausal structure a comparative study of Romance languages*. Oxford: Oxford University Press.

Zeijlstra, H. (2004). *Sentential negation and negative concord*. Netherlands Graduate School of Linguistics.

Zeijlstra, H. (2017). Universal quantifier PPIs. *Glossa: a journal of general linguistics 2*(1), 91.

Zeijlstra, H. (2022). *Negation and negative dependencies*. Oxford: Oxford University Press.

Zhan, F. and C. Sun (2013). A copula analysis of *shì* in the Chinese cleft construction. *Language and Linguistics 14*(4), 755–789.

Züfle, M. and R. Katzir (2022). Reasoning about stored representations in semantics using the typology of lexicalized quantifiers. In D. Gutzmann and S. Repp (Eds.), *Proceedings of Sinn und Bedeutung 26*, pp. 923–944.

תקציר

עבודת הדוקטור עוסקת בשונות הבין-לשונית בשימוש במבני טווח הפוך כגון *All that glitters is not gold*. במבנים אלה שלילת הפסוקית לוקחת טווח רחב על כמת כולל המופיע לפניה במשפט. מבני "כל... לא" אלו עומדים בניגוד למבני טווח שקוף, כגון *Not all that glitters is gold*, המביעים את יחסי הטווח דרך מבנה השטח.

לצורך הבחנה בין צורה למשמעות, העבודה מאמצת את המוסכמות הבאות. מבנה "כל... לא" הוא תבנית מבנית שבה כמת כולל בעמדת נושא מופיע לפני סמן השלילה בתוך אותה פסוקית, זאת ללא קשר לפירוש הסמנטי של מבנה זה בשפה או בהקשר נתונים. לעומת זאת, טענות 'לא כל ה-X הם Y' הן טענות שבהן השלילה לוקחת טווח רחב על כמת כולל, זאת במנותק מהאופן שבו המשמעות הזאת מובעת מבחינה תחבירית.

לאור העובדה שטווח שקוף מסייע לעיבוד ולרכישת שפה, וכן כי עקרונות של חסכוניות תחבירית מנבאים העדפה למזעור אי התאמות בין צורה לוגית (LF) לצורה פונטית (PF), הציפייה היא שתהיה העדפה אוניברסלית לטווח שקוף.

אולם, האם קיימות ראיות בין-לשוניות לכך ששפות אכן מעדיפות טווח שקוף? ניתן לצפות שאם קיימת הטיה כזו, שפות יעדיפו באופן גורף מבנים בעלי טווח שקוף וישתמשו במבני טווח הפוך רק כמוצא אחרון. עם זאת, מחקר זה מראה כי התמונה הבין-לשונית מורכבת לאין שיעור: מבני טווח הפוך אינם רק נפוצים, אלא שפות רבות אף מעדיפות אותם על פני מבנים חלופיים בעלי טווח שקוף.

כדי ליישב את הסתירה לכאורה בין ההשערה בדבר הטיה אוניברסלית לטווח שקוף לבין השכיחות של מבני טווח הפוך, מחקר זה בוחן אם ההעדפה לטווח שקוף פועלת ככוח אוניברסלי חלש—כוח המשפיע באופן שיטתי על שינוי שפה לאורך זמן אך אינו חזק מספיק כדי להתגבר על לחצים מנוגדים ברמה הסינכרונית.

ההשערה המרכזית של מחקר זה היא כדלקמן:

(1) **השערת הטווח השקוף**

מבנה השפה מושפע מהטיה חלשה להבעת יחסי הטווח באופן שקוף.

לאור ההשפעה החלשה הצפויה של ההטיה לטווח שקוף, בדיקת השערת הטווח השקוף מצריכה מדגם שפות רחב ומגוון. מכיוון שמבני טווח הפוך מתועדים רק לעתים רחוקות בספרי דקדוק, מחקר זה מסתמך על איסוף נתונים בין-לשוניים ראשוניים. למטרה זו, ננקטת גישה מבוססת קורפוס, המנתחת תרגומים של הברית החדשה ב-217 שפות ממשפחות שפות ומאזורים לשוניים מגוונים. בניגוד למחקרים קודמים, שחקרו תופעות טווח בעיקר דרך שיפוטי דקדוקיות, גישה זו נעזרת בקורפוס מקביל גדול כדי לבחון באופן שיטתי את התפוצה של מבני טווח הפוך ושקוף על פני מגוון שפות. כדי להבטיח את תוקפם של הממצאים, נערכו ניתוחים סטטיסטיים תוך בקרה על הטיות גנאלוגיות ואזוריות.

מאגר הנתונים נבנה על בסיס תרגומים של הברית החדשה ממקורות רבים, תוך מתן דגש על איסוף תרגומים היסטוריים, כולל ספרים סרוקים וכתבי יד. טקסטים היסטוריים היו זמינים עבור 21 שפות גרמאניות ורומאניות, והללו פותחים צוהר להתפתחויות דיאכרוניות באינטראקציה בין שלילה לכמת כולל.

סיווג המבנים השונים מתבסס על תכונותיהם התחביריות הנראות לעין, תוך הבחנה בין חמש אסטרטגיות מבניות להבעת 'לא כל ה-X הם Y'. כדי להבטיח שהמבנים שזוהו באמת משקפים דפוסים דקדוקיים הקיימים בשפת היעד ולא תוצרי תרגום, ניתוח הנתונים נותן עדיפות למקרים שבהם המתרגמים חורגים מטקסט המקור וכן מאמת את הממצאים על ידי בחינת מספר תרגומים נפרדים לכל שפה.

במטרה לבחון את השערת הטווח השקוף, מחקר זה מנסח שתי השערות ספציפיות יותר הניתנות לבדיקה אמפירית, כמתואר ב-(2) וב-(3). ההשערה הסינכרונית (2) גורסת כי על אף שמבני טווח הפוך נפוצים בשפות העולם, התפוצה שלהן מוגבלת חלקית על ידי ההטיה לטווח שקוף. כתוצאה מכך, הם מופיעים בשכיחות נמוכה יותר בשפות העולם ממה שניתן היה לצפות תחת השערת האפס של היעדר הטיה כזו.

כדי לבדוק את הניבוי הזה, יש לקבוע בסיס השוואה שאליו ניתן להשוות את השכיחות הנצפית של מבני טווח הפוך בשפות העולם. לשם כך, עלינו ראשית לשאול מדוע שפות משתמשות במבני טווח הפוך מלכתחילה. אני טוען ששכיחותם הבין-לשונית נובעת בראש ובראשונה מהעובדה שברוב המוחלט של השפות, שימוש במבנה השלילה הסטנדרטית (המבנה הבסיסי ששפה משתמשת בו כדי לשלול פרדיקטים פועליים) מוביל לטווח הפוך. כדי להימנע מטווח הפוך, רוב השפות צריכות להשתמש במבנה שלילה אחר או לאפשר סדר מילים חלופי שהופך את יחסי הטווח לשקופים.

קו מחשבה זה מוביל לתחזית הבאה:

(2) **השערה סינכרונית**

מבני טווח הפוך מופיעים בשכיחות נמוכה יותר מבחינה בין-לשונית ממה שניתן היה לצפות בהתבסס על מיקומם היחסי של הנושא וסמן שלילת הפסוקית בשפות העולם.

תחת השערת האפס של היעדר הטיה לטווח שקוף, משפטי 'לא כל ה-X הם Y' אמורים להיות דומים למשפטי שלילה אחרים שבהם הנושא אינו אלמנט נושא טווח. כלומר, אם מבנה השלילה הסטנדרטית של שפה נתונה יוצר סדר מילים מסוים במשפטים שליליים רגילים—כגון "העוגיות לא נשרפו"—היינו מצפים שאותו מבנה ישמש גם כאשר הנושא הוא כמת כולל, וכך היה נוצר מבנה טווח הפוך: "כל העוגיות לא נשרפו". לעומת זאת, אם קיימת הטיה לטווח שקוף, שפות אמורות להפגין נטייה לחרוג מדפוס השלילה הסטנדרטית ולהשתמש באסטרטגיות חלופיות—כגון שלילת רכיב, למשל "לא כל העוגיות נשרפו"—זאת במטרה להביע באופן שקוף את יחסי הטווח במבנה השטח.

הממצאים מצביעים על כך ששפות רבות נמנעות באופן שיטתי משימוש במבני טווח הפוך, אפילו במחיר סטייה מהדפוסים המבניים הרגילים. ממצאים אלו תומכים בהשערה הסינכרונית. שפות מסוימות מפתחות מבני שלילה יעודיים שתפקידם העיקרי הוא לאייך כמת כולל בעמדת נושא. לעומת זאת, שפות אחרות מסבות מבנים קיימים לצורך שמירה על טווח שקוף—לרוב מבנים שתפקידם העיקרי קשור למבנה המסר.

התוצאות הכמותיות מספקות תמיכה חלקית להשערה הסינכרונית. במטרה לשלוט בהשפעות גנאלוגיות ואזוריות, הניתוח בוצע בנפרד עבור כל מקרו-אזור, תוך הבטחת עצמאות גנאלוגית. באפריקה ובאירואסיה, מבני טווח הפוך שכיחים פחות מהצפוי באופן מובהק ביחס לדפוסי השלילה הסטנדרטית. הדבר מצביע על כך שההטיה לטווח שקוף משפיעה על התפוצה הנצפית. עם זאת, לא נמצאה השפעה מובהקת עבור צפון אמריקה, דרום אמריקה או פפונזיה, ככל הנראה בגלל מגבלות גודל המדגם. ניתן להסיק מכך שעל אף שההעדפה לטווח שקוף ממלאה תפקיד בעיצוב מבנה השפה, השפעתה חלשה יחסית.

בנוסף, המחקר מעלה את ההשערה הדיאכרונית הבאה:

(3) **השערה דיאכרונית**

א. בשפה נתונה, השכיחות היחסית של מבני טווח שקוף נוטה לעלות עם הזמן.

ב. לעומת זאת, מבני טווח הפוך נעשים שכיחים יותר רק כאשר אסטרטגיית הטווח השקוף השלטת יוצאת משימוש עקב שינויים דקדוקיים כלליים יותר, והדבר דוחף את השפה להסתמך על מבנה השלילה הסטנדרטית.

ההשערה הדיאכרונית נתמכת באופן מובהק על ידי נתונים היסטוריים משפות גרמאניות ורומאניות. המחקר מזהה תהליך מחזורי שבו מבני טווח הפוך דועכים בעקבות התפתחות מבנים חלופיים בעלי טווח שקוף:

(4) **מסלול דעיכה של מבני טווח הפוך**

שלב א': מבני טווח הפוך שכיחים ולעיתים אף מהווים את האסטרטגיה השלטת להבעת 'לא כל ה-X הם Y'.

שלב ב': השפה מפתחת אסטרטגיית טווח שקוף חדשה. עם הזמן, שכיחות המבנה החדש עולה, בעוד ששכיחות מבנה הטווח ההפוך הישן יורדת.

שלב ג': מבנה הטווח ההפוך יוצא מכלל שימוש או נעשה נדיר ומוגבל בעיקר לביטויים שגורים.

לעומת זאת, מבני טווח הפוך נוצרים בעיקר כאשר אסטרטגיית הטווח השקוף השלטת יוצאת משימוש עקב שינויים דקדוקיים שאינם ספציפיים למשפטי 'לא כל ה-X הם Y', כגון שינויים בסדר המילים או גרמטיקליזציה של כמת כולל חדש:

(5) **מסלול הופעת מבני טווח הפוך**

שלב א': מבני טווח הפוך נדירים או אינם נמצאים בשימוש.

שלב ב': אסטרטגיית הטווח השקוף השלטת יוצאת משימוש עקב שינויים דקדוקיים כלליים יותר.

שלב ג': השפה נאלצת להסתמך על מבנה השלילה הסטנדרטית, והדבר לרוב מוביל לטווח הפוך.

אחד הממצאים המרכזיים של מחקר זה הוא שהשונות הבין-לשונית בזמינות של פירוש טווח הפוך אינה ניתנת לחיזוי באופן מלא בהתבסס על המאפיינים הדקדוקיים של השפה או על מצאי אסטרטגיות הטווח השקוף שלה. במקום זאת, כל שפה נמצאת במצב סינכרוני המשקף את המסלול הדיאכרוני הייחודי לה. חשוב לציין שהעיתוי של שינויים דקדוקיים ממלא תפקיד מכריע ביצירת התפוצה הסינכרונית דרך אפקט ההתמדה, שלפיו מבנים ישנים יותר נשארים בשימוש גם לאחר שצצות חלופות חדשות.

ממצאים אלו תורמים להבנתנו כיצד אילוצים תחביריים רכים (ברי הפרה) משפיעים על מבנה השפה. בעוד שאפקט ההתמדה עשוי לטשטש את ההשפעה הסינכרונית של אילוצים אלה, השפעתם מתגלה באסימטריות בתפוצה הבין-לשונית ובמסלולים דיאכרוניים חוזרים. באמצעות שילוב של נקודות מבט בין-לשוניות ודיאכרוניות, מחקר זה מדגים שהטיות אוניברסליות יכולות לעצב את התפתחות השפה בטווח הארוך מבלי לייצר בהכרח תוצאות נראות לעין בכל שפה כשלעצמה.

מעבר לכך, מחקר זה שופך אור על אופיה של תחרות דיאכרונית בממשק בין התחביר לסמנטיקה ומראה שהיא יכולה להתרחש גם בין מבנים מורכבים הנבדלים באופן ניכר זה מזה במבנה התחבירי שלהם. בעוד שרוב המודלים של תחרות סינכרונית המבוססים על אילוצים מניחים שקבוצת המועמדים מוגבלת ל-PF-ים שנגזרו מאותו LF, הראיות המוצגות כאן מצביעות על כך שתחרות דיאכרונית אינה כרוכה במגבלה זו. במקום זאת, מבנים המבטאים את אותה משמעות יכולים לחסום זה את זה במרוצת הזמן, אפילו אם הם נגזרים מ-LF-ים שונים.

לסיכום, עבודת הדוקטור מספקת ראיות אמפיריות לכך שההעדפה לטווח שקוף פועלת ככוח אוניברסלי חלש המשפיע באופן מתון הן על תפוצות סינכרוניות והן על שינוי דיאכרוני, אם כי אפקט ההתמדה ושינויים מחזוריים מחלישים את ההשפעות האלה.

עבודת הדוקטור מורכבת מחמישה חלקים. חלק א' מניח את היסודות התיאורטיים ומתווה את ההשערות העיקריות (פרקים 1-3). חלק ב' מציג את הגישה המתודולוגית, המבוססת על קורפוס מקביל של תרגומי הברית החדשה, ובוחן את מהימנותה (פרקים 4-6). חלק ג' עוסק בהשערה הסינכרונית תוך מיפוי האסטרטגיות התחביריות הקיימות בשפות שונות, ניתוח כמותי של תפוצתן ובחינת מסלולי התפתחות של מבני טווח שקוף (פרקים 7-9). חלק ד' מתמקד בהשערה הדיאכרונית, מתאר שינויים היסטוריים בשפות הגרמאניות והרומאניות ומזהה מסלולים מחזוריים בהתפתחות של אסטרטגיות טווח הפוך ושקוף (פרקים 10-11). לבסוף, חלק ה' מסכם את הממצאים הללו ומדגיש את החשיבות של שילוב בין נקודות מבט בין-לשוניות ודיאכרוניות בחקר הטיות דקדוקיות אוניברסליות (פרק 12).

עבודה זו נעשתה בהדרכתה של פרופ' נורה בונה.

טווח הפוך במבני "כל...לא":
חקירה בין-לשונית ודיאכרונית

חיבור לשם קבלת תואר דוקטור לפילוסופיה
מאת

עמרי עמירז

THE HEBREW
UNIVERSITY
OF JERUSALEM

הוגש לסנט האוניברסיטה העברית בירושלים

מרץ 2025